

TABLE BXII.γ.193. (cont.)

Characteristic	<i>Providencia rustigianii</i>	<i>Providencia stuartii</i>	<i>Rahnella aquatilis</i>	<i>Salmonella bongori</i>	<i>Salmonella enterica</i> subsp. <i>enterica</i> ^c	<i>Salmonella enterica</i> subsp. <i>arizonae</i> ^c	<i>Salmonella enterica</i> subsp. <i>diarizonae</i> ^c	<i>Salmonella enterica</i> subsp. <i>houstenae</i> ^c	<i>Salmonella enterica</i> subsp. <i>indica</i> ^c	<i>Salmonella enterica</i> subsp. <i>salamae</i> ^c	<i>Salmonella serovar</i> <i>Choleraesuis</i> ^c	<i>Salmonella serovar</i> <i>Gallinarum</i> ^c	<i>Salmonella serovar</i> <i>Paratyphi A</i> ^c	<i>Salmonella serovar</i> <i>Pullorum</i> ^c	<i>Salmonella serovar</i> <i>Typhi</i> ^c	<i>Serratia marcescens</i>
Indole production	98	98	0	0	1	1	2	0	0	2	0	0	0	0	0	1
Methyl red	65	100	88	100	100	100	100	100	100	100	100	100	100	90	100	20
Voges-Proskauer	0	0	100	0	0	0	0	0	0	0	0	0	0	0	0	98
Citrate (Simmons)	15	93	94	94	95	99	98	98	89	100	25	0	0	0	0	98
Hydrogen sulfide (TSI)	0	0	0	100	95	99	99	100	100	100	50	100	10	90	97	0
Urea hydrolysis	0	30	0	0	1	0	0	2	0	0	0	0	0	0	0	15
Phenylalanine deaminase	100	95	95	0	0	0	0	0	0	0	0	0	0	0	0	0
Lysine decarboxylase	0	0	0	100	98	99	99	100	100	100	95	90	0	100	98	99
Arginine dihydrolase	0	0	0	94	70	70	70	70	67	90	55	10	15	10	3	0
Ornithine decarboxylase	0	0	0	100	97	99	99	100	100	100	100	1	95	95	0	99
Motility	30	85	6	100	95	99	99	98	100	98	95	0	95	0	97	97
Gelatin hydrolysis (22°C)	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	90
Growth in KCN	100	100	0	100	0	1	1	95	0	0	0	0	0	0	0	95
Malonate utilization	0	0	100	0	0	95	95	0	0	95	0	0	0	0	0	3
Esculin hydrolysis	0	0	100	0	5	1	1	0	0	15	0	0	0	0	0	95
Tartrate, Jordan's	50	90	6	0	90	5	20	65	100	50	85	100	0	0	100	75
Acetate utilization	25	75	6	100	90	90	75	70	89	95	1	0	0	0	0	50
Lipase (corn oil)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	98
DNase (25°C)	0	10	0	0	2	2	2	0	0	0	0	10	0	0	0	98
Nitrate oxidized to nitrite	100	100	100	100	100	100	100	100	100	100	98	100	100	100	100	98
Oxidase, Kovacs	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ONPG test	0	10	100	94	2	100	92	0	44	15	0	0	0	0	0	95
Yellow pigment	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
D-Glucose, acid	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
D-Glucose, gas	35	0	98	94	96	99	99	100	100	100	95	0	99	90	0	55
<i>Fermentation of:</i>																
Adonitol	0	5	0	0	0	0	0	5	0	0	0	0	0	0	0	40
L-Arabinose	0	1	100	94	99	99	99	100	100	100	0	80	100	100	2	0
D-Arabitol	0	0	0	0	0	1	1	5	0	0	1	0	0	0	0	0
Cellobiose	0	5	100	0	5	1	1	50	0	0	0	10	5	5	0	5
Dulcitol	0	0	88	94	96	0	1	0	67	90	5	90	90	0	0	0
Erythritol	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1
Glycerol	5	50	13	0	5	10	10	0	33	25	0	0	10	0	20	95
myo-Inositol	0	95	0	0	35	0	0	0	0	5	0	0	0	0	0	75
Lactose	0	2	100	0	1	15	85	0	22	1	0	0	0	0	1	2
Maltose	0	1	94	100	97	98	98	100	100	100	95	90	95	5	97	96
D-Mannitol	0	10	100	100	100	100	100	98	100	100	98	100	100	100	100	99
D-Mannose	100	100	100	100	100	100	100	100	100	95	95	100	100	100	100	99
Melibiose	0	0	100	94	95	95	95	100	89	8	45	0	95	0	100	0
α-Methyl-D-glucoside	0	0	0	0	2	1	1	0	0	8	0	0	0	0	0	0
Mucate	0	0	30	88	90	90	30	0	89	96	0	50	0	0	0	0
Raffinose	0	7	94	0	2	1	1	0	0	0	1	10	0	1	0	2
L-Rhamnose	0	0	94	88	95	99	99	98	100	100	100	10	100	100	0	0
Salicin	0	2	100	0	0	0	0	60	0	5	0	0	0	0	0	95
D-Sorbitol	0	1	94	100	95	99	99	100	0	100	90	1	95	10	99	99
Sucrose	35	50	100	0	1	1	5	0	0	1	0	0	0	0	0	99
Trehalose	0	98	100	100	99	99	99	100	100	100	0	50	100	90	100	99
D-Xylose	0	7	94	100	97	100	100	100	100	100	98	70	0	90	82	7

^aEach number is the percentage of positive reactions after 2 d at 36°C unless otherwise indicated (gelatin liquefaction and deoxyribonuclease; *Photobacterium luminescens*).

^bTable taken from Farmer, 1999.

^cIn a Request for an Opinion published in 1987, Le Minor and Popoff proposed replacement of the type species of *Salmonella* (*Salmonella choleraesuis* subsp. *choleraesuis*) with *Salmonella enterica* as the former was considered to be a source of confusion. Although the Request was denied by the Judicial Commission, their proposal resulted in an alternative naming convention which has found widespread endorsement in the public health community. This matter was revisited in July 2002 by the Judicial Commission during the IUMS Congress in response to several new Requests for an Opinion and will likely result in a decision to replace the type strain *Salmonella choleraesuis* subsp. *choleraesuis* with *Salmonella enterica* subsp. *enterica* LT2, while preserving the former rather than placing it on the list of rejected names. We view the six subspecies of *Salmonella choleraesuis* as deprecated. Readers are also advised that the names *Salmonella enteritidis*, *Salmonella paratyphi*, *Salmonella typhi*, and *Salmonella typhimurium* are synonyms of *Salmonella enterica* subsp. *enterica* and refer to specific serovars. These names have not been deprecated at this time as they remain in use by some public health reporting agencies.

(continued)

TABLE BXII.γ.193. (cont.)

Characteristic	<i>Serratia entomophila</i>	<i>Serratia ficaria</i>	<i>Serratia fonticola</i>	<i>Serratia liquefaciens</i>	<i>Serratia marcescens</i> biogroup 1	<i>Serratia odorifera</i> biogroup 1	<i>Serratia odorifera</i> biogroup 2	<i>Serratia plymuthica</i>	<i>Serratia rubidaea</i>	<i>Shigella dysenteriae</i>	<i>Shigella boydii</i>	<i>Shigella flexneri</i>	<i>Shigella sonnei</i>	<i>Tatumella physcos</i>	<i>Trabulsietia guamensis</i>	<i>Xenorhabdus nematophilus</i>
Indole production	0	0	0	1	0	60	50	0	0	45	25	50	0	0	40	40
Methyl red	20	75	100	93	100	100	60	94	20	99	100	100	100	0	100	0
Voges-Proskauer	100	75	9	93	60	50	100	80	100	0	0	0	0	5	0	0
Citrate (Simmons)	100	100	91	90	30	100	97	75	95	0	0	0	0	2	88	0
Hydrogen sulfide (TSI)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	100	0
Urea hydrolysis	0	0	13	3	0	5	0	0	2	0	0	0	0	0	0	0
Phenylalanine deaminase	0	0	0	0	0	0	0	0	0	0	0	0	0	90	0	0
Lysine decarboxylase	0	0	100	95	55	100	94	0	55	0	0	0	0	0	100	0
Arginine dihydrolase	0	0	0	0	4	0	0	0	0	2	18	5	2	0	50	0
Ornithine decarboxylase	0	0	97	95	65	100	0	0	0	0	2	0	98	0	100	0
Motility	100	100	91	95	17	100	100	50	85	0	0	0	0	0	100	100
Gelatin hydrolysis (22°C)	100	100	0	90	30	95	94	60	90	0	0	0	0	0	0	80
Growth in KCN	100	55	70	90	70	60	19	30	25	0	0	100	0	0	100	0
Malonate utilization	0	0	88	2	0	0	0	0	94	100	0	0	0	0	0	0
Esculin hydrolysis	100	100	100	97	96	95	40	81	94	0	0	0	0	0	40	0
Tartrate, Jordan's	100	17	58	75	50	100	100	100	70	75	50	30	90	0	50	60
Acetate utilization	80	40	15	40	4	60	65	55	80	0	0	8	0	0	88	0
Lipase (corn oil)	20	77	0	85	75	35	65	70	99	0	0	0	0	0	0	0
DNase (25°C)	100	100	0	85	82	100	100	100	99	0	0	0	0	0	0	20
Nitrate oxidized to nitrite	100	92	100	100	83	100	100	100	100	99	100	99	100	98	100	20
Oxidase, Kovacs	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ONPG test	100	100	100	93	75	100	100	70	100	30	10	1	90	0	100	0
Yellow pigment	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	60
D-Glucose, acid	100	100	100	100	100	100	100	100	100	0	100	100	100	100	100	80
D-Glucose, gas	0	0	79	75	0	0	13	40	30	0	0	3	0	0	100	0
<i>Fermentation of:</i>																
Adonitol	0	0	100	5	30	50	55	0	99	0	0	0	0	0	0	0
L-Arabinose	0	100	100	98	0	100	100	100	100	45	94	60	95	0	100	0
D-Arabitol	60	100	100	0	0	0	0	0	85	0	0	1	0	0	0	0
Cellobiose	0	100	6	5	4	100	100	88	94	0	0	0	5	0	100	0
Dulcitol	0	0	91	0	0	0	0	0	0	0	5	1	0	0	0	0
Erythritol	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0
Glycerol	0	0	88	95	92	40	50	50	20	10	50	10	15	7	0	0
myo-Inositol	0	55	30	60	30	100	100	50	20	0	0	0	0	0	0	0
Lactose	0	15	97	10	4	70	97	80	100	0	1	1	2	0	0	0
Maltose	100	100	97	98	70	100	100	94	99	15	20	30	90	0	100	0
D-Mannitol	100	100	100	100	96	100	97	100	100	100	97	95	99	0	100	0
D-Mannose	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	80
Melibiose	0	40	98	75	0	100	96	93	99	0	15	55	25	25	0	0
α-Methyl-D-glucoside	0	8	91	5	0	0	0	70	1	0	0	0	0	0	0	0
Mucate	0	0	0	0	0	5	0	0	0	0	0	0	10	0	100	0
Raffinose	0	70	100	85	0	100	7	94	99	0	0	40	3	11	0	0
L-Rhamnose	0	35	76	15	0	95	94	0	1	30	1	5	75	0	100	0
Salicin	100	100	100	97	92	98	45	94	99	0	0	0	0	55	13	0
D-Sorbitol	0	100	100	95	92	100	100	65	1	30	43	29	2	0	100	0
Sucrose	100	100	21	98	100	100	0	100	99	1	0	1	1	98	0	0
Trehalose	100	100	100	100	100	100	100	100	100	90	85	65	100	93	100	0
D-Xylose	40	100	85	100	0	100	100	94	99	4	11	2	2	9	100	0

^aEach number is the percentage of positive reactions after 2 d at 36°C unless otherwise indicated (gelatin liquefaction and deoxyribonuclease; *Photorhabdus luminescens*).

^bTable taken from Farmer, 1999.

^cIn a Request for an Opinion published in 1987, Le Minor and Popoff proposed replacement of the type species of *Salmonella* (*Salmonella choleraesuis* subsp. *choleraesuis*) with *Salmonella enterica* as the former was considered to be a source of confusion. Although the Request was denied by the Judicial Commission, their proposal resulted in an alternative naming convention which has found widespread endorsement in the public health community. This matter was revisited in July 2002 by the Judicial Commission during the IUMS Congress in response to several new Requests for an Opinion and will likely result in a decision to replace the type strain *Salmonella choleraesuis* subsp. *choleraesuis* with *Salmonella enterica* subsp. *enterica* LT2, while preserving the former rather than placing it on the list of rejected names. We view the six subspecies of *Salmonella choleraesuis* as deprecated. Readers are also advised that the names *Salmonella enteritidis*, *Salmonella paratyphi*, *Salmonella typhi*, and *Salmonella typhimurium* are synonyms of *Salmonella enterica* subsp. *enterica* and refer to specific serovars. These names have not been deprecated at this time as they remain in use by some public health reporting agencies.

(continued)

TABLE BXII.γ.193. (cont.)

Characteristic	<i>Yersinia pestis</i>	<i>Yersinia aldovae</i>	<i>Yersinia bercovieri</i>	<i>Yersinia enterocolitica</i>	<i>Yersinia frederiksenii</i>	<i>Yersinia intermedia</i>	<i>Yersinia kristensenii</i>	<i>Yersinia mollaretii</i>	<i>Yersinia pseudotuberculosis</i>	<i>Yersinia rohdei</i>	<i>Yersinia ruckeri</i>	<i>Yokenella regensburgi</i> (<i>Koserella trabulsii</i>)	Enteric Group 58	Enteric Group 60	Enteric Group 68	Enteric Group 69
Indole production	0	0	0	50	100	100	30	0	0	0	0	0	0	0	0	0
Methyl red	80	80	100	97	100	100	92	100	100	62	97	100	100	100	100	0
Voges-Proskauer	0	0	0	2	0	5	0	0	0	0	10	0	0	0	50	100
Citrate (Simmons)	0	0	0	0	15	5	0	0	0	0	0	92	85	0	0	100
Hydrogen sulfide (TSI)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Urea hydrolysis	5	60	60	75	70	80	77	20	95	62	0	0	70	50	0	0
Phenylalanine deaminase	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Lysine decarboxylase	0	0	0	0	0	0	0	0	0	0	50	100	100	0	0	0
Arginine dihydrolase	0	0	0	0	0	0	0	0	0	0	5	8	0	0	0	100
Ornithine decarboxylase	0	40	80	95	95	100	92	80	0	25	100	100	85	100	0	100
Motility	0	0	0	2	5	5	5	0	0	0	0	100	100	75	0	100
Gelatin hydrolysis (22°C)	0	0	0	0	0	0	0	0	0	0	30	0	0	0	0	0
Growth in KCN	0	0	0	2	0	10	0	0	0	0	15	92	100	0	100	100
Malonate utilization	0	0	0	0	0	5	0	0	0	0	0	0	85	100	0	100
Esculin hydrolysis	50	0	20	25	85	100	0	0	95	0	0	67	0	0	0	100
Tartrate, Jordan's	0	100	100	85	55	88	40	100	50	100	30	0	60	75	0	0
Acetate utilization	0	0	0	15	15	18	8	0	0	0	0	25	45	0	0	25
Lipase (corn oil)	0	0	0	55	55	12	0	0	0	0	30	0	0	0	0	0
DNase (25°C)	0	0	0	5	0	0	0	0	0	0	0	0	0	0	100	0
Nitrate oxidized to nitrite	85	100	100	98	100	94	100	100	95	88	75	100	100	100	100	100
Oxidase, Kovacs	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ONPG test	50	0	80	95	100	90	70	20	70	50	50	100	100	100	0	100
Yellow pigment	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
D-Glucose, acid	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
D-Glucose, gas	0	0	0	5	40	18	23	0	0	0	5	100	85	100	0	100
<i>Fermentation of:</i>																
Adonitol	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
L-Arabinose	100	60	100	98	100	100	77	100	50	100	5	100	100	25	0	100
D-Arabitol	0	0	0	40	100	45	45	0	0	0	0	0	0	0	0	0
Cellobiose	0	0	100	75	100	96	100	100	0	25	5	100	100	0	0	100
Dulcitol	0	0	0	0	0	0	0	0	0	0	0	0	85	0	0	100
Erythritol	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Glycerol	50	0	0	90	85	60	70	20	50	38	30	0	30	75	50	0
myo-Inositol	0	0	0	30	20	15	15	0	0	0	0	0	0	0	0	0
Lactose	0	0	20	5	40	35	8	40	0	0	0	0	30	0	0	100
Maltose	80	0	100	75	100	100	100	60	95	0	95	100	100	0	50	100
D-Mannitol	97	80	100	98	100	100	100	100	100	100	100	100	100	50	100	100
D-Mannose	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
Melibiose	20	0	0	1	0	80	0	0	70	50	0	92	0	0	0	100
α-Methyl-D-glucoside	0	0	0	0	0	77	0	0	0	0	0	0	55	0	0	100
Mucate	0	0	0	0	5	6	0	0	0	0	0	0	0	0	0	100
Raffinose	0	0	0	5	30	45	0	0	15	62	5	25	0	0	0	100
L-Rhamnose	1	0	0	1	99	100	0	0	70	0	0	100	100	75	0	100
Salicin	70	0	20	20	92	100	15	20	25	0	0	8	100	0	50	100
D-Sorbitol	50	60	100	99	100	100	100	100	0	100	50	0	100	0	0	100
Sucrose	0	20	100	95	100	100	0	100	0	100	0	0	0	0	100	25
Trehalose	100	80	100	98	100	100	100	100	100	100	95	100	100	100	100	100
D-Xylose	90	40	100	70	100	100	85	60	100	38	0	100	100	0	0	100

^aEach number is the percentage of positive reactions after 2 d at 36°C unless otherwise indicated (gelatin liquefaction and deoxyribonuclease; *Phototuberculosis luminescens*).

^bTable taken from Farmer, 1999.

^cIn a Request for an Opinion published in 1987, Le Minor and Popoff proposed replacement of the type species of *Salmonella* (*Salmonella choleraesuis* subsp. *choleraesuis*) with *Salmonella enterica* as the former was considered to be a source of confusion. Although the Request was denied by the Judicial Commission, their proposal resulted in an alternative naming convention which has found widespread endorsement in the public health community. This matter was revisited in July 2002 by the Judicial Commission during the IUMS Congress in response to several new Requests for an Opinion and will likely result in a decision to replace the type strain *Salmonella choleraesuis* subsp. *choleraesuis* with *Salmonella enterica* subsp. *enterica* LT2, while preserving the former rather than placing it on the list of rejected names. We view the six subspecies of *Salmonella choleraesuis* as deprecated. Readers are also advised that the names *Salmonella enteritidis*, *Salmonella paratyphi*, *Salmonella typhi*, and *Salmonella typhimurium* are synonyms of *Salmonella enterica* subsp. *enterica* and refer to specific serovars. These names have not been deprecated at this time as they remain in use by some public health reporting agencies.

TABLE BXII.γ.194. Screening tests for genera and species of *Enterobacteriaceae* often isolated from human clinical specimens^a

Organism (genus, species, or serovar)	Test or property ^{b, c, d}
<i>Salmonella</i>	Lactose ⁻ , sucrose ⁻ , H ₂ S ⁺ , O1 phage ^{+c} , MUCAP ⁺ , agglutinates in polyvalent serum ^d , typical colonies on media selective/differential for <i>Salmonella</i> (brilliant green agar, SS agar, Rambach agar, etc.), lysed by the <i>Salmonella</i> specific bacteriophage ^c , often antibiotic resistant
<i>Salmonella</i> Typhi	H ₂ S ⁺ (trace amount only), agglutinates in group D serum
<i>Shigella</i>	Nonmotile, lysine ⁻ , gas ⁻ , agglutinates in polyvalent serum, biochemically inactive, often antibiotic resistant, molecular test: <i>phoE</i> ⁺
<i>Shigella dysenteriae</i>	Agglutinates in group A serum, D-mannitol ⁻
<i>Shigella dysenteriae</i> O1	Catalase ⁻ , agglutinates in O1 serum, Shiga toxin ⁺
<i>Shigella flexneri</i>	Agglutinates in group B serum, D-mannitol ⁺
<i>Shigella boydii</i>	Agglutinates in group C serum, D-mannitol ⁺
<i>Shigella sonnei</i>	Agglutinates in group D serum, D-mannitol ⁺
<i>Escherichia coli</i>	Extremely variable biochemically, indole ⁺ , MUG ⁺ , grows at 44.5°C, sometimes antibiotic resistant, molecular test: <i>phoE</i> ⁺
<i>Escherichia coli</i> O157:H7	Colorless colonies on sorbitol-MacConkey agar, MUG ⁻ , D-sorbitol ⁻ (or delayed), agglutinates in O157 serum and H7 serum
<i>Yersinia</i>	Grows on CIN agar; often more active biochemically at 25°C than 36°C; motile at 25°C, nonmotile at 36°C; urea ⁺
<i>Yersinia enterocolitica</i> (pathogenic serogroups)	CR-MOX ⁺ , pyrazinamidase ⁻ , salicin ⁻ , esculin ⁻ , agglutinates in O sera: 3; 4, 32; 5, 27; 8; 9; 13a, 13b; 18; 20; or 21
<i>Yersinia enterocolitica</i> O3 (a pathogenic serogroup)	D-Xylose ⁻ , agglutinates in O3 serum, tiny colonies at 24 h on plating media
<i>Yersinia enterocolitica</i> nonpathogenic serogroups	CR-MOX ⁻ , pyrazinamidase ⁺ , salicin ⁺ , esculin ⁺ , no agglutination in O sera: 3; 4, 32; 5, 27; 8; 9; 13a, 13b; 18; 20 or 21
<i>Citrobacter</i>	Citrate ⁺ , lysine decarboxylase ⁻ , often grows on CIN agar, strong characteristic odor
<i>Hafnia</i>	Lysed by <i>Hafnia</i> -specific bacteriophage ^c , often more active biochemically at 25°C than 36°C
<i>Klebsiella</i>	Mucoid colonies, encapsulated cells, nonmotile, lysine ⁺ , very active biochemically, ferments most sugars, VP ⁺ , malonate ⁺ , resistant to carbenicillin and ampicillin
<i>Enterobacter</i>	Variable biochemically, citrate ⁺ , VP ⁺ , resistant to cephalothin
<i>Serratia</i>	DNase ⁺ , gelatinase ⁺ , lipase ⁺ , resistant to colistin and cephalothin
<i>Serratia marcescens</i>	L-arabinose ⁻
<i>Serratia</i> , other species	L-arabinose ⁺
<i>Proteus-Providencia-Morganella</i>	Phenylalanine ⁺ , tyrosine hydrolysis ⁺ , often urea ⁺ , resistant to colistin
<i>Proteus</i>	Swarms on blood agar, pungent odor, H ₂ S ⁺ , gelatin ⁺ , lipase ⁺
<i>Proteus mirabilis</i>	Urea ⁺ , indole ⁻ , ornithine decarboxylase ⁺ , maltose ⁻
<i>Proteus vulgaris</i>	Urea ⁺ , indole ⁺ , ornithine decarboxylase ⁻ , maltose ⁺
<i>Providencia</i>	No swarming, H ₂ S ⁻ , ornithine decarboxylase ⁻ , lipase ⁻
<i>Morganella</i>	Very inactive biochemically, no swarming, citrate ⁻ , H ₂ S ⁻ , ornithine decarboxylase ⁺ , gelatin ⁻ , lipase ⁻ , urea ⁺

^aTable taken from Farmer, 1999.^bAbbreviations: CIN, cefsulodin-irgasan-novobiocin agar (a plating medium selective for *Yersinia*); CR-MOX, Congo red, magnesium oxalate agar (a differential medium useful for distinguishing pathogenic from nonpathogenic strains of *Yersinia*); MUCAP, 4-methyl-umbelliferyl caprylate (a genus-specific test for *Salmonella*); MUG, 4-methyl-umbelliferyl-β-D-glucuronidase; ONPG, o-nitrophenyl-β-D-galactopyranoside; *phoE*, a test done by PCR that is sensitive and specific for *E. coli*/*Shigella*; VP, Voges-Proskauer.^cThis table gives only the general properties of the genera, species, and serogroups; exceptions occur. See Table BXII.γ.193 for full biochemical reactions. The properties listed for a genus or group of genera generally apply for each of its species, and the properties listed for a species generally apply for each of its serogroups.^dBiochemical test results are given as percentages in Table BXII.γ.193. The serological tests refer to slide agglutination in group or individual antisera (O1, O3, etc.) for *Salmonella*, *Shigella*, *E. coli*, or *Yersinia*, respectively.^eThese are bacteriophage tests useful for identification.

Taxonomy and Nomenclature Compared with the 1st edition of *Bergey's Manual of Systematic Bacteriology* and the 9th edition of *Bergey's Manual of Determinative Bacteriology*, the present volume contains nomenclatural changes, new genera, and new species. In Table BXII.γ.192, the authors' proposed classification is compared with that in the previous two *Manuals*. Nomenclatural synonyms exist for a number of species, as shown in Table BXII.γ.192.

Arsenophonus A second species has provisionally been added to the genus as *Arsenophonus triatominarum*. The *candidatus* status was used because this species has only been recognized on the basis of its 16S rDNA sequence, and has not been isolated or characterized biochemically (Hypsa and Dale, 1997).

Candidatus Phlomobacter fragariae The designation "Candidatus" is given to presumed new species that have not been cultivated. One such organism has been described whose 16S rDNA sequences place it within the family *Enterobacteriaceae* (Zreik

et al., 1998). It has been named, *Candidatus Phlomobacter fragariae*. It is restricted to the phloem of strawberries and is associated with marginal chlorosis of strawberry (Zreik et al., 1998).

Citrobacter *Citrobacter koseri*, *Citrobacter diversus*, and *Levinea malonatica* are all synonyms for the same species. In response to a request for an opinion (Frederiksen, 1990), the Judicial Commission issued the opinion that *C. diversus* was a *nomina rejicienda* (rejected name). Therefore, the correct name is *C. koseri*. Eight new *Citrobacter* species were recently described (Brenner et al., 1993; Schauer et al., 1995). *Citrobacter farmeri* was formerly known as a biogroup of *C. amalonaticus*. The remaining 6 named and 2 unnamed species were previously included in *C. freundii* or had been previously identified as atypical citrobacteria.

Enterobacter *Enterobacter cancerogenus* (formerly *Erwinia cancerogena*) was shown to be a senior subjective synonym of *Enterobacter taylorae* and is therefore the correct name for this species (Grimont and Ageron, 1989). In several recently proposed classifi-

TABLE BXII.γ.195. Intrinsic antimicrobial resistance in some of the common *Enterobacteriaceae*^a

Genus/Species	Most strains are resistant to
<i>Buttiauxella</i> species	Cephalothin
<i>Cedecea</i> species	Polymyxins, ampicillin, cephalothin
<i>Citrobacter amalonaticus</i>	Ampicillin
<i>Citrobacter freundii</i>	Cephalothin
<i>Citrobacter diversus</i>	Cephalothin, carbenicillin
<i>Edwardsiella tarda</i>	Colistin
<i>Enterobacter cloacae</i>	Cephalothin
<i>Enterobacter aerogenes</i>	Cephalothin
Many other <i>Enterobacter</i> species	Cephalothin
<i>Escherichia hermannii</i>	Ampicillin, carbenicillin
<i>Ewingella americana</i>	Cephalothin
<i>Hafnia alvei</i>	Cephalothin
<i>Klebsiella pneumoniae</i>	Ampicillin, carbenicillin
<i>Kluyvera ascorbata</i>	Ampicillin
<i>Kluyvera cryocrescens</i>	Ampicillin
<i>Proteus mirabilis</i>	Polymyxins, tetracycline, nitrofurantoin
<i>Proteus vulgaris</i>	Polymyxins, ampicillin, nitrofurantoin, tetracycline
<i>Morganella morganii</i>	Polymyxins, ampicillin, cephalothin
<i>Providencia rettgeri</i>	Polymyxins, cephalothin, nitrofurantoin, tetracycline
Other <i>Providencia</i> species ^b	Polymyxins, nitrofurantoin
<i>Serratia marcescens</i> ^c	Polymyxins, cephalothin, nitrofurantoin
<i>Serratia fonticola</i>	Ampicillin, carbenicillin, cephalothin
Other <i>Serratia</i> species	Polymyxins ^d , cephalothin

^aTable taken from Farmer, 1999.^bMost strains of *Providencia stuartii* are also resistant to cephalothin and tetracycline.^c*Serratia marcescens* can also be resistant to ampicillin, carbenicillin, streptomycin, and tetracycline.^dResistance to polymyxins is common in *Serratia* species, but some strains have zones of 10–12 mm or larger.

cations, *Enterobacter agglomerans* and other species in the “*E. agglomerans* group” have been classified in the genus *Pantoea*. The *E. agglomerans* group was extremely heterogeneous and a number of species have been proposed for strains previously included in this species (see discussions of *Erwinia*, *Leclercia*, and *Pantoea* below and Table BXII.γ.192). *E. aerogenes* is genetically and phenotypically closer to *K. pneumoniae* than to *Enterobacter cloacae* (Brenner et al., 1972c; Steigerwalt et al., 1976; Bascomb et al., 1971; Grimont and Grimont, 1992). The names “*Klebsiella aerogenes*” and *Klebsiella mobilis* have been used for *E. aerogenes*. The type strain of *E. cloacae* is not representative of the species. If it is maintained, typical strains identified in clinical laboratories might not be called *E. cloacae* (Grimont and Grimont, 1992). This problem is dealt with in detail in the chapter on *Enterobacter*. Recently obtained biochemical test data and DNA–DNA hybridization data indicate that *Enterobacter intermedius* is a senior subjective synonym for the recently described species *Kluyvera cochleae* (A.G. Steigerwalt and J.J. Farmer, unpublished data).

Erwinia Plant isolates of erwiniae have been mainly studied by phytomicrobiologists and phytopathologists. The media, biochemical and other phenotypic tests used for their isolation, enrichment, cultivation, and identification are quite different from those used for other *Enterobacteriaceae*. The 35–37°C incubation temperature used for most other *Enterobacteriaceae* is near, at, or above the maximum growth temperatures of erwiniae. The isolation of erwiniae from humans or animals is rarely reported. It

is not known, however, if they are actually rarely occurring, whether their seeming lack of occurrence reflects improper isolation and enrichment procedures, or, if they are isolated, their identification fails. A study using optimal isolation procedures and an optimum incubation temperature would help to resolve this problem. Also needed is a study to characterize all erwiniae by tests and methods used for other *Enterobacteriaceae*. Erwiniae have been shown to be quite diverse on the basis of DNA relatedness studies (Gardner and Kado, 1972; Brenner et al., 1973b; Brenner et al., 1973b, and now 16S rRNA sequence studies (Kwon et al., 1997). Several recently proposed classifications have expanded the genus *Pantoea* to include the *Herbicola* or *Herbicola-Lathyrus* group of the genus *Erwinia*. Similarly, there have been proposals to change the circumscription of the genus *Pectobacterium* to include a number of species formerly contained in the genus *Erwinia*, and the new genus *Brenneria* was recently described for other species formerly in *Erwinia* (see Table BXII.γ.192 and chapters on *Erwinia*, *Pantoea*, *Pectobacterium*, and *Brenneria*).

Escherichia and Shigella The four nomenclatures of *Shigella* and *E. coli* are a single genomospecies on the basis of DNA relatedness (Brenner et al., 1972a, 1973a). *Shigella* and *E. coli* strains are often extremely difficult to separate biochemically since there are aerogenic (gas-producing) shigellae and lactose-negative, anaerogenic, nonmotile *E. coli*. Such strains can cause a dysentery-like diarrhea, so pathogenicity does not provide definitive separation. Shigellae can be considered metabolically inactive biogroups of *E. coli*. Based mainly on molecular data, it has been argued that these organisms do not represent distinct species, but a continuum of closely related lines of descent from a common ancestor. In this *Manual*, they are classified as distinct species because of the ease of communication these names provide in medical microbiology and because of the resistance and confusion that would be caused by reclassification. However, the original usage implying that shigellae were pathogenic and *E. coli* was not is certainly not true. Tamura et al. (1986) proposed an alternative classification. They defined a new genus *Leclercia*, with the single species *L. adecarboxylata*, which was originally named *Escherichia adecarboxylata*.

Hafnia *Hafnia alvei* remains the only named species in this genus. DNA studies revealed two separate DNA relatedness groups within *H. alvei*, with approximately one-half of the strains in each relatedness group (Steigerwalt et al., 1976). A second species was not designated since no single biochemical test or series of tests served to unequivocally separate the two relatedness groups (F.W. Brenner and J.J. Farmer III, personal communication). “*Hafnia protea*” is an illegitimate name for *Obesumbacterium proteus*. There were two biogroups within *O. proteus*. Biogroup 2 is *O. proteus*, whereas biogroup 1 is a metabolically inactive biogroup of *H. alvei* that has become adapted to the brewery environment.

Klebsiella In 1984, Orskov proposed a classification in which *Klebsiella ozaenae* and *Klebsiella rhinoscleromatis* would be classified as subspecies of *K. pneumoniae* (Orskov, 1984a). While there is no doubt that these 3 nomenclatures are the same genomospecies on the basis of DNA relatedness studies (Brenner, Steigerwalt, and Fanning, 1972), many laboratories still accept their original classification and report them as separate species. *Klebsiella ornithinolytica* was described by Sakazaki et al. (1989a) for strains previously determined to be a biogroup of *Klebsiella planticola* by Farmer et al., (1985a). These strains have been variously referred to as ornithine-positive *K. oxytoca*, NIH (Japan) Group 12, CDC

TABLE BXII.γ.196. Additional biochemical reactions of *Enterobacteriaceae*^{a, b, c}

Species	Nitrate reductase	Tetrathionate reductase	D-Galacturonate	2-keto-Gluconate	γ-Glutamyl transferase
<i>Citrobacter amalonaticus</i>		+		+	+
<i>Citrobacter freundii</i>	+, A	+	+	+	+
<i>Citrobacter koseri</i>		—		+	+
<i>Edwardsiella hoshinae</i>				—	
<i>Edwardsiella tarda</i>	+, B	+		—	—
<i>Erwinia carotovora</i>	+, A	—	+	—	+
<i>Enterobacter aerogenes</i>	+, A	—	+	+	+
<i>Enterobacter cloacae</i>	+, A	—	d	+	+
<i>Enterobacter gergoviae</i>				+	+
<i>Enterobacter sakazakii</i>				+	
<i>Escherichia coli</i>	+, A	—		—	+
<i>Hafnia alvei</i>	d, A or B	d	+	+	+
<i>Klebsiella oxytoca</i>	d, A	d		+	+
<i>Klebsiella pneumoniae</i> subsp. <i>ozaenae</i>		—		+	+
<i>Klebsiella pneumoniae</i> subsp. <i>pneumoniae</i>	+, A	—		+	+
<i>Klebsiella pneumoniae</i> subsp. <i>rhinoscleromatis</i>		—		d	—
<i>Morganella morganii</i>	d, A	+		—	+
<i>Pantoea agglomerans</i>	+, A	—	+	+	+
<i>Proteus mirabilis</i>	+, A	+		—	+
<i>Proteus vulgaris</i>	+, A	+		—	+
<i>Providencia alcalifaciens</i>	+, B	+		—	+
<i>Providencia rettgeri</i>	+, A	+		d	+
<i>Providencia stuartii</i>	+, A	+		—	+
<i>Salmonella enterica</i> subsp. <i>arizonae</i>	+, A	+	—	—	—
<i>Salmonella enterica</i> subsp. <i>diarizonae</i>	+, A	+	+	—	+
<i>Salmonella enterica</i> subsp. <i>enterica</i>	+, A	+	—	—	+
<i>Salmonella enterica</i> subsp. <i>houtenae</i>	+, A	+	+	—	+
<i>Salmonella enterica</i> subsp. <i>salamae</i>	+, A	+	+	—	+
<i>Serratia ficaria</i>				+	+
<i>Serratia liquefaciens</i>	+, A	+	+	+	+
<i>Serratia marcescens</i>	+, A	d	+	+	+
<i>Serratia odorifera</i>	+, A	—		+	+
<i>Serratia plymuthica</i>	+, A	—	+	+	+
<i>Serratia rubidaea</i>	+, A	—	+	+	+
<i>Shigella boydii</i>	+, A	—		—	d
<i>Shigella dysenteriae</i>		—		—	d
<i>Shigella flexneri</i>	+, A	—		—	d
<i>Shigella sonnei</i>	+, A	—		—	—
<i>Yersinia enterocolitica</i>	+, B	d		d	+
<i>Yersinia pestis</i>	d, B	d		—	—
<i>Yersinia pseudotuberculosis</i>	+, B	—		—	+
<i>Yersinia frederiksenii</i>	+, B	+			
<i>Yersinia intermedia</i>	+, A or B	+			
<i>Yersinia kristensenii</i>	+, B	d			

^aSymbols: +, 90% or more of strains are positive; d, 10.0–89.9% positive; —, 0–9.9% positive; blank space, not available; A, type A nitrate reductase; B, type B nitrate reductase.

^bThe nitrate reductase test was incubated at 32°C; all other tests were incubated at 35–37°C. The γ-glutamyl transferase test was read at 24 h; all other test results were read at 48 h.

^cData compiled from: Pichinoty et al. (1969), Grimont (1977b), Richard (1977), Grimont et al. (1978a), Le Minor et al. (1979), Bercovier et al. (1980a), Bercovier et al. (1980b), Brenner et al. (1980c), Giammanco et al. (1980), Ursing et al. (1980a), and Buissonière et al. (1981).

Klebsiella Group 47 (indole-positive, ornithine-positive); and indole-positive, ornithine-positive biogroup of *Klebsiella planticola*. Laboratories agreed on the phenotypic properties of this organism, but they disagreed on whether DNA relatedness data indicated that it should be retained as a biogroup of *K. planticola* or that it should be a separate species.

Kluyvera The recently described species *K. cochleae* (Müller et al., 1996) appears to be a junior subjective synonym for *Enterobacter intermedius* (A.G. Steigerwalt and J.J. Farmer, unpublished data). In this *Manual*, this species is not included as one of the species of *Kluyvera*, but see *Enterobacter intermedius*.

Obesumbacterium proteus. See *Hafnia* above.

Pantoea In this *Manual*, the genus *Pantoea* includes species previously contained in the *Enterobacter agglomerans* complex (including *Erwinia herbicola*, *Erwinia lathyri*, *Erwinia ananas*, *Erwinia*

uredovora, *Erwinia milletiae*, and *Erwinia stewartii*). Future proposals may expand its definition to include additional species to be defined from some of the remaining *Enterobacter agglomerans* DNA hybridization groups that have not yet been classified (Brenner et al., 1984a; Grimont and Grimont, 1992). See sections on *Enterobacter* and *Erwinia* above.

Photorhabdus* and *Xenorhabdus Substantial differences in biochemical reactions have been reported by different laboratories (Grimont et al., 1984b; Akhurst and Boemare, 1988; Farmer et al., 1989). Many of the differences are probably due to slow and weak reactions and to differences in media. In the proposed classification of Akhurst and Boemare, *Xenorhabdus beddingii* and *Xenorhabdus bovienii*, are classified as species (Akhurst and Boemare, 1988). They were originally described as subspecies of *Xenorhabdus nematophilus*. Four additional genomospecies within

Photorhabdus luminescens (*Xenorhabdus luminescens*) were not named because they are not phenotypically separable from *P. luminescens* (Farmer et al., 1989).

Proteus The type strain of *Proteus vulgaris* is in a DNA relatedness group that includes only one other strain out of 36 tested. The remaining strains are in four new genomospecies, which include essentially all isolates currently identified as *P. vulgaris* (Brenner et al., 1995). A request for an opinion to replace the type strain was submitted (Brenner et al., 1995), and was recently granted (H. Trüper, personal communication). MacDonell and Colwell (1985) recommended that *Plesiomonas shigelloides* be classified in the genus *Proteus* because its 5S rRNA sequence is closely related to that of *P. mirabilis*. They did not formally describe the new combination ("*Proteus shigelloides*"). Such a change is not supported by phenotypic characteristics or by DNA relatedness and the new combination has not been used.

Salmonella The classification and nomenclature of salmonellae has long been a source of confusion, even to specialists. Much of the confusion was due to the practice of naming serotypes (serovars) and equating them with species. All phylogenetic data indicate that the evolution of the genus *Salmonella* is a continuum, but several recent classifications recognize two distinct species of *Salmonella*, one of which contains six subspecies (Reeves et al., 1989a; Brenner and McWhorter-Murlin, 1998). The type species of the genus is *Salmonella choleraesuis*. Unfortunately, there is also a serotype *Choleraesuis* that, until recently was treated as a species. For this reason Le Minor and Popoff (1987) requested an opinion to replace the name *S. choleraesuis* with "*Salmonella enterica*". This request was not approved by the Judicial Commission (Wayne, 1994; Judicial Commission, 1991), although the wording of the refusal seemed to leave the door open for another request to remedy the confusion caused by the name *S. choleraesuis*. This request has now been resubmitted by Euzéby (1999), who also requested that "*Salmonella typhi*" be designated as a species. In the interim, the WHO International Collaborating Centre for *Salmonella* located at the Pasteur Institute in Paris, the National *Salmonella* Center located at the Centers for Disease Control and Prevention in Atlanta, and many other reference laboratories and their constituents have begun to use *S. enterica*, so both names are currently in use. In one proposed classification there are 6 subspecies of *S. choleraesuis*: subsp. *enterica* (or subspecies I or subspecies 1), for those biochemically typical serotypes previously placed in subgenus I; subsp. *salamae* (or subspecies II or subspecies 2), for biochemically atypical serotypes previously in subgenus II; subsp. *arizonae* (or subspecies IIIa or subspecies 3a) for the monophasic strains previously in "*Arizona hinshawii*" or subgenus III; subspecies subsp. *diarizonae* (or subspecies IIIb or subspecies 3b) for the diphasic strains previously in "*A. hinshawii*" or subgenus III; subsp. *houtenae* (or subspecies IV or subspecies 4) for the biochemically atypical strains previously in subgenus IV; and subsp. *indica* (or subspecies VI or subspecies 6) for more recently defined atypical strains that differ from the biochemical patterns of subspecies I-IV. The second species is *Salmonella bongori*, a small number of biochemically unique serotypes that have been classified as subsp. *bongori* (or subspecies V or subspecies 5) but are considered to be separate species in the proposed classification of Reeves et al., (1989a) and in this *Manual*. The practice in medical bacteriology is to use names for the serotypes in subsp. *enterica*. For example, the serotypes formerly reported as *S. typhi* and *S. typhimurium* would now be reported as *S. choleraesuis* (or "*S. enterica*") subsp. *enterica* (or I or 1) serotype Typhi or Typhimurium. Note that the se-

rotype name is capitalized and nonitalicized. Alternatively these serotypes could be reported simply as *Salmonella* serotype Typhi and *Salmonella* serotype Typhimurium. Serotypes in the other subspecies are reported similarly. Some laboratories choose to use names (where the serotypes are named) and some choose to use the antigenic formulae. Serotypes in subsp. *arizonae* and subsp. *diarizonae* are not named. Therefore a typical report, would be *Salmonella* serotype 60:k:z (= *Salmonella enterica* subsp. *diarizonae* serotype 60:k:z). *Salmonella enterica* subsp. *indica* serotype Brookfield has the antigenic formula 66:z41:-. Any of the following reports would be accurate: *Salmonella* serotype Brookfield, *Salmonella* subsp. *indica* serotype Brookfield, *Salmonella* subsp. V Brookfield, as would any of the above designations with the antigenic formula used instead of the name Brookfield. Much of the literature still uses the "serotype as species" classification and uses the Linnean system of writing the genus and species in italics. Examples include *Salmonella typhi*, *Salmonella typhimurium*, and *Salmonella enteritidis*.

Yokenella *Yokenella*, with its single species *Y. regensburgi*, was described by Kosako et al. in 1984. The genus *Koserella* with its single species *K. trabulsii*, was described by Hickman-Brenner et al. in 1985a. *Y. regensburgi* and *K. trabulsii* were shown to be subjective synonyms by Kosako et al. in 1987. Other synonyms for this organism are *Hafnia* hybridization group 3, Enteric Group 45, NIH (Japan) biogroup 9. Even though *K. trabulsii* had priority by virtue of its earlier alphabetical appearance on Validation List No. 17 in the *International Journal of Systematic Bacteriology*, both groups of investigators agreed that *Yokenella* should be the genus name.

Unnamed Enteric Groups A number of groups belonging to the family *Enterobacteriaceae* have been characterized in the Enteric Reference Laboratory at the CDC but not given genus and species names. Four Enteric Groups are briefly described below. Their biochemical profiles are included in Table BXII.γ.193.

Enteric Group 58 Enteric Group 58 consists of strains that were first recognized in 1981 and described in 1985 (Farmer et al., 1985a). The first five isolates were from human clinical specimens; four from wounds (foot, ankle, leg, and hip), and one from feces. Since the original report in 1985, the number of isolates at CDC has grown to 21, including an isolate from a case of bacteremia. Enteric Group 58 has the general properties of the family *Enterobacteriaceae*.

Enteric Group 60 Enteric Group 60 consists of four strains first recognized in 1981 and described in 1985 (Farmer et al., 1985a). All of the strains were from human clinical specimens (three from urine and one from sputum). Enteric Group 60 is inactive biochemically and was originally thought to be most like *Morganella morganii*. However, it is sensitive to colistin and tyrosine-negative, reactions incompatible with *Morganella*. Enteric Group 60 has the general properties of the family *Enterobacteriaceae*. Its closest phylogenetic relatives are not yet known.

Enteric Group 68 Enteric Group 68 consists of a small group of strains isolated from human urine that were first recognized in 1981 and described in 1985 (Farmer et al., 1985a). The group is positive for DNase, but otherwise quite different from *Serratia*. Enteric Group 68 also has the general properties of the family *Enterobacteriaceae*. Its closest phylogenetic relatives are not yet known.

Enteric Group 69 Enteric Group 69 is the name given to a group of strains isolated from refrigerated beef carcasses (Farmer et al., 1985a). The strains were phenotypically similar to *E. sa-*

kazakii in most biochemical reactions, including production of a yellow pigment. However, they were only 43% related to *E. sakazakii* by DNA–DNA hybridization. Recent DNA relatedness data

by Kosako et al. (1996) indicated that *Enterobacter kobei* is the closest relative of Enteric Group 69.

Genus I. *Escherichia* Castellani and Chalmers 1919, 941T^{AL}

FLEMMING SCHEUTZ AND NANCY A. STROCKBINE

Esch.er.i'chi.a. M.L. fem. n. *Escherichia* named after Theodor Escherich, who isolated the type species of the genus.

Straight cylindrical rods, $1.1\text{--}1.5 \times 2.0\text{--}6.0 \mu\text{m}$, occurring singly or in pairs. Conform to the general definition of the family *Enterobacteriaceae*. Gram negative. Motile by peritrichous flagella or nonmotile. Aerobic and facultatively anaerobic having both a respiratory and a fermentative type of metabolism, but anaerogenic biotypes occur. Oxidase negative. Chemoorganotrophic. **Both acid and gas are formed from most fermentable carbohydrates, but *D*-inositol is not utilized and *D*-adonitol is utilized only by *Escherichia fergusonii*. Lactose is fermented by most strains of *Escherichia coli*, but fermentation may be delayed or absent in *Escherichia blattae*, *Escherichia hermannii*, *Escherichia fergusonii*, and *Escherichia vulneris*. Do not grow in KCN (with the exception of *E. hermannii* and a small proportion of *E. vulneris*). Usually do not produce H_2S .** *E. coli* occur naturally in the lower part of the intestine of warm-blooded animals, *E. blattae* in the hind-gut of cockroaches, and *E. fergusonii*, *E. hermannii*, and *E. vulneris* are found in the intestine, as well as extraintestinal sites of warm-blooded animals. Seven copies of the *rrn* operon with genes coding for 16S, 23S, and 5S rRNA are present on the chromosome of *E. coli*. Comparative sequence analysis between the genes for 16S rRNA of *E. coli*, *E. vulneris*, and *E. hermannii* and homologous genes from all eubacteria places *E. coli* and *E. vulneris* together in a tightly related cluster with shigellae, and *E. hermannii* between *Salmonella* spp. and *Citrobacter freundii* (Cilia et al., 1996). Based on 16S rRNA sequencing, escherichiae belong in the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 48–59.

Type species: *Escherichia coli* (Migula 1895) Castellani and Chalmers 1919, 941 (*Bacillus coli* Migula 1895, 27.)

FURTHER DESCRIPTIVE INFORMATION

Cell morphology Escherichiae are straight, cylindrical, Gram-negative rods with rounded ends that are $1.1\text{--}1.5 \mu\text{m}$ in diameter and $2.0\text{--}6.0 \mu\text{m}$ in length. They occur singly or in pairs and can be motile by peritrichous flagella or nonmotile. Fig. BXII.γ.191 shows negatively stained preparations of each of the *Escherichia* species. See Nanninga (1985) for a comprehensive treatment of the ultrastructure of *E. coli*.

Phylogenetic and systematic treatment The genus consists of five species: *E. coli*, *E. hermannii*, *E. fergusonii*, *E. vulneris*, and *E. blattae*. Biochemical reactions that will help in differentiating between the species of *Escherichia* are listed in Table BXII.γ.197.

Biochemical reactions Escherichiae produce strong acids and usually gas from the fermentation of *D*-glucose (positive in the Methyl Red test) and do not produce acetyl-methyl carbinol (acetoin) (negative in the Voges–Proskauer test). Sodium acetate is frequently used as a sole carbon source, except by *E. blattae* and a majority of *E. vulneris* strains. Citrate (Simmons' citrate agar)

cannot be used by *E. coli* and *E. vulneris*, whereas a smaller proportion of *E. fergusonii* and *E. hermannii* exhibit immediate or delayed use of this substrate. Growth on Simmons' citrate agar by *E. blattae* is variable and probably strain specific.

Lysine is decarboxylated by the majority of strains. Exceptions include "metabolically inactive" *E. coli* strains, the majority of enteroinvasive *E. coli* strains (EIEC), and *E. hermannii*. Ornithine is decarboxylated by all species except by *E. vulneris* and a little less than half of *E. coli* strains. Indole is produced by all species except *E. blattae* and *E. vulneris*.

Other tests that help in differentiating between the species of *Escherichia* include growth in potassium cyanide, malonate utilization, and acid production from *D*-adonitol, *D*-arabitol, cellobiose, dulcitol, lactose, *D*-mannitol, melibiose, *D*-sorbitol, and mucate. See Table BXII.γ.197.

Molecular data Findings from the comparison of 16S rDNA sequences performed with strains of *E. coli*, *Salmonella* spp., and *C. freundii* have shown a close phylogenetic relatedness between these bacteria (Ahmad et al., 1990; Cilia et al., 1996; Chang et al., 1997). Analysis of 16S rDNA sequences separates the two *Salmonella* species (*S. enterica* and *S. bongori*) from the complex of *E. coli* and *Shigella*, and shows that *E. hermannii* is more closely related to *Salmonella enterica* and *C. freundii* (Christensen et al., 1998). This analysis is unable to separate inter-operon variation of *E. coli* from strains of *E. coli* and from *Shigella*. Patterns of sequence heterogeneity in strains from the *E. coli* Collection of Reference (ECOR) (Ochman and Selander, 1984) have been located at regions VI and V6 of cloned 16S rRNA genes (Martinez-Murcia et al., 1999).

Average DNA relatedness assessed by DNA–DNA hybridization among *Escherichia* species ranges from 29% to 94% (Table BXII.γ.198). DNAs from different strains of *E. coli* are closely related (average, 84%; Table BXII.γ.198). With the exception of *S. boydii* serotype 13, the DNAs of *E. coli* and the four *Shigella* species show such a high degree of relatedness (average 80–87%, except *S. boydii* type 13, which is about 65%) that these species should be considered as a single species (Brenner et al., 1973a). The distinction between these bacteria prevails, however, for reasons of historical/medical precedent and to avoid confusion in the literature and with existing surveillance systems.

Based on complete sequencing of the K-12 strain MG1655 (Blattner et al., 1997), it is estimated that the *E. coli* lineage diverged from the *Salmonella* lineage some 100 million years ago (Lawrence and Ochman, 1998). This is a little less than the 120–160 million years estimated by calibration of the rate of 16S rRNA evolution in bacteria (Ochman and Wilson, 1987). The chromosome of *E. coli* strain MG1655 is 4,639,221 bp and contains 4,288 open reading frames (ORFs). Approximately 18% of these ORFs represent genes that have been acquired and have persisted

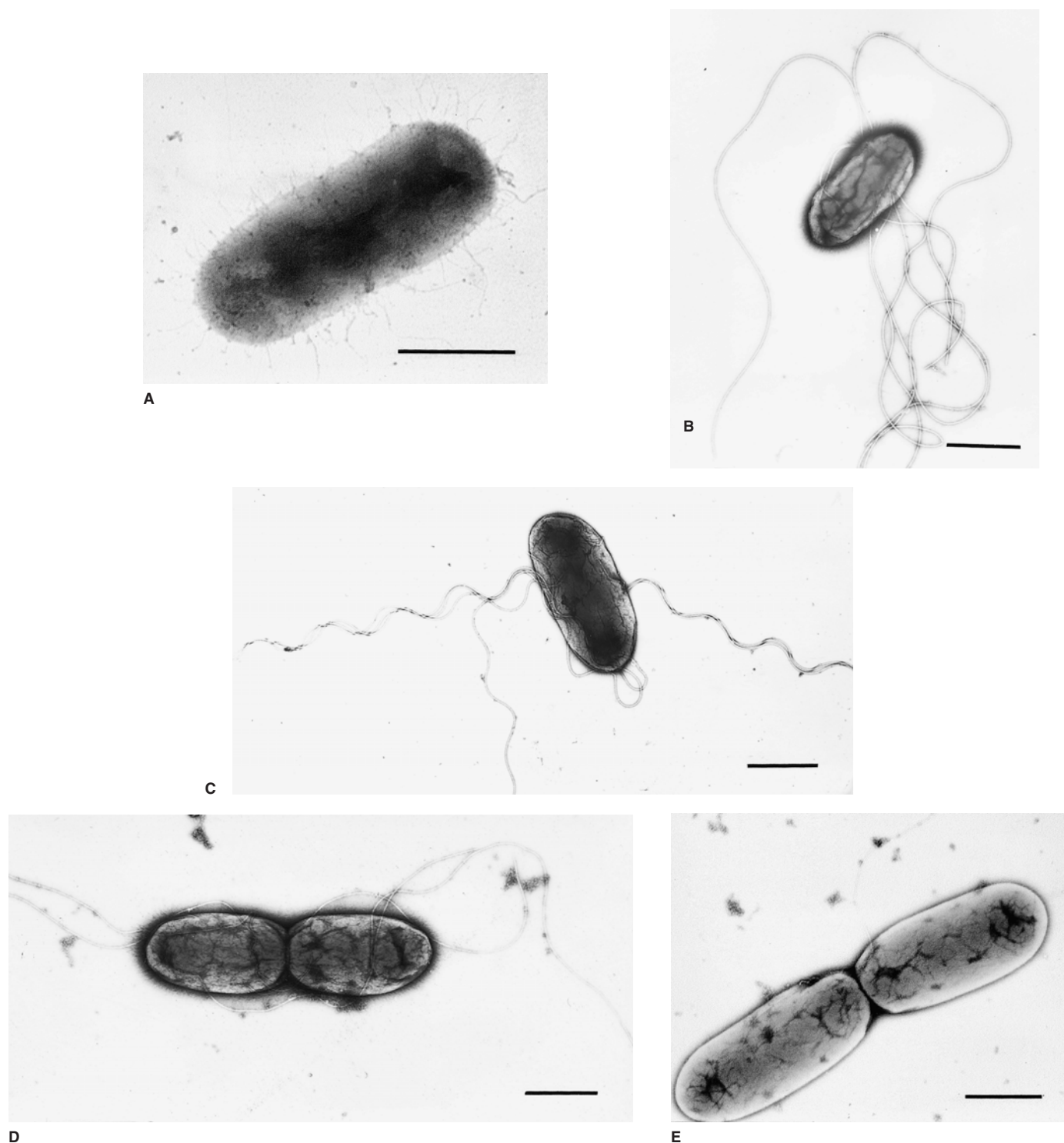


FIGURE BXII.γ.191. Electron micrographs of *E. blattae* strain 2928-78 (A), *E. coli* O157:H7 strain EDL933 (B), *E. fergusonii* strain 2460-89 (C), *E. hermannii* strain 2456-88 (D), and *E. vulneris* strain 2485-88 (E) prepared by negatively staining in 0.5% (w/v) uranyl acetate. Bar = 1000 nm. (Micrograph courtesy of Charles D. Humphrey, CDC.)

since divergence. This is similar to estimates based on analysis of codon usage, indicating that 16% of sequenced genes arose through horizontal transfer (Médigue et al., 1991). The ability of individual strains and lineages to acquire foreign DNA results in great heterogeneity in both individual genes and in the size of the *E. coli* chromosome. This is exemplified by the unusual

serotype O157:H7, the genome of which is 5.5 Mb in size, 859 Kb larger than that of the laboratory strain K-12 (Hayashi et al., 2001; Perna et al., 2001). The genome sizes of 14 *E. coli* strains from the five major subgroups in the ECOR (Ochman and Selander, 1984) range from 4.66–5.30 Mb (Bergthorsson and Ochman, 1995), and the uropathogenic *E. coli* strain J96 genome is

TABLE BXII.γ.197. Differentiation of the five species of *Escherichia*^{a,b}

Test	<i>E. coli</i>	<i>E. coli</i> (metabolically inactive strains)	<i>E. blattae</i>	<i>E. fergusonii</i>	<i>E. hermannii</i>	<i>E. vulneris</i>
Indole	+	[+]	—	+	+	—
Citrate, Simmons	—	—	d	[—]	— ^c	—
Lysine decarboxylase	+	d	+	+	— ^d	[+]
Ornithine decarboxylase	d	[—]	+	+	+	—
Motility	+	—	— ^e	+	+	+
KCN, growth	—	—	—	—	+	[—]
Malonate utilization	—	—	+	d	—	[+]
D-Glucose, gas	+	—	+	+	+	+
<i>Acid production from:</i>						
D-Adonitol	—	—	—	+	—	—
D-Arabitol	—	—	—	+	—	—
Cellobiose	—	—	—	+	+	+
Dulcitol	d	d	—	d	[—]	—
Lactose	+	[—]	—	— ^f	d	[—] ^f
D-Mannitol	+	+	—	+	+	+
Melibiose	[+]	d	—	—	—	+
D-Sorbitol	+	d	—	—	—	—
Mucate	+	d	d	—	+	[+]
Acetate utilization	+	d	—	+	[+]	d
Yellow pigmentation	—	—	—	—	+	d

^aData compiled from references Farmer (1999), Cowan et al. (1995), Holt et al. (1994), and Richard (1989). Reactions for indole for *E. fergusonii* and melibiose for *E. coli* differ slightly in these references. The reactions listed in this table are supported by our own unpublished data.

^bSymbols: —, 0–10% positive; [—], 11–25% positive; d, 26–75% positive; [+], 76–89% positive; +, 90–100% positive. Results are for 48 h incubation at 36° ± 1°C.

^cDelayed positive in approximately a fifth of *E. hermannii* strains.

^dDelayed positive in a third of *E. hermannii* strains.

^e75% of *E. blattae* strains will become motile after incubation of more than 2 d.

^fDelayed positive in approximately two thirds of *E. fergusonii* and *E. vulneris* strains.

TABLE BXII.γ.198. DNA relatedness among escherichiae^a

Labeled DNA from	Average percent relatedness				
	<i>E. coli</i>	<i>E. blattae</i>	<i>E. fergusonii</i>	<i>E. hermannii</i>	<i>E. vulneris</i>
<i>E. coli</i>	84	42	64	38	43
<i>E. blattae</i>		90		39	29
<i>E. fergusonii</i>	57		94	59	
<i>E. hermannii</i>	43	32		89	33
<i>E. vulneris</i>	39	29	33	36	78

^aData compiled from Ewing, 1986b.

reported to be 5.12 Mb (Melkerson-Watson et al., 2000) as estimated by pulsed field gel electrophoresis (PFGE).

Of the five species in the genus *Escherichia*, *E. coli* is the most studied. Because the amount of knowledge about the other four species is limited, the descriptive information to follow applies to *E. coli*, unless indicated otherwise.

Cell wall composition The chemical composition and molecular structure of the cell wall of *E. coli* has been extensively studied and is described in detail by Neidhardt and Umbarger (1996) and Park (1996). The structural rigidity of the cell wall is provided by the murein sacculus, which consists mainly of a single monomolecular sheet of murein, a complex polymer composed of roughly equal amounts of polysaccharides (*N*-acetylglucosamine [GlcNAc] and *N*-acetylmuramic acid [MurNAc]) and peptides (L-alanine, D-glutamic acid, L-*meso*-diaminopimelic acid (DAP), and D-alanine). Linear chains of alternating units of GlcNAc and MurNAc are linked together by β-1→4 glycosidic bonds, and short chains of the above peptides in alternating D and L optical isomers are attached to the sugars through amide linkages to the carboxyl groups of each muramic acid. Adjacent glycan strands are cross-linked to each other through the peptide side chains to create one giant molecule that provides structural

support to the cell. Notable features of the murein of *E. coli* include the presence of a small percentage of peptide chains that either lack the D-alanine or terminate in an additional D-alanine and the absence of amidation involving the carboxyl groups of glutamic acid and DAP. A molecule of lipoprotein is also attached to about every tenth muropeptide.

Outer membrane The outer membrane of an average *E. coli* cell contains over a million molecules of lipopolysaccharide (LPS), which consist of three covalently linked domains: (1) lipid A (endotoxin), (2) the core region of phosphorylated nonrepeating oligosaccharides, and (3) the O antigen polymer of immunogenic repeating oligosaccharides (1–40 units).

Fine structure

Flagella Motile organisms of the genus typically possess 5–10 flagella per cell, which are randomly situated around the cell surface (peritrichous flagellation). The flagellar filament is about 20 nm in diameter and may be up to 20 μm long. It consists of subunits of a single protein, flagellin, which is encoded by the *flaC* gene. Fifty-three antigenically distinct types of flagellin have been described (Ørskov and Ørskov, 1984a). Electron microscopic studies of these antigenically distinct types of flagellar

filaments revealed differences in surface structure; six unique flagellar morphotypes have been described (Lawn et al., 1977). Unlike *Salmonella*, most *E. coli* strains have only one flagellin gene and do not undergo phase variation. Exceptions have been described (Ratiner, 1967, 1982, 1999). See Macnab (1996) for a complete description of the structure and genes involved in motility.

Fimbriae In addition to the proteinaceous flagella, most strains have fimbriae (pili) or fibrillar proteins often extending in great numbers from the bacterial surface and far out into the surrounding medium. A typical *E. coli* K-12 cell contains 100–500 type 1 fimbriae arranged peritrichously, each with a diameter of approximately 7 nm and a length of 0.2–2.0 μ m. More than 30 different fimbriae have been described in *E. coli*, which commonly expresses more than one type at a time. The characteristics, biogenesis, and classification schemes of fimbriae are reviewed by Low et al. (1996). Fimbriae have historically been classified by phenotypic properties. One widely used scheme classifies fimbriae according to their adhesive properties for red blood cells from different host species in the presence of mannosides. By this method, two main types of fimbriae are recognized: mannose-sensitive (MS) fimbriae, which are unable to agglutinate red blood cells in the presence of α -D-mannose, and mannose-resistant (MR) fimbriae, which are able to agglutinate red blood cells in the presence of this sugar. MS fimbriae, which include the so-called type 1 fimbriae (pili), are found in the majority of *E. coli* strains and comprise a group of more or less serologically related antigens. Because they are expressed by pathogens as well as commensal organisms, their role in virulence has been difficult to establish. Evidence for the role of type 1 fimbriae in virulence is reviewed by Abraham and Jaiswal (1997). Type 1 fimbriae mediate avid bacterial attachment to mucosal surfaces, to noncellular host constituents, and to various inflammatory cells. They bind to certain oligomannoside-containing glycoproteins present on mucosal surfaces, including the Tamm-Horsfall glycoprotein, which is synthesized in the kidney and present in urinary slime (Ørskov et al., 1980a); fibronectin, a glycoprotein that is a member of a family of proteins found in the extracellular matrix (ECM), plasma, and other body fluids; and laminin, a glycoprotein present in basement membranes (Kukkonen et al., 1993). The genes involved in the synthesis and regulation of type 1 fimbriae are located on the chromosome. Expression of type 1 fimbriae is subject to being turned on or off (phase variation) as a result of the inversion of a 314 base-pair fragment of DNA containing the promoter region of the gene encoding the major fimbrial subunit (*fimA*). Expression of type 1 fimbriae is influenced by environmental and growth conditions and is controlled by global regulatory factors such as leucine-responsive regulatory protein (Lrp), integration host factor (IHF) and histone-like protein H-NS. See Abraham and Jaiswal (1997) and Low et al. (1996) for a review of the environmental factors and genes involved in synthesis and regulation of type 1 fimbriae.

MR fimbriae are serologically diverse (Ørskov et al., 1980b, 1982; Ørskov and Ørskov, 1990) and often function as virulence factors to mediate adherence that is species- and organ-specific. The genes for these proteins may be located on plasmids or on the chromosome. When located chromosomally, they often cluster together with other virulence genes in regions of the chromosome referred to as pathogenicity islands (PAIs). More than any other virulence factor, the MR pili of *E. coli* illustrate the

species' capacity to adapt to the receptor-specific epithelial cells of certain hosts primarily through horizontal acquisition of gene cassettes on plasmids, phages, or other mobile DNA elements that will allow for colonization. Evidence for the horizontal transfer of fimbrial genes in *E. coli* comes from the remarkable similarity between the genetic organization of its fimbrial operons and those of other members of the family *Enterobacteriaceae* and from the strikingly low C + G content and different codon usage pattern among the fimbrial genes compared to those observed overall among other genes on the *E. coli* chromosome.

Fimbriae may also be classified based on their morphology. One group consists of thick, rod-shaped fimbriae with a diameter of 7 nm (range 3.4–8 nm), a length of 0.5–2 μ m, and an axial hole diameter of 2.0–2.5 nm. Fimbriae with these dimensions are represented by the rigid pyelonephritis (P), sialic acid (S), type 1, F6 (987P), colonization factor antigen I (CFA/I), coli surface antigen 1 (CS1) and CS2 pili, and by the bundle-forming CS8 (CFA/III) and CS21 (longus), the latter with homology to the type 4 fimbrial family. Enteropathogenic *E. coli* (EPEC) is known to produce a type 4 fimbria called the bundle-forming pilus (BFP). Another flexible, bundle-forming fimbrial structure of 2–3 nm diameter, designated aggregative adherence fimbriae I (AAF/I), shows no homology to the type 4 class of fimbriae. Together with another recently described AAF/II fimbria, it exhibits sequence and organizational resemblance to the Dr family. Another group consisting of thinner, more flexible fibrillae with a width of 2–5 nm and a length of 0.5–2.0 μ m is represented by F4 (K88), F5 (K99), F41, and CS3. Helical fibrillae, where two fibrillae are arranged in a helix, are represented by CS5 and CS7.

Nonfimbrial and related adhesins Bacterial adherence may also be mediated by adhesions that are afimbrial (AFA) or nonfimbrial (NFA). Some of these proteins form larger multimers that aggregate around the bacterial cell as an amorphous structure reminiscent of capsular K antigens. Afimbrial adhesins are found in both uropathogenic *E. coli* (UPEC) and in diffusely adhering *E. coli* (DAEC) and represented by the Afa/Dr family consisting of Dr (previously referred to as O75X) and Dr-II (*drb*), the F1845 pilus (*daa*), and AFAI-IV (*afa*). Dr fimbriae and related adhesins recognize different epitopes of the Dr blood group antigen (Nowicki et al., 1990) and bind to a complement-regulatory protein, the common receptor decay accelerating factor (DAF). *E. coli* strains expressing Dr fimbriae are able to enter epithelial cells by interacting with DAF (Goluszko et al., 1997). Other adhesins such as the M agglutinin and AIDA-I adhesin, a plasmid-encoded outer membrane protein involved in diffuse adherence of certain types of *E. coli* (Benz and Schmidt, 1989), are commonly present.

Colonial and cultural characteristics; life cycles Depending on the degree of polymerization of the O antigen polysaccharide, the phenotypes of strains growing on agar media are described as smooth (S) or rough (R). S forms, which usually grow on nutrient agar as convex, glistening, moist, gray colonies (2–3 mm diameter) with a defined edge or in fluid medium as turbid growth, have developed polysaccharide side chains, while R forms, which usually grow as flat, dry, dull, wrinkled colonies (1–5 mm diameter) with a blurred edge on agar and agglutinate spontaneously in fluid media, have lost their polysaccharide side chains by mutation (Lüderitz et al., 1966). There are intermediate forms between these extremes. Mucoid and slime-producing forms occur. *E. hermannii* is yellow-pigmented, as are half of the described *E. vulneris* strains. See Raetz (1996) and Hull

(1997) for a discussion of the chemical structure, biosynthesis, and biological/virulence properties of LPS.

Nutrition and growth conditions Of the range of temperatures, pH values, water activities, and pressures over which bacterial growth can occur, *E. coli* strains survive and grow over the mid-range (15–45°C) of these environmental conditions. Most strains can grow over a temperature range of approximately 40°C. The normal temperature range for balanced growth extends from 21° to 37°C; however, strains that can grow at temperatures as low as 7.5–7.8°C (Shaw et al., 1971) and as high as 49°C (Herendeen et al., 1979) have been described. A minimum growth temperature for *E. vulneris* of 1.6°C (0.8–2.6°C) was reported for a strain isolated from refrigerated meat (Ridell and Korkeala, 1997). *E. coli* is neutrophilic and will grow over the mid-range of pH, from about pH 5.0 to 9.0 (Ingraham and Marr, 1996).

Metabolism and metabolic pathways Glucose and other carbohydrates are fermented with the production of pyruvate, which is further converted into lactic, acetic, and formic acids. Part of the formic acid is split by a complex hydrogenlyase system into equal amounts of CO₂ and H₂.

Phylogeny The establishment of the *Escherichia coli* Collection of Reference (ECOR) in 1984 (Ochman and Selander, 1984) and subsequent studies of the strains in ECOR and comparisons with other *E. coli* strains have contributed substantially to our understanding of the evolution and population structure of *E. coli*. ECOR is a collection of 72 strains from humans and 16 other mammalian species from various geographical areas that have been grouped into five main groups—A, B (comprising subgroups B1 and B2), C, D, and E—according to their electrophoretic types and enzyme allele (allozyme) profiles, based on the results of multilocus enzyme electrophoresis (MLEE) (Selander et al., 1986). The original data for 35 enzymes (Selander et al., 1987) have been expanded to include allozymes of four esterase loci (Goullet and Picard, 1989), and, based on allelic variation at 38 enzyme-encoding loci, multilocus genotypes have been used to construct a dendrogram based on the neighbor joining algorithm (Herzer et al., 1990), demonstrating the clonal structure of the species. The phylogenetic groups are distinguishable but not identical by random amplified polymorphic DNA (RAPD), restriction fragment length polymorphism (RFLP) of *rrn* genes (ribotyping) (Desjardins et al., 1995), and to a lesser extent by repetitive-element PCR (rep-PCR) fingerprinting using ERIC2 and BOXA1R primers (Johnson and O'Bryan, 2000). Generating a phylogenetic tree of the ECOR and 15 O157 strains by fluorescent amplified-fragment length polymorphism (FAFLP) demonstrated close correlation with the MLEE groups of ECOR and placed the STEC/VTEC O157 strains on an outlier branch (Arnold et al., 1999). Thus, there is sufficient evidence that the ECOR strains broadly represent genotypic variation in clonal groups of *E. coli* in spite of the fact that many isolates are commensal forms from healthy carriers and that only 11 out of the 72 strains are from human disease: 1 strain from a case of asymptomatic bacteriuria, 4 from acute cystitis, and 6 from acute pyelonephritis. Other MLEE studies have showed clonal relationships of diarrheagenic *E. coli*, and a collection of 78 diarrheagenic *E. coli* (DEC) representing 15 clonal groups has been established (Whittam et al., 1993). A similar collection of STEC/VTEC strains has been collected by the STEC Center based at the National Food Safety and Toxicology Center at Michigan State University and is designed to facilitate research on the Shiga/Verocytotoxin producing *E. coli* by providing a standard

reference collection of well-characterized strains and a central, on-line accessible database.*

Numerous studies have indicated that *E. coli* and the four named species of *Shigella* should be regarded as being one species (Brenner et al., 1972a, b, 1973a; Goullet, 1980; Ochman et al., 1983; Whittam et al., 1983; Hartl and Dykhuizen, 1984; Karaolis et al., 1994; Stevenson et al., 1994; Whittam, 1996; Pupo et al., 1997). Findings from MLEE studies combined with those from *mdh* (malate dehydrogenase) housekeeping gene sequence studies (Pupo et al., 1997) and ribotyping (Rolland et al., 1998) confirm that the genus *Shigella* comprises a group of closely related pathogenic *E. coli* strains and indicate that *Shigella*, EIEC, and other diarrheagenic *E. coli* strains do not have a single evolutionary origin, but are derived from different ancestral strains many times. Furthermore, pathogenic strains belonging to pathogenic groups of EIEC, EPEC, and ETEC were found to be closely related to ECOR group A strains by MLEE and have *mdh* sequences identical to five ECOR strains from group A, which is thought to represent commensal strains. It has been suggested that any *E. coli* strain may acquire virulence factors from numerous sources, including plasmids, bacteriophages, and other mobile DNA elements from a large pool of strain-specific genes whose origin could be outside the species boundaries, and that this is how a commensal form is turned into a pathogenic form (Pupo et al., 1997; Hurtado and Rodriguez-Valera, 1999; Donnenberg and Whittam, 2001). In their analysis, Lawrence and Ochman (1998) surmised that about 10% of the *E. coli* K-12 genome consists of genes that were acquired in over 200 events of lateral gene transfer, which occurred subsequent to the divergence of *E. coli* and *Salmonella* some 100 million years ago. While mutations have contributed substantially to the heterogeneity of *E. coli*, the importance of these recombinational events should not be underestimated, and transfer of smaller or bigger segments of DNA between different clonal lineages and other species has probably contributed more to the evolution of *E. coli* than anyone could have imagined. Ongoing and future studies will be directed toward increasing our understanding of the principles that govern gene patterns and the relations between individual traits in terms of their significance for virulence and host adaptation. The combined efforts should contribute to a general IDEA (Index of Diversity in Evolution and Adaptation) that will bridge the phylogenetic approach based on the clonal concept on one side and the horizontal transfer of gene cassettes on the other side.

Mutants, plasmids, phages, and bacteriocins The fact that *E. coli* mutants can easily be produced in the laboratory has substantially contributed to our understanding of many genetic mechanisms, ranging from the characterization and function of an individual gene to the description of complex operons. Plasmids carrying resistance genes (R plasmids) have been used as vectors and introduced into both laboratory strains and wild-type strains, and phages and other mobile elements such as transposons have been used widely in both research and applied biotechnology. Virulence genes are often found on plasmids (see pathogenicity section below). *E. coli* strains produce a variety of secreted antibiologically active polypeptides, bacteriocins, and microcins, which have the ability to kill or inhibit competing bacterial strains. Colicins, encoded by plasmids of *E. coli*, act on other

*Editorial Note: At the time of publication, this information could be obtained at <http://www.shigatox.net>.

E. coli or closely related bacteria that do not carry that particular Col plasmid. Small lipoproteins are important components of the secretory apparatus, which facilitates release of colicin and plasmid- and phage-encoded proteins across the outer membrane. For additional information on *E. coli* mutants, plasmids, and phages see Campbell (1996), Bachmann (1996), Helinski et al. (1996), and Harwood (1993).

Antigenic structure

O antigens The main aspect of this analysis is the O antigen determination based on antigenicity of the LPS; O group designations run from O1 to O173 but O groups O31, O47, O67, O72, O93, O94, and O122 have been removed. Also included are the provisional O groups OX3 and OX7 listed by Ewing (1986b), which will receive the designations O174 and O175, respectively. An additional six new O groups representing STEC/VTEC strains are currently being investigated and will receive O designations O176 through O181 (Scheutz, unpublished data). Subtypes exist within most O groups and are designated ab, ac, etc., e.g., O128ab and O128ac. Many of these O antigens cross-react with other O antigens and to some extent to K antigens within *E. coli*, with other members of the genus and with other enterobacteria. Recent molecular typing using primers just outside the O antigen gene cluster (*rfb*) of 148 representative O groups observed unique amplified fragments for each O group with sizes ranging from 1.7 to 20 kb (Coimbra et al., 2000). Subsequent *Mbo*II digestion of PCR-amplified products resulted in clearly identifiable and reproducible O patterns for the great majority of O groups with a variation of band numbers for each pattern ranging from 5 to 25. Computer analysis identified a total of 147 O patterns and allowed subdivision of 13 O groups. However, two or more O groups shared a pattern among 13 other O patterns. The restriction method (*rfb*-RFLP) is more rapid and may prove to be more sensitive than conventional serotyping since 100% of strains are typeable, particularly those that are O rough or nonagglutinating. Additionally, it should facilitate the typing of strains outside the existing O antigen scheme, which is restricted to include only O groups of clinical, epidemiological, or scientific relevance. The success and general application of such a typing scheme will require international collaboration to develop standardized methods for generating, comparing, and maintaining a database of O patterns.

K antigens The K antigens are the acidic capsular polysaccharide (CPS) antigens. K antigens may be separated into two distinct groups designated group I and group II. Group I antigens, which are composed of high-molecular mass (>100 kDa) CPS, are only found in strains with O groups O8, O9, O20, and O101, and are expressed at both 18° and 37°C. Group I antigens are subdivided according to the absence (IA) or presence (IB) of amino sugars on their CPS. The CPSs of group IA antigens share structural identity or resemblance to those from *Klebsiella* spp., whereas the CPSs of group IB antigens share no structural resemblance to those from other bacteria. Representative strains expressing group IA antigens do not contain the *rol* (*cll*) gene encoding the regulator of lipopolysaccharide O-chain length, whereas a similar subset of strains expressing group IB antigens contains the *rol* gene (Dodgson et al., 1996). Portions of the CPSs of some group I antigens are attached to the lipid A-core in a form that has been designated K_{LPS}, which will behave similarly to the traditional O antigens. A good example of a group I K_{LPS}

is K84, which may be operationally defined as an O antigen and was originally designated as O93.

Group II antigens, which are composed of low molecular mass (<50 kDa) CPS, are found primarily in strains with O groups that are associated with extraintestinal disease. The CPSs of many group II antigens have structural resemblance or near identity to those from Gram-positive bacteria. These antigens differ widely in composition and structural features and may be divided into subgroups based on their acidic components. Twenty to fifty percent of the CPS chains are bound to phospholipids. They were originally thought to be temperature dependent, i. e., only expressed at 37°C. However, K2, K3, K10, K11, K19, K54/K96, and K98, which are tentatively classified as group I/II antigens (Finke et al., 1990), show no temperature regulation of their capsules and, like group I antigens, do not depend on an elevated CMP-KDO concentration for capsule expression. Based on genetic data, a subset of the group I/II antigens (K3, K10, and K54/K96) has been designated group III antigens (Pearce and Roberts, 1995). A number of K antigens are closely related (indicated below by “~”) or identical (indicated below by “=”). The CPSs of some group IB antigens are structurally identical to the side chains of O antigens and are only considered as K antigens when co-expressed with another authentic O antigen. The following 60 different K antigens are recognized: K1, K2a/ac, K3, K4, K5, K6, K7 (=K56), K8, K9 (=O104), K10, K11, K12 (=K82), K13 (~K20 and ~K23), K14, K15, K16, K18a, K18ab (=K22), K19, K24, K26, K27, K28, K29, K30, K31, K34, K37, K39, K40, K41, K42, K43, K44, K45, K46, K47, K49 (=O46), K50, K51, K52, K53, K54 (~K96), K55, K74, K84, K85ab/ac (=O141), K87 (=O32), K92, K93, K95, K97, K98, K100, K101, K102, K103, KX104, KX105, and KX106. The inclusion of an “X” before the number represents a temporary K antigen designation. A description of the serology, chemistry, and genetics of *E. coli* O and K antigens is given by Ørskov et al. (1977).

H antigens Flagellar or H antigens make up the third main group of serotyping antigens. A total of 56 H antigens have been described, but two, H13 and H22, have been removed as being *C. freundii*, and H50 has been withdrawn because it is identical to H10. Cross-reactions are also seen between the H antigens.

Fields et al. (1997) described a tentative molecular method for the differentiation of flagellar antigen groups in *E. coli* based on restriction fragment length polymorphisms (RFLP) in *fliC* using the restriction enzyme *Rsa*I. A wide variety of *fliC* restriction fragment patterns was reported among isolates of 53 different flagellar antigen groups; the majority of the RFLP patterns observed corresponded to a unique H antigen group. Limited numbers of RFLP patterns were observed among members of some H groups, suggesting that the sequence of *fliC* within certain H groups is fairly well conserved. Four patterns were observed among strains expressing the H7 antigen. Interestingly, the same pattern was detected among all *E. coli* strains of serotype O157:H7 and 16 of 18 of serotype O55:H7, reflecting the common lineage of these strains observed by multilocus enzyme typing (Whittam et al., 1993). Nevertheless, sequencing of a total of 20 H7 *fliC* genes, representing 10 different serotypes (Reid et al., 1999; Wang et al., 2000), revealed a notable polymorphism in the *fliC* gene and identified 10 sequences with differences ranging from 0.06% to 3.12%. Recently, a collection of reference strains representing 48 H types was resolved into 62 patterns (F types) using *Hha*I restriction of the *fliC* gene (Machado et al., 2000). A single F type was associated with each of 39 H types and more than one

F type was associated with the other nine H types. Antigenically related H12 and H45 gave a single F type. The determination of *HhaI*-*fliC* F types could allow deduction of all H types and subdivision of some of these. The two above-mentioned molecular typing methods hold promise of a rapid, more refined and specific typing scheme for the H antigens, which may also be helpful in determining phylogenetic relatedness between different clones of *E. coli*. Furthermore, molecular typing has the advantage of allowing typing of nonmotile strains or of strains that do not (sufficiently) express the immunoreactive H antigen, and is likely to expand the present number of significantly different H types. The observed polymorphism in a single determinant, such as the H7 *fliC* gene, stresses the importance of solid and extensive validation of molecular typing with reference to the existing serotyping scheme, and calls for caution in the interpretation of patterns obtained by DNA fingerprinting methods.

Serotyping Subdivision of *E. coli* can be carried out in many ways, but serotyping remains one of the most useful ways to subdivide the species on a global basis. This typing method is based on the many antigenic differences found in structures on the bacterial surface. A serotype is recorded in the following way: O18ac:K1:H7 or O111:H2 (the latter antigenic formula indicates that K antigens are not present in the strain). MR fimbriae, which are present only in some, often pathogenic, serotypes, can also be used for the serological characterization (Ørskov et al., 1977, 1980b; Ørskov and Ørskov, 1990) in which case the complete serotype is recorded as O4:K3:H5; F13 or O147:H19; F4ac. Serotyping procedures are described in Gross and Rowe (1985) and Ørskov and Ørskov (1984a).

Even though complete serotyping involving the many known O, K, H, and F antigens has been carried out in only a very few laboratories, it is well known that the existing number of serotypes is very high.

Antibiotic or drug sensitivity Like other Gram-negative bacteria, *E. coli* is intrinsically resistant to hydrophobic antibiotics, such as macrolides, novobiocins, rifamycins, actinomycin D, and fusidic acid (Nikaido, 1996). The structure of the outer membrane of *E. coli* and its role in mediating intrinsic resistance to these molecules was reviewed by Nikaido (1996). Resistance to these compounds is attributed, in part, to the low permeability of the outer membrane bilayer to lipophilic solutes; however, active efflux mechanisms may have a synergistic effect on resistance in certain cases (Nikaido, 1996).

Acquired resistance to aminoglycosides, beta-lactams, chloramphenicol, macrolides, sulfonamides, tetracycline, and trimethoprim has been described for *E. coli* strains (reviewed by Quintiliani and Courvalin, 1995). Acquired resistance can develop by four distinct mechanisms: alteration of the target site, enzymatic detoxification of the antibiotic, decreased drug accumulation, and bypass of an antibiotic-sensitive step. The first three mechanisms may be mediated by chromosomal mutations or the acquisition of plasmids carrying resistance genes. The fourth mechanism is primarily attributable to the horizontal transfer of antibiotic resistance genes on a plasmid or transposon. The biochemical mechanisms and genetic basis of acquired resistance to antimicrobial agents of clinical importance was discussed by Quintiliani and Courvalin (1995). Genetic methods for the detection of antibacterial resistance genes were reviewed by Tenover et al. (1995).

Pathogenicity *E. coli* is a natural and essential part of the bacterial flora in the gut of humans and animals. Most *E. coli* strains are nonpathogenic and reside harmlessly in the colon; however, certain serotypes or clones play an important role in both intestinal and extraintestinal diseases. The diverse pathogenesis of this bacterium in apparently healthy individuals is largely attributable to its possession of a variety of specific virulence factors. In hosts with compromised defenses, *E. coli* can also be an excellent opportunistic pathogen.

***E. COLI* IN HUMAN INTESTINAL DISEASES** *E. coli* strains isolated from intestinal diseases have been grouped into at least six different main categories based on epidemiological evidence, phenotypic traits, clinical features of the disease they produce, and specific virulence factors. The currently recognized categories of diarrheagenic *E. coli* include enteropathogenic *E. coli* (EPEC) (actually a subgroup of attaching and effacing *E. coli* (A/EEC) defined as *eae* positive *E. coli* belonging to both the classical EPEC serotypes and nonclassical EPEC serotypes), enterotoxigenic *E. coli* (ETEC), enteroinvasive *E. coli* (EIEC), enteroaggregative *E. coli* (EAggEC), diffusely adherent *E. coli* (DAEC), and Shiga toxin-producing *E. coli* (STEC), which are also referred to as Vero cytotoxin-producing *E. coli* (VTEC). These categories are reviewed below with emphasis on their virulence factors.

ENTEROPATHOGENIC *E. COLI* (EPEC) Enteropathogenic *E. coli* was the first category of diarrheagenic *E. coli* to be recognized. The term *enteropathogenic E. coli* (EPEC) was originally used to refer to strains belonging to a limited number of O groups that were epidemiologically associated with infantile diarrhea (Neter et al., 1955). This rather imprecise definition, which allowed for the inclusion of a heterogeneous group of pathogens, was used for decades and became increasingly problematic as groups of *E. coli* that could produce diarrheal disease by the production of enterotoxins or invasion of intestinal epithelial cells were recognized. The confusion generated by the discovery of new pathogenic groups of *E. coli* and the findings that EPEC strains, which lacked the virulence properties of these newly recognized groups, caused disease in adult volunteers (Levine et al., 1978) prompted researchers in 1982 to define EPEC as "diarrheagenic *E. coli* belonging to serogroups epidemiologically incriminated as pathogens but whose pathogenic mechanisms have not yet been proven to be related to either heat-labile enterotoxins or heat-stable enterotoxins or *Shigella*-like invasiveness" (Edelman and Levine, 1983). As more was learned about the strains associated with infant diarrhea, the definition was refined to include only certain O:H serotypes associated with illness. Table BXII.γ.199 lists some of the O:H serotypes that have been regarded for many years as EPEC. Since 1982, advances in our understanding of the molecular aspects of EPEC pathogenesis have allowed researchers to move beyond the serologic markers that correlate with disease to develop a definition based on pathogenic characteristics. A definition adopted in 1995 identified the most important characteristics of EPEC as its ability to cause attaching and effacing (A/E) histopathology and its inability to produce Shiga toxins (Kaper, 1996). The pathogenesis of EPEC is highlighted below and has been reviewed in detail by Nataro and Kaper (1998) and Williams et al. (1997).

The first advance in understanding the pathogenesis of EPEC infection was the discovery that EPEC strains adhere to HEp-2 cells in cell culture (Cravioto et al., 1979) in a distinctive pattern termed localized adherence (LA) (Scaletsky et al., 1984). Expression of the LA phenotype in EPEC requires a plasmid re-

TABLE BXII.γ.199. O:H serotypes regarded as classical and newly recognized EPEC O:H serotypes^{a,b}

O group	H antigen ^c	Comments
O26	H ⁻ ; H11	O26:H ⁻ and O26:H11 may also be STEC/VTEC ^d (Levine et al., 1987; Scotland et al., 1990; Bitzan et al., 1991)
O55	H ⁻ ; H6; H7	O55:H7, H10 and H ⁻ may also be STEC/VTEC (Dorn et al., 1989)
O86	H ⁻ ; H8; H34	O86:H ⁻ may also be EAggEC (Albert et al., 1993b; Tsukamoto and Takeda, 1993; Schmidt et al., 1995b; Smith et al., 1997b) O86:H8 is a new <i>eae</i> - and <i>bfpA</i> -positive type isolated in Denmark (Scheut, unpublished data)
O88	H ⁻ ; H25	(Tsukamoto et al., 1992)
O103	H2	New EPEC type
O111	H ⁻ ; H2; H7; H12	O111:H ⁻ may also be STEC/VTEC (Dorn et al., 1989; Bitzan et al., 1991; Caprioli et al., 1994; Cameron et al., 1995; Allerberger et al., 1996) or EAggEC (Scotland et al., 1991, 1994; Tsukamoto and Takeda, 1993; Chan et al., 1994; Schmidt et al., 1995b; Monteiro-Neto et al., 1997; Morabito et al., 1998)
O114	H ⁻ ; H2	
O119	H ⁻ ; H2; H6	
O125ac	H ⁻ ; H6	O125 may also be EAggEC (Tsukamoto and Takeda, 1993; do Valle et al., 1997; Smith et al., 1997b)
O126	H ⁻ ; H2; H21; H27	
O127	H ⁻ ; H6; H21; H40	
O128ab	H ⁻ ; H2; H7; H12	O128:H2 may also be STEC/VTEC (Beutin et al., 1993a)
O142	H ⁻ ; H6; H34	
O145	H ⁻ ; H45	New EPEC type
O157	H ⁻ ; H8; H16; H45	New EPEC types
O158	H ⁻ ; H23	

^aData from Cravioto et al. (1979), Levine and Edelman (1984), Levine et al. (1985), Scaletsky et al., 1985, Robins-Browne (1987), Gomes et al. (1989b), Knutton et al. (1989, 1991), Scotland et al. (1989, 1992, 1996), Ørskov and Ørskov (1992), Donnenberg (1995).

^bO18:H⁻, H7, H14; O26:H34 and O44:H34 have also been listed but only in Knutton et al. (1991). O18 strains are probably not EPEC (Knutton et al., 1989; Ørskov and Ørskov, 1985). O44:H18 is now considered to belong to the group of enteroaggregative *E. coli* (Smith et al., 1994).

^cNonmotile strains of *E. coli* are regarded as descendants of motile strains that have lost their motility by mutation(s). Their original H antigen was often deduced from comparison of biochemical reactions (Kauffmann and Dupont, 1950; Staley et al., 1969).

^dAbbreviations: EPEC, enteropathogenic *E. coli*; STEC/VTEC, Shiga toxin-producing *E. coli* /Vero cytotoxin-producing *E. coli*; EAggEC, enteroaggregative *E. coli*.

ferred to as the EPEC adherence factor (EAF) plasmid (Baldini et al., 1983). A 1-kb DNA probe originally thought to encode the EPEC adherence factor (EAF) necessary for LA was cloned from this plasmid (Nataro et al., 1985a) and has been used extensively as a marker to study the prevalence of EPEC infections (Nataro et al., 1985a; Echeverria et al., 1987, 1991; Gomes et al., 1989a, b; Moyenuddin et al., 1989; Senerwa et al., 1989; Cravioto et al., 1991; Kain et al., 1991; Strockbine et al., 1992; Begaud et al., 1993). EAF plasmids have been found in many EPEC serotypes and range in size from 26 to 76 MDa (Scotland et al., 1989) but typically are 50–70 MDa. Evidence supporting a role for the EAF plasmid in pathogenesis was provided by feeding studies showing that volunteers ingesting a plasmid-cured EPEC strain developed less diarrhea than those ingesting the plasmid-containing parental strain (Levine et al., 1985). Genes at two loci are necessary for expression of the LA phenotype: a cluster of 14 genes on the EAF plasmid involved in biogenesis of the bundle forming pilus (BFP), a type-IV pilus, which includes genes encoding bundlin (*bfpA*), the major structural subunit of the type-IV pilus, a prepilin peptidase, which processes pre-bundlin to its mature form, and 12 other proteins, and *dsbA* on the chromosome (Donnenberg et al., 1997).

A hallmark of the histopathology of EPEC infections is the presence of attaching and effacing (A/E) lesions in the intestinal tract. On electron micrographs of jejunal biopsies from children infected with EPEC, the bacteria are seen intimately attached to the epithelial cells on cup-like pedestals composed of depolymerized cytoskeletal proteins (Knutton et al., 1987). Microvilli

are disrupted as a result of the cytoskeletal rearrangements and effaced by vesiculation (Fig. BXII.γ.192). The intimate attachment of EPEC is mediated by a protein known as intimin, which is a 94-kDa outer membrane protein encoded by the *eae* gene (*E. coli* attaching and effacing) (Jerse and Kaper, 1991). A 1-kb fragment of the *eae* gene referred to as CVD434 has been cloned (Jerse et al., 1990) and used to screen for attaching and effacing *E. coli* A/EEC (Bokete et al., 1997), and to characterize enteropathogenic *E. coli* (Scotland et al., 1996). Intimins belong to a growing family of proteins. In human EPEC strains, intimins α , β , γ , λ (δ not now used, although it is in the literature because it is a variant of the ϵ variant), and ϵ derivatives have been found to be serotype-specific and exhibit specific different binding affinities (Agin and Wolf, 1997). Intimins are also found in rabbit EPEC RDEC-1 strains, A/EEC strains from dogs (Beaudry et al., 1996), pigs (Zhu et al., 1995), and in *C. freundii* (Int_{CF}) and *Citrobacter rodentium* (Int_{CR}). Five strains initially identified as *Hafnia alvei* were reported to contain the *eae* gene (Albert et al., 1992), and a protein referred to as Int_{HA} has been characterized and compared to the above intimins. However, the strains are actually unusual biotypes belonging to the genus *Escherichia* (Janda et al., 1999), most likely *E. coli*. The *eae* gene is only one of many genes located on a pathogenicity island (PAI) known as the locus of enterocyte effacement (LEE) (McDaniel et al., 1995).

EPEC strains secrete at least four proteins, Esps for EPEC secreted proteins, encoded by LEE. EspA, B, D, and Tir proteins are secreted via the type III apparatus and are required for attaching and effacing activity. Protein secretion, the transcription

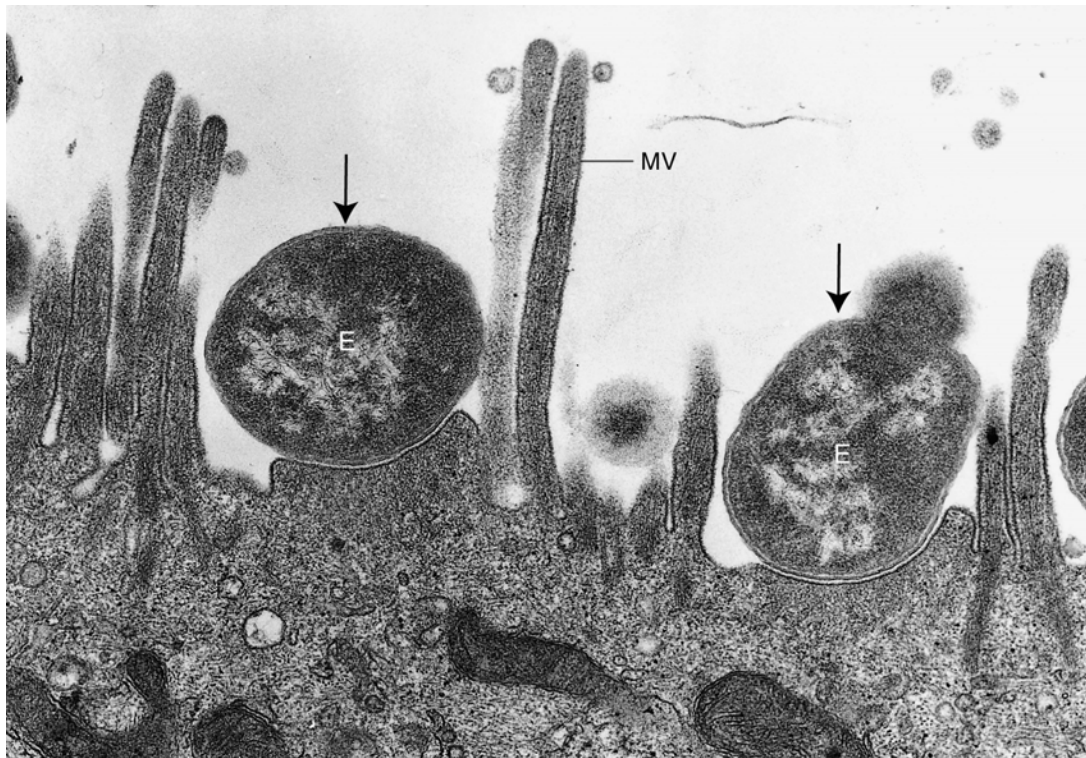


FIGURE BXII.γ.192. EM of cultured human intestinal mucosa infected with EPEC strain E2348 (*E*) for 8 h. Bacteria are seen intimately attached to the epithelial cells on cup-like pedestals composed of depolymerized short filament cytoskeletal proteins (arrows). Brush border microvilli (MV) are disrupted as a result of the cytoskeletal rearrangements and effaced by vesiculation. (Reproduced with permission from S. Knutton et al., *Infection and Immunity* 55: 69–77, 1987, ©American Society for Microbiology.)

of the intimin gene (*eae*), and the synthesis of the bundle-forming pilus are all affected by growth conditions (Haigh et al., 1995; Jarvis et al., 1995; Kenny and Finlay, 1995) and a plasmid-encoded regulatory region. The plasmid locus is either referred to as the *per* (plasmid-encoded regulator) genes (*perABCD*) (Gómez-Duarte and Kaper, 1995) or as an integrated part of the bundle-forming pilus (*bfp*) operon, genes *bfpT*, *V*, and *W* (Tobe et al., 1996).

EPEC interferes with normal pathways of signal transduction in epithelial cells. EPEC will induce tyrosine phosphorylation; phosphorylation of myosin light chain, vinculin, and α -actinin; *in vitro* release of intracellular calcium; phospholipase C activity, resulting in elevated levels of inositol phosphates and diacylglycerol, in turn activating protein kinase C; reactions that induce host cell proteins to initiate cytoskeletal rearrangement; and bacterial uptake (Rosenshine et al., 1992; Manjarrez-Hernandez et al., 1996).

EPEC is not invasive in the same way as *Shigella*, *Yersinia*, and *Salmonella* (Geyid et al., 1996), although some EPEC strains may exhibit *in vitro* invasion of epithelial cells at levels comparable to those seen in EIEC (Donnenberg et al., 1989; Pelayo et al., 1999). A 4.5-kb fragment from a large so-called EPEC plasmid of an O111:H[−] strain negative for both EAF and *eae* will confer epithelial cell invasivity and the attaching and effacing ability on a noninvasive laboratory strain (Fletcher et al., 1990, 1992). Only a smaller proportion of EPEC strains hybridize with the cloned fragment.

Of the EPEC O:H serotypes shown in Table BXII.γ.199, only 16 of the possible 175 recognized O groups are represented and

only a few H antigens, e.g., H2, H6, H7, H12, H21, and H34, occur among several of these 16 O groups. The more recently described serotypes possessing EPEC-associated virulence markers have been added to the list. Some of these are O86:H8, O88:H25, O103:H2, O127:H40, O142:H34, O145:H45, O157:H8, O157:H16, and O157:H45.

ENTEROTOXIGENIC *E. COLI* (ETEC) Enterotoxigenic *E. coli* strains are important causes of diarrhea in both humans and domestic animals. ETEC strains do not invade epithelial cells but produce one or more enterotoxins that are either heat-labile (LT), which is closely related to cholera toxin, or heat-stable (ST) (Cohen and Giannella, 1995). LT-I and ST are plasmid-encoded (LT-II is chromosomally encoded) and found alone or together, and are often associated with a limited number of O:K:H serotypes and O groups. Table BXII.γ.200 shows some of the most common serotypes from humans. These enterotoxins cause intestinal secretion either by activation of guanylate (ST) or adenylate (LT) cyclase and are subdivided based on their biological activities, receptors, and chemical and antigenic properties. STh and STp are used to indicate strains of human or porcine origin. The gene encoding another heat-stable enterotoxin called enteroaggregative heat-stable toxin 1 (EAST1), originally thought to be produced by EAggEC only, may also be present in addition to the STa gene. The different variants of enterotoxins are summarized in Table BXII.γ.201.

There is host specificity among ETEC strains causing diarrhea in humans or different species of domestic animals. This is mainly due to the specific recognition between bacterial colonization factors and the epithelial receptors during host-parasite inter-

TABLE BXII.γ.200. O:(K):H serotypes of human ETEC

O group	(K):H antigen
O6	H ⁻ K15:H16
O7	H ⁻ H18
O8	K47:H ⁻ K25:H9; K40:H9; H10; K87:H19
O9	H ⁻ K9; K84:H2
O11	H27
O15	H11; H15; H45
O17	K23:H45; H18
O20	H ⁻ H30
O21	H21
O25	H ⁻ K7:H42; H16
O27	H7; H20; H27
O29	H?
O48	H26
O55	H7
O56	H ⁻
O63	H12; H30
O64	H ⁻
O65	H12
O71	H36
O73	H45
O77	H45
O78	H ⁻ K2; H11; H12
O85	H7
O86	H2
O88	H25
O105	H?
O114	H ⁻ H21
O115	H ⁻ H21; H40; H51
O119	H6
O126	H ⁻ H9; H12
O128ac	H7; H12; H19; H21
O133	H16
O138	K81
O139	H28
O141	H ⁻ H4
O147	H?
O148	H28
O149	H4; H10; H19
O153	H10
O159	H ⁻ H4; H5; H12; H20; H21; H27; H34; H37
O166	H27
O167	H5
O?	H2; H10; H28; K39:H32

TABLE BXII.γ.201. Variants of enterotoxins found in ETEC strains^a

Toxin type	Subtypes		Comments
<i>LT: Heat-labile enterotoxins</i>			
LT-I	LTp (LTp-I)	LTh (LTh-I)	Associated with disease in both humans and animals
LT-II	LT-IIa	LT-IIb	No specific association with disease. Rare in human isolates
<i>ST: Heat-stable enterotoxins</i>			
STa (STI)	STp (STIa)	STh (STIb)	Produced by ETEC and several other Gram-negative bacteria. In ETEC, ST is often found together with the genes for EAST1.
STb (STII)			Induces histological damage in the intestinal epithelium. Most often found in porcine ETEC.

^aSuffixes commonly used are: h, human variant; p, porcine variant.

action. Most of the fimbriae found in ETEC are typically MR fimbriae. In human ETEC strains at least 21 different surface structures called CS, for coli surface antigens, or CFA, for colonization factor antigens, usually plasmid encoded, have been described (Gaastra and Svennerholm, 1996). In animals, the most common fimbriae are F4 (which was originally described as K88), F5 (originally named K99), F6 (originally named 987P), F17, F41, F42, CS31A, F141, and F165. These fimbriae are found in enterotoxigenic strains from newborn piglets, pigs, lambs, and newborn calves. A close association between the EAST1 toxin and CS31A, which is related to F4, among pathogenic bovine *E. coli* has been suggested (Bertin et al., 1998).

More recently, the fimbrial F18 antigen (provisionally designated F107, 2134P, or 8813) has been found in both porcine ETEC and STEC/VTEC strains. It is characteristic that the fimbriae usually are found in a limited number of serotypes. The antigenic variants of F18 fimbriae (F18ab and F18ac) are biologically distinct. F18ab fimbriae are expressed poorly both *in vitro* and *in vivo* and are frequently linked with the production of Stx2e/VT2e and O group O139, while F18ac are more efficiently expressed *in vitro* and *in vivo* and most often are linked with enterotoxin (STa, STb) production, and O groups O141 and O157.

ENTEROINVASIVE *E. COLI* (EIEC) Enteroinvasive *E. coli* are very similar to *Shigella*. Like *Shigella*, they are capable of invading and multiplying in the intestinal epithelial cells of the distal large bowel in humans. Approximately two thirds of EIEC strains are lactose negative and virtually all are lysine negative. Multiple chromosomal and plasmid genes are associated with virulence (Acheson and Keusch, 1995). EIEC are restricted to a very limited number of serotypes, most of which are nonmotile (Table BXII.γ.202).

ENTEROAGGREGATIVE *E. COLI* (EAggEC) Enterotoxigenic *E. coli* are characterized by a distinct aggregative adherence (AA) pattern to HEp-2 cells *in vitro* first described by Nataro et al. (1987). This pattern is distinguished by the prominent autoagglutination of bacterial cells to each other, to the surface of the HEp-2 cells, as well as to the glass cover slip in a characteristic layering best described as a "stacked brick" configuration. The AA pattern is plasmid-mediated (Nataro et al., 1985b) and was suspected to be a putative agent of diarrheal disease as early as 1988 (Vial et al., 1988). Volunteer studies with EAggEC have indicated that certain types will cause diarrhea and other enteric symptoms including borborygmia and cramps (summarized in

TABLE BXII.γ.202. O:H serotypes of EIEC

O group	H antigen
O28ac	H ⁻
O29	H ⁻
O112ac	H ⁻
O115	H ⁻
O121	H ⁻
O124	H ⁻ ; H7; H30; H32
O135	H ⁻
O136	H ⁻
O143	H ⁻
O144	H ⁻ ; H25
O152	H ⁻
O159	H2
O164	H ⁻
O167	H ⁻ ; H4; H5
O173	H ⁻

Nataro, 1995), and epidemiological studies have implicated EAggEC as a cause of travelers' diarrhea (Brook et al., 1994; Scotland et al., 1994). Only about half of the case-control studies that have been carried out find significantly higher isolation rates in cases than in controls (summarized by Law and Chart, 1998). Most recently, EAggEC have been associated with acute and chronic diarrhea and abdominal colic in young children in Germany (Huppertz et al., 1997), and with four outbreaks of gastroenteritis in the UK (Smith et al., 1997b). A DNA probe, referred to as CVD432, has been cloned from the plasmid of strain O42, serotype O44:H18 (formerly an EPEC O:H serotype) and used to identify EAggEC (Baudry et al., 1990). A plasmid encoded flexible, bundle-forming fimbrial structure, designated aggregative adherence fimbria I (AAF/I), is encoded by two regions (Nataro et al., 1992, 1993). Region 2 encodes a transcriptional activator of the AraC family of DNA-binding proteins (Nataro et al., 1993). At least one toxin similar to the heat-stable toxin of ETEC, ST designated EAST1, has been identified in an EAggEC strain (Savarino et al., 1991, 1993). The gene encoding EAST1 has been shown to be broadly distributed among diarrheagenic *E. coli*: 100% of 75 O157:H7 STEC/VTEC, 41% of 227 EAggEC, 41% of 149 ETEC, 22% of 65 EPEC, and 38% of 47 *E. coli* from asymptomatic children hybridized with the EAST1 DNA probe (SS126) (Savarino et al., 1996). Despite the common occurrence of the EAST1 gene in many diarrheagenic *E. coli* groups, outbreaks had not been attributed to EAST1-only-producing *E. coli* until recently, when an outbreak in Japan was associated with an O166:H? strain positive for the EAST1 gene (Nishikawa et al., 1999). In Spain, a case-control study suggested that EAST1 positive *E. coli* strains are associated with diarrheal diseases in Spanish children, whereas EAggEC strains are not (Vila et al., 1998). The significance of these findings remains to be established. Many serotypes have been observed and some of the most commonly reported serotypes are listed in Table BXII.γ.203.

DIFFUSELY ADHERENT *E. COLI* (DAEC) Diffusely adherent *E. coli* are defined by the presence of the diffuse adherence (DA) pattern of *E. coli* strains to HEp-2 cells (Scaletsky et al., 1984; Nataro et al., 1985b). A surface fimbriae designated F1845 confers the DA phenotype, and a DNA probe has been cloned (Bilge et al., 1989). Another adhesin (designated AIDA-I) has also been associated with DA of *E. coli* of serotype O126:H27 (Benz and Schmidt, 1989). The role of DAEC in diarrhea is unclear.

TABLE BXII.γ.203. O:H serotypes of the most frequently reported enteroaggregative *E. coli* (EAggEC)^a

O group	H antigen	References listing these serotypes as EAggEC
O3	H2	Albert et al. (1993b)
O15	H18	Vial et al., 1988, Albert et al. (1993b), Tsukamoto and Takeda (1993), Scotland et al. (1994)
O44	H ⁻ ; H18	Scotland et al. (1991, 1994), Tsukamoto and Takeda (1993), Smith et al. (1994), Schmidt et al. (1995b)
O86	H ⁻	Albert et al. (1993b), Tsukamoto and Takeda (1993), Schmidt et al. (1995b), Smith et al. (1997b)
O111	H12; H21	Scotland et al. (1991, 1994), Tsukamoto and Takeda (1993), Chan et al. (1994), Schmidt et al. (1995b), Monteiro-Neto et al. (1997), Morabito et al. (1998)
O125	H9; H21	Tsukamoto and Takeda (1993), do Valle et al. (1997), Smith et al. (1997b)

^aMany other serotypes of EAggEC have been published by several authors.

SHIGA TOXIN-PRODUCING *E. COLI* (STEC) OR VERO CYTOTOXIN-PRODUCING *E. COLI* (VTEC) Shiga toxin-producing *E. coli* or Vero cytotoxin-producing *E. coli* strains are characterized by their ability to produce either one or both of at least two antigenically distinct, usually bacteriophage-mediated cytotoxins referred to as Stx1 or VT1 (first described as Shiga-like toxin I, SLTI) and Stx2 or VT2 (first described as Shiga-like toxin II, SLTII). Whereas STEC/VTEC refers to all *E. coli* strains that produce Stx/VT in culture supernatants (Konowalchuk et al., 1977, 1978), the term *enterohemorrhagic E. coli* (EHEC) has been used to refer to strains that have the same clinical and pathogenic features associated with the prototype organism *E. coli* O157:H7 (Levine, 1987). In practice, EHEC is used to describe a subgroup of STEC/VTEC that causes hemorrhagic colitis.

Shiga toxins or Vero cytotoxins belong to the Shiga toxin family, comprises the following members: Shiga toxin, which is produced by *Shigella dysenteriae* type 1, Stx1/VT1, Stx2/VT2, and Stx2e/VT2e variants (Stxv/VT2v). Stx/VT genes, products, and synonyms are summarized in Table BXII.γ.204. Stx2e/VT2e, one of the variants, is produced by STEC/VTEC strains causing edema disease (Marques et al., 1987), a usually fatal disease, in weanling pigs and referred to as Stx2e/VT2e. Unlike all other Stxs/VTs, this is not cytotoxic to HeLa cells and binds to a different receptor, Gb4. The functional receptor for human Stx/VT is the glycolipid globotriosyl ceramide Gb3 (galactose-α-(1-4)-galactose-β-(1-4)-glucose ceramide) (Lingwood et al., 1987; Waddell et al., 1988) found in human renal endothelial cells (Obrig et al., 1993).

Stxs/VTs inhibit protein synthesis by depurination of adenine in 28S rRNA (N-glycosidases), thus inhibiting the elongation factor 1 (EF-1)-dependent aminoacyl-tRNA binding to 60S ribosomal subunits (Endo et al., 1988). In Vero cells the result is cell death by apoptosis (Inward et al., 1995).

NOMENCLATURE OF SHIGA TOXINS/VEROCYTOTOXINS The recognition and investigation of cytotoxin-producing *E. coli* infections by several groups has resulted in the use of differing systems of nomenclatures for the toxins produced by these bacteria. In 1994, O'Brien et al. (1994) developed a proposal for rationalizing the nomenclature of the *E. coli* cytotoxins. In their proposal, they recommended guidelines for classifying and designating members of the toxin family and stated the toxins shall be referred to by two alternate but interchangeable names, Shiga-like toxins (SLT) and Vero cytotoxins (VT). Two years later, a proposal to simplify the Shiga-like toxin nomenclature was published by Calderwood et al. (1996). In this proposal, the word "like" was omitted and the toxins and genes were renamed to reflect their relationship to Shiga toxin, the prototype toxin for the family. To avoid confusion in the literature, it was suggested that cross-reference to existing VT nomenclature could be used. While the omission of the word "like" has received general acceptance in the scientific community, strong arguments for maintaining the existing phenotype nomenclatures for *E. coli* cytotoxins were immediately put forward (Karmali et al., 1996), and the two systems of nomenclature are still being widely used. Toxins included in the Shiga toxin/Verocytotoxin family share the following properties: DNA sequence homology and operon structure (A subunit gene immediately upstream of the B subunit gene); polypeptide subunit structure (five B subunits to one A subunit in the mature holotoxin); enzymatic activity (N-glycosidases); binding to specific glycolipid receptors and biological properties, including enterotoxicity in ligated rabbit ileal loops;

TABLE BXII-γ.204. Designation of *stx/vtx* genes, their products (Stx/VT)^a, and their previous designations

Toxin type ^b	Toxin gene	Toxin	Prototype organisms (reference)	Serotype of prototype organism	Previous designation or synonym for toxin gene	Previous designation or synonym for toxin
1	<i>stx1/vtx1</i>	Stx1/VT1	EDL933 (O'Brien et al., 1984) H19 (Konowalchuk et al., 1977) H30 (Konowalchuk et al., 1977) PMK1 (Mariani-Kurkdjian et al., 1993)	O157:H7 O26:H11 O26:H11 O103:H2	<i>stx1</i>	SLT-I
	<i>stx1-O103/vtx1-O103</i>	Stx1-O103/VT1-O103				
	<i>stx1-O111-PH/vtx1-O111-PH</i> <i>stx1-OX3/vtx1-OX3</i> <i>stx1-O48/vtx1-O48</i> <i>stx1-O111-CB168/vtx1-O111-CB168</i>	Stx1-O111-PH/VT1-O111-PH Stx1-OX3/VT1-OX3 Stx1-O48/VT1-O48 Stx1-O111-CB168/VT1-O111-CB168	PH (Paton et al., 1993a) 131/3 (Paton et al., 1995a) 94C (Paton et al., 1995a) CB168 (Paton et al., 1995a)	O111:H ⁻ OX3:H8 O48:H21 O111:H ⁻	<i>stx1/PH</i> <i>stx1/OX3</i> <i>stx1/O48</i> <i>stx1/CB</i>	SLT-I/PH SLT-I/OX3 SLT-I/O48 SLT-I/CB
2	<i>stx2/vtx2</i>	Stx2/VT2	EDL933 (O'Brien et al., 1984) E32511 (Schmitt et al., 1991)	O157:H7 O157:H ⁻	<i>stx2</i>	SLT-II
2c	<i>stx2c/vtx2c</i>	Stx2c/VT2c			<i>stx2c</i>	Stx2c/VT2v SLT-IIc VT2v(pKTN1054) VT2v(pKTN1050)
	<i>stx2-O22/vtx2-O22</i> <i>stx2-O157-TK51/vtx2-O157-TK51</i>	Stx2-O22/VT2-O22 Stx2-O157-TK51/VT2-O157-TK51	KY-O19 (Lin et al., 1993a) TK-51 (Lin et al., 1993a)	O22:H ⁻ O157:H7	<i>stx2</i>	SLT-II/OX3 SLT-II/OX3b SLT-II/O48 SLT-II/O111/VT2d-O111 ^c Stx2d/VT2d-Ount Stx2-NV206
	<i>stx2-OX3/a-031/vtx2-OX3/a-031</i> <i>stx2-OX3/b-031/vtx2-OX3/b-031</i>	Stx2-OX3/a-031/VT2-OX3/a-031 Stx2-OX3/b-031/VT2-OX3/b-031	031 (Paton et al., 1992) 031 (Paton et al., 1993b)	OX3:H21 OX3:H21	<i>stx2</i>	SLT-II/OX3 SLT-II/OX3b SLT-II/O48 SLT-II/O111/VT2d-O111 ^c Stx2d/VT2d-Ount Stx2-NV206
2d ^d	<i>stx2-O48/vtx2-O48</i> <i>stx2-O111-PH/vtx2-O111-PH</i> <i>stx2-O118/vtx2-O118</i> <i>stx2-O6/vtx2-O6</i> <i>stx2d/vtx2d</i> <i>stx2d1/vtx2d1</i>	Stx2-O48/VT2-O48 Stx2-O111-PH/VT2-O111-PH Stx2-O118/VT2-O118 ^d Stx2-O6/VT2-O6 Stx2d/VT2d Stx2d1/VT2d1	94C (Paton et al., 1995b) CB168 (Paton et al., 1995b) EH250 (Pierard et al., 1998) NV206 (Bertin et al., 2001) B2F1 (Ito et al., 1990) B2F1 (Ito et al., 1990)	O48:H21 O111:H ⁻ O118:H12 O6:H10 O91:H21 O91:H21	<i>stx2</i>	SLT-II/OX3 SLT-II/OX3b SLT-II/O48 SLT-II/O111/VT2d-O111 ^c Stx2d/VT2d-Ount Stx2-NV206 SLT-IIvh Stx2vha/VT2vha Stx2vh-a/VT2vh-a VT2v-a SLT-IIvha Stx2vhb/VT2vhb Stx2vh-b/VT2vh-b VT2v-b SLT-IIvhb SLT-IIv SLT-IIva ^e SLT-IIe/VTe VT2vp VT2vp1 Stx2ev/VT2ev Stx2vp2/VT2vp2 VTev SLT-IIvhc SLT-IId/VT2d ^d
2e	<i>stx2e/vtx2e</i>	Stx2e/VT2e	412 (Gyles et al., 1988) S1191 (Weinstein et al., 1988)	O139:K12:H1 O139:H1	<i>stx2e</i>	SLT-IIv SLT-IIva SLT-IIe
2f ^f	<i>stx2f/vtx2f</i>	Stx2f/VT2f	H.1.8 (Gannon et al., 1990) T4/97 (Schmidt et al., 2000)	O128:H2 O128:H2	<i>stx2f</i>	VT2vp VT2vp1 Stx2ev/VT2ev Stx2vp2/VT2vp2 VTev SLT-IIvhc SLT-IId/VT2d ^d

^aData compiled from references Bastian et al. (1998), Bertin et al. (2001), Calderwood et al. (1996), Gannon et al. (1990), Ito et al. (1990), W.M. Johnson et al. (1991), Karmali et al. (1996), Kokai-Kum et al. (2000), Lin et al. (1993a, b), Takeda et al. (1993), and Tyler et al. (1991).

^bToxin types are defined according to antigenic variability, differences in toxicity for tissue culture cells and/or animals, their capacity to be activated by mouse elastase, and differences in DNA or amino acid sequences. There are several toxins suffixed by d in the literature: The Stx2d/VT2d toxins of O91:H21 (Melton-Celsa and O'Brien, 1998), the VT2d (= Stx2-O group/strain designation and/or year) variants by Paton et al. (1992, 1993a, 1995b), Pierard et al. (1998), and the SLT-IId/VT2d (= Stx2f/VT2f) toxin produced by strain H.1.8 (serotype O128:H2) as proposed by Gyles (1994). We support the use of the d designation for activatable Stx2/VT2 toxins as proposed by Melton-Celsa and O'Brien, (1998).

^cStx2-O118/VT2-O118 (formerly known as VT2d-Ount) is expected to be nonactivatable based on analysis of the nucleotide sequence (Denis Piérard and Angela Melton-Celsa; unpublished); the original strain has been retyped as O118:H12 (Lohar Beutin and Flemming Scheut, unpublished).

^dThis designation has also been referred to for the Stx2f/VT2f toxin produced by strain H.1.8 (serotype O128:H2).

^eThe nucleotide sequence of the former *stx2ev/vtx2ev* of strain H.1.8 (serotype O128:H2) is nearly identical to the recently published *stx2f/vtx2f* found in strain T4/97 (serotype O128:H2) from feral pigeons (Schmidt et al., 2000). As its nucleotide sequence is distinctly different from those of the other Stx2/VT2 toxins and variants as well as Stx1/VT1, we support the proposal of renaming *stx2ev/vtx2ev* as *stx2f/vtx2f*.

neurotoxicity in mice; and cytotoxicity to receptor-expressing tissue culture cell lines, such as Vero and HeLa cell lines. Classification of the toxins into major toxin types, designated with Arabic numbers, is based on differences that result in no cross-neutralization by homologous polyclonal antisera and no DNA-DNA cross-hybridization of their genes under conditions of high stringency. Toxin subtypes, designated with letters added to the type name, share cross-hybridization of their genes under high stringency but show significant differences in biologic activity, including the capacity to be activated; serologic reactivity; or receptor binding. Table BXII.γ.204 summarizes the currently reported Shiga toxin/Verocytotoxin genes and toxins, prototype organisms, and their previous designations in the literature.

The distinction between toxin types 2c and 2d in Table BXII.γ.204 is based on differences in the A subunit determining whether the toxin is activatable (Stx2d/VT2d) or nonactivatable (Stx2c/VT2c). The other Stx2/VT2 subtypes described in the literature, which have not been tested for all properties, have been tentatively placed together with toxin type 2c, primarily based on similarities in nucleotide sequences that place them in a phylogenetically related Stx2/VT2 cluster including all variants. Based on their degree of overall nucleotide sequence similarity, Stx2/VT2 toxins fall into two phylogenetically distinct groups (Bastian et al., 1998; Piérard et al., 1998). Toxins in group 1, which include Stx2/VT2, Stx2c/VT2c (including variants Stx2-O22/VT2-O22, Stx2-O157-TK-51/VT2-O157-TK-51, Stx2-OX3/b-031/VT2-OX3/b-031, Stx2-O48/VT2-O48), and Stx2d/VT2d, share 99.1–99.2% and 95.9–98.5% nucleotide sequence similarity in their A and B subunits, respectively; while toxins in group 2, which include Stx2-OX3/a-031/VT2-OX3/a-031, Stx2-O111-PH/VT2-O111-PH, and Stx2-O118/VT2-O118, share 96.9–99.9% and 99.6–100% nucleotide sequence similarity in their A and B subunits, respectively. The similarities of individual sequences between group 1 and group 2 are 93.4–96.0% for the A subunits and 86.2–89.3% for the B subunits. The extent of, and differences between, toxicity for tissue culture cells and/or animals and their capacity to be activated are not fully established for all the types. Further analysis against all the criteria necessary to allow definitive placement with appropriate other toxin types is required. In the absence of this information, suffixes have been added after the O group of the source organisms, and, when necessary, the original strain designation. Stx2/VT2 subtype toxins found in the same original strain are suffixed “/a”, “/b” etc. as in Stx2-OX3/a-031/VT2-OX3/a-031 and Stx2-OX3/b-031/VT2-OX3/b-031.

For a brief presentation and practical use of the subtyping of Stx/VT genes with 4 Stx1/VT1 and 9 Stx2/VT2 oligonucleotide DNA probes, and 3 Stx1/VT1 and 8 Stx2/VT2 primer pairs, see Smith et al. (1993), Yamasaki et al. (1996), Piérard et al. (1997), and Bastian et al. (1998).

Like EPEC, some STEC/VTEC strains have been shown to cause attaching and effacing lesions *in vivo* (Hall et al., 1990), in animal models (Francis et al., 1986; Tzipori et al., 1986; Sherman et al., 1988), and *in vitro* (Knutton et al., 1989). Two separate groups have been cloned, sequenced, and characterized the *eae* homologue from VTEC O157:H7 (Beebakhee et al., 1992; Yu and Kaper, 1992). Homology between EPEC and STEC/VTEC sequences was 86% and 83% at the nucleotide and amino acid levels, respectively (Yu and Kaper, 1992), and the STEC/VTEC *eae* sequence was 97% homologous to the EPEC *eae* gene for the first 2200 bp and 59% homologous over the last 800 bp (Beebakhee et al., 1992). Both *eae* sequences show 50% homology to

the central region of the *Yersinia pseudotuberculosis inv* gene (Jerse et al., 1990; Beebakhee et al., 1992), and the predicted amino acid sequence of the STEC/VTEC *eae* gene share 31% identity and 51% similarity with the invasins molecule of *Yersinia pseudotuberculosis* (Yu and Kaper, 1992).

Serotype specific heterogeneity of the *eae* gene in STEC/VTEC strains O55:H7 or H[−], O111:H8 and O157:H7 or H[−], and in O groups O26, O103, and O157 has been demonstrated (Gannon et al., 1993; Louie et al., 1994).

Almost all STEC/VTEC O157:H7 strains harbor a large 60–65 MDa plasmid (Johnson et al., 1983), designated pO157, that plays a role in virulence (Karch et al., 1987) and a small plasmid of 6.6 kb found in O157:H7 STEC/VTEC strains appear to synthesize colicin D (Bradley et al., 1991). O26:H11 strains also possess at least one plasmid in the range of 55–70 MDa. Restriction enzyme patterns of plasmids from other O:H serotypes (including O5, O91, O103, O111, O121, and O127) show a notable similarity with the large plasmids in O157 and O26 strains (Levine et al., 1987). A 3.4-kb fragment from a large plasmid of O157:H7 (prototype EDL 933) has been cloned and used as a DNA probe (referred to as CVD419) to identify EHEC plasmids (Levine et al., 1987); i.e., large plasmids found in Verocytotoxin producing *E. coli* strains. DNA probing with gene probes defining the incompatibility group of plasmids indicates that the STEC/VTEC plasmids share an approximately 23-kb fragment with EPEC plasmids and that the large plasmids of both EPEC and STEC/VTEC constitute a family of transfer-deficient Inc F-IIA plasmids (Hales et al., 1992), while sequencing of pO157 reveals high homology to the *orfI* of the RepFIB replicon (Schmidt et al., 1996a).

The large plasmid of O157 encodes the EHEC-hemolysin (Ehx), which is homologous to the *E. coli* α hemolysin (Schmidt et al., 1994, 1995a), and a novel catalase-peroxidase, KatP (Brunker et al., 1996). In contrast to α hemolysin, Ehx can be detected only on blood agar plates containing washed sheep erythrocytes. The zones of hemolysis on these plates are smaller and more turbid than those caused by α hemolysin and require overnight incubation before they become visible (Beutin, 1991; Beutin et al., 1988, 1989). A role for Ehx in the pathogenesis of diarrheal disease has not been demonstrated. Because α hemolysin-producing strains are uncommon in feces, these hemolysins serve as useful phenotypic markers for the detection of the majority of STEC/VTEC organisms.

The genes encoding the Ehx constitute a typical RTX (repeats in toxin) determinant, the Ehx-operon, with the gene order CABD (Schmidt et al., 1996a). The *ehxA* gene encodes the active protein, and *ehxB* and *ehxD* share high sequence homology with other RTX transport proteins (Schmidt et al., 1995a, 1996a). Like α hemolysin, the Ehx is a highly active cytolysin of the RTX family with a similar but not identical pore-forming capacity (Schmidt et al., 1996b). The Ehx plasmid DNA probe (CVD419) covers the *ehxA* and part of the *ehxB* gene (Schmidt et al., 1995a).

Two other enterohemolysins Ehly1 and Ehly2 have been described (Beutin et al., 1993b; Stroehrer et al., 1993). Ehly1 is a 33-kDa cell-associated protein encoded by a bacteriophage, ΦC3888, found in O26:H11 STEC/VTEC. Ehly1 has no known sequence homology to any other DNA or protein sequence. The Ehly2 enterohemolysin is also encoded by a bacteriophage, ΦC3208, found in O26:H11. It is in part homologous to DNA of bacteriophageλ; but completely unrelated to Ehly1 (Beutin et al., 1993b).

Most information on the source and transmission of STEC/VTEC has been learned from outbreak investigations. Findings

from these investigations showed that most outbreaks are related to carriage of the organism in ruminants, especially cattle, which show no symptoms of disease. During the period from 1982 to 1993, at least 20 outbreaks of O157:H7 have been reported in the USA (summarized in Anonymous, 1994). These outbreaks have affected 1509 patients, resulting in the hospitalization of 346 patients, 86 cases of HUS, and 19 deaths. This increased dramatically in the following years with 13 outbreaks in 1993 and 30 outbreaks in 1994 (Armstrong et al., 1996). The largest multistate outbreak in the USA occurred in early 1993 with more than 700 illnesses and 4 deaths (Bell et al., 1994; Davis, 1994). It has been estimated that *E. coli* O157:H7 causes 73,000 illnesses annually in the United States and non-O157 STEC/VTEC, 37,000 illnesses; and that 91 deaths occur each year in the USA (Mead et al., 1999). In Canada, 15 outbreaks were reported in 1982–1987 with 242 cases, 24 cases of HUS, and 15 deaths (Karmali, 1989). The first recognized community outbreak of O157:H7 in Europe occurred in the UK in the summer of 1985 affecting at least 24 persons. Eleven patients were hospitalized and one died (Morgan et al., 1988). In England and Wales, O157:H7 was isolated from 39% of sporadic cases of hemorrhagic colitis (Smith et al., 1987) and 33% of sporadic HUS cases (Scotland et al., 1988). In an outbreak of HUS in the West Midlands, O157:H7 was isolated from 33% of cases (Taylor et al., 1986a; Willshaw et al., 2001). Subsequent outbreaks and sporadic cases in the UK have been reported (Salmon et al., 1989). Scotland has one of the highest rates of infection with O157 increasing from 1.37/100,000 of the population in 1989 (Thomas et al., 1996a) to 32.3/100,000 in 1996 (Reilly and Carter, 1997). The worst food poisoning outbreak with O157 STEC/VTEC in Scotland occurred in 1996 with 501 cases; 151 were hospitalized and 20 elderly died (Ahmed and Donaghy, 1998).

O157 STEC/VTEC has been isolated from outbreaks and from sporadic cases of diarrhea and HUS in many parts of the world: Canada, UK, Argentina, Germany, Central Europe, Chile, and Italy (summarized by Griffin, 1995; see also Chapters 2–9 in Kaper and O'Brien, 1998). Sakai City in Japan experienced the largest outbreak of O157 STEC/VTEC ever recorded in July 1996, which was part of several outbreaks that summer with an estimated number of a little less than 8,000 cases and 6 deaths (Infectious Disease Surveillance Center, Japan, 1997).

Other STEC/VTEC O:H serotypes have caused outbreaks of diarrhea and HUS: O111:H[−], O145:H[−], and O?:H19 in Japan (Kudoh et al., 1994); O26:H11 in the Czech Republic (Bielaszewska et al., 1990), USA (Brown et al., 1998), and Ireland (McMaster et al., 2001); O103:H2 in France (Mariani-Kurkdjian et al., 1993); O104:H21 in the USA (Centers for Disease Control, 1995); O111:H[−] in Australia (Cameron et al., 1995), Italy (Carprioli et al., 1994), USA (Banatvala et al., 1996; Centers for Disease Control, 2000), Spain (Blanco et al., 1996), and France (Boudailliez et al., 1997; Mariani-Kurkdjian et al., 1997); O113:H21 in Australia (Paton et al., 1999); and O119 in France (Deschenes et al., 1996). A clone of sorbitol fermenting O157:H7 STEC/VTEC has been isolated from patients with diarrhea and HUS (Gunzer et al., 1992; Karch et al., 1993) and caused an outbreak in Germany (Ammon et al., 1999). Studies in Europe indicate that non-O157 STEC/VTEC strains are increasing in frequency as a cause of hemolytic-uremic syndrome (HUS) and found much more commonly in children with diarrhea (Verwey et al., 1999; Scheutz et al., 2001; Tozzi et al., 2001).

Among the over 400 STEC/VTEC serotypes, and apart from O157:H[−] and O157:H7, those in O groups O26, O103, O111,

and O145 are most commonly isolated from humans worldwide. These, along with strains that have caused outbreaks, are clearly recognized as pathogens. Table BXII.γ.205 shows the non-O157 STEC/VTEC serotypes that have been isolated from humans.

E. COLI IN HUMAN EXTRAINTESTINAL INFECTIONS

EXTRAINTESTINAL PATHOGENIC *E. COLI* (ExPEC) Extraintestinal pathogenic *E. coli* (ExPEC) are *E. coli* strains that possess currently recognized extraintestinal virulence factors or have been demonstrated to possess enhanced virulence in an appropriate animal model (Russo and Johnson, 2000). ExPEC strains primarily belong to pathogenic clones of a limited number of O:K:H serotypes (Ørskov and Ørskov, 1975), usually with MR fimbriae (P and S fimbriae), siderophores (e.g., aerobactin), host defense-avoidance mechanisms such as capsules (Ørskov and Ørskov, 1977; Ørskov et al., 1982), O antigens, and serum resistance and toxins (often α-hemolysin). ExPEC is common in all age groups and may occur at almost any extraintestinal site. The most common infections include urinary tract infections (UTIs) ranging from uncomplicated to febrile to invasive, pyelonephritis, neonatal, and postneurosurgical meningitis and septicemia. This group is epidemiologically and phylogenetically distinct from commensal and intestinal strains of *E. coli* (Picard et al., 1999). Virulence genes are often located on pathogenicity islands (PAIs), which have the tendency to delete with high frequencies or may undergo duplications and amplifications. They are often associated with tRNA loci, which may represent target sites for the chromosomal integration of these elements (Hacker et al., 1997). Many produce toxins that can lyse erythrocytes of different mammalian species. The best characterized of these is α-hemolysin, which is often produced by strains causing UTIs (J.R. Johnson, 1991) and is believed to play an important role. α-Hemolysin is secreted and can be demonstrated in culture fluid filtrates (Beutin, 1991). Hemolytic colonies can also be identified by the clear zone of hemolysis produced on blood agar plates after 3–4 h of incubation.

UROPATHOGENIC *E. COLI* (UPEC) This somewhat misleading acronym (clones or virulence factors are not syndrome-specific) has been used to refer to the majority of specific clonal groups of uropathogenic *E. coli* isolated from UTIs including pyelonephritis. They are characterized by a number of virulence factors that together play a role in their pathogenesis. First, UPEC is dominated by a limited number of O groups with O groups O1, O2, O4, O6, O7, O18ac, O75, O16, and O15 as the most commonly isolated (Ørskov and Ørskov, 1985). These strains are also represented by a limited number of K antigens: K1, K2, K3, K5, K12, and K13. Common serotypes include O1:K1:H7, O2:K1:H4, O4:K12:H1, O4:K12:H5, O6:K2:H1, O6:K5:H1, O6:K13:H1 (cystitis), O16:K1:H6, and O18ac:K5:H7. Furthermore, the majority of UPECs express P fimbriae of F types F7 through F16 and/or S fimbriae.

NEONATAL MENINGITIS *E. COLI* (NMEC) Neonatal meningitis *E. coli* is frequently associated with O groups O7, O18ac, O1, and O6 that have the K1 antigen identical to the capsule of *N. meningitidis* type B (Sarff et al., 1975). O83:K1 strains are also common but apparently only in Europe (Ørskov and Ørskov, 1985). One of the most commonly isolated types is an S fimbriated clone of serotype O18ac:K1:H7.

***E. COLI* IN ANIMAL INFECTIONS** As is the case in humans, certain strains of *E. coli* can cause disease in animals. In farm animals, *E. coli* strains are associated with a variety of pathological

TABLE BXII.γ.205. Serotypes of non-O157 STEC/VTEC isolated from humans^{a,b,c}

Serotype	Serotype	Serotype	Serotype	Serotype	Serotype	Serotype	Serotype	Serotype	Serotype
O1:H⁻	O8:H21	O25:K2:H2	O52:H23	O83:H1	O103:H18	O114:H4	O126:H20	O146:H11	O169:H ⁻
O1:H1	O8:H25	O25:H14	O52:H25	O84:H⁻	O103:H21	O114:H48	O126:H21	O146:H14	O171:H⁻
O1:H2	O9ab:H⁻	O26:H⁻	O54:H21	O84:H2	O103:H25	O114:H?	O126:H27	O146:H21	O171:H2
O1:H7	O9:H7	O26:H2	O55:H⁻	O84:H20	O103:HNT	O115:H10	O127	O146:H28	O172:H⁻
O1:H20	O9:H21	O26:H8	O55:H6	O85:H⁻	O104:H⁻	O115:H18	O128:H⁻	O148:H28	O172:H?
O1:HNT	O11:H⁻	O26:H11	O55:H7	O85:H10	O104:H2	O116:H ⁻	O128ab:H2	O150:H ⁻	O173:H2
O2:H ⁻	O11:H2	O26:H12	O55:H9	O85:H23	O104:H7	O116:H4	O128:H7	O150:H8	O174:H⁻ ^d
O2:H1	O11:H8	O26:H32	O55:H10	O86:H⁻	O104:H16	O116:H10	O128:H8	O150:H10	O174:H2^d
O2:K1:H2	O11:H49	O26:H46	O55:H19	O86:H10	O104:H21	O116:H19	O128:H10	O152:H4	O174:H8 ^d
O2:H5	O12:H ⁻	O27:H ⁻	O55:H?	O86:H40	O105ac:H18	O117:H ⁻	O128:H12	O153:H2	O174:H21^d
O2:H6	O14:H⁻	O27:H30	O60:H ⁻	O87:H16	O105:H19	O117:H4	O128:H25	O153:H11	O175:H16 ^e
O2:H7	O15:H⁻	O28ab:H ⁻	O64:H25	O88:H ⁻	O105:H20	O117:H7	O128:H31	O153:H12	O176:H ⁻ ^f
O2:H11	O15:H2	O28:H25	O65:H16	O88:H25	O106	O117:K1:H7	O128:H45	O153:H21	OX177:H⁻ ^f
O2:H27	O15:H8	O28:H35	O68:H ⁻	O89:H ⁻	O107:H27	O117:H8	O129:H ⁻	O153:H25	OX177:H11 ^f
O2:H29	O15:H27	O30:H2	O69:H⁻	O90:H ⁻	O109:H2	O117:H19	O130:H11	O153:H30	OX178:H7 ^f
O2:H44	O16:H⁻	O30:H21	O69:H11	O91:H⁻	O109:H16	O117:H28	O131:H4	O153:H33	OX179:H8 ^f
O3:H10	O16:H6	O30:H23	O70:H11	O91:H4	O110:H ⁻	O118:H ⁻	O132:H ⁻	O154:H⁻	OX181:H15 ^f
O4:H ⁻	O16:H21	O37:H41	O71:H ⁻	O91:H10	O110:H19	O118:H2	O133:H ⁻	O154:H4	OX181:H49 ^e
O4:H5	O17:H18	O38:H21	O73:H34	O91:H14	O110:H28	O118:H12	O133:H53	O154:H19/20	ONT:H ⁻
O4:H10	O17:H41	O38:H26	O74	O91:H15	O111:H⁻	O118:H16	O134:H25	O156:H ⁻	ONT:H2
O4:H40	O18:H⁻	O39:H4	O75:H⁻	O91:H21	O111:H2	O118:H30	O137:H6	O156:H4	ONT:H8
O5:H⁻	O18:H7	O39:H8	O75:H1	O91:H40	O111:H7	O119:H ⁻	O137:H41	O156:H7	ONT:H18
O5:H16	O18:H12	O39:H28	O75:H5	O91:HNT	O111:H8	O119:H5	O138:H2	O156:H25	ONT:H19
O6:H⁻	O18:H15	O40:H2	O75:H8	O92:H3	O111:H11	O119:H6	O141:H ⁻	O156:H27	ONT:H21
O6:H1	O18:H?	O40:H8	O76:H7	O92:H11	O111:H21	O119:H25	O141:H2	O156:HNT	ONT:H25
O6:H2	O20:H ⁻	O41:H2	O76:H19	O95:H ⁻	O111:H30	O120:H19	O141:H8	O160:H?	ONT:H41
O6:H4	O20:H7	O41:H26	O77:H⁻	O96:H10	O111:H34	O121:H⁻	O142	O161:H⁻	ONT:H47
O6:H12	O20:H19	O44	O77:H4	O98:H⁻	O111:H40	O121:H8	O143:H ⁻	O162:H4	ONT:K39:H48
O6:H28	O21:H5	O45:H ⁻	O77:H7	O98:H8	O111:H49	O121:H11	O144:H ⁻	O163:H ⁻	Orough:H ⁻
O6:H29	O21:H8	O45:H2	O77:H18	O100:H25	O111:H?	O121:H19	O145:H⁻	O163:H19	Orough:H2
O6:H31	O21:H?	O45:H7	O77:H41	O100:H32	O112ab:H2	O123:H19	O145:H4	O163:H25	Orough:H5
O6:H34	O22:H ⁻	O46:H2	O78:H ⁻	O101:H⁻	O112:H19	O123:H49	O145:H8	O165:H⁻	Orough:K1:H6
O6:H49	O22:H1	O46:H31	O79:H7	O101:H9	O112:H21	O124:H ⁻	O145:H16	O165:H10	Orough:K1:H7
O7:H4	O22:H5	O46:H38	O79:H14	O102:H6	O113:H2	O125:H ⁻	O145:H25	O165:H19	Orough:H11
O7:H8	O22:H8	O48:H21	O79:H23	O103:H⁻	O113:H4	O125:H8	O145:H26	O165:H21	Orough:H16
O8:H ⁻	O22:H16	O49:H⁻	O80:H ⁻	O103:H2	O113:H5	O125:H?	O145:H28	O165:H25	Orough:H18
O8:H2	O22:H40	O49:H10	O81:H?	O103:H4	O113:H7	O126:H ⁻	O145:H46	O166:H12	Orough:H20
O8:H9	O23:H7	O50:H ⁻	O82:H ⁻	O103:H6	O113:H21	O126:H2	O145:HNT	O166:H15	Orough:H21
O8:H11	O23:H16	O50:H7	O82:H5	O103:H7	O113:H32	O126:H8	O146:H ⁻	O166:H28	Orough:H28
O8:H14	O23:H21	O51:H49	O82:H8	O103:H11	O113:H53	O126:H11	L	O168:H ⁻	Orough:H46
O8:H19	O25:H ⁻	O52:H19	O83:H ⁻						

^aData from Scheutz et al. (2001 and unpublished results); Blanco et al. (2001), WHO (1999).^bSerotypes in bold represent strains isolated from patients with HUS.^cAn updated list of STEC, with literature references, can be found at <http://www.microbionet.com.au/frames/feature/vtec/brief01.html>^dFormerly known as OX3.^eFormerly known as OX7.^fProvisional designation for new O antigens.

conditions, which include colibacillary diarrhea, colibacillary toxemia in pigs, systemic colibacillosis, coliform mastitis, and UTIs. Colibacillary diarrhea is an acute diarrheal disease due to ETEC infection, which occurs primarily in 1–3-d-old calves, lambs, and piglets. A limited number of O groups are represented among these ETEC strains. In England and Wales, the most common O groups of *E. coli* isolates from pigs with diarrhea are O149, O8, O158, O147, and O157 (Wray et al., 1993). Colibacillary toxemia in pigs can take several forms: shock in weaner syndrome, hemorrhagic enteritis, and edema disease. These disease syndromes are also attributable to *E. coli* belonging to a small number of O groups (O8, O45, O138, O139, O141, and O149). Rapid absorption of endotoxin from the bowel is hypothesized to play a role in the pathogenesis of the shock in weaner syndrome and hemorrhagic enteritis, while the toxin Stx2e/VT2e, which is produced by many of the strains having the above specified O groups, has been shown to play a role in the pathogenesis of edema

disease (Macleod et al., 1991). Systemic colibacillosis occurs when septicemic strains of *E. coli* pass through the intestinal or respiratory mucosa into the bloodstream of calves, lambs, and poultry. Once they enter the bloodstream, they can cause either a generalized infection or a localized infection, such as meningitis and/or arthritis in calves and lambs or air sacculitis and pericarditis in poultry. *E. coli* strains are also an important cause of mastitis in cows. Endotoxin is believed to play a role in the inflammatory response observed during this disease.

The roles of various adherence mechanisms and toxins in the pathogenesis of the infections were reviewed by Wray and Woodward (1997). Fimbrial antigens and putative colonization factors associated with strains of *E. coli* causing disease in animals cited include F1, F4 (K88), F5 (K99), F41, F6 (987P), F17, F18, CS31A, F165, M326, C1213, F42, F11, curli, type IV pilins, and Nfa. Toxins produced by *E. coli* causing disease in animals include the heat-labile enterotoxins LTI and LTII, heat-stable enterotoxins STa

and STb, cytotoxic necrotizing factors 1 and 2 (CNF1 and CNF2), and Shiga/Verocytotoxins.

Other typing methods For a description of other methods for subdivision of *E. coli*, i.e., phage typing, colicin typing, biotyping, typing by outer membrane protein (OMP) pattern, typing by antibiotic resistance patterns, and typing by direct hemagglutination, see Ørskov and Ørskov (1984a) and Sussman (1985). Phage typing is very useful for certain antigens because antisera are difficult to produce. This is particularly true for K1 (Gross et al., 1977), K3, K5, K7, K12, K13 (Nimmich et al., 1992), and K95 (Nimmich, 1994). A phage-typing scheme for STEC/VTEC O157:H7 established in 1987 (Ahmed et al., 1987) and extended in 1990 (Khakhria et al., 1990) has proven very useful in the epidemiological surveillance of STEC/VTEC O157 (Frost et al., 1993; Saari et al., 2001) infections and is applicable even in low-technology laboratories. In general, phage typing of O157 should be supplemented with one of the molecular typing methods mentioned below. Because of their discriminatory power, speed, use of commercially available reagents and equipment, and amenability to automation and electronic networking, molecular subtyping methods have become very popular for subtyping of *E. coli*, particularly strains involved in causing outbreaks of food-borne disease. Molecular subtyping methods for *E. coli* O157:H7 and other foodborne bacterial pathogens were reviewed by Barrett (1997). Some recently described methods include macrorestriction endonuclease analysis with PFGE (Preston et al., 2000; Zhang et al., 2000; Swaminathan et al., 2001); detection of insertion sequences and characterization of virulence genes by DNA probes or PCR (Thompson et al., 1998; Zhang et al., 2000); detection of amplified fragment length polymorphisms (Iyoda et al., 1999); computer identification by rRNA gene restriction patterns (Machado et al., 1998); and the analysis of randomly amplified polymorphic DNA (Hopkins and Hilton, 2001). In 1996, a molecular subtyping network in the United States, designated PulseNet, for the electronic comparison of DNA fingerprints generated by macrorestriction endonuclease analysis with PFGE was developed to subtype *E. coli* O157:H7. PulseNet has proved an exceptionally valuable tool for detecting outbreaks of *E. coli* O157:H7 infection. The development of standardized laboratory and data analysis protocols and their successful use in providing surveillance for *E. coli* O157:H7 and other foodborne bacterial pathogens was reviewed by Swaminathan et al. (2001).

ENRICHMENT AND ISOLATION PROCEDURES

Many simple agar media can be used for isolation. Media used for selective isolation from feces usually contain substances that partly or completely inhibit growth of bacteria other than *Enterobacteriaceae* (tetrathionate, deoxycholate, bile salts, etc.). The addition of Maranil (dodecylbenzolsulfonate) at a concentration of 0.005% will inhibit swarming of *Proteus* organisms. For details, see Edwards and Ewing (1972) or Kauffmann (1966) or any catalogue from one of the medium-producing companies. At Statens Serum Institut, Copenhagen, a medium developed in the media department of the institute, bromothymol blue (BTB) agar, is used.¹

1. Bromothymol blue agar (selective for *Enterobacteriaceae*). Combine the following ingredients: peptone (Orthana Ltd., Copenhagen), 10.0 g; NaCl, 5.0 g; yeast extract (Oxoid), 5.0 g; and distilled water, 1000 ml. The pH is adjusted to 8.0, agar powder is added, and the preparation is autoclaved at 120°C for 20 min. The following components are then added aseptically from sterile stock solutions: Maranil solution [Paste A75 (dodecylbenzolsulfonate), Henkel, Germany], 1.0 ml; sodium thiosul-

MAINTENANCE PROCEDURES

E. coli strains can be kept viable for many years in beef extract agar stabs (tightly closed, e.g., by corks soaked in melted paraffin wax) or on Dorset egg medium. Cultures are initially incubated at 37°C followed by storage in the dark at room temperature (20–22°C). After a few weeks or months, such cultures often contain many mutational forms such as R forms and acapsular forms; consequently, we prefer to store important cultures in beef broth containing 10% glycerol at –80°C. Screw-capped vials are used for easy access.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Kilian and Bülow (1976) found that a very high percentage (97%) of *Escherichia coli* and the majority (57%) of *Shigella* strains, exclusively among the *Enterobacteriaceae*, produce β -glucuronidase (GUD). Prolonged incubation of 28 h increases positivity to 99.5% (Rice et al., 1990), which is in accordance with the presence of the *uidA* (GUD) gene in all *E. coli* strains (McDaniels et al., 1996). This test (referred to as the GUD-, PGUA-, MUG-, GUR-, or GLUase-HR test) therefore is very suitable as a screening test for *E. coli*, with the unfortunate exception of most STEC/VTEC O157 strains, which are phenotypically negative. None of the other four species of the genus *Escherichia* are positive for this enzyme (Rice et al., 1991). Both genotypic and phenotypic assays for glutamate decarboxylase (GAD) used by environmental scientists have been described as highly specific for *E. coli*. Unfortunately, these tests have only recently been shown to exhibit the same specificity on a smaller collection of pathogenic isolates of *E. coli* (Grant et al., 2001).

DIFFERENTIATION OF THE GENUS *ESCHERICHIA* FROM OTHER GENERA

See Table BXII.γ.193 of the family *Enterobacteriaceae* for characteristics that can be used to differentiate this genus from other genera of the family.

TAXONOMIC COMMENTS

The identification of *Escherichia* strains seldom causes problems; however, many studies have shown that “*Escherichia* is a genus (or species) made up of phenotypically variable strains” (Farmer and Brenner, 1977). DNA–DNA hybridization studies have been an invaluable tool for solving problems in this field. The genus *Shigella* is closely related to *Escherichia*, and only historical reasons make it acceptable that these two genera are not united. Several typical EIEC types have been found that have pathogenic traits that are similar to those of *Shigella*. The Sereny test (Séryny, 1957), which demonstrates the capacity to cause keratoconjunctivitis in the guinea pig, typical of *Shigella* strains, is also found in these special *Escherichia* strains. Day et al. (1981) described a tissue culture technique that can be used as a substitute for the Sereny test. Typically, such dysentery-associated *E. coli* strains have O antigens that are closely related or identical to *Shigella* O antigens. Brenner et al. (1972a), by DNA reassociation studies, found species-level relatedness between *Shigella* strains and these special *Escherichia* strains, as well as nonpathogenic *E. coli* strains.

fate (50% solution), 2.0 ml; bromothymol blue (Riedel de Haen, Germany; 1.0% solution), 10.0 ml; lactose (33% solution), 27 ml; and glucose (33% solution), 1.2 ml. The pH is adjusted to 7.7–7.8. To obtain optimum results, the amount of glucose must be adjusted for every new batch of yeast extract, peptone, and agar. This medium is very useful for differentiation of lactose-fermenting colonies based on their color.

Not unexpectedly, many strains are phenotypically intermediate between *Escherichia* and *Shigella*, but for obvious reasons a special taxonomic status for such strains is not warranted. In the older literature the name *Alkalescens-Dispar* is used, but, as stated by Brenner (1978), this group is virtually indistinguishable from *E. coli* strains and is, in fact, a biogroup of *E. coli* that is anaerogenic, lactose-negative (or delayed), and nonmotile.

While most or all characters that classically have been used for definition of the genus *Escherichia* are chromosomally determined, several traits that are not characteristic of *Escherichia* have been found in otherwise typical *Escherichia* strains. Lautrop et al. (1971) and Layne et al. (1971) described H₂S-positive strains of *Escherichia*, and this character was plasmid-determined. It is not known which selective forces account for the simultaneous isolation of H₂S-positive *Escherichia* strains in different parts of the world.

Other "forbidden" phenotypic traits have similarly been described in *Escherichia*, many of which are plasmid-determined. Ørskov et al. (1961) found many urease-producing strains among typical serotypes from piglet diarrhea. Wachsmuth et al. (1979) demonstrated the plasmid-determined nature of a similar urease-positive phenotype in human *E. coli* strains. Citrate-utilizing *E. coli* strains were described by Washington and Timm (1976) and were found to be plasmid determined in similar strains by Sato et al. (1978). Carbon dioxide-dependent cultures can be found (Eykn and Phillips, 1978). A citrate-positive, malonate-positive biogroup and a biogroup negative in these reactions were described (Burgess et al., 1973).

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DIFFERENTIATION OF THE SPECIES OF THE GENUS *ESCHERICHIA*

Characteristics useful in distinguishing the five species of *Escherichia* are given in Table BXII.γ.193 of the family *Enterobacteriaceae* and in Table BXII.γ.197 of the genus *Escherichia*.

List of species of the genus *Escherichia*

1. ***Escherichia coli*** (Migula 1895) Castellani and Chalmers 1919, 941^{AL} (*Bacillus coli* Migula 1895, 27.)
co'li. Gr. n. *colon* large intestine, colon; M.L. gen. n. *coli* of the colon.

The characteristics are as described for the genus and as listed in Table BXII.γ.193 of the family *Enterobacteriaceae*. Occurs naturally in the lower part of the intestine of warm-blooded animals, and as intestinal (some foodborne) and extraintestinal pathogens of humans and animals.

The mol% G + C of the DNA is: 48.5–52.1 (*T_m*).

Type strain: ATCC 11775, CCM 5172, CIP 54.8, DSM 30083, IAM 12119, NCDO 1989, NCTC 9001. Serotype O1:K1(L1):H7.

GenBank accession number (16S rRNA): X80725.

Additional Remarks: Other sequences are listed in Table BXII.γ.206.

2. ***Escherichia blattae*** Burgess, McDermott and Whiting 1973, 4^{AL}
blat' tae. L. fem. n. *blatta* cockroach; L. gen. n. *blattae* of the cockroach.

The characteristics are as described for the genus and as listed in Table BXII.γ.193 of the family *Enterobacteriaceae*. *E. blattae* was isolated from the hindgut of healthy cockroaches, *Blatta orientalis*, in England (Burgess et al., 1973) and on Easter Island (Nogrady and Aubert, personal com-

TABLE BXII.γ.206. *rrn* operon sequences of *Escherichia* strains

Source and strain ^{a, b}	EMBL ^c	Method ^d
<i>Escherichia coli</i> :		
	J01859	rRNA
	J01695	<i>rrnB</i>
	V00348	<i>rrnB</i>
(PK3)	X80731	PCR
(MC4100)	X80732	PCR
CIP (ATCC 11775 ^T)	X80725	PCR
ATCC 25922	X80724	PCR
(K-12)	M87049	<i>rrnA</i>
(K-12)	U00006	<i>rrnB</i>
(K-12)	L10328	<i>rrnC</i>
(K-12)	U18997	<i>rrnD</i>
(K-12)	U00006	<i>rrnE</i>
	M29364	<i>rrnG</i>
(K-12)	D15061	<i>rrnH</i>
BioM	X80733	PCR
(PK3)	X80721	<i>rrnA</i>
(PK3)	X80722	<i>rrnB</i>
(PK3)	X80723	<i>rrnC</i>
(PK3)	X80727	<i>rrnD</i>
(PK3)	X80728	<i>rrnE</i>
(PK3)	X80729	<i>rrnG</i>
(PK3)	X80730	<i>rrnH</i>
<i>Escherichia fergusonii</i> :		
ATCC 35469	AF530475	NA ^e
<i>Escherichia hermannii</i> :		
BioM	X80675	rRNA
<i>Escherichia vulneris</i> :		
CIP (ATCC 33821 ^T)	X80734	PCR

^aSome strain numbers have been lost; sequences were most probably obtained using *E. coli* K-12.

^bBacterial collection from which each strain is deposited: ATCC (American Type Culture Collection); BioM (BioMérieux, Marcy l'Étoile, France), CIP (Collection de l'Institut Pasteur).

^cAccession numbers under which sequence is available.

^dMethod by which each sequence has been obtained: rRNA (total rRNA sequenced using reverse transcriptase), PCR (total PCR products sequenced using T7-DNA polymerase); *rrnX* (sequence of a single operon, X).

^eNot available.

munication). It appears as two biotypes, one of which is citrate and malonate positive, the other negative, and it is the only species within *Escherichia* that is gluconate positive (Burgess et al., 1973). *E. blattae* has not been associated with disease either in humans or in cockroaches.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 29907, CDC 9005-74, DSM 4481, NCTC 12127.

GenBank accession number (16S rRNA): X87025.

3. ***Escherichia fergusonii*** Farmer, Fanning, Davis, O'Hara, Riddle, Hickman-Brenner, Asbury, Lowery and Brenner 1985c, 223^{VP} (Effective publication: Farmer, Fanning, Davis, O'Hara, Riddle, Hickman-Brenner, Asbury, Lowery and Brenner 1985b, 77.)*

fer.gu.so'ni.i. M.L. masc. (substantive) *fergusonii* coined to honor the American microbiologist William W. Ferguson, who made many contributions to enteric bacteriology and was one of the first to show the role of certain strains of *E. coli* in infantile diarrhea (Farmer et al., 1985b).

The characteristics are as described for the genus and as listed in Table BXII.γ.193 of the family *Enterobacteriaceae*.

Has been isolated from human clinical specimens (stool, urine, blood, and an abdominal wound), the feces of captive raptors belonging to the order *Falconiformes* or *Strigiformes* (Bangert et al., 1988), and from unspecified sites for other warm-blooded animals.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 35469, CDC 0568-73.

Additional Remarks: Other sequences are listed in Table BXII.γ.206.

4. ***Escherichia hermannii*** Brenner, Davis, Steigerwalt, Riddle, McWhorter, Allen, Farmer, Saitoh and Fanning 1983a, 438^{VP} (Effective publication: Brenner, Davis, Steigerwalt, Riddle, McWhorter, Allen, Farmer, Saitoh and Fanning 1982a, 705.)*
- her.man'ni.i.* M.L. *hermannii* of Hermann, named in honor of George J. Hermann, former chief of the Enteric Section at the CDC, for his many contributions to enteric bacteriology, and Lloyd G. Herman, formerly of the Environmental Services Branch, National Institutes of Health, Bethesda, MD, for his contributions to the study of yellow-pigmented bacteria (Brenner et al., 1982a).

The characteristics are as described for the genus and as listed in Table BXII.γ.193 of the family *Enterobacteriaceae*. Those that together distinguish it from most other members of this family include growth in the presence of KCN, fermentation of cellobiose, and production of yellow pigment. Has been isolated from human clinical specimens (wounds, sputum, lung, stool, blood, and spinal fluid) and recently from the sludge of an industrial wastewater treatment plant (Kiernicka et al., 1999). The sludge isolate shows promise for bioremediation; it grows in and degrades high concentrations of chlorobenzene.

The mol% G + C of the DNA is: 53–58 (*T_m*).

Type strain: ATCC 33650, CDC 980-72, DSM 4560.

Additional Remarks: Other sequences are listed in Table BXII.γ.206.

5. ***Escherichia vulneris*** Brenner, McWhorter, Leete Knutson and Steigerwalt 1983d, 438^{VP} (Effective publication: Brenner, McWhorter, Leete Knutson and Steigerwalt 1982b, 1137.)*

vul.ne'is. L. n. *vulnus* a wound; L. gen. n. *vulneris* of a wound; *Escherichia vulneris* the *Escherichia* of a wound.

The characteristics are as described for the genus and as listed in Table BXII.γ.193 of the family *Enterobacteriaceae* (Brenner et al., 1982b). Has been isolated from human clinical specimens, primarily wounds, the majority of which occurred on the arms or legs, but also blood, throat, sputum, vagina, urine, and stool, and other warm-blooded animals. The type species was isolated from the intestine of a cowbird in Michigan, USA.

The mol% G + C of the DNA is: 58.5–58.7 (*T_m*).

Type strain: ATCC 33821, CDC 875-72, DSM 4564, NIH 580.

GenBank accession number (16S rRNA): X80734.

Additional Remarks: Other sequences are listed in Table BXII.γ.206.

*Editorial Note: This species was formerly known as Enteric Group 11.

*Editorial Note: This species was formerly known as Enteric Group 1.

*Editorial Note: This species was formerly known as Enteric Group 10.

Genus II. **Alterococcus** Shieh and Jean 1999, 341^{VP} (Effective publication: Shieh and Jean 1998, 644)

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Al.te.ro.coc'cus. L. *alter* another; Gr. n. *coccus* a grain or berry; M.L. masc. n. *Alterococcus* another coccus.

Spherical cells 0.8–0.9 µm in diameter, occurring singly or in pairs. **Motile by means of a single flagellum**. Gram negative. **Facultatively anaerobic**, capable of aerobic and anaerobic fermentation. Chemoheterotrophic. **Oxidase and catalase positive**. Optimum temperature for growth, 45°C. No growth occurs at 30°C or 60°C. **Halophilic**; growth occurs in the presence of 1–3% NaCl (optimum, 2.0%), but not 0% or 5% NaCl. The major cellular fatty acid is *anteiso*-15-carbon acid (C_{15:0 anteiso}). Cause agar liquefaction. Habitat: coastal hot springs.

The mol% G + C of the DNA is: 65.5–67.0.

Type species: Alterococcus agarolyticus Shieh and Jean 1999, 341 (Effective publication: Shieh and Jean 1998, 644)

FURTHER DESCRIPTIVE INFORMATION

Butyrate and propionate are formed during both aerobic and anaerobic growth in PY semisolid medium¹. Butyrate with propionate and/or formate is produced during both aerobic and anaerobic growth in PYG broth².

ENRICHMENT AND ISOLATION PROCEDURES

Water samples are decimally diluted with sterile NaCl-MOPSO buffer³ and spread on PY agar. The plates are incubated aerobically at 50°C in the dark for 3–7 d. Colonies that cause agar liquefaction on the incubated plates are purified by successive streaking.

MAINTENANCE PROCEDURES

Early stationary-phase cultures grown in PY broth are inoculated into sterile 60% seawater at a ratio of 0.25 ml in 5 ml (1:20) and stored for two or three months at 45°C.

DIFFERENTIATION OF THE GENUS *ALTEROCOCCUS* FROM OTHER GENERA

The genus is the only one known to contain Gram-negative, halophilic, thermophilic bacteria that degrade agar and grow both aerobically and anaerobically. The only other Gram-negative genera that contain heterotrophic, halophilic, thermophilic bacteria are *Rhodothermus*, *Thermotoga*, *Thermosipho*, and *Spirochaeta*. *Rhodothermus* contains obligately aerobic rods, whereas *Alterococcus* contains cocci. The genera *Thermotoga* and *Thermosipho* contain strictly anaerobic rods, whereas *Alterococcus* is facultatively anaerobic. *Spirochaeta* contains facultatively anaerobic species, but the cells are spiral, not coccoid.

TAXONOMIC COMMENTS

16S rDNA-based phylogenetic analysis of the type strain of *Alterococcus agarolyticus* has indicated that the species is most closely related to members of the family *Enterobacteriaceae*. In this edition of the *Manual*, the genus is placed in this family, which belongs to the order *Enterobacteriales* in the class *Gammaproteobacteria*.

List of species of the genus *Alterococcus*

1. ***Alterococcus agarolyticus*** Shieh and Jean 1999, 341^{VP} (Effective publication: Shieh and Jean 1998, 644.)

a.gar.o.ly'ti.cus. Malayan n. *agar* agar, a complex gelling polysaccharide from marine red algae; Gr. adj. *lyticum* dissolving; M.L. adj. *agarolyticus* agar-dissolving.

The characteristics are as described for the genus and as listed in Table BXII.γ.207. Young colonies on agar media are white, circular, and opaque. Glucose, cellobiose, galactose, lactose, sucrose, trehalose, and xylose are fermented. Sensitive to ampicillin, chloramphenicol, erythromycin, penicillin G, and tetracycline.

Isolated from two hot springs in the intertidal zone of Luta Island, Taiwan.

The mol% G + C of the DNA is: 65.8.

Type strain: ADT3, CCRC 19135.

GenBank accession number (16S rRNA): AF075271.

1. PY broth is composed of (g/l deionized water): Bacto Peptone (Difco), 4.0; Bacto yeast extract (Difco), 2.0; NaCl, 20.0; MgSO₄·7H₂O, 0.5; CaCl₂, 0.01; and 3-(N-morpholino)-2-hydroxypropanesulfonic acid (MOPSO; Sigma), 4.5. This broth medium is adjusted to pH 7.0. Bacto agar (Difco) is added to this medium at 5 and 15 g/l for the preparation of PY semisolid and plating media, respectively.

2. PYG medium is composed of two parts. Part 1 consists of the following ingredients dissolved in 900 ml of distilled water: Bacto Peptone (Difco), 4.0 g; Bacto yeast extract (Difco), 2.0 g; NaCl, 20.0 g; MgSO₄·7H₂O, 0.5 g; CaCl₂, 0.01g; and Tris buffer (Sigma), 6.0 g. The pH is adjusted to 7.8. Part 2 consists of glucose (5.0 g) dissolved in 100 ml of distilled water. Parts 1 and 2 are autoclaved separately and combined aseptically after cooling to room temperature.

3. NaCl-MOPSO buffer consists of (g/l deionized water): NaCl, 25.0 and MOPSO (N-morpholino)-2-hydroxypropanesulfonic acid, 0.45; pH 7.0.

TABLE BXII.γ.207. Phenotypic characteristics of *Alterococcus agarolyticus*^{a,b}

Characteristic	Reaction
<i>Fermentation of:</i>	
Adonitol, D-arabinose, dulcitol, myo-inositol, mannitol, melibiose, sorbitol	–
Cellobiose, galactose, glucose, lactose, sucrose, trehalose, xylose	+
Mannose	W
<i>Motility</i>	+
Swarming	–
Luminescence	–
Agarase activity	+
Hydrolysis of starch	W
Hydrolysis of casein, DNA, gelatin, fats	–
Catalase, oxidase	+
Arginine dihydrolase, lysine decarboxylase, ornithine decarboxylase	–
<i>Growth at:</i>	
30°C	–
38°C	W
40–56°C	+
58°C	W
60°C	–
<i>Growth in presence of NaCl levels of:</i>	
0%	–
0.50%	W
1.0–3.5%	+
4.00%	W
5.00%	–

^aSymbols: +, positive; –, negative; W, weakly positive.

^bData from Shieh and Jean (1998).

Genus III. Arsenophonus Gherna, Werren, Weisburg, Cote, Woese, Mandelco and Brenner 1991, 564^{VP}

JOHN H. WERREN

Ar.se.no.pho' nus. Gr. n. *arsen* a male; Gr. suff. *phonus* slayer; N.L. masc. n. *Arsenophonus* male-killer.

Cells are Gram-negative, nonmotile, nonsporeforming, nonflagellated, long to highly filamentous rods, which divide by septation. **The bacteria infect insect tissues.** The type species can be cultivated on cell-free media. Colonies are mucoid, gray-white, round, and convex with entire edges. Enzymatically digested proteins best serve as nitrogen sources. Does not utilize KNO₃, (NH₄)₂SO₄, complete defined amino acid mixtures, or acid-hydrolyzed peptones as nitrogen sources. Sugars sucrose, fructose, and D-glucose are utilized as primary carbon sources. Weak growth occurs with maltose, trehalose, cellobiose, and D-xylose. Acid is produced with D-glucose, fructose, and sucrose. Growth is negative with L-arabinose, glycerol, dulcitol, lactose, D-mannitol, raffinose, and inositol. Positive for catalase and gelatin liquefaction; negative for Voges-Proskauer, methyl red, nitrate reduction, indole, oxidase, hydrogen sulfide, o-nitrophenyl-β-D-galactopyranoside, lysine and ornithine decarboxylase, urease, and arginine dehydrolase. Cells grow at pH 6.2–8.7 (optimum pH range 7.4–8.0), and a temperature range of 15–35°C (30°C optimum).

There is currently only one validly named species in the genus

(and an additional *Candidatus* species). However, **the genus may be widespread in insects**, based upon recent evidence. Therefore, the genus description may well change as additional information becomes available.

The mol% G + C of the DNA is: 39.5.

Type species: *Arsenophonus nasoniae* Gherna Werren, Weisburg, Cote, Woese, Mandelco and Brenner 1991, 564.

FURTHER DESCRIPTIVE INFORMATION

General biology The genus *Arsenophonus* was originally described from a single bacterial species that is the causative agent of the “son-killer” trait in the parasitic wasp *Nasonia vitripennis* (insect order *Hymenoptera*). Subsequently, a second species has been proposed for the genus (*Candidatus Arsenophonus triatominarum*), isolated from a reduviid bug (insect order *Heteroptera*). Despite being found in divergent insects, the two are similar in morphology, patterns of infection of host insect tissues, and 16S rDNA sequence (97.8% similarity). Biochemical characterization of “*Candidatus Arsenophonus triatominarum*” has not yet

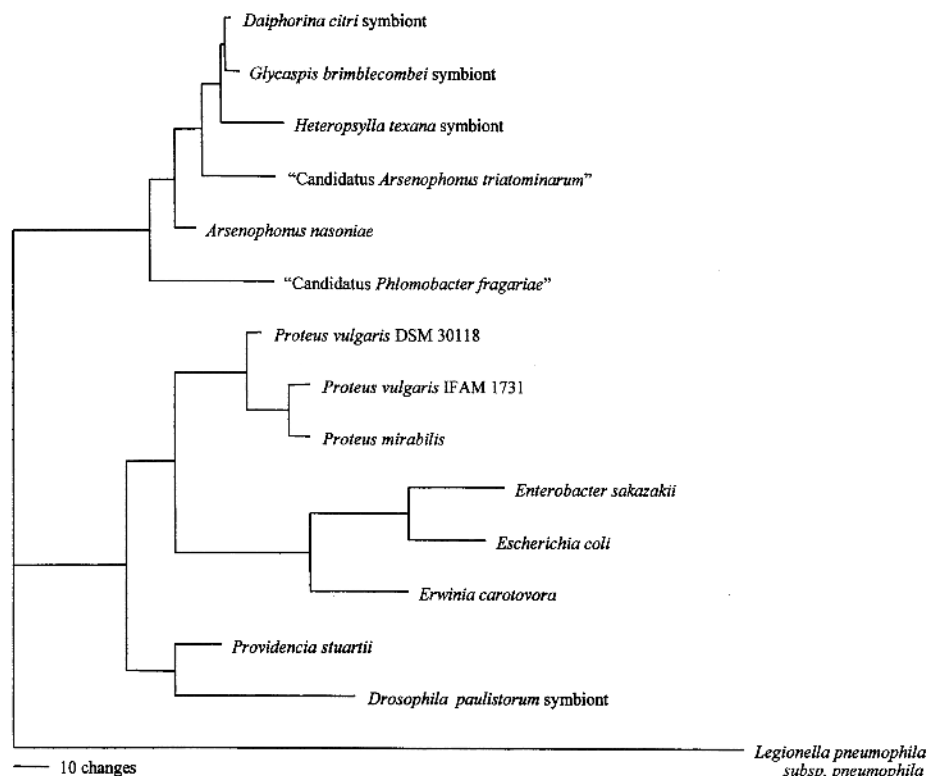


FIGURE BXII.γ.193. A phylogenetic tree based on 16S rDNA sequences is presented based on a parsimony analysis using a heuristic search and stepwise addition (D.L. Swofford, 1996, PAUP 4.0 Sinauer, Sunderland, MA) with *Legionella pneumophila* defined as the outgroup. Partial 16S rDNA nucleotide sequences (1476 sites) of *Arsenophonus nasoniae*, *Candidatus A. triatominarum*, other endosymbiotic and free-living bacteria were analyzed. Names of the insect host species are shown for the psyllid secondary symbionts and the unidentified bacterium from *D. paulistorum*. In all other cases the bacterial genus and species are shown. Numbers above nodes are bootstrap percentages from 1000 replicate searches. The sequence data for *Daiphorina citri* secondary symbiont was provided by T. Fukatsu (Subandiyah et al., 2000). Other sequences were obtained from the NCBI database (accession numbers AF263561, AF263562, U91786, M90801, U91515, X07652, AJ233425, X07652, AF008582, AB004746, AB035924, AJ233411, AF008581, U20273, and X73402).

been performed; nor are its effects on the host insect known. *A. nasoniae* can be grown on cell-free media; "*Candidatus A. triatominarum*" has not yet been grown in cell-free culture, but can be maintained on insect tissue culture.

Phylogenetic analysis using 16S rDNA sequences (see below) indicates that closely related bacteria are secondary symbionts in several species of psyllid insects (insect order *Homoptera*), although the taxonomic status of these bacteria has not yet been resolved. Taken together, the results suggest that *Arsenophonus* (and related bacteria) may have a wide distribution as infectious agents of insects. Another closely related bacterium, *Candidatus Phlomobacter fragariae*, causes disease in strawberry (Zreik et al., 1998).

Phylogenetic analyses Based on 16S rDNA phylogeny *Arsenophonus* is a lineage in the *Gamma*proteobacteria (Gherna et al., 1991; Hypsa and Dale, 1997), within the family *Enterobacteriaceae*. The phylogenetic analysis places *A. nasoniae* and the *Candidatus A. triatominarum* as close relatives (97.8% similarity) (Fig. BXII.γ.193). Other closely related organisms (Fig. BXII.γ.193) are secondary symbionts found in psyllid insects (98.6–97.6% similarity). Although they are closely related, the specific phylogenetic relationships between *A. nasoniae*, *Candidatus A. tri-*

tominarum, and the psyllid secondary symbionts are not clearly resolved. These may eventually be placed in the genus *Arsenophonus* when further characterization is completed. *A. nasoniae*, *Candidatus A. triatominarum*, and the three psyllid secondary endosymbionts form a monophyletic clade with *Candidatus Phlomobacter fragariae* (97.0% similarity to *A. nasoniae*). This monophyletic group is strongly supported by various phylogenetic analyses. *Candidatus Phlomobacter fragariae* is found within the phloem of strawberry and is associated with strawberry marginal chlorosis disease (Zreik et al., 1998). Plant phloem associated bacteria can be vectored by phloem feeding insects such as psyllids and aphids, which may suggest a biological basis for its phylogenetic similarity to psyllid secondary endosymbionts. The apparent monophyly of this bacterium with *Arsenophonus* and the psyllid secondary endosymbionts (Subandiyah et al., 2000; Thao et al., 2000b) suggests that separate generic status for this bacterium may not be warranted. However, additional sequences and biological characterization are needed to resolve the issue. The closest known bacteria outside of the "*Arsenophonus* group" include *Proteus vulgaris*, *Proteus mirabilis*, and *Providencia stuartii* (94.2–94.3% similarity).

List of species of the genus *Arsenophonus*

1. ***Arsenophonus nasoniae*** Gherna, Werren, Weisburg, Cote, Woese, Mandelco and Brenner 1991, 564^{VP}
na.so'ni.ae. N.L. n. *Nasonia* a genus of parasitoid wasps; *nasoniae* of the genus *Nasonia*.

The cells are nonmotile nonsporeforming, long rods to highly filamentous ($0.40\text{--}0.57 \times 6.9\text{--}10.0 \mu\text{m}$) in culture. Filamentous cells can also be found in insect tissues. Biochemical features are presented in the genus description. *A. nasoniae* is the causative agent of the "son-killer" trait in the parasitic wasp *N. vitripennis*. *Nasonia* wasps parasitize the pupae of various dipterans (flies). They sting the host pupa and lay their eggs underneath the puparial wall. The insects hatch and feed upon the host fly to complete development. In female wasps infected with the bacterium, male eggs fail to hatch, whereas female eggs develop normally (Werren et al., 1986). Genetic experiments indicate that the bacterium acts by blocking the development of unfertilized eggs, which normally develop into males in this haplodiploid species. The son-killer bacterium is found in *N. vitripennis* populations throughout North America at frequencies around 10%, and has also been detected in the sibling species *N. longicornis* (Balas et al., 1996).

The bacterium is introduced into the dipteran host pupa during stinging by the infected female, where it replicates and is ingested by the feeding wasp larvae (Huger et al., 1985). As a result, the bacteria can be transmitted maternally from an infected female to her developing larvae, and horizontally when more than one female parasitizes the same host. The bacteria initially invade the midgut epithelium and subsequently spread to various tissues, including brain, fat body, muscles, eyes, and hemocytes. Massive infections of tissues can be observed in adults without major negative effects on insect viability. In insect tissues, the bacteria are pleomorphic, with rods being the most common morphology. Both male and female larvae can become infected by host feeding during larval development. Female reproductive tracts are often infected with the bacteria, whereas infections are not observed in male reproductive tissues. In adult wasps, drop-like bacterial masses can be discharged from the midgut epithelium into the gut lumen,

suggesting a possible additional fecal route for infectious transmission. The mechanism by which *A. nasoniae* causes lethality of male eggs is unknown. Experiments indicate that unfertilized eggs, which normally develop into males in the haplodiploid insect host, fail to develop when the female is infected. Levels of male embryo mortality can vary in different infected females (Balas et al., 1996).

A. nasoniae can be isolated from infected insects by surface sterilization of the insect followed by maceration of the insect onto GC media with Kellogg's supplement, and serial streaking of the macerated material. Characteristic gray-white mucoid colonies will develop in 2–4 d at 26°C. Bacteria can also be isolated by antiseptic removal of hemolymph from fly pupae parasitized by infected *Nasonia* females, and inoculation of the hemolymph onto GC media with Kellogg's supplement.

The type strain grows poorly or does not grow on many standard media. It can be grown on brain–heart infusion broth (Difco no. 0037) or solid media. However, improved growth was found on a GC media base (Difco no. 0289) supplemented with Kellogg's additive (Kellogg et al., 1963). Even on this medium, bacterial growth is slow relative to many culturable bacteria, and visible colonies may take 2–3 d at 26°C. Growth is best on solid media. Broth cultures show elevated frequencies of filamentous bacteria relative to agar base solid media. The type strain was isolated from an *N. vitripennis* son-killer strain collected near Salt Lake City, Utah.

The mol% G + C of the DNA is: 39.5 (T_m).

Type strain: SKI4, ATCC 49151.

GenBank accession number (16S rRNA): M90801.

2. ***Candidatus Arsenophonus triatominarum*** Hypsa and Dale 1997, 1143.

tri'a.to'min.a.rum. N.L. n. *Triatominae* subfamily of reduviid bugs; *triatominarum* of the *Triatominae*.

Cells are nonmotile, nonsporeforming, nonflagellated highly filamentous rods ($1\text{--}1.5 \times >15 \mu\text{m}$), and divide by septation. Bacteria grow within the cytoplasm of cells of the reduviid bug *Triatoma infestans*. Phylogenetic analysis of 16S ribosomal gene sequence indicates a close phylogenetic re-

lationship to *A. nasoniae* (97.8% similarity), as do general morphological characteristics and the pattern of infection within tissues of their respective hosts. The bacterium cannot be grown on cell-free minimal medium, but can be maintained in *A. albopictus* cell culture. Biochemical characteristics of the bacterium have not yet been determined.

"*Candidatus A. triatominarum*" was isolated from the hemolymph of *Triatoma infestans*, a reduviid bug also known to be a vector of *Trypanosoma cruzi*, the causative agent of Chagas disease in humans. Within infected *T. infestans*, heavy infections can be found in neural ganglia, visceral

muscles, salivary glands, nephrocytes, testes, ovaries, and dorsal vessels. In contrast to *A. nasoniae*, the bacterium could not be grown on a variety of tested cell-free media, including media upon which *A. nasoniae* can be cultured (Hypsa and Dale, 1997). Phenotypic effects of the bacterium on *T. infestans* are unknown. Strain T11 was isolated from hemolymph of *T. infestans* individuals from a laboratory colony, and cultured in *Aedes albopictus* cell line C6/36.

The mol% G + C of the DNA is: unknown.

Deposited strain: T11.

GenBank accession number (16S rRNA): U91786.

Genus IV. *Brenneria* Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 394.)

LYSIANE HAUBEN AND JEAN SWINGS

Bren. ne'ri.a. M.L. fem. n. *Brenneria* named after Don J. Brenner.

Cells are 0.5–1.0 × 1.3–3.0 µm; have rounded ends; occur singly or rarely in pairs. Gram negative. Motile by peritrichous flagella. Facultatively anaerobic, but anaerobic growth by some species is weak. Optimum temperature, 27–30°C; maximum temperature for growth is 40°C. Oxidase negative. Catalase positive. Acid is produced from fructose, D-galactose, D-glucose, D-mannose, salicin, and sucrose but not from adonitol or dulcitol. Do not possess arginine decarboxylases, arginine dihydrolase, lysine decarboxylases, ornithine decarboxylases, or starch hydrolase.

The species of the genus *Brenneria* comprise a distinct phylogenetic group, as determined by 16S rRNA gene sequence comparisons, and have 12 characteristic signature nucleotides (Table BXII.γ.223 of the genus *Erwinia*).

Brenneria species cause diseases on trees (e.g., deciduous trees and walnut trees), which include blights, cankers, wilts, necrosis, and rots. Ingress by the pathogen generally occurs through natural openings and wounds. More details on disease symptoms are described for each species.

The mol% G + C of the DNA is: 50.1–56.1 (*T_m*, Bd).

Type species: Brenneria salicis (Day 1924) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 394) (*Erwinia salicis* (Day 1924) Chester 1939, 406; "*Bacterium salicis*" Day 1924, 14.)

FURTHER DESCRIPTIVE INFORMATION

Metabolic features are the same as for the genus *Erwinia*.

Pectate lyases are produced by strains of *B. rubrifaciens* (Gardner and Kado, 1976).

Fermentation end products from D-glucose are CO₂ and different combinations of succinate, lactate, formate, and acetate; some form 2,3-butanediol and some ethanol (White and Starr, 1971). Starch is not hydrolyzed beyond dextrins.

16S rDNA sequence analyses of the species of the genus *Brenneria* by Kwon et al. (1997) and Hauben et al. (1998a) are in good agreement except for the type strain of *B. salicis*, and place the genus *Brenneria* within the *Enterobacteriaceae*, closely related to the genera *Pectobacterium*, *Erwinia*, *Pantoea*, and *Enterobacter* (Fig. BXII.γ.201 and Table BXII.γ.223 of the genus *Erwinia*). The gene sequences of the type strains of *B. salicis* sequenced by Kwon et al. (1997) (ATCC 15712) and Hauben et al. (1998a) (LMG 2698) differ in 73 nucleotides. A genomic fingerprinting study by AFLP of 77 *B. salicis* strains revealed the profile of strain LMG 2698 to be highly similar (>80% similarity) to the ones of other

authentic *B. salicis* strains, indicating that strain LMG 2698 is an authentic *B. salicis* (Hauben et al., 1998b).

Virulent or temperate phages have been isolated, characterized, and reported to be active against strains of *B. nigrifluens* and *B. rubrifaciens* (Zeitoun and Wilson, 1969).

Antisera prepared against live or heat-killed cells, nonpurified or purified immunogens, have been used for the differentiation or identification of all *Brenneria* species (De Kam, 1976; Schaad, 1979).

ENRICHMENT AND ISOLATION PROCEDURES

The isolation procedure is the same as for the genus *Erwinia*. The affected plant material is washed in tap water, then in sterile water, and dried by paper toweling. Surface sterilization (3 min in 1:10 dilution of 5.25% active sodium hypochlorite) is sometimes detrimental for isolation. Affected tissue is removed from a young lesion or the edge of older necrotic areas by a sterile scalpel; the tissue is crushed in sterile water, saline, or buffer solution, and is streaked onto a solid medium.

The isolation of some *Brenneria* species can be facilitated by use of selective-differential media, but such media are usually not necessary. *B. nigrifluens*, *B. quercina*, and *B. rubrifaciens* will grow on MS medium (Miller and Schroth, 1972) and produce characteristic colonies. A soluble pink pigment is produced by *B. rubrifaciens*.

MAINTENANCE PROCEDURES

Stock cultures of *Brenneria* species should be grown on standard media of choice at 25–30°C until good growth occurs. The cultures can be maintained for short-term storage in a refrigerator (4–5°C).

Long-term preservation is the same as for the genus *Erwinia*. The bacteria can be successfully stored as lyophilized cultures, usually suspended in a filter-sterilized mixture of 200 ml of horse serum (Oxoid SR035C) to which 1675 mg of nutrient broth (Oxoid) and 20 g of glucose in 67 ml of distilled water is added. Strains have also been stored in liquid nitrogen and in glycerol at –80°C (broth + 15% glycerol).

DIFFERENTIATION OF THE GENUS *BRENNERIA* FROM OTHER GENERA

The differential characteristics of the species of *Brenneria* are given in Tables BXII.γ.208 and BXII.γ.209.

TABLE BXII.γ.208. Diagnostic reactions for *Brenneria* species^a

Characteristic	1. <i>B. salicis</i>	2. <i>B. alni</i>	3. <i>B. nigrifluens</i>	4. <i>B. paradisiaca</i>	5. <i>B. quercina</i>	6. <i>B. rubrifaciens</i>
Indole	—	—	—	+	—	—
Methyl red	nd	+	+	nd	—	nd
Growth factors required	nd	—	—	nd	+	nd
Endoglucanase activity	nd	+	—	nd	+	nd
Esculin hydrolase	nd	—	+	nd	—	nd
β-Galactosidase	+	—	v	+	v	+
Pectinase	nd	—	—	+	—	nd
Urease	—	+	v	—	—	—
<i>Acid production from:</i>						
Arabinose	—	+	+	+	—	+
Esculin	—	nd	+	nd	—	—
Maltose	nd	+	—	—	—	+
Melibiose	—	—	+	+	—	nd
Raffinose	+	—	+	+	—	—
Sorbitol	nd	—	+	—	v	—
Trehalose	nd	+	+	nd	—	nd
Xylose	—	+	+	+	—	—
<i>Utilization of carbon sources:</i>						
L-Arabinose	nd	nd	+	+	—	+
Citrate	+	—	v	+	+	+
Galacturonic acid	—	nd	—	+	—	—
Maltose	nd	nd	—	nd	+	—
Raffinose	+	nd	+	nd	—	—
L-Rhamnose	nd	nd	nd	+	nd	—
Sorbitol	—	nd	+	nd	—	—
Trehalose	nd	nd	+	+	—	—
Xylose	—	nd	—	+	—	—
<i>Utilization of nitrogen sources:</i>						
Acetamide	nd	nd	—	nd	—	+
Anthranilic acid	+	nd	—	nd	—	nd
Glycocyamine	nd	nd	+	nd	—	+
L-Serine	+	nd	+	+	—	+
Threonine	+	nd	—	+	+	+
Tryptamine	—	nd	—	+	—	—
Xanthin	+	nd	+	—	nd	+
<i>Sensitivity toward:</i>						
Streptomycin	nd	nd	—	+	nd	nd

^aFor symbols see standard definitions; nd, not determined.**TAXONOMIC COMMENTS**

The species of the genus *Brenneria* were formerly classified in the genus *Erwinia*. We refer to the taxonomic comments of the genus *Erwinia* for a discussion of this issue.

Characteristics that differentiate *Brenneria* from the genera

Pectobacterium, *Pantoea*, and *Erwinia* are given in Table BXII.γ.223 of the *Erwinia* chapter. As it is very difficult to differentiate them phenotypically, genomic methods are recommended for differentiation.

List of species of the genus Brenneria

1. ***Brenneria salicis*** (Day 1924) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 394) (*Erwinia salicis* (Day 1924) Chester 1939, 406; "*Bacterium salicis*" Day 1924, 14.)

sa' li. cis. L. n. *salix* the willow; L. gen. n. *salicis* of the willow.

The characteristics are as given for the genus and as listed in Tables BXII.γ.208 and BXII.γ.209.

Grows poorly on nutrient agar but moderately well on YDC agar (0.5% yeast extract, 1% dextrose, 3% CaCO₃) or on glucose nutrient agar. Colonies on 0.5% starch potato agar (pH 6.5) are yellowish in 2–3 d. A bright yellow pigment is produced on autoclaved potato tissue. Craters form around colonies on the pectate gel of Paton (1959).

A PCR based identification and detection method was developed with primers Es1a (5'-GCGGCGGACGGGTGA-GTAAA-3') and Es4b (5'-CTAGCCTGTCAGTTTGAATG-CT-3'), annealing at 64°C, derived from the 16S rDNA sequence of *B. salicis* (Hauben et al., 1998b).

Antisera have been used for the identification of *B. salicis* (De Kam, 1976; 1986) and soluble antigens were detected in the leaves of *Salix alba* (De Kam, 1982).

Causes a vascular wilt (watermark disease) of *Salix* species. The pathogen occurs mainly in the xylem vessels of the host plant. Infected willows show wilted, dried, brown-colored leaves and a watery, transparent color of the wood. Occasionally the whole tree is killed.

The mol% G + C of the DNA is: 51.3–51.5 (*T_m*, Bd.).

Type strain: ATCC 15712, ICMP 1587, LMG 2698, NCPPB 447.

GenBank accession number (16S rRNA): U80210 and Z96097.

2. ***Brenneria alni*** (Surico, Mugnai, Pastorelli, Giovannetti and Stead 1996) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 395) (*Erwinia alni* Surico, Mugnai, Pastorelli, Giovannetti and Stead 1996, 725.)

TABLE BXII.γ-209. Additional reactions for *Brenneria* species^a

Characteristic	1. <i>B. salicis</i>	2. <i>B. alni</i>	3. <i>B. nigrifluens</i>	4. <i>B. paradisiaca</i>	5. <i>B. quercina</i>	6. <i>B. rubrifaciens</i>
Growth in 5% NaCl	nd	+	+	nd	+	nd
Growth at 37°C	nd	+	+	nd	+	nd
NO ₃ ⁻ → NO ₂ ⁻	nd	-	-	nd	-	nd
H ₂ S from cysteine	nd	+	+	nd	+	nd
Acetoin	nd	+	+	nd	+	nd
Oxidase	-	-	-	-	-	-
Catalase	+	+	+	+	+	+
Arginine dihydrolase	-	-	-	-	-	-
Caseinase	-	-	-	-	-	-
Arginine decarboxylase	-	-	-	-	-	-
Aspartase	+	nd	+	nd	nd	nd
Gelatinase	nd	-	-	nd	-	nd
Lysine decarboxylase	-	-	-	-	-	-
Ornithine decarboxylase	-	-	-	-	-	-
Phenylalanine deaminase	-	-	-	nd	-	-
Starch hydrolase	-	-	-	-	-	-
<i>Acid production from:</i>						
N-Acetylglucosamine	+	nd	+	+	+	+
Adonitol	-	-	-	-	-	-
Amygdalin	+	-	+	nd	+	+
Arabitrol	-	nd	-	-	-	-
Arbutine	+	nd	+	+	+	+
Cellobiose	nd	nd	nd	nd	-	-
Dextrin	-	nd	-	nd	-	-
Dulcitol	-	-	-	-	-	-
Erythritol	-	nd	-	-	-	-
Erythrose	-	-	nd	-	nd	nd
Fructose	+	+	+	+	+	+
DL-Fucose	-	nd	-	-	-	-
D-Galactose	+	+	+	+	+	+
Gentiobiose	nd	nd	nd	+	nd	nd
D-Glucose	+	+	+	+	+	+
Gluconate	nd	nd	nd	+	nd	nd
Glycerol	nd	+	+	nd	+	nd
Glycogen	-	nd	-	-	-	-
Inulin	-	nd	-	nd	-	-
Lactose	-	-	nd	-	-	-
D-Lyxose	nd	nd	nd	-	nd	nd
D-Mannitol	+	+	+	nd	+	+
D-Mannose	+	+	+	+	+	+
α-Methylmannoside	-	nd	-	-	-	-
α-D-Melezitose	-	-	nd	-	-	-
β-Methylglucoside	+	nd	+	+	+	+
β-Methylxyloside	-	nd	-	-	-	-
Rhamnose	+	nd	+	+	+	+
Ribose	+	nd	+	+	+	+
Salicin	+	+	+	+	+	+
Sorbose	-	nd	-	-	-	-
Starch	-	nd	-	-	-	-
Sucrose	+	+	+	+	+	+
Tagatose	-	nd	-	-	-	-
Xylitol	nd	nd	nd	-	nd	nd
<i>Utilization of carbon sources:</i>						
Adipate	-	nd	-	-	-	-
Adonitol	-	nd	-	nd	v	-
Amygdalin	nd	nd	-	nd	nd	nd
D-Arabinose	-	nd	nd	nd	nd	nd
D-Arabitrol	-	nd	-	-	-	-
Arbutin	+	nd	+	+	+	+
Benzoate	-	nd	-	-	-	-
Betaine	-	nd	-	-	-	-
Butanate	nd	nd	-	nd	nd	nd
Butanol	-	nd	nd	-	-	-
Cellobiose	nd	nd	-	nd	-	nd
Dextrin	-	nd	-	nd	-	-
Dulcitol	-	nd	-	nd	-	-
meso-Erythritol	-	nd	-	nd	-	-
Esculin	+	nd	+	nd	+	+
Ethylene glycol	-	nd	-	nd	-	-
Fructose	+	nd	+	+	+	+
Fumarate	+	nd	+	+	+	+
D-Galactose	+	nd	+	+	+	+
Gallate	-	nd	-	-	-	-

(continued)

TABLE BXII.γ-209. (cont.)

Characteristic	1. <i>B. salicis</i>	2. <i>B. alni</i>	3. <i>B. nigrifluens</i>	4. <i>B. paradisiaca</i>	5. <i>B. quercina</i>	6. <i>B. rubrifaciens</i>
Gluconate	+	nd	+	+	+	+
Glucose	+	nd	+	+	+	+
Glucuronic acid	nd	nd	+	nd	nd	nd
L-Glutaminic acid	—	nd	nd	nd	nd	—
Glycerol	+	nd	+	+	+	+
Glycogen	—		—	nd	—	—
Glycol	—	nd	nd	nd	nd	nd
α-Ketoglutarate	nd	nd	nd	+	nd	nd
Lactose	—	nd	—	nd	—	—
Lactulose	—	nd	—	nd	nd	—
Lyxose	—	nd	—	nd	—	—
Malate	+	nd	+	+	+	+
Malonate	nd		—	—	nd	—
Malonic acid	—	nd	nd	nd	—	—
D-Mannitol	+	nd	+	+	+	+
Mannose	+	nd	+	+	+	+
Melibiose	+	nd	+	+	nd	nd
α-D-Melezitose	—	nd	—	nd	—	—
Methanol	—	nd	—	—	—	—
α-Methylglucoside	nd	nd	nd	—	nd	nd
β-Methylglucoside	+	nd	+	+	+	+
Mucate	nd	nd	nd	+	nd	nd
Naphthalene	—		—	nd	—	—
Oxalate	—	nd	—	—	—	—
Pectinic acid	—	nd	nd	nd	nd	—
Propanol	—	nd	—	nd	—	—
Propionate	—	nd	—	—	—	—
Ribose	+	nd	+	+	+	+
Salicin	+	nd	+	+	+	+
Sorbinic acid	—	nd	—	nd	—	—
Sorbose	—	nd	—	—	—	—
Starch	—	nd	—	nd	—	—
Succinate	+	nd	+	+	+	+
Sucrose	+	nd	+	nd	+	+
Tartrate	nd	nd	nd	—	nd	nd
Triacetine	nd	nd	nd	—	nd	nd
Xylitol	—	nd	—	—	—	—
<i>Utilization of nitrogen sources:</i>						
Adenine	nd	nd	nd	+	nd	nd
Alanine	+	nd	+	+	+	+
Allantoine	+	nd	+	+	+	+
Ammonium chloride	+	nd	+	+	+	+
Arginine	+	nd	+	+	+	+
Asparagine	+	nd	nd	nd	+	+
Asparaginic acid	+	nd	+	+	+	+
Betaine	—	nd	—	—	—	—
Carnosine	nd	nd	nd	+	nd	nd
Choline	—	nd	—	—	—	—
Citrulline	+	nd	+	+	+	+
Cysteamine	—	nd	—	—	—	—
Creatine	nd	nd	nd	+	nd	nd
Ethanolamine	nd	nd	+	nd	+	+
Glucosamine	+	nd	+	+	+	+
L-Glutaminic acid	+	nd	+	+	+	+
Glutathion	+	nd	+	+	+	+
Glycine	+	nd	+	+	+	+
Glycylglycine	+	nd	+	+	+	+
Guanine	nd	nd	nd	—	nd	nd
Histidine	+	nd	+	+	+	+
Hydroxyproline	—	nd	—	—	—	—
Isoleucine	+	nd	nd	nd	+	+
Kynureninic acid	—	nd	—	—	—	—
Leucine	+	nd	+	+	+	+
L-Methionine	+	nd	+	+	+	+
Nicotinic acid	nd	nd	nd	—	nd	nd
Octopine	nd	nd	nd	+	nd	nd
Ornithine	nd	nd	nd	+	nd	nd
Phenylalanine	+	nd	+	+	+	+
Quinolinic acid	—	nd	—	—	—	—
Sarcosine	—	nd	—	—	—	—
Spermidine	—	nd	—	—	—	—
Spermine	—	nd	—	—	—	—
Taurine	nd	nd	nd	—	nd	nd

(continued)

TABLE BXII.γ.209. (cont.)

Characteristic	1. <i>B. salicis</i>	2. <i>B. alni</i>	3. <i>B. nigrifluens</i>	4. <i>B. paradisiaca</i>	5. <i>B. quercina</i>	6. <i>B. rubrifaciens</i>
Thymine	—	nd	—	nd	—	—
Trigonelline	—	nd	—	—	—	—
L-Tryptophan	+	nd	nd	+	+	+
Tyrosine	+	nd	+	+	+	+
Ureum	+	nd	+	+	+	+
Valine	+	nd	+	+	+	+
<i>Sensitivity toward:</i>						
Amoxycillin	nd	nd	+	+	+	+
Ampicillin	nd	nd	+	+	+	+
Bacitracin	—	nd	—	nd	—	—
Carbenicillin	+	nd	+	+	+	+
Cephalexin	+	nd	+	+	+	+
Cephaloridine	+	nd	+	+	+	+
Cephalotine	+	nd	+	+	nd	+
Chloramphenicol	+	nd	+	+	+	+
Cloxacyclin	—	nd	—	nd	—	—
Colistine sulfate	—	nd	—	nd	—	—
Doxycyclin	+	nd	+	nd	+	+
Erythromycin	—	nd	—	nd	—	—
Framycetine	+	nd	+	+	nd	+
Furazolidone	+	nd	+	+	+	+
Fusidinic acid	—	nd	—	—	—	—
Gentamicin	—	nd	—	nd	—	—
Kanamycin	+	nd	nd	+	v	nd
Lincomycin	—	nd	—	nd	nd	—
Methicillin	—	nd	—	—	—	—
Nalidixinic acid	+	nd	+	+	+	+
Neomycin	nd	nd	—	nd	—	nd
Nitrofurantoin	+	nd	+	nd	+	+
Oxytetracycline	+	nd	+	+	+	+
Penicillin	nd	nd	+	nd	+	+
Polymyxin	nd	nd	—	nd	nd	—
Spectinomycin	—	nd	—	nd	nd	nd
Sulfafurazol	—	nd	—	—	—	—
Novobiocine	nd	nd	+	nd	nd	nd
Tetracycline	+	nd	+	+	+	+

^aFor symbols see standard definitions; nd, not determined.

al' ni. L. fem. n. *alnus* alder; L. gen. n. *alni* of alder, referring to the plant from which the organism was first isolated.

The characteristics are as given for the genus and as listed in Tables BXII.γ.208 and BXII.γ.209.

Abundant growth occurs in nutrient broth and on nutrient agar, YDC (0.5% yeast extract, 1% dextrose, 3% CaCO₃), Miller-Schroth medium. Colonies on NSA (nutrient agar containing 5% sucrose) are umbonate with translucent margins, whitish, and 2–2.5 mm in diameter after 5 days at 27°C.

Causes bark canker disease of alder (*Alnus* sp.). At first, diseased alder plants have small necrotic cankers in the bark of the trunk and in the bark of branches, twigs, and suckers. These cankers are slightly sunken, dark brown, and irregularly circular and appear to be water soaked. They usually develop at lenticels or are localized at nodes with dead twigs or on leaf scars. As the infection progresses, the necrosis spreads laterally, becoming deeper and reaching the cambium and sometimes the first layers of wood, which then turn brown. Cankers may eventually girdle and kill a branch or tree.

The mol% G + C of the DNA is: 50.1–50.7 (*T_m*).

Type strain: ICMP 12481, NCPPB 3934.

GenBank accession number (16S rRNA): AJ223468.

3. **Brenneria nigrifluens** (Wilson, Starr and Berger 1957) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben,

Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 395) (*Erwinia nigrifluens* Wilson, Starr and Berger 1957, 673.)

ni.gri.flu'ens. L. adj. *niger* nigra black; L. v. *fluo* flow; M.L. part. adj. *nigrifluens* black flowing.

The characteristics are as described for the genus and as listed in Tables BXII.γ.208 and BXII.γ.209.

Colonies on Bacto-EMB (Difco) agar are dark violet with a green metallic sheen. Craters form around colonies on the polypectate medium (1 liter nearly boiling distilled water, 1 ml of bromothymol blue, 6 ml of 10% CaCl₂·H₂O, 2 g of sodium polypectate) of Hildebrand (1971). Growth media should contain yeast extract and should be at pH 7–8.

Causes a bark necrosis of the Persian walnut (*Juglans regia*). The disease is visible as dark brown necrotic areas on the trunks or large branches. The lesions extend well beyond the discolored bark. Exudates may be produced. The lesions are superficial, but may penetrate enough to kill the phloem.

The mol% G + C of the DNA is: 56.1 (*T_m*, Bd).

Type strain: ATCC 13028, ICMP 1578, LMG 2694.

GenBank accession number (16S rRNA): U80203, Z96095.

4. **Brenneria paradisiaca** (Fernández-Borrero and López-Duque 1970) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publica-

tion: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 396) (*Erwinia paradisiaca* Fernández-Borrero and López-Duque 1970, 22.)

pa.ra.di.si.a' ca. M.L. n. generic name *Musa paradisiaca* banana.

The characteristics are as given for the genus and as listed in Tables BXII.γ.208 and BXII.γ.209.

Causes rhizome rot of roots and soft rot of green fruit of *Musa paradisiaca*.

The mol% G + C of the DNA is: 54.7 (T_m , Bd).

Type strain: ATCC 33242, LMG 2542, NCPPB 2511.

GenBank accession number (16S rRNA): Z96096.

5. **Brenneria quercina** (Hildebrand and Schroth 1967) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 395) (*Erwinia quercina* Hildebrand and Schroth 1967, 253.)

quer.ci' na. L. n. *quercus* oak; L. suff. *ina* belonging to; M.L. part. adj. *quercina* oak-belonging.

The characteristics are as given for the genus and as listed in Tables BXII.γ.208 and BXII.γ.209.

Growth on PGPC agar (60 g homogenized potato, 10 g glucose, 10 g peptone, 1 g CaCO₃, 1 liter distilled water) is luxuriant, and, after 24 h, colonies are white, circular, and raised with entire margins. Craters form around colonies on polypectate gel of Hildebrand (1971), but no peptolytic activity was observed in potato or carrot tissue.

Small amounts of gas are produced (possibly from peptone) in a glucose peptone medium and in PGPC.

Causes copious oozing of sap from acorns ("drippy nut") and shoot blight of oak: *Quercus agrifolia* and *Q. wislizeni*. Acorns may rot somewhat, and cups may ooze after nutfall. Oozing takes place in late summer when temperatures average about 29°C. Superficially rots onion (but not potato) slices and induces profuse lateral root development in 3–4 days on slices of carrot, turnip, or beet.

The mol% G + C of the DNA is: from 54.6–55.1 (T_m , Bd).

Type strain: ATCC 29281, ICMP 1845, LMG 2724, NCPPB 1852.

GenBank accession number (16S rRNA): AJ223469.

6. **Brenneria rubrifaciens** (Wilson, Zeitoun and Fredrickson 1967) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 395) (*Erwinia rubrifaciens* Wilson, Zeitoun and Fredrickson 1967, 621.)

rub.ri.fac'i.ens. L. adj. *ruber* red; L. v. *facio* make; M.L. part. adj. *rubrifaciens* red-producing.

The characteristics are as given for the genus and as listed in Tables BXII.γ.208 and BXII.γ.209.

Produces a pink soluble pigment when grown on yeast extract glucose chalk agar (YDC, 0.5% yeast extract, 1% dextrose, 3% calcium carbonate (Dye, 1968). The structure of a 3-(3-carboxymethyl-4,5-dihydro-4,5-dioxo-2-pyrrolyl)-4,5,6-trihydroxy-pyridine-2-acetic acid (rubrifacine) is suggested (Feistner et al., 1984) as the main component of the pigment.

Grows poorly on nutrient agar, but well on YDC on which colonies are cream to yellow, low convex, smooth, shining with entire margins. Craters form around colonies on the polypectate gel B and C of Hildebrand (1971), but no peptolytic activity was observed on vegetable tissue.

Causes a phloem necrosis of Persian walnut trees (*Juglans regia*). Dark brown to black necrotic streaks from along the inner bark and outer sapwood. In places, the necrosis develops outward to the periderm, which breaks, and a dark exudate emerges.

The mol% G + C of the DNA is: 52.0–52.6 (T_m , Bd).

Type strain: ATCC 29291, ICMP 1915, LMG 2709, NCPPB 2020.

GenBank accession number (16S rRNA): U80207 and Z96098.

Genus V. *Buchnera* Munson, Baumann and Kinsey 1991b, 566^{VP}

PAUL BAUMANN, LINDA BAUMANN AND NANCY A. MORAN

Buch.ne'ra. M.L. fem. n. *Buchnera* named for Paul Buchner, German biologist who made extensive contributions to the study of endosymbiosis.

Round or slightly oval cells, 2–5 µm in diameter. **Gram-negative** cell wall, lack flagella. Cells divide by binary fission. Do not form resting stages or endospores. **Found in bacteriocytes** of aphids in vesicles derived from the host membranes. Capable of respiration. **Cannot be cultivated** outside the aphid host. Essential for the survival of the aphid.

The mol% G + C of the DNA is: 26–27.

Type species: ***Buchnera aphidicola*** Munson, Baumann and Kinsey 1991b, 567.

FURTHER DESCRIPTIVE INFORMATION

Cell structure and life cycle During their reproductive phase most aphids contain within their body cavity a bilobed structure called a bacteriome consisting of 60–90 polyploid cells called bacteriocytes (Douglas and Dixon, 1987). These cells are filled with host-derived vesicles (symbiosomes) containing *Buchnera*

(Fig. BXII.γ.194) (Hinde 1971b; Griffiths and Beck, 1973; McLean and Houk, 1973; Akhtar and van Emden, 1994). This organism is spherical or oval in shape having a diameter of 2–5 µm and a cell wall resembling that of Gram-negative bacteria (Fig. BXII.γ.195). A thin line indicative of the peptidoglycan layer has been detected between the two unit membranes (Houk et al., 1977). The presence of peptidoglycan is also indicated by chemical analysis and by the alteration of cell wall structure observed upon addition of penicillin to the aphid diet (Griffiths and Beck, 1974; Houk et al., 1977).

In their most active reproductive stage, aphids are females that reproduce by parthenogenesis. The embryos develop within the mother, which gives birth to live young (Blackman, 1987). A representative aphid, *Schizaphis graminum*, when born weighs 24 mg and contains 2×10^5 cells of *Buchnera*. In 9–10 d, it reaches its maximum weight of 540 mg and contains 5.6×10^6 endo-

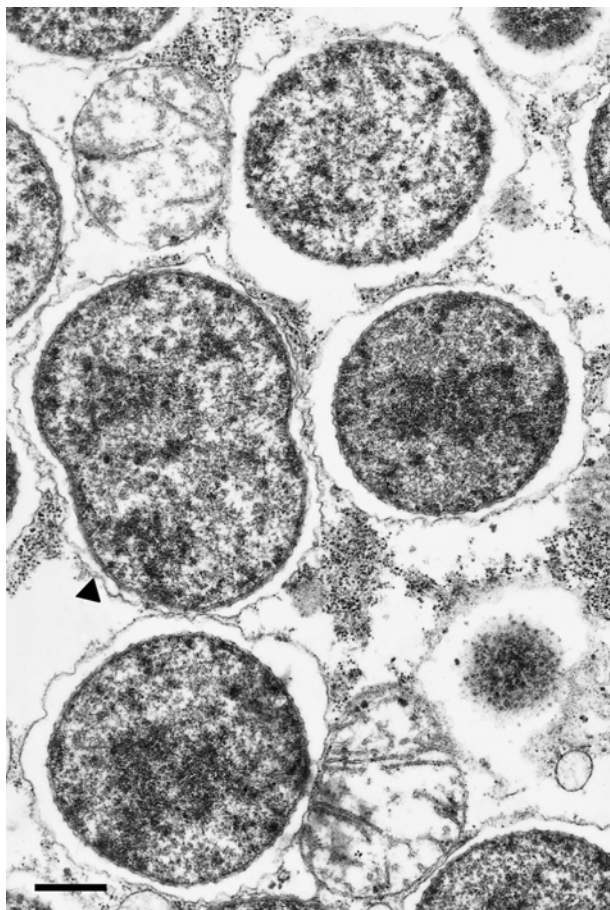


FIGURE BXII.γ.194. Electron micrograph of *Buchnera* within aphid bacteriocytes. Arrow indicates host-derived vesicle membrane. Bar = 1 μm. (Printed with permission of M. Kinsey and D.L. McLean.)

symbionts (Baumann and Baumann, 1994). The increase in the total number of *Buchnera* parallels the increase in the weight of the aphid. The endosymbionts are partitioned between the maternal and the embryonic bacteriocytes. In mature aphids, most of the endosymbionts are found in the embryos (Humphreys and Douglas, 1997). Halfway through the reproductive period the number of maternal bacteriocytes undergoes a sharp decrease (Douglas and Dixon, 1987) and there is degradation of the endosymbionts (Hinde, 1971a; Griffiths and Beck, 1973).

Aphids may also form sexual forms; the females deposit eggs that overwinter and hatch in the spring. *Buchnera* is transmitted maternally to the embryos and the eggs. The mechanism of transmission is complex and has not been extensively studied (Hinde, 1971a; Blackman, 1987; Brough and Dixon, 1990).

Phylogenetic analyses Based on 16S rDNA phylogeny, *Buchnera* is a lineage within the *Gamma*proteobacteria (Unterman et al., 1989; Munson et al., 1991a; Moran et al., 1993; van Ham et al., 1997). The closest known organisms (Fig. BXII.γ.196) are the nonculturable *Wigglesworthia glossinidia* (endosymbionts of tsetse flies) and endosymbionts of carpenter ants as well as members of the *Enterobacteriaceae* (Aksoy, 1995b; Aksoy et al., 1995; Schröder et al., 1996). These organisms constitute four separate lineages and their relationship to each other is not clearly resolved based on 16S rDNA.

Fig. BXII.γ.197 presents the results of a phylogenetic analysis

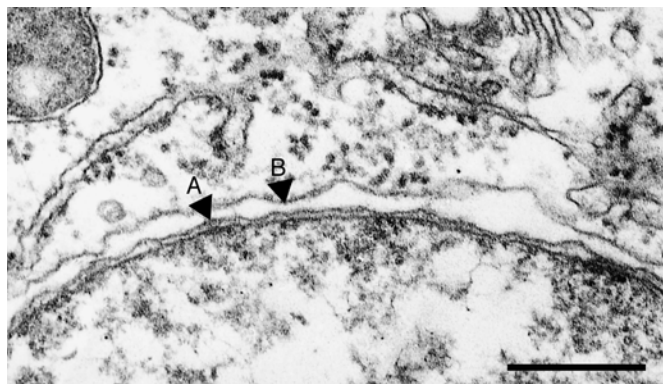


FIGURE BXII.γ.195. Electron micrograph of *Buchnera* showing the Gram-negative cell wall (A) and the vesicle membrane (B). Bar = 0.5 μm. (Printed with permission of M. Kinsey and D.L. McLean.)

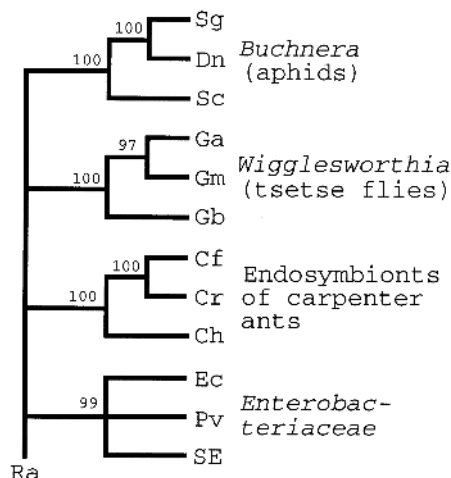


FIGURE BXII.γ.196. Phylogenetic trees resulting from parsimony analyses (D.L. Swofford, 1993, PAUP 3.1.1, Champaign, IL) using 16S rDNA nucleotide sequences (1189 sites) of *Buchnera* and other closely related insect endosymbionts as well as free-living bacteria. Numbers above nodes are bootstrap percentages from parsimony searches (1000 replicates). Data for *Wigglesworthia glossinidia* from Aksoy (1995b); the following abbreviations designate the insect hosts (in parentheses): Ga (*Glossina austeni*), Gm (*G. morsitans*), Gb (*G. brevipalpis*). Data for carpenter ant endosymbionts from Schröder et al. (1996); hosts: Cf (*Camponotus floridanus*), Cr (*C. rufipes*), Ch (*C. herculeanus*); SE (secondary endosymbiont of the aphid *Acyrtosiphon pisum*); Ra (*Ruminobacter amylophilus*).

primarily restricted to *Buchnera*. Most of the characterized endosymbionts are from aphids belonging to the family *Aphididae*. Based on 16S rDNA gene sequences (Fig. BXII.γ.197a), *Buchnera* forms one cluster within which a subcluster of endosymbionts from aphids of *Aphididae* is clearly differentiated. An additional cluster consists of endosymbionts from the aphids *Melaphis rhois* (Mr) and *Schlechtendalia chinensis* (Sc), both of which are members of the tribe Melaphidini in the family *Pemphigidae*. Relationships between the remaining endosymbionts from this family as well as from two other aphid families are not clearly resolved using 16S rDNA gene sequences. Using a portion of the chromosomal *trpB*, relationships among endosymbionts of some members of the *Aphididae* and especially within the aphid genus *Uro-*

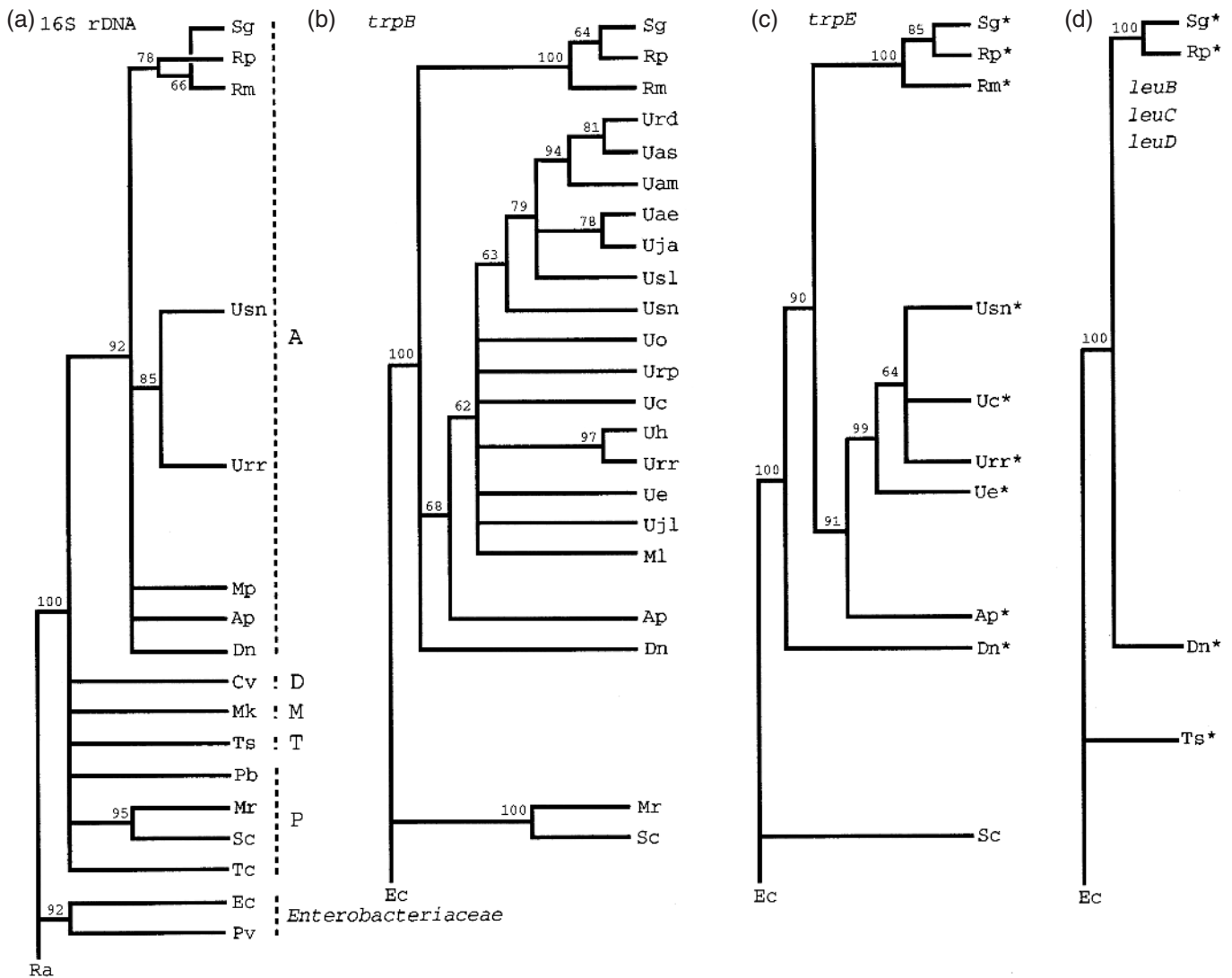


FIGURE BXII.γ.197. Phylogenetic trees resulting from parsimony analyses (D.L. Swofford, 1993, PAUP 3.1.1, Champaign, IL) using *Buchnera* (a) 16S rDNA (1475 sites), (b) *trpB* (679 sites), (c) *trpE* (1623 sites), and (d) *leuB*, *leuC*, and *leuD* (total 3140 sites). *, plasmid-associated genes; numbers above nodes are bootstrap percentages from parsimony searches (1000 replicates). Abbreviations designate insect hosts in the order from top to bottom (given in parentheses): *Sg* (*Schizaphis graminum*), *Rp* (*Rhopalosiphum padi*), *Rm* (*R. maidis*), *Urd* (*Uroleucon rudbeckiae*), *Uas* (*U. astronomus*), *Uam* (*U. ambrosiae*), *Uae* (*U. aeneum*), *Uja* (*U. jaceiae*), *Usl* (*U. solidaginis*), *Usn* (*U. sonchi*), *Uo* (*U. obscurum*), *Urp* (*U. rapunculoidis*), *Uc* (*U. caligatum*), *Uh* (*U. helianthicola*), *Urr* (*U. rurale*), *Ue* (*U. erigeronense*), *Ujl* (*U. jaceicola*), *MI* (*Macrosiphoniella ludoviciana*), *Mp* (*Myzus persicae*), *Ap* (*Acyrtosiphon pisum*), *Dn* (*Diuraphis noxia*), *Cv* (*Chaitophorus viminalis*), *Mk* (*Mindarus kinseyi*), *Ts* (*Thelaxes suberi*), *Pb* (*Pemphigus betae*), *Mr* (*Melaphis rhois*), *Sc* (*Schlechtendalia chinensis*), *Tc* (*Tetraneura caerulescens*). Species of the Enterobacteriaceae: *Ec* (*E. coli*), *Pv* (*P. vulgaris*). Dashed lines in 4a designate aphid species within one family: A, Aphididae, D, Drepanosiphidae, M, Mindaridae, T, Thelaxidae, P, Pemphigidae. Data for *Ts* and *Tc* in 4a and 4d from van Ham et al. (1997); for *Rp* in 4d from Bracho et al. (1995); for other sources of data for *Buchnera* see text.

leucon could be resolved (Fig. BXII.γ.197b). These relationships are in broad agreement with more limited studies involving *Buchnera* plasmid-associated *trpE* and plasmid-associated *leuBCD* (Fig. BXII.γ.197c and d) (Bracho et al., 1995; Rouhbachsh et al., 1996, 1997; Baumann et al., 1997, 1999; van Ham et al., 1997). In addition, the phylogenetic relationships based on *trpB* agree with the phylogeny reconstructed for the same species of *Uroleucon* using nuclear and mitochondrial sequences from the aphids (Moran et al., 1999). This pattern of congruence of phylogenies derived from chromosomal genes, plasmid genes, and host genes is strong support for a completely vertical mode of evolution, with no transfer of either bacteria or plasmids among lineages of hosts (Moran and Baumann, 1994). An implication of the

congruence is that bacterial and host ancestors representing corresponding nodes on the phylogenies occurred contemporaneously, allowing dates inferred from fossil aphids to be extended to ancestral *Buchnera* (Moran et al., 1993). Phylogenetic analysis and the rate of 16S rDNA change inferred from fossil aphids has suggested that the *Buchnera*-aphid association is the result of a single infection of an aphid ancestor that occurred 150–250 million years ago. In *Buchnera* the rate of 16S rDNA evolutionary change is about twice that of free-living bacteria and about 36 times greater than in homologous regions of the host 18S rDNA (Moran, 1996; Moran et al., 1995). The increased rate of change in endosymbionts compared to free-living bacteria extends to genes other than 16S rDNA (Moran, 1996).

Nutrition and metabolism Insects, like other animals, are thought to require preformed ten essential amino acids (Dadd, 1985; Douglas, 1998). Plant sap (phloem), the diet of aphids, is rich in carbohydrates but is deficient in nitrogenous compounds (Dadd, 1985; Sandström and Pettersson, 1994). It has been thought that one of the functions of *Buchnera* is the synthesis of the essential amino acids and their provision to the aphid host (Dadd, 1985; Douglas, 1998). Some species of aphids are able to grow on synthetic diets even in the absence of some of the essential amino acids. The incorporation of antibiotics into such diets results in the elimination of the endosymbionts and the failure of the aphid to grow and reproduce. There is some sparing effect when essential amino acids are included in the antibiotic-containing diet. These results have been interpreted as being consistent with the synthesis of essential amino acids by the endosymbionts (Douglas, 1998). The most detailed experiments involve the essential amino acid tryptophan. Douglas and Prosser (1992) have shown there is a sparing effect of tryptophan on aphid growth in chlortetracycline-containing synthetic diet. In addition, these investigators detected the enzyme tryptophan synthase in *Buchnera*. Activity was absent in aphids treated with chlortetracycline that lacked the endosymbionts.

Using synthetic diets containing radiolabeled amino acids, it has been shown that the synthesis of the essential amino acids arginine, threonine, isoleucine, and lysine was eliminated by the inclusion of rifampicin in the diet (Liadouze et al., 1996). Similarly, experiments using radiolabeled sulfate showed that endosymbionts reduce sulfur and incorporate it into methionine and cysteine, which are made available to aphid tissue (Douglas, 1988). Sasaki et al. (1991) showed that treatment of aphids with rifampicin eliminated the incorporation of dietary [^{15}N]-glutamine into the essential amino acids arginine, histidine, isoleucine and/or leucine, phenylalanine, threonine, and valine.

Glutamine is one of the predominant amino acids in phloem (Sasaki et al., 1990; Sandström and Pettersson, 1994). Sasaki and Ishikawa (1995) have found that it is the predominant amino acid in aphid hemolymph. These investigators have shown that bacteriocytes take up glutamine and convert it to glutamate. Subsequently glutamate is taken up by *Buchnera*. The isolated endosymbionts incorporate the nitrogen from this compound into the essential amino acids isoleucine, leucine, valine, and phenylalanine as well as into a number of other nonessential amino acids.

Whitehead and Douglas (1993) isolated vesicles containing *Buchnera* (symbiosomes). The symbiosomes readily took up acetic, glutamic, and aspartic acid as well as tricarboxylic acid cycle intermediates and oxidized them to CO_2 . Oxygen consumption was also detected and was greatly reduced by KCN. These results indicate a respiratory metabolism.

Genetics *Buchnera* contains a single copy of the genes encoding for rRNAs that are organized into two transcription units consisting of 1) 16S rRNA (GenBank accession number M63246) and 2) tRNA^{Glu}-23S rRNA-5S rRNA (GenBank accession number U09230) (Baumann et al., 1995). A single copy of these genes is consistent with the slow growth rate of the endosymbiont (Baumann et al., 1995). Upstream of the rRNA genes are sequences that are readily recognized as -35 and -10 regions of a putative promoter.

The genome size of *Buchnera* from the aphid *Acyrthosiphon pisum* has been found to be 657 kb (Charles and Ishikawa, 1999). Approximately 113 kb of the genome of *Buchnera* from the aphid *S. graminum*, 12 kb from the endosymbiont of *Diuraphis noxia*, 11

kb from the endosymbiont of *S. chinensis*, and 6 kb from the endosymbiont of *A. pisum* have been sequenced (Ohtaka et al., 1992; Baumann et al., 1995; Sato and Ishikawa, 1997b; Baumann and Baumann, 1998; Clark et al., 1998a, 1998b; Thao and Baumann, 1998). The gene content obtained from the nucleotide sequence is an indication of the metabolic capacity of an organism. In *Buchnera* from *S. graminum* 109 open reading frames were detected. Of these, 89 corresponded to known genes and the remaining 20 had similarities to open reading frames on the *E. coli* genome, which have no known function (Blattner et al., 1997). *Buchnera* contains *dnaA*, encoding a protein that initiates bidirectional chromosome replication, and *ftsZ*, encoding a protein involved in septum formation during cell division (Lai and Baumann, 1992; Baumann and Baumann, 1998). Among other genes that were found are those encoding proteins for peptidoglycan synthesis, cell division, DNA replication, DNA transcription, ribosomal proteins, amino acid tRNA synthases, ATP synthase, electron transport, protein secretion, and glycolysis. In addition, genes for three tRNAs were detected. Genes encoding homologs of proteins involved in the *E. coli* heat shock response (*groEL*, *groES*, *htrA*, *dnaK*, *dnaJ*) and the cold shock response (*hscA*, *hscB*) were also detected (Ohtaka et al., 1992; Hassan et al., 1996; Sato and Ishikawa, 1997b; Clark et al., 1998b). In addition, some of the genes encoding enzymes for the biosynthesis of aromatic amino acids (shikimate pathway, tryptophan branch), branched chain amino acids (isoleucine, valine, leucine), lysine, cysteine, and serine, as well as genes for the complete pathway of histidine biosynthesis, were also found. The presence of genes for enzymes of amino acid biosynthesis is in marked contrast to such fastidious organisms as *Mycoplasma genitalium* and *Borrelia burgdorferi* in which the genes encoding enzymes of amino acid biosynthesis were absent (Fraser et al., 1995, 1997). Retention of amino acid biosynthetic genes may reflect the role of these pathways in the mutualistic association with the host aphids.

The overproduction of essential amino acids for the aphid host may involve modifications of some mechanisms that regulate the levels of biosynthetic enzymes (Baumann et al., 1995). One possible modification is an increase in the level of a key regulated enzyme by gene amplification. This appears to be the case in the tryptophan biosynthetic pathway of rapidly growing aphids in which the genes for the enzyme anthranilate synthase (*trpEG*) are present as tandem repeats on plasmids. In the case of *Buchnera* from *S. graminum*, *trpEG* is present as four tandem repeats that constitute a plasmid of 14.3 kb (Lai et al., 1994). The remaining genes of the tryptophan biosynthetic pathway (*trpDC[F]BA*) are chromosomal (Munson and Baumann, 1993). There are three to four copies of the *trpEG*-containing plasmid for each of the chromosomal genes. Amplification of *trpEG* is widespread in *Buchnera* of the family *Aphididae*, which is composed of rapidly growing aphids (Fig. BXII.γ.197c) (Lai et al., 1994; Rouhbachsh et al., 1996, 1997). In each case, the region upstream of *trpE* contains two or more copies of a nine-nucleotide sequence known as a DnaA box. Since the DnaA protein binds to this sequence and initiates DNA replication, this region may be a putative origin of plasmid replication. In the slow-growing aphid *Schlechtendalia chinensis*, *trpEG* is present as one copy on the *Buchnera* chromosome (Lai et al., 1995). Curiously, the *trpEG*-containing plasmids of *Buchnera* from *D. noxia* and from *U. sonchi* contain an intact copy of *trpEG* as well as numerous copies of *trpEG* pseudogenes (Lai et al., 1996; Baumann et al., 1997). There is currently no satisfactory explanation for this observation.

An independent case in which genes for amino acid biosyn-

thesis are plasmid-borne involves the leucine biosynthetic pathway. In *Buchnera* from several aphids, all of the enzymes of leucine biosynthesis (*leuABCD*) are on plasmids (pLeu) of about 8 kb (Bracho et al., 1995; van Ham et al., 1997; Baumann et al., 1999). In *Buchnera* from the aphid *S. graminum* there are about 23 copies of pLeu per endosymbiont chromosomal gene (Thao et al., 1998). Aphids from which pLeu has been sequenced are given in Fig. BXII.γ.197d. These plasmids have no similarity to the *trpEG*-containing plasmids and lack DnaA boxes. Instead they contain open reading frames (*repA1*, *repA2*) that correspond to putative proteins related to RepA of the IncFII incompatibility group plasmids, which is involved in plasmid DNA replication (Bracho et al., 1995). Also encoded on pLeu is a conserved open reading frame of unknown function. A most remarkable finding is the presence of a 1.7-kb plasmid in *Buchnera* from the aphid *Tetraneura caerulea*, which contains only *repA1* and an unidentified open reading frame (van Ham et al., 1997). It is possible that this plasmid represents a minimal-size replicon that under the right selection can be used for the amplification of other chromosomal genes.

GroEL overproduction In *Buchnera*, the chaperonin GroEL constitutes a major fraction of the total protein (Sato and Ishikawa, 1997b). In addition, GroEL is detected in the aphid hemolymph (van den Heuvel et al., 1994). Overproduction of GroEL is a characteristic of endosymbionts and pathogens in the intracellular environment (Baumann et al., 1995; Hogenhout et al., 1998). GroEL has been localized in maternal and embryonic endosymbionts by immunohistochemistry (Fukatsu and Ishikawa, 1992b). Electron micrographs indicate that the purified *Buchnera* GroEL has the characteristic double-ring appearance observed with the *E. coli* protein (Hara and Ishikawa, 1990; Filichkin et al., 1997). In addition, it has ATPase activity and in the presence of *E. coli* GroES and ATP could reconstitute denatured *Rhodospirillum rubrum* ribulose-1,5-bisphosphate carboxylase (Kakeda and Ishikawa, 1991). *Buchnera* GroEL is able to complement *E. coli* mutants (Ohtaka et al., 1992).

The *Buchnera groESL* operon organization resembles that of *E. coli* (Ohtaka et al., 1992; Hassan et al., 1996; Hogenhout et al., 1998). Upstream of *groES* are nucleotide sequences characteristic of the -35 and -10 regions of σ^{32} promoters. A message of 2.1 kb (containing both *groES* and *groEL*) is made by the endosymbiont using only this promoter (Sato and Ishikawa, 1997a). It is not understood why GroES is barely detectable in the endosymbiont, in contrast to the high quantities of GroEL (Kakeda and Ishikawa, 1991; Fukatsu and Ishikawa, 1993). The genes for σ^{32} (*rpoH*) as well as *dnaKJ* have been cloned and sequenced (Sato and Ishikawa, 1997a, b). The latter are also transcribed solely from a σ^{32} promoter. In *E. coli* as well as other organisms, transcription of the *groESL* operon and the *dnaKJ* operon are in part of the σ^{32} regulon and their synthesis is increased by heat shock (Gross, 1996). It would appear that this mode of regulation is modified in *Buchnera* (Sato and Ishikawa, 1997a, b). Synthesis of *groESL* and *dnaKJ* messenger RNA is constitutive and is not increased by heat shock. This conclusion is supported by the observation that there is no increase in the level of total GroEL in aphids shifted from a temperature of 23°C to 33°C for a period of 1 d (Baumann et al., 1996).

Pathogenicity Aphids are major pests of agricultural plants (Blackman and Eastop, 1984). Although large populations of aphids can cause plant debilitation due to nutrient consumption, perhaps the major economic effect of aphids on agriculture is

due to their transmission of plant viruses. Recently, *Buchnera*-derived GroEL has been implicated in the survival of luteoviruses in the hemolymph (van den Heuvel et al., 1994; Filichkin et al., 1997; Hogenhout et al., 1998). These viruses replicate in the plant and are ingested by aphids when they feed on phloem sap. Subsequently they are transported from the digestive tract into the hemolymph and from there into the salivary gland for transmission to plants via salivary secretions. The viruses are retained in an infective form (without replication) in the hemolymph throughout the life span of the aphid. There is evidence that the GroEL that is found in the hemolymph coats the virus particles and protects them from host defenses. A region in *Buchnera* GroEL has been identified that is essential for binding to the virus (Hogenhout et al., 1998), and similarly a portion of a viral capsid protein has been identified as the region to which the endosymbiont GroEL binds (van den Heuvel et al., 1997).

Antibiotic and drug sensitivity A variety of antibiotics have been used to eliminate endosymbionts, resulting in a reduction of weight gain of the aphid and the loss of reproductive potential. These antibiotics include chloramphenicol, chlortetracycline, neomycin, penicillin, rifampicin, and streptomycin.

Additional symbionts Many species of aphids also contain rod-shaped procaryotic endosymbionts designated as the secondary (S)-endosymbionts (Buchner, 1965; Houk and Griffiths, 1980; Fukatsu and Ishikawa, 1993). These organisms are usually absent from bacteriocytes and are found in vesicles within the sheath cells that surround the bacteriome. The S-endosymbiont from the aphid *A. pisum* is a member of the *Enterobacteriaceae* (Unterman et al., 1989). In this organism, the 16S rDNA is upstream of 23S rDNA (Unterman et al., 1989; Unterman and Baumann, 1990). The S-endosymbiont of *A. pisum* is dispensable (Chen and Purcell, 1997) since strains lacking this endosymbiont have been found in nature. The S-endosymbiont can be introduced into such strains by injection of hemolymph from strains of *A. pisum* that contain the S-endosymbiont. Some aphids may also contain a rod-shaped rickettsia closely related to *Rickettsia bellii* (Chen et al., 1996).

Although most species of aphids appear to contain bacteriocytes with *Buchnera*, some species of the tribe Cerataphidini lack both (Fukatsu and Ishikawa, 1992a; Fukatsu et al., 1994). These aphids instead contain, in their body cavity, a yeast-like symbiont that based on its 18S rDNA was found to belong to the subphylum Ascomycotina of the class Pyrenomycetes (Fukatsu and Ishikawa, 1996). These symbionts were related to those of planthoppers.

ENRICHMENT AND ISOLATION PROCEDURES

Endosymbiont-enriched preparations have been obtained using methods previously described (Ishikawa, 1982; Harrison et al., 1989; Sasaki and Ishikawa, 1995). The endosymbionts require high osmolarity for the retention of their structure. The best criteria of purity have involved examination of the purified preparations by electron microscopy, which also allows determination of whether the endosymbionts are still within host-derived vesicles.

Both the aphids and the endosymbionts have a similar mol% G + C DNA content (Ishikawa, 1987; Unterman and Baumann, 1990), and consequently endosymbiont DNA cannot be separated from host DNA by CsCl density gradient centrifugation. For the routine purification of *Buchnera* from *S. graminum*, to obtain starting material for the preparation of endosymbiont-

enriched DNA, the method of Sasaki and Ishikawa (1995) has been used. All of the reagents and equipment are kept on ice and the procedures are performed as rapidly as possible. Approximately 2–3 g (wet weight) aphids are transferred to a 1.5 cm diameter tissue grinder. Ten milliliters of buffer A¹ is added and the aphids are ground with a loose-fitting plunger for 5 min and the preparation passed through a double layer of a nylon mesh to remove large particulate material. The filtrate is brought to a volume of about 100 ml with buffer A and then quickly passed through a 100- μ m nylon filter (Spectrum Medical Industries, Inc., Houston, Texas, USA), followed by filtration through 20- μ m and 10- μ m nylon filters. Only slight vacuum pressure is applied during the last two filtration steps. The volume is made up to approximately 120 ml with buffer A and 30-ml aliquots are centrifuged in a swinging bucket rotor for 6 min at 1500 $\times$ g. The pellets are gently resuspended in 1.25 ml of buffer A and used for DNA purification (Unterman et al., 1989; Munson et al., 1991a). The DNA obtained from such preparations is endosymbiont-enriched 13- to 15-fold relative to the DNA obtained from the whole aphid (Baumann et al., 1997). Further purification of the endosymbiont preparation can be obtained by centrifugation through 10–90% Percoll gradients (Pharmacia Biotech, Uppsala, Sweden) as described by Ishikawa (1982).

MAINTENANCE PROCEDURES

Established lines of *S. graminum*, the aphid that contains the type strain of *B. aphidicola*, can be readily grown on wheat or barley at 23–24°C with 1500 foot candles illumination and a photoperiod of 16 h light and 8 h dark. Seeds are planted in rich soil, and, after 5–6 d, the seedlings are inoculated with aphids. The inoculum is obtained by cutting aphid-infested plants close to the roots and shaking them over the seedlings to dislodge the aphids. New seedlings are inoculated every 7–10 d.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Buchnera is somewhat unusual in that its rRNA genes are organized into two transcription units (Baumann et al., 1995). Consequently, 16S rDNA is not upstream of 23S rDNA. In eight strains of *Buchnera*, encompassing the diversity presented in Fig. BXII.γ.197a, *aroE* has been found to be upstream of 23S rDNA (Rouhbakhsh and Baumann, 1995). This property has been used for the identification of *Buchnera* using oligonucleotide primers complementary to *aroE* and 16S rDNA and PCR (Rouhbakhsh et al., 1994). This and other PCR-based identification methods are discussed by Rouhbakhsh et al. (1994). It should be noted that the S-endosymbiont found in some aphids has 16S rDNA upstream of the 23S rDNA (Unterman and Baumann, 1990).

DIFFERENTIATION OF THE GENUS *BUCHNERA* FROM OTHER GENERA

Since different evolutionary lineages of nonculturable endosymbionts are found within aphids, carpenter ants, and tsetse flies,

their initial putative identification is at the level of the insect host (Fig. BXII.γ.196). In *Wigglesworthia* (endosymbionts of tsetse flies), 16S rDNA is upstream of 23S rRNA and this is a useful differential character (Aksoy, 1995a, b). In the carpenter ant endosymbionts the 16S rDNA is not upstream of 23S rDNA, and it appears that the rRNA genes are organized into at least two transcription units (C. Elsishans and R. Gross, personal communication). It is not known if in carpenter ant endosymbionts *aroE* is upstream of the 23S rDNA, as is the case in *Buchnera*. The members of the *Enterobacteriaceae* are mostly free-living, readily identifiable bacteria with their rRNA genes arranged in the order 16S–23S–5S.

TAXONOMIC COMMENTS

There are over 4000 species of aphids (Blackman and Eastop, 1984), of which only 28 have been characterized by molecular methods. Consequently, our conclusions concerning *Buchnera* are based on a very small sample of aphid species. 16S rDNA has been useful for showing the monophyletic origin of the aphid symbiosis. It appears also to be useful for the differentiation of major aphid subgroups, as for example the *Aphididae* and possibly other families (Fig. BXII.γ.197a). 16S rDNA, however, is far too conserved for the elucidation of relationships between endosymbionts of closely related aphids. We have had some success with the use of a portion of *trpB* for this purpose (Fig. BXII.γ.197b). However, other informational molecules should also be tried.

The name *Buchnera aphidicola* as currently used designates the whole aphid endosymbiont lineage (Fig. BXII.γ.197a). The 16S rDNA sequence difference between *Buchnera* from *S. graminum* and *Buchnera* from *S. chinensis* (Sg and Sc in Fig. BXII.γ.197a) is approximately the same as that between *E. coli* and *Proteus vulgaris*. This clearly indicates that in subsequent studies *Buchnera* should be subdivided into new species. For this purpose, it is necessary to study a molecule that is less conserved than the 16S rDNA, and it is also necessary to know the range of variation within the endosymbionts of a single aphid species. In addition, since there is co-speciation between the host and the endosymbiont, the phylogeny of the host should also be elucidated to serve as an aid in the recognition of new endosymbiont species.

FURTHER READING

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List of species of the genus *Buchnera*

1. *Buchnera aphidicola* Munson, Baumann and Kinsey 1991b, 567^{VP}
a.ph' di.co.la. M.L. fem. n. *aphidicola* aphid dweller.

The characteristics are those described for the genus. Morphological features are depicted in Figs. BXII.γ.194 and BXII.γ.195.

The mol% G + C of the DNA is: 27 (determined from the nucleotide sequence of over 100 kb of DNA).

Type strain: Endosymbiont of the aphid *Schizaphis graminum*.

GenBank accession number (16S rRNA): M63246.

1. Buffer A contains 0.25 M sucrose, 35 mM Tris-HCl (pH 7.6), 25 mM KCl, 10 mM MgCl₂, and 1 mM dithiothreitol (Ishikawa, 1982).

Genus VI. Budvicia Aldová, Hausner, and Gabrhelová in Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985, 63^{VP}

J.J. FARMER III

Bud.vi'ci.a. L. fem. n. *Budvicia* derived from Budvicium, the Latin name of the city České Budějovice where the bacterium was first isolated.

Small, straight, rod-shaped cells, nonsporeforming, nonencapsulated, and Gram negative, conforming to the general definition of the family *Enterobacteriaceae*. Contains the enterobacterial common antigen. **Motile, producing peritrichous flagella when grown at 22°C; less motile at 36°C.** Grows on laboratory media including MacConkey agar. Grows slowly on nutrient agar, forming colonies only about 0.1 mm at 24 h, 36°C; colonies are five times larger when grown at 30°C for 24 h. More active biochemically at 25°C than 36°C. Oxidase negative, reduce nitrate to nitrite. **Positive for: H₂S production, urea hydrolysis, and fermentation of glucose, L-arabinose, L-rhamnose, D-xylose, and D-galactose.**

Negative for indole production, Voges-Proskauer, citrate utilization (Simmons), phenylalanine deaminase, lysine decarboxylase, arginine dihydrolase, ornithine decarboxylase, growth in the presence of cyanide (KCN test), malonate utilization, esculin hydrolysis, gelatin hydrolysis (22°C), lipase (corn oil), DNase, and the fermentation of sucrose, dulcitol, salicin, adonitol, *myo*-inositol, D-sorbitol, raffinose, maltose, trehalose, cellobiose, α-methyl-D-glucoside, erythritol, melibiose, glycerol, and D-mannose.

Utilize 29 of 83 carbon sources tested. **Grow at 4°, 10°, 22°, 32°, and 37°C, but not at 42°C.** Complex growth factor requirement: **absolute requirement for nicotinic acid**; other vitamins and amino acids stimulate growth. Susceptible to colistin, nalidixic acid, sulfadiazine, gentamicin, streptomycin, kanamycin, chloramphenicol, and carbenicillin (disk diffusion method on Mueller-Hinton agar); **resistant to penicillin, ampicillin, and cephalothin. Isolated from drinking and environmental water in Europe, and from fecal specimens in United States. No evidence that it is a pathogen for plants or animals.**

The mol% G + C of the DNA is: 46 ± 1 (Bouvet et al., 1985); 51–54 (Aldová et al., 1983).

Type species: *Budvicia aquatica* Aldová, Hausner, and Gabrhelová in Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985, 63.

FURTHER DESCRIPTIVE INFORMATION

Guide to the literature Since the genus name was first published in 1983, there have been a few reports in the literature. "HG group" should be included in computer literature searches to retrieve early references. A search from 1983 to May 1999 yielded only seven citations in MEDLINE and 12 citations in BIOSIS. Most are about the organism (Aldová et al., 1983, 1984; Bouvet et al., 1985), comparative taxonomic studies (Bouvet and Grimont, 1987a), or water bacteriology (Hausner et al., 1986; Schubert and Groeger-Söhn, 1998).

Sources Most isolates of *Budvicia* have been from various types of water: wells, rivers and streams, swimming pools, conduits, environmental, and sewage. Hausner et al. (1986) isolated 170 strains of *B. aquatica* from 21,300 water samples, mainly from wells and water mains. One isolate was from the digestive tract of a shrew in Spain (Bouvet et al., 1985).

B. aquatica has also been isolated from human feces in United States (Enteric Reference Laboratory, CDC, unpublished). However, there is no evidence that *B. aquatica* causes diarrhea or infections of the intestinal tract. Apparently, it has not been isolated from extraintestinal human clinical specimens.

Bacteriocin-like agents Strains of *Budvicia* produce bacteriocin-like agents that Šmarda (1987) termed aquaticins.

Isolation, identification, culture preservation, phenotypic characterization Strains of *Budvicia* are not difficult to grow, and are typical *Enterobacteriaceae* in most respects. Strains grow slowly on nutrient agar, forming colonies only about 0.1 mm at 24 h at 36°C, but about five times larger when grown at 30°C for 24 h (Bouvet et al., 1985). Similarly, the biochemical reactions may be different at the two temperatures; and strains may be more active at 25°C. Schubert and Groeger-Söhn (1998) described a method for its detection and quantification in water and sewage samples. See the chapter on the family *Enterobacteriaceae* in this *Manual* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, computer programs, and antibiotic susceptibility.

Biochemical reactions and differentiation from other *Enterobacteriaceae* Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* gives the results at 36°C for *Budvicia* strains in 47 biochemical tests normally used for identification (Farmer, 2003). There are no genus- or species-specific tests or sequences for the identification of *Budvicia*. The best approach is to do a complete set of phenotypic tests (growth on plating media, biochemical tests, antibiogram, etc.) and compare the results with all the other named organisms in the family *Enterobacteriaceae*. This approach is described in more detail in the chapter on the family *Enterobacteriaceae*. The biochemical results of a test strain can be compared to all the organisms listed in Table BXII.γ.193 of the chapter on the family *Enterobacteriaceae*. Several computer programs greatly facilitate analyzing the results. Bouvet et al. (1985) point out that strains of *B. aquatica* could be misidentified as H₂S-positive strains of *Yersinia*, *Citrobacter*, or *Salmonella*.

TAXONOMIC COMMENTS

Discovery, history, nomenclature, and type strains In 1976 the first strain of an unusual H₂S-producing strain of *Enterobacteriaceae* was isolated at the Public Health Laboratory in České Budějovice, Czechoslovakia (Aldová et al., 1983). From 1976 to 1983, 27 additional strains were isolated from water samples in three different areas of Bavaria. In 1983 Aldová et al. described this collection of strains, which had been given the vernacular name "HG group". All the data indicated that this was a new group that was distinct from all of the described genera and species of *Enterobacteriaceae*. Throughout the paper Aldová et al. (1983) used the term "HG group" for the new organism; however, at the end of the paper there was a separate section with the heading "Note", which stated that after the manuscript had been submitted DNA-DNA hybridization data became available from the laboratory of P.A.D. Grimont of the Institute Pasteur confirming that the group was a new genus of *Enterobacteriaceae*. In this note they stated: "The authors accordingly propose for this genus the name *Budvicia* (from the ancient Latin name 'Budvicium' for the city of České Budějovice where the microorganism was first isolated) and for its only species the name *Budvicia aquatica*. The epithet, as is obvious, is derived from the Latin adjective 'aquaticus'."

It is clear from this paragraph that a formal nomenclatural proposal was actually made. However, since no type strain was designated for *B. aquatica*, the names *Budvicia* and *Budvicia aquatica* were not "validly published" (see the *Bacteriological Code*, pages 28–29), and thus are illegitimate names. This situation was

apparently rectified when the names *Budvicia* Aldová, Hausner, Gabrhelová, Schindler, Petráš and Braná 1985a and *Budvicia aquatica* Aldová, Hausner, Gabrhelová, Schindler, Petráš and Braná 1985a gained standing in nomenclature and were validly published in Validation List 17 in the April 1985 issue of the IJSB (see page 223). Footnote f of Validation List 17 states that the type strain of *Budvicia aquatica* is 20186 (CNCTC 350); "Personal communication to the editor of the IJSB". In 1984, Aldová et al. had published a second paper that specifically defined this as the type strain. A sentence on page 234 of Aldová et al. (1984) stated: "As the type strain, we propose strain No. 20186 of the National Dysentery Reference Laboratory, which has been deposited at the Czechoslovak National Collection of Type Cultures, Prague under No. M 350."

However, the 1984 paper of Aldová et al. was not listed in the Literature Cited section of Validation List 17. Validation List 17 would have ended a slightly confusing matter; however, in January, 1985 there was a new set of proposals for *Budvicia* and *Budvicia aquatica*. Bouvet et al. published a paper in the *International Journal of Systematic Bacteriology* proposing *Budvicia* and *Budvicia aquatica* as a new genus and species, and their species appeared to be based on a different type strain. This latter paper made a confusing situation even more confusing. In addition, they compounded the confusion by proposing an unconventional name citation (see Nomenclatural problem 3, below). These four conflicting proposals (Aldová et al., 1983, 1984, 1985a; Bouvet et al., 1985) create four nomenclatural problems that require analysis and discussion.

Nomenclatural problem 1 Priority of *Budvicia* Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985 over *Budvicia* Aldová, Hausner, Gabrhelová, Schindler, Petráš and Braná 1985a. The first name clearly has priority because it was validly published first (as an article in the *International Journal of Systematic Bacteriology*, January 1985, pages 60–64). The second name was pub-

lished in the next issue (April 1985) in Validation List 17 (page 223).

Nomenclatural problem 2 Priority of *Budvicia aquatica* Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985 whose type strain is 20186HG01 (ATCC 35567) over *Budvicia aquatica* Aldová, Hausner, Gabrhelová, Schindler, Petráš and Braná 1985a whose type strain is 20186 (CNCTC 350). The first name has priority for the same reason given above.

Nomenclatural problem 3 Are *Budvicia aquatica* Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985 and *Budvicia aquatica* Aldová, Hausner, Gabrhelová, Schindler, Petráš and Braná 1985a objective or subjective synonyms? The two species would be objective synonyms if they have the same type strain, but would be subjective synonyms if they have different type strains. The type strains were published with different, but similar, strain numbers (20186HG01 vs 20186). Whether these strain numbers are the same or different is not clear from any of the published papers.

Nomenclatural problem 4 Author citations for *Budvicia* and *Budvicia aquatica*. In their original paper in the *International Journal of Systematic Bacteriology* Bouvet et al. (1985) stated: "To recognize the priority of three authors in the delineation of the new genus and species, we propose the following citation of names: *Budvicia* Aldová, Hausner, and Gabrhelová in Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985, and *Budvicia aquatica* Aldová, Hausner, and Gabrhelová in Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985."

It seems confusing and a danger to "stability in nomenclature" to have a name citation that disagrees with the authorship of the original paper. The conventional style for author citations as specified in the Bacteriological Code has been followed here, and the reader is referred to the irregular citation proposed by Bouvet et al. (1985). However, in the chapter on *Tatumella*, an

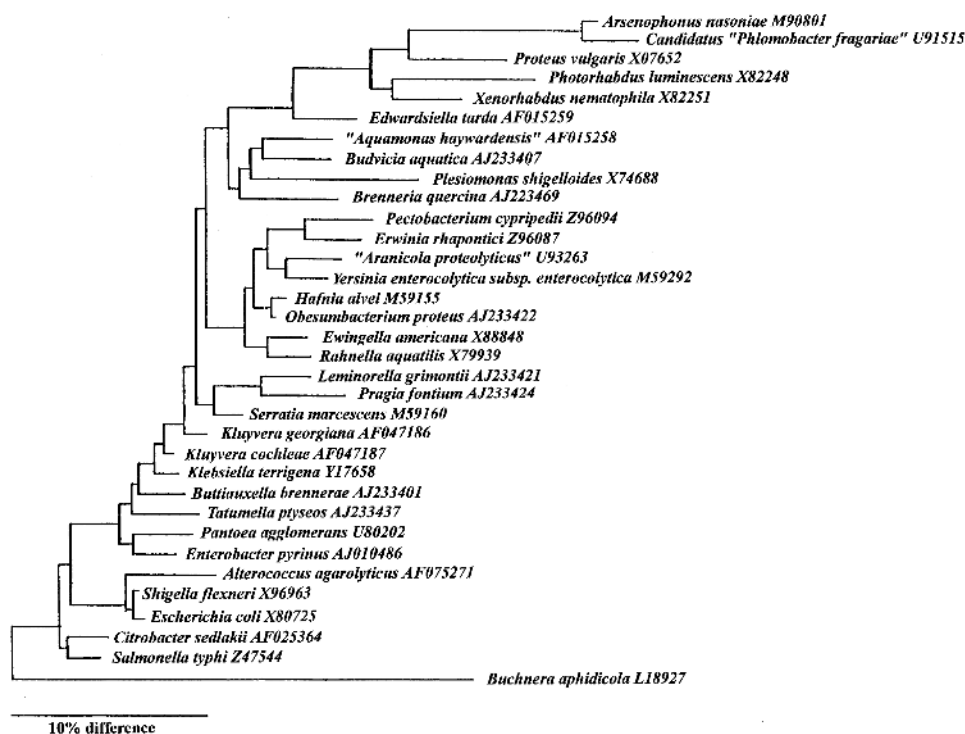


FIGURE BXII.γ.198. Relationship of *Budvicia* and several other new *Enterobacteriaceae* to other organisms in the family based on 16S rDNA sequencing data. Also, see Figure BXII.γ.189 in the family *Enterobacteriaceae*.

irregular name citation was also used. For consistency, the irregular author citation given by Bouvet et al. (1985) was used in this chapter on *Budvicia*.

Interestingly, this is a point that can cause instability in nomenclature. Because an "authors' proposal for an unorthodox name citation" is not specifically covered in the Bacteriological Code, remedial action in the form of a rule modification or Opinion of the Judicial Commission will be needed to guide future writers.

Phylogeny based on DNA–DNA hybridization and 16S rRNA sequencing Based on the DNA–DNA hybridization data *Budvicia* is a distinct and well-defined genus of *Enterobacteriaceae* with its one species *Budvicia aquatica* (Bouvet et al., 1985). None of the other genera of *Enterobacteriaceae* are closely related. Fig.BXII.γ.198 is a tree based on 16S rDNA sequencing that confirms this relationship. The organism on the same branch as *B. aquatica* is a new H₂S producing *Enterobacteriaceae* that will be

named *Aquimonas* (originally listed as *Aquamonas* in the GenBank database of sequences). This tree should be compared with the tree published by Spröer et al. (1999), which also shows the distinctness of *Budvicia*. However, this latter tree has *Pragia* and *Leminorella*, two other H₂S-producing genera of *Enterobacteriaceae*, on the same branch.

Schindler et al. (1992) agreed that *Budvicia* strains are distinct from other H₂S-producing *Enterobacteriaceae* based on SDS-PAGE protein patterns.

FURTHER READING

- Aldová, E., O. Hausner, M. Gabrhelová, J. Schindler, P. Petráš and H. Braná. 1983. A hydrogen sulfide-producing gram-negative rod from water. Zentbl. Bakterirol. Mikrobiol. Hyg. 1 Abt. Orig. A. 254: 95–108.
- Bouvet, O.M.M., P.A.D. Grimont, C. Richard, E. Aldová, O. Hausner and M. Gabrhelová. 1985. *Budvicia aquatica*, gen. nov., sp. nov.: a hydrogen sulfide-producing member of the *Enterobacteriaceae*. Int. J. Syst. Bacteriol. 35: 60–64.

List of species of the genus *Budvicia*

1. ***Budvicia aquatica*** Aldová, Hausner, and Gabrhelová in Bouvet, Grimont, Richard, Aldová, Hausner and Gabrhelová 1985, 63^{VP}
a.qua'ti.ca. L. fem. adj. *aquatica* living in water; named to show the aquatic habitat of the organism, since all but one of the original strains were isolated from water.

The characteristics are as previously described for the genus. Biochemical characteristics of the species are given in Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*. Isolated from well water, rivers and streams, swimming pools, water conduits, environmental water, sewage, the digestive tract of a shrew, and from human fecal spec-

imens in United States (Enteric Reference Laboratory, CDC, unpublished). There is no evidence that *B. aquatica* causes diarrhea, infections of the intestinal tract, or extraintestinal infections.

The mol% G + C of the DNA is: 46 ± 1 (*T_m*) (Bouvet et al., 1985); 51–54 (*T_m* and UV spectrophotometry) (Aldová et al., 1983).

Type strain: 20186HG01, ATCC 35567, Eb 13/82, CDC 0440-84, CNCTC 20186, DSM 5075.

GenBank accession number (16S rRNA): AJ233407.

Additional Remarks: The ATCC also lists two other strains ATCC 35566 (Eb 4/81, CNCTC 21930, M 325) and ATCC 51341 (85-01-010).

Genus VII. *Buttiauxella* Ferragut, Izard, Gavini, Lefebvre and Leclerc 1982, 266^{VP} (Effective publication: Ferragut, Izard, Gavini, Lefebvre and Leclerc 1981, 40)

PETER KÄMPFER

But.ti'aux'el.la. M.L. dim. ending -ella; M.L. fem. n. *Buttiauxella* named after René Buttiaux, a French microbiologist for his numerous contributions to the taxonomy of *Enterobacteriaceae*.

Straight rods, 0.5–0.7 × 2–3 μm, conforming to the general definition of the family *Enterobacteriaceae*. **Gram negative, motile by peritrichous flagella.** Facultatively anaerobic. Chemoorganotrophic. Psychrotolerant, growing at 4°C. D-Glucose is fermented with the production of acid and gas. Nitrate is reduced to nitrite. Oxidase negative, catalase positive. **Most strains are methyl red positive. The majority of strains are indole negative. Acids are produced from various carbohydrates, including L-arabinose, cellobiose, maltose, D-mannose, L-rhamnose, D-xylose, and salicin.** 16S rDNA sequence analysis of the type strain of *Buttiauxella agrestis* clearly places the genus in the family *Enterobacteriaceae* within the *Gammaproteobacteria*. Often isolated from fresh water, but also found in soil, and especially in the intestines of snails and slugs from various regions in the world.

The mol% G + C of the DNA is: 47–51.

Type species: Buttiauxella agrestis Ferragut, Izard, Gavini, Lefebvre and Leclerc 1982, 266, emend. Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 60 (Effective publication: Ferragut, Izard, Gavini, Lefebvre and Leclerc 1981, 40).

FURTHER DESCRIPTIVE INFORMATION

Based on 16S rDNA sequence analysis of the type strain, *B. agrestis* was most similar to the type strain of *Klebsiella planticola* (97.9%).

Analysis of 16S rDNA of the type strains of the seven species revealed a very high degree of phylogenetic relatedness (99.1–99.7% similarity) (Spröer et al., 1999). The phenotypic description of the genus *Buttiauxella* is largely based on the studies of Gavini et al. (1976b), Ferragut et al. (1981), Farmer (1984b), Farmer et al. (1985a), Brenner (1992a), and Müller et al. (1996). Based on DNA–DNA hybridization it was obvious that the *Buttiauxella* strains comprised seven closely related species, which were very difficult to differentiate phenotypically. The subdivision of the genus *Buttiauxella* is described below (see Taxonomic Comments).

Cells are straight rods, 0.5–0.7 × 2–3 μm, and can occur singly or in pairs. They are motile at 36°C by peritrichous flagella. *Buttiauxella* strains grow readily on many common media; no growth requirements have been described. A nutrient-rich medium (e.g., Tryptone–Soy agar, sheep-blood agar, or nutrient agar) gives best results. Optimal growth occurs at temperatures ranging from 25°C to 35°C, often yielding better growth than incubation at 37°C. The majority of strains are psychrotolerant, growing (slowly) at 4°C, but not at 41°C.

All members of the genus *Buttiauxella* produce a positive β-galactosidase reaction (ONPG test), although not all produce acid from lactose. The β-glucuronidase test is negative for all

TABLE BXII.γ.210. Characteristics of species of the genus *Buttiauxella*^{a,b,c}

Test	Method (s) ^d	<i>B. agrestis</i>	<i>B. brennerae</i>	<i>B. ferrugutiae</i>	<i>B. gaviniae</i>	<i>B. izardii</i>	<i>B. noackiae</i>	<i>B. warmboldiae</i>
Hybridization group		1	4	2	3	5	6	7
Acetate	Ut.	[+]	d	+	[+]	+	+	+
N-Acetyl-D-galactosamine	Ut.	+	+	+	d	+	+	+
N-Acetyl-L-glutamate	Ut.	+	+	+	+	+	+	+
N-Acetyl-L-glutamine	Ut.	—	—	—	[—]	—	[+]	—
N-Acetyl-glycine	Ut.	—	—	—	—	—	—	—
Adonitol	Acid., Ut.	—	d	—	d	—	—	—
D-Arabinose	Ut.	d	—	—	d	+	d	+
D-Arabitol	Acid., Ut.	—	d	—	d	—	—	—
L-Arginine dihydrolase	Alcal.	—	d	—	+	[—]	+	+
L-Arginine	Ut.	—	—	—	—	—	[—]	—
Citrate utilization (Simmons)	Ut.,	d	d	—	d	+	+	—
Dulcitol	Acid., Ut.	—	—	—	[—]	—	—	—
L-Fucose	Acid., Ut.	[+]	—	—	d	[+]	—	+
Glycerol	Acid.	d	—	—	—	—	—	—
3-Hydroxybenzoate	Ut.	—	—	—	—	—	—	—
3-Hydroxyphenylacetate	Ut.	—	—	[+]	—	—	—	—
Indole production	Kovacs	—	—	—	d	—	d	—
myo-Inositol	Acid.	—	—	—	—	—	—	+
KCN	Growth	+	+	+	[+]	+	+	—
5-Ketogluconate	Ut.	—	+	—	+	d	+	—
Lactose	Acid.	+	+	—	—	+	—	—
Lactulose	Acid., Ut.	—	—	—	—	—	—	—
L-Lysine decarboxylase	Alcal.	—	—	+	—	—	—	—
Malonate utilization	Ut.	+	+	—	+	+	+	+
Maltitol	Acid., Ut.	[+]	+	+	[+]	d	[+]	—
Melibiose	Acid.	d	+	[+]	—	d	—	—
α-Methyl-D-glucopyranoside	Acid., Ut.	—	d	[—]	—	—	—	—
3-Methyl-D-glucopyranoside	Ut.	+	d	[+]	+	[+]	[+]	—
Methyl red	Acid.	+	+	+	+	+	+	+
Mucate	Acid.	d	+	[—]	d	d	+	—
L-Ornithine decarboxylase	Alcal.	+	d	+	—	+	—	—
Palatinose	Acid., Ut.	+	+	+	+	d	+	—
Phenylacetate	Ut.	d	—	d	[—]	d	d	—
L-Phenylalanine deaminase	FeCl ₃ test	—	—	—	—	—	d	—
3-Phenylpropionate	Ut.	—	—	—	—	—	—	—
L-Proline	Ut.	[+]	+	[+]	+	+	[+]	+
Raffinose	Acid., Ut.	d	+	[—]	—	—	—	—
D-Sorbitol	Acid., Ut.	d	—	+	[—]	—	—	—
Sucrose	Acid.	—	—	—	[—]	—	—	—
D-Tagatose	Acid., Ut.	—	—	—	[—]	—	—	—
Tartrate (Jordan's)	Acid.	d	[—]	—	d	[—]	+	—
Voges-Proskauer	Acetoin	—	—	—	—	—	—	—

^aData adapted from Müller et al. (1996).^bSymbols: +, positive for 90–100% of strains; [+], positive for 75–89% of strains; d, positive for 25–74% of strains; [—], positive for 11–24% of strains; —, positive for 0–10% of strains. All reactions, unless otherwise stated, were done at 36° ± 1°C and read after 48 h.^cAll of the strains studied were Gram-negative, oxidase-negative, catalase-positive, D-glucose-fermenting, nitrate-reducing, rod-shaped organisms. All strains grew on MacConkey agar and on Endo agar at 30°C. With few exceptions, all of the strains were positive in standard tests for motility, acid and gas production from D-glucose, fermentation of D-mannitol, salicin, L-arabinose, L-rhamnose, maltose, D-xylose, trehalose, cellobiose, D-mannose, galactose, gentiobiose, D-ribose, and arbutin, esculin hydrolysis, and o-nitrophenyl-β-D-galactopyranoside (β-galactosidase) test. They were all negative for production of hydrogen sulfide on triple sugar iron, Christensen urease activity, gelatin liquefaction (at 22°C), lipase activity (Tween 80 and corn oil), DNase activity, and production of a yellow pigment, and did not ferment 2-deoxy-D-glucose, 2-deoxy-D-ribose, erythritol, α-D-fucose, D-lyxose, and xylitol.

Almost all of the strains utilized the following compounds as sole carbon and energy sources: N-acetyl-D-glucosamine, D-alanine, L-alanine, L-arabinose, arbutin, DL-asparagine, D-cellobiose, D-fructose, D-galactose, D-galacturonate, gentiobiose, D-gluconate, D-glucosamine, D-glucuronate, L-glutamine, L-glutamate, DL-glycerate, glycerol, 4-hydroxybenzoate, 2-ketogluconate, D-lactate, DL-lactate, maltose, maltotriose, mannitol, D-mannose, methyl-α-galactoside, methyl-β-galactoside, methyl-β-D-glucoside, mucate, oxaloacetate, palatinose, protocatechuate, pyruvate, quinate, L-rhamnose, D-ribose, D-saccharate, salicin, L-serine, starch, D-trehalose, and D-xylose.

Almost all of the strains were unable to utilize the following compounds as sole carbon and energy sources: acetamide, acetamidocaproate, N-acetyl-DL-methionine, N-acetyl-L-proline, cis-aconitate, trans-aconitate, adipate, β-alanine, allantoin, altrose, DL-2-aminoadipate, 2-aminobenzoate, 3-aminobenzoate, 4-aminobenzoate, DL-2-aminobutyrate, DL-3-aminobutyrate, 4-aminobutyrate, DL-2-aminoisobutyrate, DL-3-aminoisobutyrate, 5-aminovalerate, aminoxyacetate, amygdalin, anthranilate, L-arabitol, arabinonate, D-arginine, D-asparagine, azelate, benzoate, betaine, 1-butanol, 2-butanol, n-butyrate, cadaverine, caprate, caprylate, carnitine, carnosine, L-citrulline, citraconate, L-cysteinate, dextran, DL-2,4-diaminobutyrate, diaminopimelate, 2,3-diaminopropionate, dimethylglycine, m-erythritol, ethanol, ethanolamine, ethylamine, D-fucose, D-glutamate, glutarate, glycineamide, glycogen, glycolate, glycyrrhizinate, 1-hexanol, 1,6-hexandiol, hexylamine, hippurate, histamine, D-histidine, L-histidine, L-homoserine, DL-3-hydroxybutyrate, 4-hydroxybutyrate, DL-2-hydroxycaprate, DL-2-hydroxyisobutyrate, L-2-hydroxyisocaproate, 2-hydroxyisovalerate, DL-δ-hydroxylysine, 2-hydroxyphenylacetate, 4-hydroxyphenylglycine, DL-hydroxyproline, 2-hydroxyvalerate, HQ-β-glucuronide, heptanoate, indole-3-acetate, isobutyrate, isophthalate, itaconate, 2-ketoglutarate, 2-ketoisocaproate, D-leucine, L-leucine, levulinate, D-lysine, D-malate, maleate, D-mandelate, L-mandelate, D-mannoheptulose, D-melezitose, mesaconate, mesoxalate, D-methionine, L-methionine, DL-methioninesulfone, methyl-α-D-glucoside, α-methyl-D-mannoside, β-methyl-D-xyloside, L-norleucine, D-norvaline, L-norvaline, 1,8-octandiol, D-ornithine, phenoxyacetate, D-phenylalanine, phenylglycine, phenyllactate, phosphoenolpyruvate, phthalate, pimelate, poly-D-galactomannan, polygalacturonate, D-proline, propionate, protocatechuate, putrescine, salicylamide, salicylate, sarcosine, sorbate, L-sorbose, spermine, suberate, D-tartrate, meso-tartrate, tartronate, taurine, thiamine, tricarballoylate, trigonelline, tropate, tryptamine, D-tryptophan, L-tryptophan, tyramine, L-tyrosine, ureidosuccinate, n-valerate, isovalerate, L-valine, xylitol, and L-xylose.

The strains varied in their ability to utilize the following compounds as sole sources of carbon and energy, and these characteristics could not be used to differentiate hybridization groups: D-alanine, L-asparagine, L-aspartate, benzoate, fumarate, D-glucarate, D-glucosamine, glycerate, glycerophosphate, glycine, glyoxylate, 4-hydroxybenzoate, 4-hydroxyphenylacetate, inulin, DL-isocitrate, DL-isoleucine, 2-ketoglutarate, L-lactate, D-lyxose, L-lyxose, L-malate, L-mannose, phenylpyruvate, D-serine, spermidine, succinate, L-tartrate, L-threonine, and D-turanose.

B. agrestis and *B. izardii* could not be distinguished on the basis of the results of any single test. These groups had to be distinguished on the basis of their overall biochemical profiles. *B. gaviniae* and *B. noackiae* could also not be separated on the basis of the results of a single test, but could be differentiated on the basis of their N-acetyl-L-glutamine, adonitol, D-arabitol, L-fucose, mucate, and tartrate reactions. As stated in the text, these two pairs of hybridization groups exhibited the highest levels of interspecies DNA relatedness.^dUt, utilization test; Acid, acidification test; Alcal, alkalization test.

strains. Hydrogen sulfide formation, Voges–Proskauer, urease, and DNase tests are negative for all strains.

The reduction of nitrous oxide to dinitrogen has been reported in some strains of *B. agrestis* (Kaldorf et al., 1993). Various carbon sources, including carbohydrates, organic acids, and amino acids are utilized as sole sources of carbon (Gavini et al., 1976b; Ferragut et al., 1981; Müller et al., 1996; Table BXII.γ.210).

Isoprenoid quinone Q-8 is the predominant quinone type and a small amount of the menaquinone MK-8 is present in *B. agrestis* ATCC 33320^T (P. Kämpfer, unpublished results). The fatty acid composition as determined by gas chromatographic analysis of 66 strains representing all species (Kämpfer et al., 1997) differed only slightly. All strains contained the fatty acids C_{14:0}, C_{16:0}, C_{17:0 cyclo}, summed feature (C_{16:1 iso} and/or C_{14:0 3OH}), summed feature (C_{16:1 ω7c} and/or C_{15:0 iso 2OH}), and summed feature (C_{18:1 ω7c}, C_{18:1 ω9t}, and/or C_{18:1 ω12t}) (Kämpfer et al., 1997), a fatty acid profile typical for the family *Enterobacteriaceae*.

Antimicrobial susceptibility data are summarized by Freney et al. (1988). Members of the genus *Buttiauxella* (13 strains) were usually sensitive to aminoglycosides (gentamicin, tobramycin, and amikacin), doxycycline, and trimethoprim, but resistant to chloramphenicol. Susceptibility to ampicillin, aminoxycillin-clavulanic acid, and ticarcillin was observed. Cephalothin and cefoxitin were inactive. *Buttiauxella* produces enterobacterial common antigen (Böttger et al., 1987).

Pathogenicity of *Buttiauxella* for humans and animals has not been documented. Only one *B. agrestis* strain in the study of Müller et al. (1996) was from a human clinical specimen. Farmer et al. (1985a) described seven strains of Enteric Group 59 (now *Buttiauxella noackiae*) originating from sputum and a foot wound. *Buttiauxella* is widely distributed in nature, may be isolated from foods and is occasionally isolated from human sources. Although the natural habitat of *Buttiauxella* was originally thought to be water, the majority of strains have been isolated from the intestines of snails and slugs (Müller et al., 1996).

ENRICHMENT AND ISOLATION PROCEDURES

The cultivation media used for the isolation of *Buttiauxella* are those regarded useful for other members of the family *Enterobacteriaceae*. No specific selective medium has been reported for *Buttiauxella* species. Because of the relatively low clinical significance and the presence of *Buttiauxella* in various habitats, the selective isolation of *Buttiauxella* is rarely required. In most cases, organisms belonging to the genus *Buttiauxella* are isolated with differential media not inhibitory for *Enterobacteriaceae*, such as MacConkey agar, bromothymol blue lactose agar, or phenol red lactose agar. *Buttiauxella* strains grow readily on ordinary nutrient-rich media, and no growth requirements have been described.

MAINTENANCE PROCEDURES

Buttiauxella strains can be maintained in tryptone soy agar slabs or on nutrient agar when kept at room temperature in the dark. They can be preserved by storage in broth containing 10% glycerol or in calf or bovine serum at –80°C. Lyophilization seems to be the best procedure for preservation.

DIFFERENTIATION OF THE GENUS *BUTTIAUXELLA* FROM OTHER GENERA

No single distinguishing feature is useful for the differentiation of *Buttiauxella* from the other genera of *Enterobacteriaceae* (Table BXII.γ.193 in *Enterobacteriaceae*). Biochemically and based on DNA–DNA hybridization data, *Buttiauxella* are most similar to

the genus *Khuyvera*. (Gavini et al., 1983a; Farmer et al., 1985a; Müller et al., 1996); however, the levels of DNA similarity between members of the genera *Buttiauxella* and *Khuyvera* and more than 50 other species belonging to the family *Enterobacteriaceae* were between 15% and 36% (Gavini et al., 1983a; Müller et al., 1996), indicating the separate position of these two genera. In addition, the mol% G + C ratios of *Buttiauxella* and *Khuyvera* are quite different, with values ranging from 47–51% for *Buttiauxella* and 54–58% for *Khuyvera* (Ferragut et al., 1981; Gavini et al., 1983a). *Buttiauxella* and *Khuyvera* exhibit similar fatty acid patterns, but all strains of *K. ascorbata*, *K. cryocrescens*, and *K. georgiana* produced the summed feature C_{18:1 ω7c}, C_{18:1 ω9c}, and/or C_{18:1 ω12t} in higher amounts (>20% of all fatty acids) than any strain of *Buttiauxella* (Kämpfer et al., 1997). Although *K. georgiana* produced this feature also in amounts >20%, this species can be separated from all *Buttiauxella* species by its high amounts of 15:0 (>8%), which is not found in *Buttiauxella* (Kämpfer et al., 1997). Further physiological and biochemical characters that are helpful for differentiation of the two genera are given in Table BXII.γ.211.

TAXONOMIC COMMENTS

Based on a numerical taxonomic study of *Enterobacteriaceae*, Gavini et al. (1976b) defined a new group of strains within the family and gave it the vernacular name “group F”. Originally these 17 strains isolated from water and unpolluted soils were regarded to be similar to the genus *Citrobacter* because of their negative indole and Voges–Proskauer reactions and their positive citrate and methyl red tests. Subsequently, Ferragut et al. (1981) used DNA–DNA hybridization to compare strains of group F to each other and to named species of *Enterobacteriaceae*. Based on DNA similarity of 82–96% within group F and its low level of DNA similarity to other *Enterobacteriaceae*, they proposed the new genus *Buttiauxella* with one species *Buttiauxella agrestis*. The names *Buttiauxella* and *B. agrestis* were effectively published, but were not validated in the *International Journal of Systematic Bacteriology* before January 1, 1980. They did not appear on the Approved Lists of Bacterial Names (Skerman et al., 1980), but both names have now been validly published (Ferragut et al., 1982) and have standing in nomenclature. Four additional “group F” strains were only 62–66% related to *B. agrestis*. Based on phenotypic differences, three of these strains were placed in a group that was given the name Enteric Group 63, and the remaining strain was designated as Enteric Group 64 (Farmer et al., 1985a). In 1981, another group of strains, phenotypically similar to *Pantoea agglomerans* (formerly *Enterobacter agglomerans*), except for a positive arginine dihydrolase reaction, was reported and named Enteric Group 59 (Farmer et al., 1985a). Eight isolates, originating from sputum (6), a foot wound (1), and ham (1), were studied.

Between 1984 and 1988, Müller et al. (1995a, b, 1996)

TABLE BXII.γ.211. Differentiation of *Buttiauxella* and *Khuyvera*^{a,b}

Test	<i>Buttiauxella</i>	<i>Khuyvera</i>
Mol% G + C content	47–51	54–58
Indole production	–	+
Citrate (Simmons)	d	[+]
Sucrose fermentation	–	[+]
Raffinose fermentation	[–]	+
α-Methyl-D-glucose fermentation	–	+
Melibiose fermentation	d	+
N-Acetyl-L-glutamate utilization	+	–

^aData adapted from Ferragut et al. (1981), Gavini et al. (1983a), Müller et al. (1996)

^bSymbols: +, positive for 90–100% of strains; [+], positive for 75–89% of strains; d, positive for 25–74% of strains; [–], positive for 11–24% of strains; – positive for 0–10% of strains.

^cExcept for *K. cochlear*.

screened the intestinal contents of snails and slugs for the presence of *Enterobacteriaceae* and, in addition to *Rahnella* isolates and the infrequently seen species *Ewingella americana* and *Serratia fonticola*, found many strains belonging to the genera *Buttiauxella* and *Kluyvera*.

Two hundred and nineteen strains belonging to the genera *Buttiauxella* and *Kluyvera* were subjected to an extensive DNA similarity study. The results indicated that the strains belonging to the genus *Buttiauxella* comprised seven closely related hybridization groups. One of these (hybridization group 1) corresponded to *B. agrestis*. Hybridization groups 2 through 7 were described as the new species: *B. ferragutiae* (formerly Enteric Group 63), *B. gaviniae* (formerly Enteric Group 64), *B. brennerae*, *B. izardii*, *B. noackiae* (formerly Enteric Group 59), and *B. warmboldiae* (Müller et al., 1996). The levels of similarity obtained for members of the seven species were generally between 45% and 65%, a good indication that all of these groups belong to a single genus. The hybridization groups that exhibited the highest levels of intergeneric similarity were hybridization groups 1 (*B. agrestis*) and 5 (*B. izardii*) and hybridization groups 3 (*B. gaviniae*) and 6 (*B. noackiae*). The relative binding ratios obtained from DNA-DNA hybridization studies between the *Buttiauxella* species are given in Table BXII.γ.212. The levels of DNA similarity between members of the *Buttiauxella* and *Kluyvera* species were generally between 15% and 30% (Müller et al., 1996).

List of species of the genus *Buttiauxella*

1. ***Buttiauxella agrestis*** Ferragut, Izard, Gavini, Lefebvre and Leclerc 1982, 266^{VP}, emend. Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 60 (Effective publication: Ferragut, Izard, Gavini, Lefebvre and Leclerc 1981, 40.)
a.gres'tis. L. masc. n. *ager*; *agri*; adj. *agrestis* living in the fields, so named because all original strains were isolated from unpolluted soils and water.

B. agrestis was called DNA hybridization group 1 in the study of Müller et al. (1996). All 30 strains studied by Müller et al. (1996) were ≥73% DNA-related in 60°C reactions (range 73–100%). In the original description (Ferragut et al., 1981), which was based on the 17 strains originally described by Gavini et al. (1976b), it was observed that 60% of *B. agrestis* strains were malonate positive. In the study of Müller et al. (1996), 96% of the strains were malonate positive. Biochemical tests that are useful for differentiating *B. agrestis* from other *Buttiauxella* species are given in Table BXII.γ.210. The arginine dihydrolase, fucose, glycerol, lactose, melibiose, ornithine decarboxylase, palatinose, and D-sorbitol tests are helpful. Clinical significance, if any, is unknown. Isolated from mollusks, as well as water, soil, and human materials.

The mol% G + C of the DNA is: 47–50 (T_m).

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I thank Don J. Brenner for his helpful comments and critical reading of this chapter, Hans E. Müller for interesting discussions and for furnishing strains, and W. Ludwig for critical comments and his experience in 16S rDNA sequence analyses.

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Type strain: ATCC 33320, CDC 1176-81, CUETM 77-167, DSM 4586.

GenBank accession number (16S rRNA): AJ233400.

2. ***Buttiauxella brennerae*** Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 62^{VP}
bren'nerae. M.L. fem. gen. n. *brennerae* of Brenner, in honor of Frances W. Hickman-Brenner, an American microbiologist, for her contributions to the study of many genera of the *Enterobacteriaceae*.

Biochemical characteristics are shown in Table BXII.γ.210. Tests that can be used to differentiate *B. brennerae* from other *Buttiauxella* species are the L-arabinose, arginine dihydrolase, fucose, myo-inositol, 5-ketogluconate, lysine decarboxylase, and malonate. Clinical significance, if any, is unknown. Isolated from mollusks.

The mol% G + C of the DNA is: ~50 (T_m).

Type strain: ATCC 51605, DSM 9396.

GenBank accession number (16S rRNA): AJ233401.

3. ***Buttiauxella ferragutiae*** Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 60^{VP}
fer'ra.gut.i.ae. M.L. fem. gen. n. *ferragutiae* of Ferragut, in honor of Carmen Ferragut, a French microbiologist, for

TABLE BXII.γ.212. Relative binding ratios of DNAs at 60°C between the species of the genus *Buttiauxella*^{a,b}

Test	<i>B. agrestis</i>	<i>B. brennerae</i>	<i>B. ferragutiae</i>	<i>B. gaviniae</i>	<i>B. izardii</i>	<i>B. noackiae</i>	<i>B. warmboldiae</i>
<i>B. agrestis</i>	81 (73–100)	53 (34–58)	51 (46–54)	56 (51–64)	63 (57–71)	58 (52–64)	56
<i>B. brennerae</i>		85 (68–100)	50 (34–57)	59 (39–68)	53 (44–60)	55 (38–66)	50 (44–55)
<i>B. ferragutiae</i>			100	50 (44–56)	48 (43–56)	55 (52–58)	49
<i>B. gaviniae</i>				84 (72–100)	52 (49–59)	71 (67–77)	49 (42–54)
<i>B. izardii</i>					88 (66–100)	50 (43–62)	48 (45–58)
<i>B. noackiae</i>						88 (81–100)	48 (46–51)
<i>B. warmboldiae</i>							99 (98–100)

^aData adapted from Müller et al. (1996).

^bResults are expressed as percentages, figures in parentheses indicate the range of similarity.

her contribution to the study of the genus *Buttiauxella*, previously called Enteric Group 63.

Biochemical characteristics are shown in Table BXII.γ.210. Positive lysine decarboxylase and D-sorbitol tests and negative ketogluconate and malonate tests can clearly differentiate this species from all other *Buttiauxella* species. Clinical significance, if any, is unknown.

Isolated from water and soil.

The mol% G + C of the DNA is: 48–50 (T_m).

Type strain: ATCC 51602, CDC 1180-81, CUETM 78-31, DSM 9390.

GenBank accession number (16S rRNA): AJ233402.

4. *Buttiauxella gaviniae* Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 62^{VP}

ga.vin'i.ae. M.L. fem. gen. n. *gaviniae* of Gavini, in honor of Françoise Gavini, a French microbiologist, for her contributions to the study of the genus *Buttiauxella*, previously called Enteric Group 64.

Biochemical characteristics are shown in Table BXII.γ.210. Positive arginine dihydrolase, 5-ketogluconate, and palatinose reactions and negative ornithine decarboxylase and raffinose reactions are useful in differentiating *B. gaviniae* from other *Buttiauxella* species. A combination of reaction results is necessary to differentiate this species from *Buttiauxella noackiae*. Clinical significance, if any, is unknown. All but one strain were isolated from mollusks.

The mol% G + C of the DNA is: ~51 (T_m).

Type strain: ATCC 51604, DSM 9393.

5. *Buttiauxella izardii* Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 62^{VP}

iz.ard'i.i. M.L. masc. gen. n. *izardii* of Izard, in honor of Daniel Izard, a French microbiologist, for his contribution to the study of the genus *Buttiauxella*.

Biochemical characteristics are shown in Table BXII.γ.210. No single biochemical characteristic differentiates *B. izardii* from *B. agrestis*, but a combination of several characteristics differentiates these taxa. Useful differential biochemical tests are the L-arabinose, citrate, fucose, myo-inositol, ornithine decarboxylase, and raffinose. Clinical significance, if any, is unknown. Isolated from mollusks.

The mol% G + C of the DNA is: ~50 (T_m).

Type strain: ATCC 51606, DSM 9397.

GenBank accession number (16S rRNA): AJ233404.

6. *Buttiauxella noackiae* Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 62^{VP}

no.ack'i.ae. M.L. fem. gen. n. *noackiae* of Noack, in honor of Katrin Noack, who phenotypically characterized the *Buttiauxella* strains, previously called Enteric Group 59.

Biochemical characteristics of 15 strains isolated from snails are shown in Table BXII.γ.210. These characteristics are almost identical to those described for Enteric Group 59; the only exception is the result of the lactose test. All snail isolates lack a yellow pigment and are lactose negative, but seven of the eight Enteric Group 59 strains isolated from humans and food were shown to be lactose positive (Farmer et al., 1985a). It is possible that there is a correlation between the ability to split and metabolize lactose and the ability to survive in humans. Tests that are useful in differentiating *B. noackiae* from other *Buttiauxella* species are the N-acetyl-L-glutamine, L-arginine dihydrolase, melibiose, and L-ornithine decarboxylase. A combination of several characteristics is necessary to differentiate *B. noackiae* from other *Buttiauxella* species.

Isolated from mollusks, human sputum, human wounds, and food. Clinical significance, if any, is unknown.

The mol% G + C of the DNA is: ~50 (T_m).

Type strain: ATCC 51607, DSM 9401.

GenBank accession number (16S rRNA): AJ233405.

7. *Buttiauxella warmboldiae* Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 62^{VP}

warm'bold.i.ae. M.L. fem. gen. *warmboldiae* of Warmbold, in honor of Sabine Warmbold, who isolated most strains of the new *Buttiauxella* species at the Staatliches Medizinaluntersuchungsamt Braunschweig.

Biochemical characteristics are shown in Table BXII.γ.210. Myo-inositol is utilized and acid is produced by freshly isolated strains for many months, in contrast to all other species belonging to the genus *Buttiauxella*. However, these characteristics are lost after some years of storage; therefore, they are not constitutive. Other biochemical tests that are useful in differentiating *B. warmboldiae* from other *Buttiauxella* species are the L-arabinose, arginine dihydrolase, citrate, fucose, KCN, malonate, maltitol, ornithine decarboxylase, and palatinose. This species is the species that is the most distant from all other *Buttiauxella* species and has the lowest level of DNA–DNA similarity to other *Buttiauxella* species. Clinical significance, if any, is unknown. Isolated from snails.

The mol% G + C of the DNA is: ~52 (T_m).

Type strain: ATCC 51608, DSM 9404.

GenBank accession number (16S rRNA): AJ233406.

Genus VIII. *Calymmatobacterium** Aragão and Vianna 1913, 221^{AL}

GEORGE H. BROWNELL

Ca.lym.ma.to.bac.te'ri.um. Gr. n. *calymma* mantle, sheath; Gr. dim. neut. n. *bakterion* a small rod; M.L. neut. n. *Calymmatobacterium* the sheathed rodlet.

Pleomorphic rods, 0.5–1.5 × 1.0–2.0 μm, with rounded ends. Occur singly or in clusters. **The cells exhibit single or bipolar condensation of chromatin. Capsules are present.** Gram negative. Nonmotile. The exudate from infected tissues, when stained by

Wright's stain or by Giemsa stain, demonstrates **characteristic intracellular organisms in the cytoplasm of large mononuclear phagocytes** ("Donovan bodies"). Can be cultivated *in vivo* in the yolk sac of embryonated chicken eggs, peripheral blood mononuclear cells, and human epithelial (HEP-2) monolayers, or *in vitro* on special egg yolk-containing media; has not been reproducibly cultivated in bacteriologic media. Optimum growth temperature, 37°C. Pathogenic for humans, causing **Donovanosis (granuloma inguinale)**.

*Editorial Note: The type species of the genus *Calymmatobacterium* has been transferred to *Klebsiella* as *Klebsiella granulomatis* (Aragão and Vianna 1913) Carter, Bowden, Bastian, Myers, Sriprakash and Kemp 1999.

The mol% G + C of the DNA is: not known.

Type species: *Calymmatobacterium granulomatis* Aragão and Vianna 1913, 221.

FURTHER DESCRIPTIVE INFORMATION

In diseased tissue smears stained by Wright's method, *C. granulomatis* occurs within the cytoplasm of large mononuclear monocytes as blue to purple pleomorphic rods surrounded by pink capsules (Fig. BXII.γ.199). The organism may occasionally be observed free in extracellular spaces. The single or bipolar condensation of chromatin gives rise to characteristic "safety-pin" forms. The ultrastructure of the intracellular organisms has been described by Davis and Collins (1969), Dodson et al. (1973),

Kuberski et al. (1980), Chandra et al. (1989), Chandra and Jain (1991), and Kharsany et al. (1997). Electron micrographs reveal encapsulated bacilliforms with characteristic Gram-negative cell walls. The presence of fimbriae-like structures have been reported by some investigators: Kuberski et al. (1980), Chandra et al. (1989), and Chandra and Jain (1991).

Calymmatobacterium granulomatis has been clinically proven to be the causal agent of granuloma inguinale (Dienst et al., 1938). The disease is often referred to as donovanosis because initial lesions have been diagnosed in skin areas other than the genital region. Most researchers support the contention that donovanosis is not necessarily a venereal disease but is an infection resulting from intimate contamination and poor hygiene. Infection

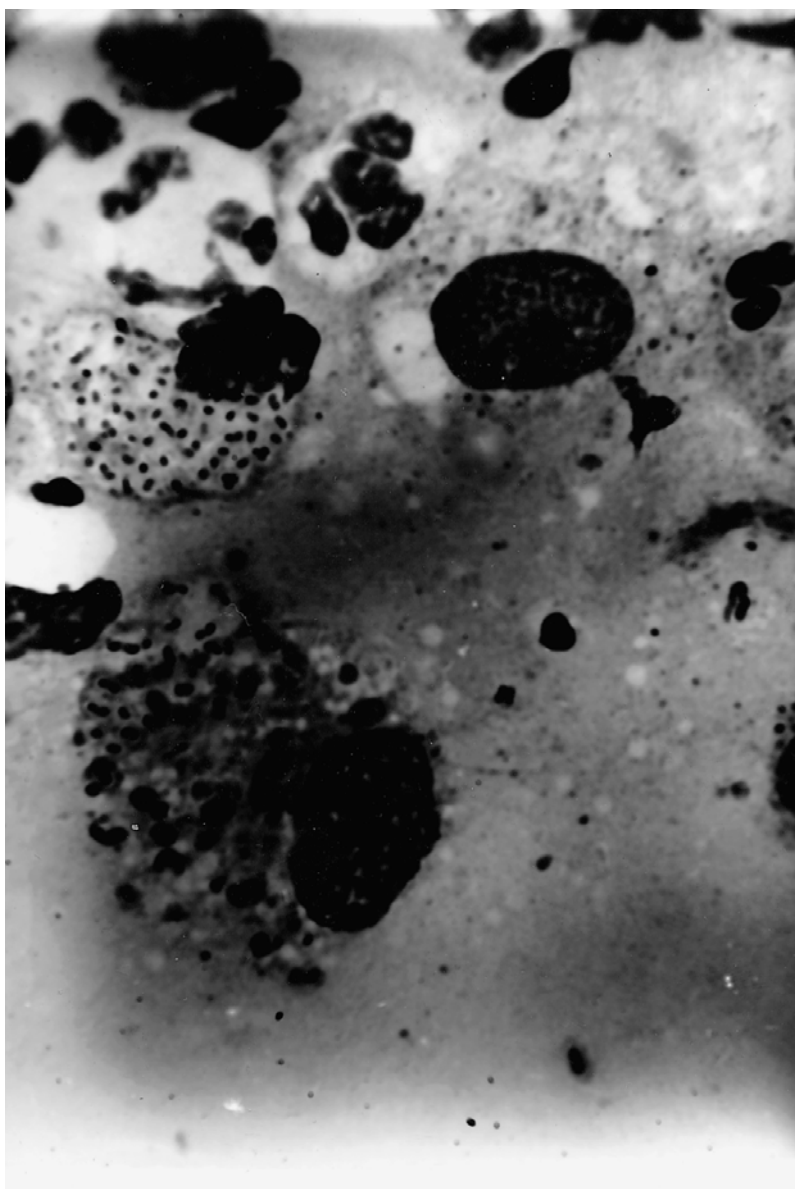


FIGURE BXII.γ.199. Large mononuclear phagocytes filled with *C. granulomatis*. Wright's stain ($\times 675$). (Reproduced with permission from R.B. Dienst and G.H. Brownell in M.P. Starr et al. (Eds.) *The Prokaryotes: a Handbook on Habitats, Isolation and Identification of Bacteria*. p. 1410, 1981, ©Springer-Verlag, New York.)

is usually seen as a chronic, granulomatous, genital ulcerative disease. Donovanosis is an uncommon disease found in specific geographical regions including northern and central Australia, New Guinea, parts of central and south Africa, southeast India, the Caribbean, and parts of South America. The organism is pathogenic only for humans and infection cannot be produced in laboratory animals.

There are no protective antibodies produced by a patient infected with *C. granulomatis*. Once the infection occurs, the disease persists chronically and may spread through the lymphatics to all tissue unless treated with antibiotics. The patient does produce specific sensitizing antibodies as shown by skin testing (Chen et al., 1949). Antibodies can be detected in the serum of patients by complement-fixation procedures (Anderson et al., 1945b; Dulaney and Packer, 1947; Goldberg et al., 1953). The test antigens used in these techniques have included pus from granulomatous lesions, whole or ruptured *C. granulomatis*, or boiled or extracted egg yolk medium following growth of the organism. Immunological studies first noted a relationship between strains of *C. granulomatis* and *Klebsiella pneumoniae* (Packer and Goldberg, 1950). An indirect immunofluorescence test for granuloma inguinale was reported by Freinkel et al. (1992). The antigen consisted of paraffin-embedded tissue sections from lesions containing Donovan bodies. Unabsorbed sera at dilutions of 1:160 were reported to be 100% sensitive and 98% specific. After absorption with *K. pneumoniae* antigen, sera from proven granuloma inguinale patients remained reactive but at lower titers. Antigens obtained from newly reported culture methods well undoubtedly lead to more effective serological tests. (Carter et al., 1997; Kharsany et al., 1997).

ENRICHMENT AND ISOLATION PROCEDURES

At present, the only sources for *C. granulomatis* are the lesions of donovanosis, although new culture methods may soon provide reference strains. An organism was reportedly isolated from human feces (Goldberg, 1962) that had antigenic similarities to *C. granulomatis*. Early reports of the cultivation of *C. granulomatis* in cell-free media include the yolk of embryonated eggs (Anderson, 1944), fresh yolk medium (Dienst et al., 1948), Locke-yolk Dulaney slants (Dulaney et al., 1948), and egg yolk replaced by lactalbumin hydrolysate (Goldberg, 1959). Reports of the successful culturing of *C. granulomatis* in peripheral blood mononuclear cells (Kharsany et al., 1997) or, more conveniently, human epithelial (HEp-2) cells (Carter et al., 1997), should provide the materials that can provide fundamental information about this species that to date is unknown.

The following procedure is given by Morse (1980). The ulcerative lesions are cleansed with sterile, saline-soaked gauze before obtaining samples, in order to decrease contamination and remove tissue debris. Samples of tissue are removed by scraping or by means of a biopsy punch from beneath the border of the lesion, and small cleansed pieces of tissue are minced into small particles. Inoculation is made into the yolk sacs of 5-day-old embryonated eggs. After incubation for 72 h, the organisms can be detected in the yolk sac fluid.

C. granulomatis can also be isolated and grown *in vitro*. For example, a pure culture was isolated by Dienst et al. (1948) by inoculating fresh egg medium exudate aspirated from a pseudobubo of a patient. This isolate was reported to be subcultured and maintained in the same culture medium, and examination of the subcultures revealed large numbers of encapsulated organisms consistent with the morphology of *C. granulomatis*. Dienst

et al. (1948) indicated that several factors were important for isolation and cultivation: (a) maintenance of a low oxidation-reduction potential, (b) the requirement for a growth factor found in egg yolk, and (c) use of semisolid media containing 0.12% agar.

Dulaney slants have also been used for isolation and cultivation of *C. granulomatis* (Dulaney et al., 1948). After inoculation of lesion material onto a Dulaney slant, Locke's fluid is added to cover three-quarters of the slant, and the tubes are then incubated in a vertical position for 48–72 h (Morse, 1980). A semi-synthetic medium has been devised by Goldberg (1959) for cultivation of laboratory strains of *C. granulomatis*. In this medium, the requirement for egg yolk is replaced by lactalbumin hydrolysate or by papaic digest of soy meal USP.

Recent efforts to culture *C. granulomatis* from biopsy material using these cell-free growth media suggest that they are unreliable. The recommended procedure is therefore the use of fresh monocytes (Kharsany et al., 1996, 1997) or the technically less demanding HEP-2 cell line. The latter employs standard *Chlamydia* culturing procedure (Carter et al., 1997).

Because diagnosis is still based on the observations of *C. granulomatis* in stained biopsy specimens and because of unreliable culture methods, this organism is a prime candidate for PCR primer targeting identification. Using PCR primers designed to target the *phoE* gene encoding for porin protein among *klebsiellae* and other enterobacteria, Bastian and Bowden (1996) reported amplification of *phoE* gene fragments from biopsy material showing Donovan bodies. The amplified sequences showed close correlation to those from *K. pneumoniae*, *K. rhinoscleromatis*, and *K. ozaenae*, and considerable divergence from corresponding sequences of other enterobacteria analyzed. Using the *phoE* gene primers in combination with primers for *Klebsiella scrA* (sucrose transport gene), reportedly absent in *C. granulomatis*, Carter et al. (1997) obtained *phoE*-positive and *scrA*-negative amplification from original clinical swab material as well as HEP-2 cell cultured isolates. Specific *C. granulomatis* primers have not yet been reported.

DIFFERENTIATION OF THE GENUS *CALYMMATOBACTERIUM* FROM OTHER GENERA

As of this printing, the examination of diseased tissue smears stained by Wright's blood stain or Giemsa stain remains the simplest procedure for identification for *C. granulomatis*. The characteristic appearance of the intracellular organisms (Fig. BXII.γ.199) is specific for the diagnosis of donovanosis. Species-specific PCR primers as well as immunofluorescence tests should soon be available.

TAXONOMIC COMMENTS

The coccobacillary microorganisms first observed by Donovan (1905) were frequently referred to as "Donovan bodies" when seen in tissue smears from patients with granulomatis lesions in the inguinal region. The Donovan bodies were later called *Calymmatobacterium granulomatis* by Aragão and Vianna (1913). When Anderson et al. (1945a) first isolated the organisms by yolk sac inoculation, they termed the etiologic agent of granuloma inguinale "*Donovania granulomatis*"; however, the name "*Donovania*" did not have priority over *Calymmatobacterium*.

In the first edition of *Bergey's Manual of Systematic Bacteriology*, the genus was not assigned to any family. It has been suggested that it should be placed in the family *Enterobacteriaceae* (Rake, 1948), and this relationship is supported (based on very limited

PCR-generated fragment analysis with “*Klebsiella*-like” sequences) by Bastian and Bowden (1996). Kharsany et al. (1997) observed no cross-reactivity with sera from patients showing Donovan bodies and laboratory reference strains of *K. pneumoniae*, *K. oxytoca*, or *Enterobacter aerogenes*. Thus, the taxonomic relationships of *Calymmatobacterium* to other bacterial genera are not yet understood,

and in the present edition of the *Manual* it seems desirable not to ally the genus with any established family, although it must be noted that the type strain was recently transferred to the genus *Klebsiella* as *Klebsiella granulomatis* (Carter et al., 1999).

List of species of the genus Calymmatobacterium

1. *Calymmatobacterium granulomatis* Aragão and Vianna 1913, 221^{AL}
gran.u.lo' ma.tis. L. dim. n. *granulum* a small grain; Gr. suff. -oma a swelling or tumor; M.L. n. *granuloma* a granuloma; M.L. gen. n. *granulomatis* of a granuloma.

The characteristics are as described for the genus and as depicted in Fig. BXII.γ.199.

The mol% G + C of the DNA is: not known.

Type strain: no strain extant.

Genus IX. *Cedecea* Grimont, Grimont, Farmer and Asbury 1981a, 325^{VP} (Enteric Group 15
Farmer, Grimont, Grimont and Asbury 1980b, 295)

J.J. FARMER III

Ce.de' ce.a. M.L. fem. n. *Cedecea* formed from the abbreviation CDC. The name was coined by P.A.D. Grimont and F. Grimont for the Centers for Disease Control, Atlanta, Georgia, where the organisms were originally recognized as a new group and named Enteric Group 15.

Rod-shaped cells 0.6–0.7 × 1.3–1.9 μm, conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. Motile, with five to nine peritrichous flagella. Grow at 15°, 20°, and 37°C. Facultatively anaerobic. **Catalase positive, strong and rapid.** Oxidase negative. Nonpigmented. Reduce nitrate to nitrite.

Positive for methyl red, Voges–Proskauer, citrate utilization (Simmons), motility at 36°C, growth in the presence of cyanide (KCN test), malonate utilization, D-glucose fermentation, and the fermentation of D-mannitol, salicin, maltose, trehalose, cellobiose, melibiose, D-arabitol, D-mannose, and D-galactose. Many strains produce visible gas during fermentation. Lipase (corn oil, Tween 40, Tween 60, Tween 80, and tributyrin) is positive, but gelatin hydrolysis, DNase, chitinase, polygalacturonase, and amylase are negative. Utilize 24 of 97 carbon sources.

Negative for indole production, H₂S production (TSI), urea hydrolysis, phenylalanine deaminase, lysine decarboxylase and the fermentation of dulcitol, adonitol, *myo*-inositol, D-sorbitol, L-arabinose, L-rhamnose, α-methyl-D-glucoside, erythritol, melibiose, glycerol, and mucate.

Susceptible to nalidixic acid, sulfadiazine, trimethoprim, gentamicin, streptomycin, kanamycin, tobramycin, amikacin, tetracycline, minocycline, chloramphenicol, carbenicillin, and furantoin (disk diffusion method on Mueller-Hinton agar); **resistant to colistin, polymyxin, penicillin, ampicillin, and cephalothin.**

Isolated from human clinical specimens that are normally sterile such as blood, urine, gallbladder, and lung tissue; also isolated from other clinical specimens such as throat, sputum, ulcers, and wounds. **An opportunistic pathogen that occasionally causes extraintestinal human infections**, or colonizes body surfaces. Isolated from water, ticks, and insects. A **rarely isolated genus of a *Enterobacteriaceae***. No 16S rRNA sequences of *Cedecea* strains have been reported.

The mol% G + C of the DNA is: 48–52.

Type species: *Cedecea davisae* Grimont, Grimont, Farmer and Asbury 1981a, 325.

FURTHER DESCRIPTIVE INFORMATION

Literature Since the genus was described in 1981 there have been a few reports in the literature; 11 reports cataloged in

MEDLINE and 27 cataloged in BIOSIS. These have described the genus in human clinical specimens and infections, its isolation from natural sources, and its basic physiology-metabolism.

Infections in humans Most of the original isolates of *Cedecea* in the CDC collection were from human respiratory tract specimens, so it was difficult to assess clinical significance (Farmer et al., 1980b). However, a few isolates were from body sites that are normally sterile. Since the original description in 1981, there have been several case reports that suggest clinical significance. The first case of bacteremia due to a strain of *Cedecea* was caused by *Cedecea neteri* (Farmer et al., 1982) and led to the naming of this organism, which was previously without a scientific name (“*Cedecea* species 4”). Other cases of bacteremia have now been described (Perkins, et al., 1986; Aguilera et al., 1995). Several other cases have been described in which a *Cedecea* strain was isolated from a body site that is normally sterile, or thought to be clinically significant (Bae et al., 1975; Hansen and Glupczynski, 1984; Coudron and Markowitz, 1987; Anon et al., 1993). Infections due to *Cedecea* have usually been in elderly hospital patients with debilitating conditions such as heart disease, diabetes, alcoholism, and renal insufficiency. A few isolates have been from human feces, but there is no evidence that any *Cedecea* species causes diarrhea or intestinal infections.

Possible origin of the *Cedecea* strains in human clinical specimens Originally there was little information on this point. Strains of *Cedecea* have now been isolated from well water, ticks, and insects (Jang and Nishijimi, 1990; Kaaya and Okech, 1990; Pellegrini et al., 1992), which may be natural reservoirs and lead to exposure of humans. Berkowitz and Metchock (1995) described bacteria in the feces of hospitalized children that are resistant to third-generation cephalosporins. Antibiotic usage is common in hospitals and selects for strains that have intrinsic or acquired resistance. Since *Cedecea* strains have high intrinsic resistance to penicillin, ampicillin, and the cephalosporin antibiotic cephalothin, they probably have a selective advantage in the feces of hospitalized patients. The reduction of the normal host flora by antibiotic usage could favor the growth and selection

of *Cedecea* strains, which might then colonize certain body sites, and under the right conditions cause infection.

Additional studies and more case reports are needed to better define the pathogenic potential of the five species in the genus. The clinical significance of a *Cedecea* is probably as an infrequent colonizer or infrequent opportunistic pathogen, particularly in older people who are debilitated. More information is needed on ecology, epidemiology, and how humans are exposed to and acquire the organism.

Bacteriology Strains of *Cedecea* grow well on media normally used in enteric bacteriology. Colonies on nutrient agar are about 1.5 mm (24 h, 37°C). They are typical *Enterobacteriaceae* in most of their properties. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media, and methods for biochemical testing, identification, and antibiotic susceptibility. Also see Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*, which gives the percentage positive for the five *Cedecea* species on 47 biochemical tests done at the Enteric Reference Laboratories at the Centers for Disease Control (CDC) with standard media and methods (Farmer, 1995). These tests and computer analysis have proved useful for identification and for differentiating strains of *Cedecea* from other species in the family. Table BXII.γ.213 lists phenotypic tests that are useful for differentiating and identifying the five species of *Cedecea*. Additional descriptive material, including the results for 97 carbon source utilization tests, can be found in the original description of the genus (Grimont et al., 1981a). Chester and Moskowitz (1987) found that *Cedecea* strains produce catalase, and that the catalase reaction was very strong and rapid and could be useful as a screening test.

TAXONOMIC COMMENTS

History and discovery *Cedecea* was proposed as a new genus in *Enterobacteriaceae* in 1980 (Farmer et al., 1980b; Grimont et al., 1981a). In 1977 it was first recognized and named "Enteric Group 15" as a diagnostic culture was being studied at the CDC. The laboratory's STRAIN MATCHER computer program indicated a group of over a dozen other cultures that were very similar to

the diagnostic culture, and all had been reported "unidentified". The strains were lipase-positive (corn oil) and resistant to the antibiotics colistin and cephalothin. Among *Enterobacteriaceae*, these properties are unique to the genus *Serratia*, but the new group differed from *Serratia* because it was negative for DNase and gelatin hydrolysis. Originally, Enteric Group 15 was thought to be a uniform group of strains that would probably be a new species of *Serratia*, intermediate between the "typical" *Serratia fonticola*, which is negative for all three of these tests (Farmer et al., 1980b). Strains of Enteric Group 15 were sent to the Grimonts in France (Farmer et al., 1980b; Grimont et al., 1981a), who used phenotypic characterization and DNA-DNA hybridization (SI nuclease method) to characterize them. Fifteen strains of Enteric Group 15 were more closely related (Fig. BXII.γ.200) to each other (32–100%) than to strains of the six named *Serratia* species (6–10%) or to other *Enterobacteriaceae* (1–23%). Originally, Enteric Group 15 was defined as a single group of 17 strains, but further study indicated five different DNA hybridization groups (Fig. BXII.γ.200) that were also phenotypically distinct (Table BXII.γ.213). In the original paper proposing the genus *Cedecea*, the two largest hybridization groups were named *C. davisae* and *C. lapagei*. The name *Cedecea neteri* was later given to DNA hybridization group 4 (Farmer et al., 1982). The original vernacular names "*Cedecea* species 3" and "*Cedecea* species 5" are still being used for the two other DNA hybridization groups until more strains are available, or there is a compelling reason to name them (see Other Organisms, below).

FURTHER READING

- Farmer, J.J., III, N.K. Sheth, J.A. Hudzinski, H.D. Rose and M.F. Asbury. 1982. Bacteremia due to *Cedecea neteri* sp. nov. J. Clin. Microbiol. 16: 775–778.
- Grimont, P.A.D., F. Grimont, J.J. Farmer, III and M.A. Asbury. 1981. *Cedecea davisae*, gen. nov., sp. nov. and *Cedecea lapagei*, sp. nov., new *Enterobacteriaceae* from clinical specimens. Int. J. Syst. Bacteriol. 31: 317–326.
- Perkins, S.R., T.A. Beckett and C.M. Bump. 1986. *Cedecea davisae* bacteremia. J. Clin. Microbiol. 24: 675–676.

List of species of the genus *Cedecea*

1. ***Cedecea davisae*** Grimont, Grimont, Farmer and Asbury 1981a, 325^{VP}
da'vi.sae. M.L. gen. n. *davisae* named to honor Betty Davis, the American bacteriologist of the Enteric Bacteriology Lab-

oratories, Centers for Disease Control and Prevention, Atlanta, Georgia, who made many contributions to the biochemical and serological identification of *Enterobacteriaceae* and *Vibrionaceae*.

TABLE BXII.γ.213. Differentiation of the three named and two unnamed *Cedecea* species

Test or property	<i>Cedecea davisae</i>	<i>Cedecea lapagei</i>	<i>Cedecea neteri</i>	<i>Cedecea</i> species 3	<i>Cedecea</i> species 5
Ornithine decarboxylase	95 ^a	0	0	0	50
Sucrose fermentation	100	0	100	50	100
D-Sorbitol fermentation	0	0	100	0	100
Raffinose fermentation	10	0	0	100	100
D-Xylose fermentation	100	0	100	100	100
Melibiose fermentation	0	0	0	100	100
Malonate utilization	91	100	100	0	0
Thiamin required for growth ^b	100	0	0	0	0
Growth at 5°C ^b	100	0	0	0	0

^aFor the first seven tests each number gives the percentage positive after 2 d incubation at 36°C and is based on the data summarized by Farmer (1999). The vast majority of these positive reactions occur within 24 h. Reactions that become positive after 2 d are not considered.

^bThe results for thiamin requirement and growth at 5°C are from Grimont et al. (1981a).

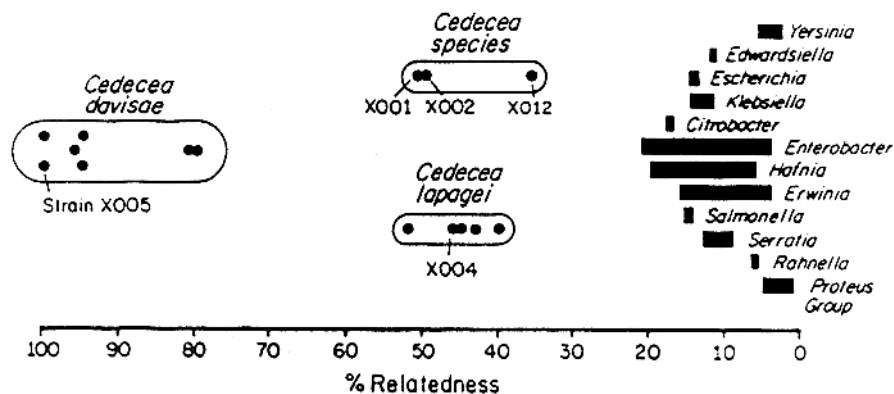


FIGURE BXII.γ.200. Relatedness by DNA–DNA hybridization of the *Cedecea davisae* type strain (X005) to 14 other *Cedecea* strains and to other genera of Enterobacteriaceae. Strain X004 is the type strain of *Cedecea lapagei*; X002 is the type strain of *Cedecea neteri*; X001 is the reference strain of “*Cedecea* species 3”; and X012 is the reference strain of “*Cedecea* species 5”. (Redrawn with permission from J.J. Farmer III, P.A.D. Grimont, F. Grimont, and M.A. Asbury.)

The characteristics are as given for the genus. All strains require thiamin (0.01 µg/ml) for growth in minimal media and one strain also requires *p*-aminobenzoic acid (Grimont et al., 1981a). Most strains grow at 5°C, unlike the other *Cedecea* species. See Table BXII.γ.193 in the chapter on the family Enterobacteriaceae for a more complete phenotypic description based on the results of 47 biochemical tests. One strain produces a strong distinct aroma, the “odor of *Serratia odorifera*”, which is apparently due to the pyrazine compound 3-sec-butyl-2-methoxypyrazine (Gallois and Grimont, 1985).

Isolated from human clinical specimens including blood, gallbladder, eye, urine, sputum, throat, wound, feces; probably a rare opportunistic pathogen or colonizer. Perkins et al. (1986) described a case of bacteremia in a 70-year-old woman with a history of heart disease and a diagnosis of bronchitis and chronic obstructive pulmonary disease. Other case reports have described pneumonia in an elderly diabetic with a heart condition (Bae et al., 1975) and a scrotal abscess in a patient with alcoholic liver disease (Bae and Sureka, 1983). Isolated from the cockroaches *Blatta germania* and *Blatta orientalis* (Pellegrini et al., 1992). The 16S rRNA sequence has not been determined.

The mol% G + C of the DNA is: 49–50 (T_m).

Type strain: ATCC 33431, CDC 3278-77, CIP 80.34.

Additional Remarks: The American Type Culture Collection includes four other human strains; two from sputum and one each from gallbladder and wound.

2. *Cedecea lapagei* Grimont, Grimont, Farmer and Asbury 1981a, 325^{VP}

la.pa'ge.i. M.L. gen. n. *lapagei* named to honor the late Stephen Lapage, a British bacteriologist who made many contributions to the Enterobacteriaceae, bacterial systematics, and particularly as an editor of the Bacteriological Code.

The characteristics are as given for the genus. See Table BXII.γ.193 in the chapter on the family Enterobacteriaceae for a more complete phenotypic description based on the results of 47 biochemical tests. All five strains of *C. lapagei* are highly related by DNA–DNA hybridization, but strain 1554-

75 had higher divergence values and differed from the other strains in phenotype (Grimont et al., 1981a).

Isolated from human respiratory tract (throat and sputum) and from ticks. Coudron and Markowitz (1987) described an isolate from lung tissue in a case report, but concluded that its etiological role was not proven. Appears to be a tick pathogen. Brun and Texeira (1992a, b) described genital tract infections of the tick species *Boophilus microplus* that resulted in “engorged females”. The 16S rRNA sequence has not been determined.

The mol% G + C of the DNA is: 48–52 (T_m).

Type strain: ATCC 33432, CDC 0485-76, CIP 80.35.

Additional Remarks: The American Type Culture Collection includes four other strains, all from human sputum.

3. *Cedecea neteri* Farmer, Sheth, Hudzinski, Rose and Asbury 1983, 438^{VP} (Effective publication: Farmer, Sheth, Hudzinski, Rose and Asbury 1982, 777.)

ne'te.ri. M.L. gen. n. *neteri* named to honor Erwin Neter, an American physician and microbiologist who made many contributions to our knowledge of the family Enterobacteriaceae, particularly the role of this family in human disease.

The characteristics are as given for the genus. Also see Table BXII.γ.193 in the chapter on the family Enterobacteriaceae for a more complete phenotypic description based on the results of 47 biochemical tests.

Isolated from human blood, sputum, and wound. Farmer et al. (1982) described a case of bacteremia with possible endocarditis in a 62-year-old man with vascular heart disease and recurrent fever and chills of 4 d duration. Anon et al. (1993) described a case of peritonitis after vigorous abdominal surgery in which *Cedecea neteri* was isolated along with *Escherichia vulneris*. Aguilera et al. (1995) described a case of bacteremia in a patient with systemic lupus erythematosus.

The mol% G + C of the DNA is: not reported.

Type strain: ATCC 33855, CDC 0621-75.

Additional Remarks: The American Type Culture Collection includes two other strains, one from human blood (the case report of Farmer et al., 1982) and one from sputum.

Other Organisms

1. "Cedecea species 3"

The CDC collection has three strains of this organism, from blood, gallbladder, and sputum.

The mol% G + C of the DNA is: not reported.

Deposited strain: X001, CDC 4853-73.

2. "Cedecea species 5"

The CDC collection has only one strain of this organism, from a human toe.

The mol% G + C of the DNA is: not reported.

Deposited strain: X012, CDC 3699-73.

The CDC collection has at least 11 other cultures from a variety of human clinical specimens that were reported "Cedecea species" because they had high computer identification scores as "genus *Cedecea*" and had characteristics typical for the genus. However, the phenotypic properties of these strains differed from the five *Cedecea* species described above. The data on the two unnamed *Cedecea* species and these 11 "Cedecea species" clearly indicate that there are additional species of *Cedecea*. Future studies are needed to better define this collection. There have been two case reports describing *Cedecea* strains that are phenotypically different from the five named species. Hansen and Glupczynski

(1984) described a *Cedecea* isolate from a cutaneous ulcer with a purulent discharge in a 79-year-old diabetic with arthritis of both lower limbs. Cultures were positive on days 2, 5, and 43. The isolate was most like *Cedecea davisae* except that it was positive for acetate utilization and D-sorbitol fermentation. The isolate was lost before it could be confirmed by a reference laboratory. Magnum and Radisch (1982) described a *Cedecea* isolate from the postmortem heart blood of a 76-year-old alcoholic man with decubiti, anemia, metabolic acidosis, renal insufficiency, and urinary incontinence. This strain was most like *Cedecea* species 3 except it was Voges-Proskauer positive, fermented sucrose, and grew on acetate agar.

Two additional *Cedecea* species have been described, but they were given the vernacular names of "Cedecea species 3" and "Cedecea species 5" rather than scientific names (Farmer et al., 1980b; Grimont et al., 1981a). Thus, the two organisms do not have standing in nomenclature, but appear to be unnamed species of *Cedecea* based on DNA-DNA hybridization and phenotypic differences (Table BXII.γ.213). See Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* for a more complete phenotypic description of this organism based on the results of 47 biochemical tests.

Genus X. *Citrobacter* Werkman and Gillen 1932, 173^{AL}

WILHELM FREDERIKSEN

Cit.ro.bac'ter. L. n. *citrus* lemon; M.L. n. *bacter* a small rod; M.L. masc. n. *Citrobacter* a citrate-utilizing rod.

Straight rods, ~1.0 µm × 2.0–6.0 µm. Occur singly and in pairs. Conform to the general definition of the family *Enterobacteriaceae*. Usually not encapsulated. Gram negative. Usually **motile by peritrichous flagella**. Facultatively anaerobic, having both a respiratory and a fermentative type of metabolism. Grow readily on ordinary media. Colonies on nutrient agar are generally 2–4 mm in diameter, smooth, low convex, moist, translucent or opaque, and gray with a shiny surface and entire edge. Mucoid or rough forms may occur occasionally. Oxidase negative. Catalase positive. Chemoorganotrophic. **Citrate can be utilized as a sole carbon source by most strains. Lysine is not decarboxylated.** Alginate and pectate are not decomposed. D-glucose is fermented with the production of acid and gas. The methyl red test is positive; the Voges-Proskauer test is negative. Occur in the feces of humans and some animals; probably normal intestinal inhabitants. Sometimes pathogenic and often isolated from clinical specimens as opportunistic pathogens. Can also be found in soil, water, sewage, and food.

The mol% G + C of the DNA is: 50–52 (*T_m*).

Type species: *Citrobacter freundii* (Braak 1928) Werkman and Gillen 1932, 173 (*Bacterium freundii* Braak 1928, 140.)

FURTHER DESCRIPTIVE INFORMATION

Members of *Citrobacter* may or may not ferment lactose promptly but nearly always produce β-galactosidase. L-arabinose, maltose, L-rhamnose, trehalose, D-xylose, D-mannitol, and D-sorbitol are fermented rapidly by the majority of strains. Erythritol and myo-inositol are rarely attacked.

Ornithine is decarboxylated by almost all strains of *C. koseri*, *C. amalonaticus*, *C. farmeri*, *C. braakii*, *C. sedlakii*, and *C. rodentium*, but not by *C. freundii*, *C. youngae*, and *C. werkmanii*. *C. koseri* and *C. rodentium* do not grow in media containing potassium cyanide (KCN) in contrast to the other species.

Strains of *C. koseri* ferment D-adonitol and D-arabitol, but strains of the other species do not. Malonate is utilized as a sole carbon source by most strains of *C. koseri*, *C. werkmanii*, *C. sedlakii*, and *C. rodentium*, but can be used by less than 15% of the strains of the other species.

The majority of strains of *C. freundii*, *C. youngae*, *C. braakii*, and *C. werkmanii* produce abundant H₂S in the butt of Kligler iron agar and triple-sugar iron agar.

Indole is produced by *C. koseri*, *C. amalonaticus*, *C. farmeri*, and *C. sedlakii* with few exceptions, but few strains of the other species give a positive indole test.

Based on 16S rRNA gene sequence, the genus *Citrobacter* clusters in the family *Enterobacteriaceae* within the *Gamma*proteobacteria. Its nearest relatives are members of the genera *Salmonella* and *Pantoea*.

DNA-DNA hybridization experiments (Brenner et al., 1993) using the hydroxyapatite method showed intraspecific relatedness almost always above 70%, with 5% or less divergence within related sequences, and interspecific relatedness from ~30% to ~70%. *C. farmeri* was closely related to *C. amalonaticus*; *C. youngae*, *C. braakii*, and *C. werkmanii* were related to each other and to *C. freundii* at the 50–70% level. The other interspecies relations were on a lower level (~30% to ~55%).

Some strains of *C. koseri* produce bacteriocins, and many can be induced to produce bacteriophages (Hamon et al., 1974; Markel et al., 1975).

The antigenic structure was developed around the so-called Bethesda-Ballerup group that was considered close to *Salmonella* (Kauffmann and Møller, 1940; West and Edwards, 1954). After its inclusion in *Citrobacter freundii*, the group formed the basis for the elucidation of the O- and H-antigen structure of this species, and its relation to other taxa, especially within the genus *Salmonella*; see Ewing (1986a) for a detailed description. The Vi antigen of *Salmonella* serovar Typhi (*S. typhi* or *S. enterica*, subsp. *enterica*, serovar Typhi) can be found in certain strains of *C. freundii*.

The antigenic structure of *C. koseri* was studied by several groups (Gross and Rowe, 1974, 1983; Popoff and Richard, 1975; Sourek and Aldová, 1976) resulting in three different O- and H-antigen serotyping systems. Sourek and Aldová (1976) also established an O-antigen scheme for *C. amalonaticus*, which they extended in 1988 (Sourek and Aldová, 1988). Van Oye et al. (1975) found that many strains of *C. amalonaticus* reacted in different *Shigella* sera.

Miki et al. (1996) reexamined the 90 reference strains listed in the scheme for *C. freundii* by West and Edwards (1954). They found that 40 of these strains belonged to *C. youngae* as defined by Brenner et al. (1993), 25 to *C. braakii*, 13 to *C. werkmanii*, and only three to the redefined *C. freundii*. Nine O-group-29 strains formerly allocated to the Ballerup group were all identified as *C. braakii*.

With modern typing systems there seems to be less place for serotyping of citrobacters for epidemiological purposes. Both multilocus enzyme electrophoresis (Woods et al., 1992), ribotyping (Papasian et al., 1996; El Harrif-Heraud et al., 1997), PCR methods (Woods et al., 1992; El Harrif-Heraud et al., 1997), and pulsed field gel electrophoresis (Papasian et al., 1996) have been applied, often to elucidate possible outbreaks of *C. koseri* meningitis/cerebral abscesses.

A 32-kDa outer membrane protein (OMP) was found to be a marker of virulence for *C. koseri* (Kline et al., 1988), and possibly a factor of virulence (Li et al., 1990).

Antibiotic sensitivity Strains of *Citrobacter* are naturally susceptible to sulfonamides, trimethoprim, aminoglycosides, chloramphenicol, tetracycline, nalidixic acid, fluoroquinolones, nitrofurantoin, polymyxins, and fosfomycin. Like other enterobacteria they are resistant to erythromycin and other macrolides, lincosamides, fusidic acid, and vancomycin. Resistance to one or more of the former group of antibiotics may be acquired, depending on circumstances, especially antibiotic usage policy in a broad sense.

Citrobacter strains are usually extremely susceptible to the fluoroquinolones. However, like many other enterobacteria they are prone to develop resistance, although this seems to occur rarely. The majority of strains are susceptible to the aminoglycosides, although less so to streptomycin than to newer aminoglycosides.

Almost all strains of *Citrobacter* produce a β -lactamase. *C. freundii* and *C. koseri* differ in the type of β -lactamase produced, *C. freundii* being resistant to cephalosporins and susceptible to carboxypenicillins, while *C. koseri* behave in the reverse way. Resistance to a broad range of newer β -lactam antibiotics may appear in both species; however, imipenem seems to remain active. See

Frederiksen and Søgaard (1992) for a broader review of antibiotics and *Citrobacter*. El Harrif-Heraud et al. (1997) described a nosocomial outbreak where six identical strains of *C. koseri* produced an extended spectrum β -lactamase (ESBL), leaving susceptibility only to imipenem, latamoxef, and some combinations with clavulanic acid. The strains harbored a plasmid mediating a SHV-4 type lactamase.

Members of the genus *Citrobacter* occur not only in feces of humans and animals with no disorder but also in water, sewage, soil, and food. Isolates of *C. freundii* recovered from fish were compared with strains from various other sources by Toranzo et al. (1994), and a possible role as a fish pathogen discussed. *Citrobacters* are found in clinical bacteriology not only in stools but also in urine, sputum, and specimens from bacteremia, meningitis, otitis media, wounds, abscesses, the throat, and autopsies. Their role often seems to be that of an opportunistic pathogen. Cases of neonatal meningitis caused by *C. koseri* have been reported (Gwynn and George, 1973; Kline, 1988). Khashe and Janda (1996) found that an iron-scavenging mechanism such as induction of high molecular mass proteins (72–83 kDa) could be a virulence factor for *C. koseri*.

Although *C. freundii* was once considered an enteropathogen, it seems rather to be a normal inhabitant of the intestine (Sakazaki et al., 1960), but the role in diarrhea is dubious (Sedlák, 1973; Lipsky et al., 1980). In most instances the organism is probably a normal intestinal inhabitant, but some strains produce enterotoxins like many other enterobacteria, and may then act as intestinal pathogens (Guarino et al., 1987, 1989). Septicemia and other disseminated infections, e.g., osteomyelitis, occur with low frequency (Frederiksen and Søgaard, 1992).

C. rodentium has been found only in rodents; it was shown to be the cause of transmissible murine colonic hyperplasia, a disease of laboratory mice (Schauer and Falkow, 1993; Schauer et al., 1995).

Ecology As intestinal inhabitants of humans and animals, citrobacters are excreted into the environment and found in sewage, water, and soil. Apart from this, nothing is known about the ecological role of *Citrobacter* species in the environment.

Phylogenetic position On the basis of their 16S rDNA sequences, citrobacteria group within the *Gammaproteobacteria*, with *Escherichia*, *Erwinia*, *Salmonella*, and *Serratia* species as their nearest relatives.

ENRICHMENT AND ISOLATION PROCEDURES

The majority of *Citrobacter* strains can grow in liquid enrichment media such as selenite broth and tetrathionate broth and on selective isolation media such as *Salmonella-Shigella* agar, deoxycholate-citrate agar, brilliant green agar, and bismuth sulfite agar. Colonies that ferment lactose slowly often resemble *Salmonella* colonies.

MAINTENANCE PROCEDURES

Stock cultures of *Citrobacter* strains may be maintained at room temperature in a semisolid medium containing 1.0% Bacto-casitone (Difco), 0.3% yeast extract, 0.5% NaCl, and 0.3% agar, pH 7.0. The cultures remain viable for up to a year without subculturing if they are sealed with a rubber stopper or a cork that has been soaked in hot paraffin wax. Strains may be preserved indefinitely by lyophilization. They can also be maintained for years as stab cultures in sealed tubes of meat extract agar.

DIFFERENTIATION OF THE GENUS *CITROBACTER* FROM OTHER GENERA

The differentiation of *Salmonella* from lactose-negative *Citrobacter* proved difficult ever since the "Bethesda-Ballerup" group was recognized as belonging in the genus *Citrobacter*. The most useful tests to distinguish among these lactose-negative strains are lysine decarboxylase and growth in KCN medium. The greatest difficulties arise with the rare lactose/o-nitrophenyl- β -galactosidase (ONPG) positive *Salmonella* strains, the ONPG-negative *Citrobacter* strains (also rare), and, in addition, when H₂S positive *E. coli* are encountered. Table BXII.γ.214 gives some tests useful for differentiation and correct identification of such strains.

TAXONOMIC COMMENTS

The genus *Citrobacter* was proposed by Werkman and Gillen (1932) for the citrate-utilizing "coli-aerogenes intermediates". The organisms were since described under several names, and *C. freundii* was called "*Escherichia freundii*" by Yale (1939a). The role of citrobacters as possible pathogens was first noticed by Kauffmann and Møller (1940), who described an organism called "*Salmonella ballerup*" that is presently classified in *C. freundii*. Later this biogroup of organisms was removed from the genus *Salmonella* and was called the Ballerup group (Bruner et al., 1949; Harhoff, 1949). Independently of the Ballerup group of organisms, Edwards et al. (1948) and Moran and Bruner (1949) studied a group of bacteria characterized by Barnes and Cherry (1946) and referred to it as the Bethesda group of bacteria. West and Edwards (1954) found that organisms of both the Bethesda and Ballerup groups were biochemically and serologically indistinguishable and combined the two groups into the Bethesda-Ballerup group. Moreover, West and Edwards (1954) and Møller (1954) called attention to the close biochemical relationship between members of the Bethesda-Ballerup group and strains of "*E. freundii*". Accordingly, Kauffmann (1954) reclassified the Bethesda-Ballerup group into "*E. freundii*", and later revived the genus *Citrobacter* for "*E. freundii*" (Kauffmann, 1956).

Frederiksen (1970) described a new species that he named *Citrobacter koseri*. Young et al. (1971) described a new genus, *Levinea*, which contained two species, *L. malonatica* and *L. amaltonatica*. Ewing and Davis (1972a) published a paper on *Citrobacter diversus*, using the specific epithet "*diversum*" of Werkman and Gillen (1932) for a group of organisms that they considered to belong in *Citrobacter*. It became apparent that *C. koseri*, *L. malonatica*, and *C. diversus* were probably different names for the same species. DNA-DNA hybridization studies by Crosa et al. (1974) showed that *C. diversus* and *L. malonatica* belonged to one hybridization group and that this group and *C. freundii* and *L. amaltonatica* were all related with binding ratios around 50–60%.

Thus, they all could be considered to belong in the genus *Citrobacter*, with no need for the genus *Levinea*.

Crosa et al. (1974) suggested moving *L. amaltonatica* to *Citrobacter*, and Sakazaki et al. (1976), as a result of numerical taxonomy, suggested placing *L. amaltonatica* in *Citrobacter* as a species separate from *C. freundii*. The name *C. amaltonaticus* was formally proposed by Brenner and Farmer (1981, 1982). Macierevitz (1966) studied a group of organisms that were H₂S-negative and ornithine decarboxylase-positive and proposed the name "*Padlewskia*", without designating a specific epithet for the organisms of this genus. From the biochemical characteristics described it was probable that "*Padlewskia*" organisms and *C. amaltonaticus* were identical. This was confirmed when strains from Macierevitz and from Young were compared by the author (unpublished observations).

Werkman and Gillen (1932) also proposed the species name "*Citrobacter intermedium*" (sic) for four of their strains. Frederiksen (1970) showed that the only extant strain, ATCC 6750, was a typical *C. freundii*. The name "*Citrobacter intermedium*", therefore, was not included on the Approved Lists of Bacterial Names in 1980, and has no nomenclatural standing.

For a time there were three names—*C. koseri*, *L. malonatica*, and *C. diversus*—for the same species, all of which were on the Lists of Approved names. Frederiksen (1990) requested that the name *C. diversus* (Werkman and Gillen, 1932) be placed on the list of rejected names, because it was incorrectly used by Ewing and Davis (1972a), as the organism they described differed in at least eight characteristics from the organism described by Werkman and Gillen as "*Citrobacter diversum*" (sic), and is thus a nomen dubium. This request was granted by the Judicial Commission (1993). The epithet *malonatica* is a junior synonym and should not be used.

The concept of what should be called *C. freundii* was changed when Brenner et al. (1993) published a study showing that a number of atypical *Citrobacter* strains (considered *C. freundii* or *Citrobacter* species) could be arranged in nine genomospecies apart from *C. koseri* and *C. amaltonaticus*. They named five new species: *C. farmeri* for *C. amaltonaticus* biovar 1, and *C. youngae*, *C. braakii*, *C. werkmanii*, and *C. sedlakii* for strains that had been considered to be *C. freundii* or atypical *C. freundii*. Only 9 of 66 strains were shown genotypically to belong in *C. freundii*. The remaining 57 strains were allocated to seven genomospecies that could all be differentiated phenotypically. Four of them were named (*C. youngae*, *C. braakii*, *C. werkmanii*, and *C. sedlakii*), whereas three (number 9, 10, and 11) were not named, as they contained only three strains each.

Genomospecies 9, containing strains isolated from rodents, was subsequently named *C. rodentium* by Schauer et al. (1995), who studied three additional strains isolated from mouse intestine that were previously called *C. freundii* biotype 4280 (Barthold et al., 1976).

Janda et al. (1994) examined 235 *Citrobacter* strains. Within what was called a *C. freundii* complex, 37% were found to be *C. freundii*, 24% to be *C. youngae*, 13% *C. braakii*, and 6% *C. werkmanii*.

The recent study by Miki et al. (1996) showed that the test strains from the West and Edwards (1954) scheme for the Bethesda-Ballerup group mainly belonged in *C. youngae* (40 strains), *C. braakii* (25 strains), *C. werkmanii* (13 strains), and genomospecies 10 (six strains). Only three strains could be allocated to *C. freundii* in the new sense. Nine strains of the O-29 group ("Ballerup", in which the Vi antigen may occur) were found to belong in *C. braakii*.

TABLE BXII.γ.214. The differentiation of H₂S positive *Citrobacter* from ONPG positive *Salmonella* and H₂S positive *Escherichia coli*^a

Characteristic	<i>Citrobacter</i>	<i>Salmonella</i>	<i>E. coli</i>
Indole	d	—	+
Citrate	+	+	—
KCN (growth)	+	—	—
Lysine	—	+	+
Ornithine	d	+	d
Sucrose	d	—	d
Cellobiose	d	—	—
ONPG	+	+	+
H ₂ S (iron agar)	+	+	+

^aFor symbols, see standard definitions.

The definition of *C. freundii* has thus changed; the new species and *C. freundii* can be differentiated phenotypically and genotypically, but there are no good “key” tests to characterize them; H₂S-positive strains are found in several species, and so are indole-positive strains. The relative distribution within the *C. freundii* complex evidently depends on how the strains included had been selected: strains sent to reference laboratories, or “nonse-

lected” strains from clinical microbiology laboratories (some of the strains used by Janda et al., 1994). There is evidently a need for a thorough characterization of a large unselected collection of *Citrobacter* strains, both phenotypically and genotypically, in order to evaluate the relative importance of the species of the genus as now defined by Brenner and his co-workers.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *CITROBACTER*

The differential characteristics of the species of *Citrobacter* are indicated in Table BXII.γ.215. Table BXII.γ.216 lists other char-

acteristics of the species. See also Tables BXII.γ.193, BXII.γ.194, BXII.γ.195, and BXII.γ.196 in the chapter on *Enterobacteriaceae*.

List of species of the genus *Citrobacter*

1. **Citrobacter freundii** (Braak 1928) Werkman and Gillen 1932, 173^{AL} (*Bacterium freundii* Braak 1928, 140.)
freun' di.i. M.L. gen. n. *freundii* of Freund, named after A. Freund, the bacteriologist who first observed that trimethylene glycol was a product of fermentation.
The morphology is as given for the genus. Usually motile. Usually not encapsulated, although encapsulated strains may occur.
The colony morphology is similar to that of *Escherichia coli*.
Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216. Ewing (1986a) listed only 2.1% of strains as indole positive, but among the nine strains included in the study of Brenner et al. (1993), 38% were positive. Janda et al. (1994) found two indole-positive strains among 60 strains allocated to *C. freundii* sensu Brenner et al. (1993). The urease test is positive for most strains if Christensen's method is used, but usually negative if tested for the preformed enzyme.
Found in humans and animals including mammals, birds, reptiles, amphibians, and fish. Also found in soil, water, sewage, and food. Often found in clinical specimens

- such as urine, throat, sputum, blood, and wound swabs as an opportunistic or secondary pathogen.
The mol% G + C of the DNA is: 50–51 (*T_m*).
Type strain: ATCC 8090, DSM 30039, IFO 12681, NCTC 9750.
GenBank accession number (16S rRNA): AJ233408.
2. **Citrobacter amalonaticus** (Young, Kenton, Hobbs and Moody 1971) Brenner and Farmer 1982, 266^{VP} (Effective publication: Brenner and Farmer 1981, 1140) (*Levinea amal-onatica* Young, Kenton, Hobbs and Moody 1971, 58.)
a.ma.lo.na' ti.cus. Gr. prefix *a* not; M.L. adj. *malonaticus* pertaining to malonate; M.L. adj. *amalonaticus* not pertaining to malonate (i.e., not able to utilize malonate).
The morphology is as given for the genus. Motile. Not encapsulated.
Colonies on nutrient agar are translucent to opaque, resembling those of *E. coli*.
Physiological and biochemical characteristics are indicated in Tables BXII.γ.215 and BXII.γ.216. Some strains are late gelatin liquefiers.
Found in the feces of humans and animals and in soil,

TABLE BXII.γ.215. Conventional tests useful in differentiating *Citrobacter* species (adapted from Brenner et al., 1993)^a

Characteristic	1. <i>C. freundii</i>	2. <i>C. amalonaticus</i>	3. <i>C. braakii</i>	4. <i>C. farmeri</i>	5. <i>C. gillenii</i>	6. <i>C. koseri</i>	7. <i>C. murlinae</i>	8. <i>C. rodentium</i>	9. <i>C. sedlakii</i>	10. <i>C. werkmanii</i>	11. <i>C. youngae</i>
Indole	d ^b	+	d	+	—	+	+	—	+	—	d
Citrate—Simmons	d	+	d ^c	d ^c	d ^c	+	+	— ^c	d ^c	+	d ^c
H ₂ S in iron agar	d ^b	d ^d	d	—	d	—	d	—	—	+	d
Ornithine	—	+	+	+	—	+	—	+	+	—	—
KCN growth	d ^c	+	+	+	+	—	+	—	+	+	+
Malonate	d	d ^d	—	—	+	+	—	+	+	+	—
<i>Acid from:</i>											
Sucrose	+	d	—	+	d	d	d	—	—	—	d
Melibiose	+	—	d ^c	+	d	—	d	—	+	—	—
Raffinose	d ^c	—	—	+	—	—	d	—	—	—	—
Dulcitol	d	—	d	—	—	d	+	—	+	—	d
Adonitol	—	—	—	—	—	+	—	—	—	—	—
D-Arabitol	—	—	—	—	—	+	—	—	—	—	—

^aFor symbols see standard definitions.
^bPrevious reports (e.g., Ewing, 1986a), on many strains had indole, — (2.1%); and H₂S, + (93.1%).
^cLate positive reactions may occur in some tests in addition, that may change — to d and d to +.
^dPrevious reports (e.g., Young et al., 1971) had H₂S, —; malonate, —.

TABLE BXII.γ.216. Carbon source utilization reactions of *Citrobacter* species (% positive reactions at 48 h) (adapted from Brenner et al., 1993)^a

Carbon source	1. <i>C. freundii</i>	2. <i>C. amalonaticus</i>	3. <i>C. braakii</i>	4. <i>C. farmeri</i>	5. <i>C. gillenii</i>	6. <i>C. koseri</i>	7. <i>C. murliniae</i>	8. <i>C. rodentium</i>	9. <i>C. sedlakii</i>	10. <i>C. werkmanii</i>	11. <i>C. youngae</i>
<i>cis</i> -Aconitate	88	93	89	57	33	100	67	0	100	100	78
<i>trans</i> -Aconitate	13	0	0	57	0	25	0	0	50	83	0
Adonitol	0	0	0	0	0	100	0	0	0	0	0
D-Alanine	100	100	100	71	67	100	100	100	100	100	100
4-Aminobutyrate	25	0	17	7	0	0	33	0	50	0	22
5-Aminovalerate	38	33	50	7	0	6	67	0	33	100	43
D-Arabitol	0	0	0	0	0	100	0	0	0	0	0
Benzoate	0	73	0	93	0	0	0	0	100	0	0
Caprate	0	0	0	0	0	0	0	0	0	0	0
D-Cellobiose	88	100	94	100	67	94	100	100	100	67	78
<i>m</i> -Coumarate	88	0	100	0	33	0	0	0	67	100	96
Dulcitol	13	0	33	0	0	44	100	0	100	0	87
Esculin	0	0	6	0	0	0	0	0	17	0	0
Ethanolamine	0	33	0	14	0	19	0	0	17	17	9
L-Fucose	100	100	94	100	67	100	100	33	100	100	91
Gentiobiose	88	100	89	93	67	88	100	0	100	67	52
Gentisate	100	100	94	93	0	100	67	67	100	100	0
L-Glutamate	75	100	94	86	0	100	100	67	83	83	83
Glycerol	100	100	100	100	100	100	100	0	100	100	100
3-Hydroxybenzoate	100	100	100	100	0	100	67	100	100	100	0
4-Hydroxybenzoate	0	100	0	100	0	0	0	0	100	0	0
3-Hydroxybutyrate	100	20	44	7	0	6	33	0	100	67	74
<i>myo</i> -Inositol	100	0	6	0	67	100	0	0	100	0	0
2-Ketogluconate	100	100	100	100	100	100	100	0	100	100	100
5-Ketogluconate	100	100	100	100	100	100	100	0	0	50	96
2-Ketoglutarate	13	7	22	7	33	6	0	0	33	33	0
DL-Lactate	100	100	100	100	100	100	100	33	100	100	100
Lactose	88	20	78	50	67	56	33	100	100	17	22
Lactulose	88	0	67	7	67	0	33	0	100	17	0
D-Lyxose	63	0	56	0	0	100	100	0	0	83	9
Malonate	0	0	0	0	0	81	0	67	67	50	0
Maltitol	25	7	44	93	0	100	0	0	0	0	0
D-Melibiose	88	7	94	100	100	0	33	0	100	0	0
1-O-CH ₃ -α-galactoside	88	0	100	100	67	0	0	0	100	0	0
1-O-CH ₃ -β-galactoside	100	7	83	14	67	100	100	100	100	100	65
3-O-CH ₃ -D-glucose	63	87	94	86	0	13	67	0	100	100	0
1-O-CH ₃ -α-D-glucoside	25	13	39	86	0	94	0	0	0	0	0
1-O-CH ₃ -β-D-glucoside	100	100	100	100	100	100	100	0	100	100	100
Palatinose	25	13	67	100	0	100	67	0	0	0	4
Phenylacetate	25	0	0	0	0	0	0	0	0	0	0
3-Phenylpropionate	75	0	83	0	0	0	0	0	0	100	96
L-Proline	100	87	100	57	33	100	100	100	100	100	87
Propionate	88	80	72	86	0	88	100	67	83	100	78
Protocatechuate	0	100	0	100	0	0	0	0	100	0	0
Putrescine	50	0	39	0	0	0	0	100	0	67	0
D-Raffinose	75	0	6	100	67	0	33	0	0	0	0
L-Sorbitol	100	87	6	100	0	0	100	0	0	83	100
Sucrose	100	0	6	100	33	44	33	0	0	0	9
D-Tagatose	13	0	0	36	0	0	0	0	0	0	4
D-Tartrate	13	0	6	0	0	0	0	0	0	100	0
L-Tartrate	0	33	17	36	0	19	33	100	17	67	22
<i>meso</i> -Tartrate	50	40	72	7	0	0	67	100	67	100	78
Tricarballoylate	100	93	89	100	0	0	100	100	83	100	4
D-Turanose	0	0	11	36	0	6	0	0	0	0	0
L-Tyrosine	75	0	72	0	0	88	100	0	0	67	74
Xylitol	0	0	0	0	0	19	0	0	0	0	0

^aAll strains (with few single exceptions) utilized the following carbon sources: N-acetyl-D-glucosamine, L-alanine, L-arabinose, L-aspartate, citrate, D-fructose, fumarate, D-galactose, D-galacturonate, D-gluconate, D-glucosamine, D-glucose, D-glucuronate, DL-glycerate, D-malate, L-malate, maltose, maltotriose, D-mannitol, D-mannose, mucate, L-rhamnose, D-ribose, D-saccharate, L-serine, D-sorbitol, succinate D-trehalose, and D-xylose. All strains (with two single exceptions) failed to utilize the following carbon sources within 96 h: L-arabitol, betaine, caprate, caprylate, i-erythritol, glutarate, histamine, L-histidine, HQ-beta-glucuronide, itaconate, D-melezitose, quinate, trigonelline, tryptamine, and tryptophan.

water, and sewage. Also found in a variety of human clinical specimens as an opportunistic pathogen.

The mol% G + C of the DNA is: 51–52 (*T_m*).

Type strain: ATCC 25405, NCTC 10805.

3. *Citrobacter braakii* Brenner, Grimont, Steigerwalt, Fanning, Ageron, and Riddle 1993, 657^{VP}

braak' i. i. M.L. gen. n. *braakii* of Braak; named after Hendrik R. Braak, a Dutch microbiologist.

Morphology and cultural characteristics as given for the genus.

Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216.

Found in human stools and isolated from animals.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 51113, CDC 80-58.

GenBank accession number (16S rRNA): AF025368.

4. **Citrobacter farmeri** Brenner, Grimont, Steigerwalt, Fanning, Ageron, and Riddle 1993, 654^{VP}
far' mer.i. M.L. gen. n. *farmeri* of Farmer; named after J.J. Farmer III, an American bacteriologist.

Morphology and cultural characteristics as given for the genus.

Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216.

Found in human stools, urine, wounds, and blood.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 51112, CDC 2991-81.

GenBank accession number (16S rRNA): AF025371.

5. **Citrobacter gillenii** Brenner, O'Hara, Grimont, Janda, Falsen, Aldová, Ageron, Schindler, Abbott and Steigerwalt 2000, 423^{VP} (Effective publication: Brenner, O'Hara, Grimont, Janda, Falsen, Aldová, Ageron, Schindler, Abbott and Steigerwalt 1999, 2623.)
gil.len'.ii. N.L. gen. n. *gillenii* of Gillen; named after George Francis Gillen, an American microbiologist.

Morphology and cultural characteristics as given for the genus. Physiological and biochemical characteristics are present in Tables BXII.γ.215 and BXII.γ.216. Found in human stool, human urine, human blood, and environment.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 51117, CCUG 30796, CDC 4693-86, CIC 106783.

6. **Citrobacter koseri** Frederiksen 1970, 93^{AL}
ko.ser.i. M.L. gen. n. *koseri* of Koser; named after Stewart A. Koser, an American bacteriologist.

The morphology is as given for the genus. Motile. Not encapsulated.

Colonies on nutrient agar are translucent to opaque, resembling those of *E. coli*. Cultures often have a distinct fecal odor.

Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216.

Found in the feces of humans and animals and in soil, water, sewage, and food. Also isolated from human clinical specimens such as urine, throat, nose, and sputum and wound swabs. Causes neonatal meningitis, often complicated with cerebral abscesses, and often fatal. Some cases occur in small outbreaks.

The mol% G + C of the DNA is: 51-52 (T_m).

Type strain: ATCC 27028.

7. **Citrobacter murlinae** Brenner, O'Hara, Grimont, Janda, Falsen, Aldová, Ageron, Schindler, Abbott and Steigerwalt 2000, 423^{VP} (Effective publication: Brenner, O'Hara, Grimont, Janda, Falsen, Aldová, Ageron, Schindler, Abbott and Steigerwalt 1999, 2623.)
mur.lin'.i.ae. N.L. gen. n. *murlinae* of Murlin; named after Alma C. McWhorter-Murlin, an American microbiologist.

Morphology and cultural characteristics as given for the genus. Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216. Found in human stool, human wound, human blood, human urine, and food.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 51118, CCUG 30797, CDC 2970-59, CIC 104556.

8. **Citrobacter rodentium** Schauer, Zabel, Pedraza, O'Hara, Steigerwalt, and Brenner 1996, 362^{VP} (Effective publication: Schauer, Zabel, Pedraza, O'Hara, Steigerwalt, and Brenner 1995, 2067.)

ro.den'ti.um. L. part. adj. used as a gen. n. *rodentium* of rodents (gnawing animals).

Morphology and cultural characteristics as given for the genus; however, four strains were nonmotile, and two showed motility only after 4 days.

Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216. Citrate utilization was late or absent.

Found in rodents only; includes strains known to produce transmissible murine colonic hyperplasia, associated with the presence of an *eaeA* gene in the isolate.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 51116, CDC 1843-73.

GenBank accession number (16S rRNA): AF025363.

9. **Citrobacter sedlakii** Brenner, Grimont, Steigerwalt, Fanning, Ageron, and Riddle 1993, 657^{VP}
sed.lak'i.i. M.L. gen. n. *sedlakii* of Sedlak; named after Jiri Sedlák, a Czechoslovakian bacteriologist.

Morphology and cultural characteristics as given for the genus.

Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216.

Found in human stools, blood, and wounds.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 51115, CDC 4696-86.

GenBank accession number (16S rRNA): AF025364.

10. **Citrobacter werkmanii** Brenner, Grimont, Steigerwalt, Fanning, Ageron, and Riddle 1993, 657^{VP}
werk'mani.i. M.L. gen. n. *werkmanii* of Werkman; named after Chester H. Werkman, an American bacteriologist.

Morphology and cultural characteristics as given for the genus.

Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216.

Found in human stools and blood, and in soil.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 51114, CDC 876-58.

GenBank accession number (16S rRNA): AF025373.

11. **Citrobacter youngae** Brenner, Grimont, Steigerwalt, Fanning, Ageron, and Riddle 1993, 654^{VP}
young'ae. M.L. gen. n. *youngae* of Young; named after Viola M. Young, an American bacteriologist.

Morphology and cultural characteristics as given for the genus.

Physiological and biochemical characteristics are presented in Tables BXII.γ.215 and BXII.γ.216.

Found in human stools, urine, and wounds, and isolated from animals and food.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 29935, CDC 460-61.

Genus XI. Edwardsiella Ewing and McWhorter 1965, 37^{AL}

RIICHI SAKAZAKI

Ed.ward.si.el'la. M.L. dim. ending -ella; M.L. fem. n. *Edwardsiella* named after the American bacteriologist P.R. Edwards (1901–1966).

Straight rods, $\sim 1.0 \times 2.0\text{--}3.0$ μm , conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. **Motile by peritrichous flagella**, but nonmotile strains may occur. Facultatively anaerobic. Growth occurs on ordinary media with small colonies (0.5–1.0 mm in diameter) after 24 h incubation. Optimum temperature, 37°C, except for *E. ictaluri*, which prefers a lower temperature. **Nicotinamide and amino acids are required for growth**. Reduce nitrate to nitrite. Ferment D-glucose with the production of acid and often gas. Also **ferment a few other carbohydrates but are inactive compared to many taxa in the family Enterobacteriaceae**. Usually resistant to colistin but susceptible to most other antibiotics, including penicillin. Frequently isolated from river fish and cold-blooded animals and their environment, particularly fresh water. Pathogenic for eels, catfish, and other animals, sometimes causing economic losses; also opportunistic pathogen and possibly rare cause of gastroenteritis for humans. A member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 53–59.

Type species: Edwardsiella tarda Ewing and McWhorter in Ewing, McWhorter, Escobar and Lubin 1965, 37.

FURTHER DESCRIPTIVE INFORMATION

Most of studies on *Edwardsiella* concentrated on *E. tarda*; little information is available on other species. Encapsulated strains possessing significant glycocalixes have not been found in members of *Edwardsiella*, but some strains may produce slime substance (Wong et al., 1989). They are afimbriate. Wong et al. (1989) and Janda et al. (1991b) reported that approximately half of strains of *E. tarda*, but not *E. hoshinae* and *E. ictaluri*, produce afimbrial mannose-resistant hemagglutination (MRHA) against guinea pig erythrocytes. Aoki and Holland (1985), who analyzed outer membrane composition of *E. tarda*, found a large number of prominent protein bands in all strains, with major bands present at 27, 35, and 46 kDa, and a cluster of three or four outer membrane proteins located in each strain at 52–56 kDa. Studies of the lipopolysaccharide of three species of *Edwardsiella* indicated that each species may be divided into different chemofoms (Nomura and Aoki, 1985). Since strains of *E. tarda* require nicotinamide and amino acids including cysteine and methionine (d'Empaire, 1969) for their growth and other species of the genus probably have similar requirements, strains of *Edwardsiella* grow less luxuriantly than other members of the *Enterobacteriaceae* and form smaller colonies on ordinary agar plates within 24 h of incubation at 37°C. In contrast with other two *Edwardsiella* species, the optimum growth temperature of *E. ictaluri* is between 25 and 30°C and growth is very slow on plating media, often requiring 2–3 d incubation at 30°C to form typical colonies 1 mm in diameter. Biochemically, *E. ictaluri* is also the least active species of *Edwardsiella*.

Plasmids ranging from 2–120 MDa are harbored in many strains of *Edwardsiella* (Lobb and Rhoades, 1987; Janda et al., 1991a). The presence of R plasmids that mediate antibiotic resistance was reported in *E. tarda* by Aoki et al. (1977). Lobb and Rhoades (1987) recognized that catfish isolates of *E. ictaluri* always harbored two plasmids of 5.7 and 4.9 kb (pCL1 and pCL2, respectively). They suggested that plasmid function may be met-

abolically important to the host bacterium, or that these plasmids may code for factors important to the virulence, or both. Moreover, Lobb et al. (1993) found that green knife fish isolates of the organisms were serologically distinct from those of catfish and harbored four plasmids with relative mobilities of 6.0, 5.7, 4.1, and 3.1 kb, of which 5.7- and 4.1-kb plasmids strongly hybridized to probes specific for pCL1 and pCL2. Hamon et al. (1969) demonstrated bacteriocin production and sensitivity within *E. tarda*.

For *E. tarda*, two independent serotyping schemes were reported by Sakazaki (1967) and Edwards and Ewing (1972). Currently, an international serotyping scheme combining the two schemes mentioned above and comprising 61 O groups and 45 H antigens has been established by Tamura et al. (1988). Lobb et al. (1993) suggested the possibility of serotyping of *E. ictaluri* as an epidemiological tool. Other typing schemes such as bacteriocin typing, bacteriophage typing, and biotyping have not been studied for *Edwardsiella*.

Strains of *Edwardsiella* are primarily resistant to colistin and polymyxin B, but are usually susceptible to other antimicrobial agents including penicillin G, ampicillin, carbenicillin, streptomycin, kanamycin, gentamicin, tetracycline, chloramphenicol, cephalosporins, sulfadiazine, and, with few exceptions, nalidixic acid (Muyembe et al., 1973; Reinhardt et al., 1985). However, Waltman et al. (1986) reported more resistance to sulfa compounds, streptomycin, and methicillin by disk diffusion testing. These differences may be linked to the source of strains tested. All three species may show large zones around penicillin-impregnated disks. This is an unusual finding for members of the *Enterobacteriaceae*. Clark et al. (1991b) found that strains of *E. tarda*, most originating from human sources, have uniform susceptibility to 22 antibiotics. Although they produced β -lactamase, all strains showed susceptibility to β -lactamase inhibitors. Members of *Edwardsiella* are apparently more susceptible to 2,4-diamino-6,7-diisopropyl pteridine (vibriostatic agent O/129, Sigma) than other *Enterobacteriaceae* (Chatelain et al., 1979; Grimont et al., 1980).

There have been no definitive studies indicating *E. tarda* as a potential cause of human intestinal diseases, but a number of papers from tropical countries such as Madagascar (Fourquet et al., 1975), Zaire (Makulu et al., 1973), Tahiti (Fourquet et al., 1975), Dominica (de Inchaustegui et al., 1976), Panama (Kourany et al., 1977), Cuba (Rakovsky and Aldová, 1965), Australia (Iveson, 1973), India (Bhat et al., 1967; Sakazaki et al., 1971), Thailand (Bockemühl et al., 1971; Ovartharnporn et al., 1986), Viet-Nam (Nguyen-Van-Ai et al., 1975), Philippines (Tocal and Mezez, 1968), Malaysia (Gilman et al., 1971), and Singapore (Tan et al., 1977) have reported an etiologic relationship of this species to diarrheal diseases. On the other hand, *E. tarda* is also found in the feces of healthy people, although the rate of the isolation is extremely low. Onogawa et al. (1976) found only one positive culture from 97,704 food handlers and 25 positive cultures from 255,896 schoolchildren. Makulu et al. (1973) found no *E. tarda* cultures among 841 healthy subjects in Zaire. Iveson (1973) suggested that the number of isolations of *E. tarda* depends upon culture methods for stool specimens, the geographic area of the

study, and the season in which the survey is performed. In those people from whose stool specimens of *E. tarda* have been isolated, the percentage of diarrheal patients has varied from 25% (Kourany et al., 1977) to more than 75% (Bockemühl et al., 1971; Makulu et al., 1973). A higher isolation rate has invariably been recognized among patients with diarrhea than among asymptomatic people (Bhat et al., 1967; Gilman et al., 1971; Makulu et al., 1973; Nguyen-Van-Ai et al., 1975). In tropical countries there is a tendency for *E. tarda* to be isolated from diarrheal stools together with well-established enteropathogens such as *Salmonella*, *Shigella*, *Vibrio cholerae*, *Vibrio parahaemolyticus*, and intestinal parasites (Bhat et al., 1967; Bockemühl et al., 1971; Sakazaki et al., 1971; Makulu et al., 1973; Kourany et al., 1977). The incidence is rare in industrialized countries, but intestinal infection with *E. tarda* was also reported by King and Adler (1964), Chatty and Gavan (1968), and Desenclos et al. (1990) in the United States, and by Vandepitte et al. (1983) in Belgium.

Although *E. tarda* may be able to cause diarrhea, the organism should not be considered as an "inherent" pathogen such as *Salmonella* and *Shigella*. Marques et al. (1984) demonstrated the invasiveness of *E. tarda* in HeLa cells. Janda et al. (1991a) confirmed this finding with HEp-2 cells. However, invasive strains of *E. tarda* give constantly negative results in the Sereny test, suggesting that they have no KcpA-like locus as is found in *Shigella* (Ullah and Arai, 1983b; Marques et al., 1984). Janda and Abbott, (1993a) suggested that HEp-2 invasion by *E. tarda* was a microfilament-dependent process. Furthermore, they suggested the cell-associated hemolysin, which was originally demonstrated by Watson and White (1979), as a second major virulence marker. The hemolysin is not released in sufficient amounts under usual growth conditions. Activity of the hemolysin is enhanced by Ca^{2+} and Mg^{2+} and may play a role in infection through epithelial cell destruction leading to an inflammatory infiltrate in the intestinal mucosa, or it could destroy villus cells or disturb intestinal absorption function resulting in diarrhea. In addition to these, Ullah and Arai (1983a) and Janda et al. (1991b) suggested that mannose-resistant hemagglutinins, serum resistance, a dermatotoxin, and chondroitinase activity may be related to human pathogenicity of *E. tarda*. These associations are independent of serovars or source of strains. Whether intestinal infection with *E. tarda* in humans is linked to strains possessing these virulence factors remains to be determined. Bockemühl et al. (1983) reported that some *E. tarda* strains produced a heat-stable enterotoxin (ST) similar to that produced by enterotoxigenic *Escherichia coli*. However, the enterotoxigenic activity was recognized only in culture supernatants concentrated by ultrafiltration. Janda et al. (1991a) also tested *E. tarda* for ST-like activity, but all strains were negative.

Extraintestinal infections with *E. tarda* are well documented, although these occur only rarely in most industrialized countries. It causes meningitis, endocarditis, bacteremia, osteomyelitis, urinary tract infections, or wound infection (Gonzalez and Ruffolo, 1966; Field et al., 1967; Chatty and Gavan, 1968; Okubadejo and Alausa, 1968; Sonnenwirth and Kallus, 1968; Jordan and Hadley, 1969; Pankey and Seshul, 1969; Bockemühl et al., 1971; Sachs et al., 1974; Koshi and Lalitha, 1976; Le Frock et al., 1976; Clarridge et al., 1980; Rao et al., 1981; Sechter et al., 1983; Maserati et al., 1985; Martinez, 1987; Vohora and Torrijos, 1988; Wilson et al., 1989a; Vartian and Septimus, 1990; Hargreaves and Lucey, 1990; Coutlee et al., 1992; Walton et al., 1993; Zighelboim et al., 1992). In most cases of extraintestinal infection, *E. tarda* has been an opportunist in persons with underlying diseases or in conditions

with predisposing factors. Infants and the aged tend to be most susceptible to infections with *E. tarda*. *E. tarda* appears to cause serious infection in patients with preexisting liver disease or in those with altered levels of available iron. In cases of wound infection, patients who tend to develop severe infection often have a recent history of fishing, swimming, or injury caused by marine-related items. In extraintestinal infections with *E. tarda*, cell-associated hemolysin is also suggested as a virulence factor, in which a possible function of this hemolysin could be the acquisition of host iron through lysis of erythrocyte to release hemoglobin stores.

E. tarda has been infrequently isolated from warm-blooded animals such as dogs, pigs, cattle, monkeys, rats, panthers, skunks, seals, sea lions, and birds (Ewing et al., 1965; Wallace et al., 1966; Arambulo et al., 1967; Chamoiseau, 1967; Sakazaki, 1967; Tocal and Mezez, 1968; d'Empaire, 1969; Otis and Behler, 1973; White et al., 1973; Owens et al., 1974; Nguyen-Van-Ai et al., 1975; Kourany et al., 1977; Coles et al., 1978). *E. tarda* was associated with diseases in some of those animals, but its pathogenic role in warm-blooded animals is unknown.

Two ecological groups of hosts, cold-blooded animals and fish, could be considered as reservoirs of *E. tarda*. Since Sakazaki and Murata (1962) first suggested *E. tarda* as a normal intestinal inhabitant of snakes, it has been recognized that a wide range of reptiles and amphibians including snakes, crocodiles, alligators, toads, lizards, frogs, and turtles, are possible natural reservoirs for *E. tarda* (Wallace et al., 1966; d'Empaire, 1969; Jackson et al., 1969; White et al., 1969, 1973; Iveson, 1971; Makulu et al., 1973; Meyer and Bullock, 1973; Otis and Behler, 1973; Sharma et al., 1974; Van Der Waaij et al., 1974; de Inchaustegui et al., 1976; Roggendorf and Müller, 1976; Bartlett et al., 1977; Kourany et al., 1977; Wyatt et al., 1979; Tan et al., 1978). On the other hand, Van Damme and Vandepitte (1980) reported the isolation of *E. tarda* from various kinds of river fish in Zaire. It is considered that river fish and their environment seem to constitute the natural habitat of *E. tarda* and to be the most possible source of human infection, at least in tropical countries. Vandepitte et al. (1983) described protracted diarrhea in an infant associated with *E. tarda*, which was possibly introduced from a tropical aquarium fish in the home of the patient. Trust and Bartlett (1974) reported the presence of *E. tarda* in water samples from aquariums containing goldfish or tropical ornamental fish. Wyatt et al. (1979) recognized *E. tarda* in freshwater catfish and in their environment. Human wound infection with this organism resulting from swimming or diving accidents was reported by Chatty and Gavan (1968) and Clarridge et al. (1980).

E. tarda can cause outbreaks of "red disease" in pond-cultured eels (Hoshina, 1962; Wakabayashi and Egusa, 1973) or "emphysematous putrefactive disease" in channel catfish (Meyer and Bullock, 1973). Kusuda et al. (1977) reported *E. tarda* as causing an outbreak in cultured sea beam. Amandi et al. (1982) showed that *E. tarda* is often isolated from salmonid fish and other fish species in the Pacific Northwest of the United States, and that it is pathogenic for chinook salmon, steelhead and rainbow trout, as well as channel catfish.

E. hoshinae is associated with animals but only eight isolates were originally reported (Grimont et al., 1980). From their paper, the presently known habitats of *E. hoshinae* include reptiles, birds, and water. Although two human isolates of this species were included in their report, those came from feces of individuals without diarrhea; thus there is no evidence that *E. hoshinae* can cause human disease. Another species of *Edwardsiella*, *E. ic-*

taluri, is the causative agent responsible for acute septicemia of catfish. Since the first publication by Hawke et al. (1981), this disease has become a significant problem in the aquaculture of channel catfish in the southeastern United States (Waltman et al., 1986). Shotts et al. (1986) suggested that pathogenesis of *E. ictaluri* may proceed through gastrointestinal tract infection. Enteric septicemia caused by *E. ictaluri* occurs when the temperature ranges from 22–28°C and progresses rapidly, resulting in significant mortality. The isolation of *E. ictaluri* from nonictalurid fish, such as the tropical danio or green knife fish, was also documented (Waltman et al., 1985). Janda et al. (1991b) reported that *E. ictaluri* lacked the invasiveness and cell-associated hemolysin recognized in *E. tarda*, and the lower frequency of these two activities in *E. hoshinae*.

ENRICHMENT AND ISOLATION PROCEDURES

E. tarda and *E. hoshinae* can grow on plating agar media used in enterobacteriology, such as MacConkey, xylose-lysine-deoxycholate (XLD), Hektoen enteric, deoxycholate-citrate, and Salmonella-Shigella agars. Like *Salmonella*, strains of *E. tarda* produce distinguishable black colonies on XLD and Hektoen enteric agars because of H₂S production. As they require some growth factors, their growth may be slower on those media than other enteric bacteria.

Makulu et al. (1973) and Wyatt et al. (1979) reported that tetrathionate or selenite broths as enrichment media may be successful for the isolation of *E. tarda*. Iveson (1971, 1973), who carried out a comparative study to evaluate enrichment media to isolate *E. tarda* from the cloacal contents of tiger snakes and human feces, showed that strontium chloride B broth¹ was the most satisfactory for isolating *E. tarda*, as well as *Salmonella*. After 24 h of incubation at 37°C or 43°C, plates of deoxycholate citrate agar are streaked from the enrichment culture.

On the other hand, Wyatt et al. (1979) evaluated several media for the isolation of *E. tarda* from catfish and the environmental water and found that the most effective isolation of *E. tarda* was obtained with selective enrichment in double-strength SS broth² followed by plating on single-strength SS agar. They pointed out that salmonellae were not isolated with this method. A 1-ml amount of lactose broth pre-enrichment is transferred to the double-strength SS broth. After incubation at 35°C for 24 h, cultures are streaked on an SS agar plate.

Muyembe et al. (1973) pointed out that resistance of *E. tarda* to 10 µg/ml colistin can be used in isolation. On the isolation agar plate containing 10 µg/ml colistin, most enteric bacteria are inhibited, with the exception of *Serratia*, *Proteus*, *Providencia*, *Morganella*, *Cedecea*, and some strains of *Yersinia* that are colistin-resistant. Little information is available on the isolation of *E. ictaluri*. Agar medium containing colistin is also effective for the isolation of *E. ictaluri*. The optimum growth temperature of *E. ictaluri* is between 25 and 30°C and growth is slow, often requiring 2–3 days incubation to form colonies 1 mm in diameter.

1. Strontium chloride B broth (g/l): Bacto-tryptone (Difco), 5.0; NaCl, 8.0; K₂HPO₄, 1.0; and SrCl₂, 34.0. Sterilization is done by heating at 100°C for 30 min (final pH, 5.0–5.5).

2. Double-strength SS broth (g/l): SS agar, dehydrated, 120, and glucose 5. The mixture is stirred without heat to dissolve the ingredients except agar. The agar is removed by filtration through a Whatman no. 1 filter. The broth is heated at 100°C for 5 min and dispensed to 10-ml quantities into a tube (18 × 180 mm). The tubes are steamed for 30 min.

MAINTENANCE PROCEDURES

Strains of *Edwardsiella* usually survive well at least for 5 years without subculture when the culture is stabbed into a tube with semisolid medium. The tube is tightly sealed with a rubber stopper or with paraffin-coated corks and kept at room temperature in the dark without transfer. A semisolid agar containing 1% Bacto-casitone (Difco), 0.3% yeast extract, 0.5% NaCl, and 0.3% agar (pH 7.2) is an excellent choice for this purpose. However, important cultures should also be preserved at –70°C. Freeze-drying can also be used for long-term preservation.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

L-Pyrrolidonyl aminopeptidase test Dissolve 10 mg of L-pyrrolidonyl-β-naphthylamide (Sigma) in 10 ml of 95% ethanol. Dip sterile cotton swabs into the solution and rotate each swab several times with firm pressure on the inside wall of the tube to remove excess fluid. Dry the swabs overnight at 37°C and store at –20°C. For testing, make a heavy suspension (MacFarland no. 1) of the test culture into 0.2–0.3 ml of 0.1 M phosphate buffer (pH 7.4), then dip the reagent swab into the suspension and incubate at 37°C for 10 min. Finally, add a drop of cinnamaldehyde solution (*p*-dimethylaminocinnamaldehyde [Sigma] 0.5 g; glacial acetic acid 3.5 ml; ethylene glycol-1-methylester 6.0 ml; sodium lauryl sulfate 3.5 g; and distilled water 90 ml; store in a screw-capped brown container in a refrigerator). A positive result is shown by a pink or red color.

β-Galactosidase and β-glucuronidase tests Dissolve 20 mg each of 4-methylumbelliferyl-β-D-galactopyranoside and 4-methylumbelliferyl-β-D-glucuronide (Sigma) in a small volume of dimethyl sulfoxide, and bring the volume to 10 ml with 0.1 M phosphate buffer (pH 7.4). Dip sterile cotton swabs into the solution and rotate each swab several times with firm pressure on the inside wall of the tube to remove excess fluid. Dry the swabs overnight at 37°C and store at –20°C. For testing, make a heavy suspension (MacFarland no. 1) of the test culture in 0.2–0.3 ml of 0.1 M phosphate buffer (pH 7.4). Dip the reagent cotton swab in the suspension, and incubate at 37°C for 30 min. A positive reaction is indicated by development of blue fluorescence under long-wave (360 µm) ultraviolet lamp.

DIFFERENTIATION OF THE GENUS *EDWARDSIELLA* FROM OTHER GENERA

Edwardsiella is biochemically somewhat similar to *E. coli*, *Salmonella*, *Citrobacter* spp., *Leminorella* spp., *Proteus*, and *Providencia*, but is easily differentiated based on activities of L-pyrrolidonyl aminopeptidase, β-galactosidase, β-glucuronidase, and phenylalanine (or tryptophan) deaminase, indole production, and maltose and D-xylose fermentation. Table BXII.7.217 presents differential characteristics of *Edwardsiella* and other similar members of the *Enterobacteriaceae*. In addition, susceptibilities to penicillin and colistin and to the vibriostatic agent O/129 are useful for differentiation of *Edwardsiella*. Strains of *Edwardsiella* may have zones of inhibition around a penicillin-G disk and no zone around a colistin disk, whereas many groups of the *Enterobacteriaceae* usually have the opposite patterns. *Edwardsiella* may show zones around a 10-µg disk of O/129.

TAXONOMIC COMMENTS

Edwardsiella was independently reported by several investigators early in the 1960s. Sakazaki and Murata (1962) and Sakazaki (1967) reported a new group of organisms in the family *Ente-*

TABLE BXII.γ.217. Differential characteristics of the genus *Edwardsiella* and biochemically related genera^a

Characteristic	<i>Edwardsiella</i>	<i>Citrobacter</i>	<i>Escherichia coli</i>	<i>Leminorella</i>	<i>Proteus</i> and <i>Providencia</i>	<i>Salmonella</i>
H ₂ S production	D	D	—	+	D	+
Indole production	D	D	+	—	D	—
L-Pyrrolidonyl aminopeptidase	—	+	—	—	—	—
β-Galactosidase	—	+	+	—	—	—
β-Glucuronidase	—	—	+	—	—	D
Phenylalanine deaminase	—	—	—	—	+	—
Maltose fermentation	+	+	+	—	D	+
D-Xylose fermentation	—	+	+	+	D	+
Mol% G + C of DNA	53–59	50–52	48–52	52–53	38–41	53–58

^aSymbols: +, 90–100% of strains are positive; —, 90–100% of strains are negative; D, different reaction given by different species or biogroups.

robacteriaceae that were isolated from snakes in 1959 and suggested a vernacular name “Asakusa group” for this group. King and Adler (1964) described the isolation of a culture of the organism and gave it the name “Bartholomew group”. Later, Ewing et al. (1965) suggested the scientific name *Edwardsiella tarda* for a group of organisms that had been known as biotype 1483-59 in their laboratory since 1959, and indicated that these organisms were similar to those of the Asakusa and Bartholomew groups. Although the genus *Edwardsiella* with the single species *E. tarda* was proposed by Ewing et al. (1965), there was some question about this designation based upon only phenotypic characterization. Brenner et al. (1974b) demonstrated that strains of *E. tarda* were highly related by DNA hybridization, regardless of their source. They also reported that *Edwardsiella tarda* was only 8–29% related to other genera in the family *Enterobacteriaceae* and was distinct from other members of the family.

Grimont et al. (1980) described a new biogroup designated biogroup 1 of *E. tarda* that was distinguished from biochemically typical strains by its acid production from sucrose, D-mannitol, and L-arabinose, but was closely related to the latter by DNA hybridization. Walton et al. (1993) reported the isolation of a single strain of sucrose-positive, but D-mannitol- and L-arabinose-negative *E. tarda* mimicking biogroup 1 from a patient with cholelithiasis.

The second species of the genus was recognized by Grimont et al. (1980) among the organisms formerly identified as atypical *E. tarda*, and the name *Edwardsiella hoshinae* was proposed for this species. Hawke et al. (1981) reported the third species, *Edwardsiella ictaluri*, associated with septicemia and death in catfish.

Hoshina (1962) recognized an organism as an etiologic agent of “red disease” in eels and named it “*Paracolobactrum anguillimortiferum*”. Based on the International Code of Bacterial No-

menclature 1953 and 1962, “*Paracolobactrum anguillimortiferum*” of Hoshina was a legitimate name for this species. Sakazaki and Tamura (1975) suggested a new combination “*Edwardsiella anguillimortifera*” instead of *Edwardsiella tarda*, because they considered the name *E. tarda* was a junior synonym of “*Paracolobactrum anguillimortiferum*”. Farmer et al. (1976) pointed out differences of some biochemical reactions in descriptions between Hoshina (1962) and Ewing et al. (1965), and emphasized the legitimacy and validity of *Edwardsiella tarda*. Unfortunately, no type strain of “*Paracolobactrum anguillimortiferum*” was designated by Hoshina, and strains studied by him were no longer available. In this status, “*P. anguillimortiferum*” should be considered as a doubtful name, and *E. tarda* is the name that should be used to avoid unnecessary confusion in the literature.

ACKNOWLEDGMENTS

Riichi Sakazaki, who died in 2002, spent more than forty years at the Nippon Institute of Biological Sciences. His illustrious career focused on the classification and epidemiology of human and fish pathogens in Japan, particularly pathogenic bacteria in the families *Vibrionaceae* and *Enterobacteriaceae*. His contributions to Japanese bacteriology paralleled those of Drs. Edwards and Ewing in the U.S. He discovered and described many species in the genera *Enterobacter*, *Edwardsiella*, and *Vibrio*, and he developed serotyping schemes for *V. cholerae* and *V. parahaemolyticus*, as well as for several important species of *Enterobacteriaceae*. His research was always careful and comprehensive. He was one of the first Japanese scientists to publish in English language journals, thereby making his impressive accomplishments available to the world. He was recognized as one of the world's foremost experts on the genus *Vibrio* and on many genera in *Enterobacteriaceae*. He was a member of the WHO Subcommittees on the Taxonomy of *Vibrionaceae* and on the Taxonomy of *Enterobacteriaceae* for more than thirty years. He authored chapters on these organisms in three editions of *Bergey's Manual*. He was a good friend to the Bergey's Trust, as well as to all who knew him.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *EDWARDSIELLA*

Table BXII.γ.218 presents the differential characteristics of the species and biogroups of *Edwardsiella*. Table BXII.γ.219 shows additional biochemical features of those organisms. In the H₂S production test, *E. tarda* biogroup 1 and *E. hoshinae* give a weakly

positive reaction in the butt of Kligler iron agar but are usually negative in the butt portion of triple sugar iron agar in which acid produced by sucrose fermentation interferes to combine hydrogen sulfide with Fe²⁺.

List of species of the genus *Edwardsiella*

1. ***Edwardsiella tarda*** Ewing and McWhorter in Ewing, McWhorter, Escobar and Lubin 1965, 37^{AL}
tar' da. L. fem. adj. *tarda* slow (intended meaning was “inactive,” referring to the fermentation on only a few carbohydrates compared to many other *Enterobacteriaceae*).

The characteristics are as given for the genus and listed in Tables BXII.γ.218 and BXII.γ.219.

Occurs in a wide variety of animals, rarely in the feces

of healthy people. It is an opportunistic human pathogen, which may cause wound infection and probably some cases of diarrheal diseases.

The mol% G + C of the DNA is: 55–58 (*T_m*).

Type strain: ATCC 15947, DSM 30052.

GenBank accession number (16S rRNA): AF053975 and AF015259.

TABLE BXII.γ.218. Differentiation of species of the genus *Edwardsiella*^a

Characteristic	<i>E. tarda</i> ^b	<i>E. tarda</i> ^c	<i>E. hoshinae</i>	<i>E. ictaluri</i>
Indole production	+	+	d	—
H ₂ S production (Kligler)	+	(+) ^d	(+) ^d	—
Motility	+	+	+	—
Malonate utilization	—	—	+	—
<i>Fermentation of:</i>				
L-Arabinose	—	+	d	—
Sucrose	—	+	+	—
Trehalose	—	—	+	—
D-Mannitol	—	+ ^e	+	—

^aFor symbols, see standard definitions.^bResults for most strains of *E. tarda*.^cResults for *E. tarda* biogroup 1.^dWeakly positive but may be negative in the butt of TSI agar.^eSome strains may be negative.

Additional Remarks: These sequences are not from the type strain.

2. ***Edwardsiella hoshinae*** Grimont, Grimont, Richard and Sakazaki 1981b, 216^{VP} (Effective publication: Grimont, Grimont, Richard and Sakazaki 1980, 349.)

ho.shi' nae. M.L. gen. n. *hoshinae* of Hoshina; named after the late Toshikazu Hoshina, the Japanese bacteriologist who was one of the first to describe an organism that was probably an *Edwardsiella*.

The characteristics are as described for the genus and indicated in Tables BXII.γ.218 and BXII.γ.219. Hydrogen sulfide is weakly produced in Kligler iron agar but may be negative in TSI agar.

Most isolates are from animals. There is no evidence that this species causes diarrhea.

The mol% G + C of the DNA is: 56–57 (*T_m*).

Type strain: 2-78, ATCC 33379, CIP 78-56.

3. ***Edwardsiella ictaluri*** Hawke, McWhorter, Steigerwalt and Brenner 1981, 400^{VP}
ic.ta.lu'ri. *Ictalurus* the genus name for catfish; M.L. fem. adj. *ictaluri* pertaining to catfish.

The characteristics are as described for the genus and as indicated in Tables BXII.γ.218 and BXII.γ.219.

E. ictaluri is the most fastidious of the three *Edwardsiella* species. Growth is very slow on plating media. It seems to prefer a lower temperature, although characteristic biochemical reactions are apparent at 36°C (Tables BXII.γ.218

TABLE BXII.γ.219. Biochemical characteristics of the species and biogroups of the genus *Edwardsiella*^a

Characteristic	<i>E. tarda</i> ^b	<i>E. tarda</i> ^c	<i>E. hoshinae</i>	<i>E. ictaluri</i>
Indole production	+	+	+	—
Voges-Proskauer	—	—	—	—
Citrate utilization (Simmons)	—	—	—	—
H ₂ S production (Kligler)	+	(+) ^d	(+) ^d	—
Lysine decarboxylase	+	+	+	+
Arginine dihydrolase	—	—	—	—
Ornithine decarboxylase	+	+	+	d
Phenylalanine deaminase	—	—	—	—
Gelatinase	—	—	—	—
Urease (Christensen)	—	—	—	—
β-Galactosidase	—	—	—	—
β-Glucuronidase	—	—	—	—
L-Pyrrolidonyl aminopeptidase	—	—	—	—
Lipase (Tween 80)	—	—	—	—
Deoxyribonuclease	—	—	—	—
Malonate utilization	—	—	+	—
Esculin hydrolysis	—	—	—	—
Growth in KCN medium	—	—	—	—
Gas from glucose	+	d	d	d
<i>Acid from carbohydrate:</i>				
Maltose	+	+	+	+
L-Arabinose	—	+	d	—
Sucrose	—	+	+	—
D-Mannitol	—	+ ^e	+	—
Trehalose	—	—	+	—
Salicin	—	—	d	—
D-Cellobiose, lactose, melibiose, raffinose, L-rhamnose, D-xylose, adonitol, D-arabitol, dulcitol, <i>l</i> -inositol, D-sorbitol, α-methyl-D-glucoside	—	—	—	—

^aFor symbols, see standard definitions.^bResults for most strains of *E. tarda*.^cResults for *E. tarda* biogroup 1.^dWeakly positive but may be negative in the butt of TSI agar.^eSome strains may be negative.

and BXII.γ.219). Biochemically, it is also the least active of the three *Edwardsiella* species.

The antibiotic susceptibility by disk diffusion is difficult to determine because the strains grow so poorly on Mueller-Hinton agar at 37°C. They must be incubated at 25°C instead.

Occurs as a pathogen of catfish.

The mol% G + C of the DNA is: 53 (Bd).

Type strain: SECFDL, GA 7752, ATCC 33202, CDC 1976-78.

Genus XII. *Enterobacter* *Hormaeche and Edwards 1960b, 72^{AL} Nom. Cons. Opin. 28, Jud. Comm. 1963, 38*

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En.te.ro.bac' ter. Gr. neut. n. *enteron* intestine; M.L. masc. n. *bacter* equivalent of bacterium, a small rod; M.L. masc. n. *Enterobacter* intestinal small rod.

Straight rods, 0.6–1.0 × 1.2–3.0 μm, conforming to the general definition of the family *Enterobacteriaceae*. **Motile by peritrichous flagella** (generally 4–6). Gram negative. **Facultatively anaerobic**. Growth occurs readily on ordinary media. Glucose is fermented with production of acid and gas (generally CO₂:H₂ = 2:1). Gas is not produced from glucose at 44.5°C. **Most strains give a pos-**

itive Voges-Proskauer reaction and a negative methyl red test. An alkaline reaction occurs in Simmons citrate and malonate broth. Nitrate is reduced to nitrite. H₂S is not produced from thiosulfate. Tetrathionate is not reduced. **Corn oil and tributyrin are not hydrolyzed. Gelatin, DNA, and Tween 80 are either not, or very slowly, hydrolyzed.** L-Arabinose, D-cellobiose, D-fructose,

D-galactose, D-galacturonate, gentiobiose, D-gluconate, D-glucosamine, D-glucose, D-glucuronate, 2-ketogluconate, L-malate, D-mannitol, D-mannose, D-trehalose, and D-xylose utilized by all or almost all strains, as sole source of carbon and energy. L-rhamnose utilized by all strains except *Enterobacter asburiae*. L-Arabitol, ethanolamine, itaconate, 3-phenylpropionate, L-sorbose, D-tartrate, tryptamine, and xylitol are not utilized. *meso*-Erythritol, gentisate, glutarate, and tricarballoylate not utilized except by some strains of *Enterobacter gergoviae*. D-melezitose not utilized except by some strains of *Enterobacter sakazakii*. Optimum temperature for growth is 30°C. Most clinical strains grow at 37°C; some environmental strains give erratic biochemical reactions at 37°C. Widely distributed in nature; common in man and animals.

The mol% G + C of the DNA is: 52–60 (Bd).

Type species: Enterobacter cloacae (Jordan 1890) Hormaeche and Edwards 1960b72. Nom. Cons. Opin. 28, Jud. Comm. 1963, 38 (*Bacillus cloacae* Jordan 1890, 836.)

FURTHER DESCRIPTIVE INFORMATION

Phylogenetic treatment The phylogenetic approach using sequences of the *rrs* gene (encoding 16S rRNA) does not provide sufficient resolution when closely related enterobacterial species are studied, and branching is often unreliable. However, some sequence comparisons have corroborated DNA–DNA hybridization studies. *Enterobacter aerogenes* is more closely related to *Klebsiella* species than to *E. cloacae*, the type species of the genus *Enterobacter*. Therefore, in this volume, *E. aerogenes* has been treated with the genus *Klebsiella* under the name *K. mobilis*. *Enterobacter agglomerans* and several species of the *E. agglomerans* complex cluster together and away from *E. cloacae*. *E. agglomerans* has been transferred to a new genus, *Pantoea*, and in this volume, the *E. agglomerans* complex is treated with the genus *Pantoea*. *Enterobacter intermedius* (originally named *E. intermedium*; corrected by von Graevenitz (1990)) has a *rrs* sequence identical with that of *Kluyvera cochleae*. Furthermore, DNA–DNA hybridization has shown the type strains of *E. intermedius* and *K. cochleae* to be 99% related with a thermal instability of hybridized molecules of 1.5°C (D.J. Brenner, personal communication). Therefore, *E. intermedius* is not treated in this chapter. (See the section on Other organisms in the chapter on the genus *Kluyvera*.)

The *Enterobacter* species treated in this chapter do not cluster together in an *rrs* sequence comparison. However, after *rpoB* sequence comparison, all species and subgroups in the *E. cloacae* complex composed of *E. cloacae*, *Enterobacter asburiae*, *Enterobacter hormaechei*, and the type strain of *Enterobacter dissolvens* form a cluster (P.A.D. Grimont, unpublished data). *Enterobacter cancerogenus*/*Enterobacter taylorae* and *E. agglomerans* DNA group VII (not a member of *Pantoea*) form a cluster near *Leclercia adecarboxylata*. *Enterobacter amnigenus*, *Enterobacter nimipressuralis*, and *Enterobacter kobei* constitute another cluster. *Enterobacter cowanii* clusters with *E. agglomerans* DNA group IX (not a member of *Pantoea*) (unpublished data). More work is needed to establish criteria for delineating genera and ascertaining the phylogenetic position of all *Enterobacter* species.

Nutrition and growth conditions Best results are obtained when *Enterobacter* cultures are incubated at 30°C. At 37°C, yellow pigmentation of *E. sakazakii* may be weak. *E. sakazakii* is able to grow at 44°C.

Media used in the isolation of *Enterobacter* species are similar to those used for other *Enterobacteriaceae*. On nutrient agar, *E. cloacae*, *E. gergoviae*, *E. amnigenus*, and *E. nimipressuralis* form col-

onies that are round, 2–3 mm in diameter, and slightly iridescent or flat with irregular edges (Richard, 1984). *E. sakazakii* strains forms bright yellow colonies at 25°C or pale yellow colonies at 37°C, 1–3 mm in diameter. These colonies are smooth, mucoid, or dry (Farmer et al., 1980a).

Differential media that are not inhibitory for the *Enterobacteriaceae* are often used. These are bromothymol blue-lactose agar, phenol red-lactose agar, Drigalski lactose agar, eosin methylene blue agar, or MacConkey agar. It should be remembered that some strains in a species known to produce acid from lactose might fail to do so and thus give lactose-negative colonies on differential media.

In Biotype-100 strips (BioMerieux, Craponne, France), or in a minimal medium containing ammonium sulfate as the nitrogen source, the following compounds serve as sole carbon sources for most *Enterobacter* strains: *N*-acetyl-D-glucosamine, D-alanine, L-alanine, L-arabinose, L-aspartate, D-cellobiose, D-fructose, fumarate, D-galactose, D-galacturonate, gentiobiose, D-gluconate, D-glucosamine, D-glucose, D-glucuronate, L-glutamate, glycerol, 2-ketogluconate, DL-lactate, L-malate, D-mannitol, D-mannose, L-proline, D-ribose, L-serine, succinate, D-trehalose, and D-xylose. L-rhamnose is utilized by all strains except for most *E. asburiae* strains.

Most strains cannot utilize the following substrates as sole carbon and energy sources: L-arabitol, ethanolamine, itaconate, 3-phenylpropionate, and L-sorbose. The following substrates are not utilized except by some strains of *E. gergoviae*: *meso*-erythritol, gentisate, glutarate, histamine, 3-hydroxybenzoate, and tricarballoylate. D-melezitose is not utilized, except by some strains of *E. sakazakii*.

Metabolism When pyrroloquinoline quinone (PQQ) is added, gluconate is produced from D-glucose in the presence of iodoacetate under aerobic conditions by all *Enterobacter* species by means of a glucose dehydrogenase. Only *E. gergoviae*, *E. sakazakii*, *E. cowanii*, and *E. hormaechei* genomic group (*E. cloacae* group III) can produce gluconate without added PQQ (Bouvet et al., 1989). Data are not available for *E. kobei* and *E. pyrinus*. A reducing compound, 2-ketogluconate, is also produced from gluconate by *E. gergoviae* by means of a gluconate dehydrogenase (Bouvet et al., 1989). 2,5-Diketogluconate is not produced from 2-ketogluconate (Bouvet et al., 1989).

Most *Enterobacter* species and biogroups give a positive Voges–Proskauer reaction (except *E. kobei*) and Simmons' citrate tests, motility (except some strains of *E. asburiae*); acid production from D-glucose, D-mannitol, salicin, L-arabinose, L-rhamnose (except most strains of *E. asburiae*), D-xylose, trehalose, D-cellobiose, and maltose; hydrolysis of *o*-nitrophenyl- β -D-galactoside (ONPG), and nitrate reduction. All *Enterobacter* species produce gas from D-glucose.

Most *Enterobacter* species or biogroups are negative for the following tests: H₂S production from thiosulfate, phenylalanine deaminase, tetrathionate reduction, tributyrin and corn oil hydrolysis, and β -glucuronidase.

Bacteriophage and bacteriocin typing Phage typing has been used for *E. cloacae*. Gaston (1987b) isolated 76 phages active against *E. cloacae* from sewage. Of these, 26 phages were selected after numerical taxonomic analysis of their reaction patterns on 92 *E. cloacae* strains. This system has been tried on 384 isolates, 94% of which were susceptible to at least one phage. 325 phage susceptibility patterns were observed. Reproducibility of patterns was 100% when duplicate testing was on the same day, but the

reproducibility was only 40% when duplicate testing was done after 18 months (Gaston, 1987a).

Three different schemes have been developed for typing *E. cloacae* strains by susceptibility to bacteriocins. Freitag and Friedrich (1981) were able to type 51 of 65 strains (78%) which were assigned to 23 bacteriocin types designated A through X (type Y contained non-typable isolates). Traub et al. (1982) found 9% of 256 *E. cloacae* isolates to be bacteriocinogenic. Bacteriocins were produced by 16 strains after induction by mitomycin. A total of 308 isolates from various clinical sources were studied and 79% fell into 52 bacteriocin types leaving 21.4% of the isolates untypable. Bauernfeind and Petermüller (1984) found 132 of 149 isolates to produce bacteriocins. With a set of eight bacteriocin-producing strains, typability of 134 clinical isolates was 96.3%. A total of 44 different bacteriocin types could be distinguished. Only 11 (8.2%) of the isolates fell into the largest bacteriocin type.

Antigenic structure *E. cloacae* serotyping schemes have been devised. Sakazaki and Namioka (1960) distinguished 53 O- and 56 H-antigens in agglutination tests; 170 isolates were distributed among 79 serotypes. Bacterial suspensions had to be boiled in order to be agglutinated by the O-antisera. Unfortunately, epidemiological investigations using this serotyping scheme have not been reported.

Gaston et al. (1983) devised a serotyping scheme based on heat-stable somatic (O) antigens. 28 antisera including 11 absorbed sera were used, thus defining 28 O-serogroups. Of 300 clinical isolates from 66 hospitals, 78% were typable, 11% were not agglutinated by any of the sera, and 11% were autoagglutinable in saline.

Pathogenicity *Enterobacter* species are found in the natural environment including water, sewage, vegetables, and soil. The increased prevalence of *Enterobacter* spp. as nosocomial pathogens may be due to a greater resistance to disinfectants and antimicrobial agents than that of other members of the *Enterobacteriaceae*. Contaminated medicinal agents can be sources of outbreaks. Maki and Martin (1975) showed that *E. cloacae*, *E. agglomerans* and *Serratia marcescens* multiplied better in 5% dextrose solution at 25°C than did other members of the *Enterobacteriaceae*.

The pathogenicity of *Enterobacter* spp. has been reviewed by Sanders and Sanders (1997). Most *Enterobacter* bacteremias are acquired in the hospital. *E. cloacae* predominates, followed by *E. agglomerans*, *E. sakazakii*, and others. 14–53% of bacteremias that involve *Enterobacter* spp. are polymicrobial. *Enterobacter* bacteremias occur in patients with severe underlying diseases. Cases involving previously healthy individuals can be caused by administration of a contaminated medical product. The mortality rate is similar to that due to other enteric bacilli. A study of patients undergoing cardiac surgery suggested that *E. cloacae* might be more virulent than *E. aerogenes* (*Klebsiella mobilis*). Among patients colonized by *E. cloacae*, 26% developed an infection, whereas only 7% of those colonized by *K. mobilis* developed an infection.

In the lower respiratory tract, *Enterobacter* spp. have been involved in asymptomatic colonization of respiratory secretions, purulent bronchitis, lung abscess, pneumonia, and empyema. Pneumonia occurs in lung transplant recipients, patients with severe underlying diseases, and in elderly persons who are institutionalized.

Enterobacter spp. are a common cause of nosocomial infections of surgical wounds and burns. Other infections involving the skin and soft tissues are cellulitis, fasciitis, abscesses, emphysema, and

myositis. Endocarditis due to *Enterobacter* spp. occurs in intravenous drug abusers and individuals with prosthetic valves.

Enterobacter spp. have been involved in biliary sepsis. Bacteremias after hepatic transplantation or endoscopic retrograde cholangiopancreatography, hepatic gas gangrene, fulminant emphysematous cholecystitis, acute suppurative cholangitis, and peritonitis following small intestine obstruction have been reported.

Urinary tract infections due to *Enterobacter* spp. range from asymptomatic bacteriuria to pyelonephritis and urosepsis. *Enterobacter* meningitis, ventriculitis, brain abscess, and infections near foreign bodies have been reported. *Enterobacter* spp. are infrequently the cause of postoperative endophthalmitis, but consequences are devastating (loss of vision or eye). Occasionally, severe septic arthritis, osteomyelitis, infections of multiple bones and joints in infants and children, vertebral osteomyelitis, bilateral hip infections and prosthetic hip infections due to *Enterobacter* spp. have been reported (reviewed by Sanders and Sanders, 1997).

Other species in the *E. cloacae* complex (*E. dissolvens*, *E. asburiae*, *E. hormaechei*) may have been reported as *E. cloacae* due to taxonomic uncertainties. *E. hormaechei* has been isolated from wounds, sputum, and blood (O'Hara et al., 1989). An outbreak of *E. hormaechei* infection and colonization among vulnerable, low-birth-weight premature infants has been described (Wenger et al., 1997). In our opinion, the *E. hormaechei* phenotype as described by O'Hara et al. (1989) is a biogroup of a larger genomic species. Thus, the strains that caused an outbreak in Marseille hospitals were identified as belonging to the *E. hormaechei* genomic species, although they resembled *E. cloacae* in classical biochemical tests (Davin-Regli et al., 1997). Our observation is that most *E. cloacae* that cause nosocomial infections belong to the *E. cloacae* group III/ *E. hormaechei* genomic species (unpublished data).

E. asburiae strains were isolated from clinical specimens, mostly urine, respiratory tract, feces, wounds, and blood (Brenner et al., 1986). The clinical significance of this organism is not known.

Enterobacter dissolvens has not yet been reported in clinical specimens and has only been recovered from environmental sources. *E. dissolvens* was first isolated by Rosen (1922) from diseased corn and is found in rotting cornstalks.

The role of *E. sakazakii* in pathology and as a contaminant has been reviewed by Nazarowec-White and Farber (1997). Little is known about the presence of *E. sakazakii* in the environment. *E. sakazakii* has frequently been found in powdered substitutes for breast milk in many countries together with other enteric bacteria. It has also been recovered from liquid formula and milk subjected to ultra high temperature (UHT) processing. Numerous reports of *E. sakazakii* meningitis and generalized sepsis have been reported in neonates. Powdered infant formula was implicated in several cases. Even when death did not occur, severe complications or sequelae were observed (brain abscesses, necrotizing enterocolitis, hydrocephalus). There have been only three reports of *E. sakazakii* infections in adults (an ulcer in the foot of a diabetic patient, urosepsis in an elderly patient, and bacteremia) (Nazarowec-White and Farber, 1997).

E. cancerogenus (*E. taylora*) has only rarely been associated with human infections. Septicemia and urinary tract infections in patients with underlying diseases have been observed. Wound infections after severe trauma or crush injuries have been described. Some of them were chronic or unresolving infections. An environmental source was suspected (Abbott and Janda, 1997).

E. nimipressuralis has not yet been reported in clinical specimens and has only been recovered from environmental sources.

E. amnigenus is mainly found in water, but some strains were isolated from clinical specimens such as respiratory tract, wound, or feces (Farmer et al., 1985a).

E. gergoviae occurs in a variety of environmental sources such as water, cosmetics and clinical sources in France, Africa, and the United States (Richard et al., 1976; Brenner et al., 1980b). Multiply drug resistant strains have been found in urine samples during an infection outbreak (Richard et al., 1976).

Pathogenesis Adhesive properties may be important in the establishment or maintenance of bacterial infections. Adhesins are often also hemagglutinins (HA) and may or may not be located on fimbriae. Most strains of *Enterobacter amnigenus*, *E. cloacae*, and *E. sakazakii* produce a mannose-sensitive hemagglutinin (MS-HA) associated with type-1 fimbriae, i.e., thick, channeled fimbriae having an external diameter of 7–8 nm. These fimbriae can be coated by type-1 fimbrial antiserum against *E. cloacae* 035 but not by type-1 fimbrial antiserum against *Klebsiella pneumoniae* K55/1 (Adegbola and Old, 1983b). No other hemagglutinin and fimbrial type has been observed in these species.

Seven of eight *E. gergoviae* strains produce a mannose-resistant *Klebsiella*-like hemagglutinin (MR/K-HA) that agglutinates tanned ox erythrocytes and is associated with type-3 fimbriae, i.e., thin, non-channelled fimbriae having an external diameter 4–5 nm. These fimbriae can be coated with type-3 fimbrial antiserum against *K. oxytoca* K70/1 (Adegbola and Old, 1983b). No other hemagglutinin and fimbrial type has been observed in this species.

All *E. cloacae* strains tested could adhere to and invade HEp-2 cells (Keller et al., 1998). Only two of 14 *E. cloacae* isolates adhered to the human cell line Intestine-407 (Livrelli et al., 1996).

The outer membrane protein OmpX plays a role in the invasion of rabbit ileal tissue by *E. cloacae*. Invasiveness varies directly with the level of OmpX in the outer membrane (De Kort et al., 1994).

Iron is essential for bacterial growth. In the human body, iron is complexed to carrier molecules such as transferrin (in the serum) or lactoferrin (in milk and other secretions), or sequestered within cells (in heme proteins). When growing under iron-limiting conditions, potentially pathogenic *Enterobacteriaceae* produce high affinity systems to solubilize and import the required iron. These iron-chelating compound produced are mostly of two sorts, phenolate (e.g., enterochelin) and hydroxamate (aerobactin) siderophores (Payne, 1988). All strains of *E. cloacae*, *E. gergoviae*, and *E. sakazakii* tested by Reissbrodt and Rabsch (1988) produced enterochelin. Some strains of all these species produced aerobactin. Aerobactin was first isolated from a strain of *K. mobilis* (then "*Aerobacter aerogenes*") (Gibson and Magrath, 1969). Aerobactin and cloacin DF13 bind to the same receptor sites located in the outer membrane (Van Tiel-Menkveld et al., 1982). Aerobactin is encoded by a relatively simple genetic system that has been extensively characterized. Only four genes are required for synthesis of aerobactin (Carbonetti and Williams, 1984). *E. cloacae* harbor a relatively large conjugative plasmid encoding susceptibility to cloacin DF13 as well as production and uptake of aerobactin (Krone et al., 1985).

Most clinical strains of *E. cloacae* examined have been resistant to serum bactericidal activity and could produce aerobactin and mannose-sensitive hemagglutinin (Keller et al., 1998).

Antibiotic sensitivity The antibiotic sensitivity of *Enterobacter* spp. has been reviewed by Sanders and Sanders (1997). *E. cloacae* is naturally resistant to ampicillin, cephalothin and other older cephalosporins, and cefoxitin. Most strains are sensitive to uridopenicillins and carboxypenicillins. Strains are not very sensitive to cefamandole and cefuroxime but are more sensitive to expanded-spectrum cephalosporins and aztreonam. The highest sensitivity is to carbapenems (e.g., imipenem). Most strains are sensitive to aminoglycosides and ciprofloxacin. The sensitivity to trimethoprim-sulfamethoxazole is variable.

All species of *Enterobacter* examined to date produce a chromosomally encoded group 1 β -lactamase. This enzyme is produced only at low levels (noninducible) by *E. gergoviae* and some strains of *E. sakazakii*, which explains the greater sensitivity of these strains to ampicillin, older cephalosporins, and cefoxitin. The enzyme is inducible in *E. cloacae*, *E. cancerogenus*, *E. asburiae*, and most strains of *E. sakazakii*, and these strains are resistant to ampicillin, older cephalosporins, and cefoxitin (Sanders and Sanders, 1997).

The *ampD* mutation in *E. cloacae* stably derepresses the chromosomal β -lactamase, which is then produced at high levels, causing the strains to be resistant to extended-spectrum cephalosporins, broad-spectrum penicillin, and aztreonam. These mutants remain sensitive to carbapenems and cefepime. However, additional mutations involving permeability through the outer envelope can lead to resistance to these agents. A chromosomally encoded carbapenemase has been reported (Sanders and Sanders, 1997).

Strains can acquire plasmid-borne resistance determinants encoding group 2b (TEM-1, TEM-2, or SHV-1) or group 2d (OXA-1) β -lactamases. Strains carrying such plasmids are susceptible to extended-spectrum cephalosporins and β -lactamase inhibitor- β -lactam drug combinations. Resistance to extended-spectrum cephalosporins can also be acquired with plasmids encoding group 2be β -lactamase (extended-spectrum β -lactamases).

Resistance to aminoglycosides is due to the production of one or more aminoglycoside-inactivating enzymes. The most frequently occurring enzymes are acetylating enzymes AAC(3)-II, AAC(6'), AAC(3)-III, AAC(3)-I, and AAC(3)-V. A nucleotidylating enzyme, ANT(2''), may also occur (Sanders and Sanders, 1997).

Ecology Nitrogen-fixing strains of *E. cloacae* have been isolated from the roots of dryland and wetland rice (Ladha et al., 1983). The nitrogen-fixing *E. cloacae* strains isolated by Bally et al. (1983) belong to *E. cloacae* genomic group 5 (P.A.D. Grimont, unpublished observations). A strain identified as *E. cloacae* suppresses sporangium germination in *Pythium ultimum* (the cause of seed and root diseases of a wide variety of plants). This is achieved by a competitive utilization of fatty acids present in the exudate of germinating seeds (Van Dijk and Nelson, 2000). Some strains identified as *E. cloacae* are endophytic symbionts of corn (Hinton and Bacon, 1995).

A strain identified as *E. asburiae* is a cotton endophyte that is able to colonize internal tissues of different plant species (Quadt-Hallmann and Kloepper, 1996).

E. nimipressuralis has been described as being the causal agent of wetwood in elm trees (Carter, 1945). *Enterobacter cancerogenus* (as *Erwinia cancerogena*) has been described as the causal agent of a canker disease of poplar (*Populus* species). *Enterobacter pyrinus* causes a pear brown leaf spot disease in Korea (Chung and Cho, 1993).

ENRICHMENT AND ISOLATION PROCEDURES

All media designed for the isolation of *Enterobacteriaceae* can be used for the isolation of *Enterobacter* species: MacConkey agar, Drigalski lactose agar, Hektoen agar, deoxycholate lactose citrate agar, etc. *Enterobacter* can also grow on media for general use, such as blood agar, nutrient agar, tryptic soy agar, bromocresol purple lactose agar, etc.

There are no selective media for *Enterobacter* species. The requirements for devising a selective medium are (i) a precise and stable delineation of the genus (or species, if the medium is to be species specific), (ii) known common properties within the genus (or species) which are uncommon outside the genus (or species), and (iii) a clinical, public health or special need for such selective medium. Due to pending changes in the delineation of the genus *Enterobacter* and the species *E. cloacae*, conditions (i) and (ii) are not met. Except for some epidemiological studies, the selective isolation of *Enterobacter* spp. is rarely necessary, since the presence of *Enterobacter* spp. in pluribacterial habitats (feces, throat, skin) is clinically meaningless. The public health significance of *Enterobacter* spp. in water or foods is uncertain. Isolation of *Enterobacter* spp. from clinical specimens is done either by direct plating on blood agar, tryptic soy agar or nutrient agar (e.g., pus, urine) or by plating after prior growth in tryptic soy broth or nutrient broth (blood, pus, cerebrospinal fluid).

MAINTENANCE PROCEDURES

Strains are initially grown on tryptic soy agar at their optimum temperature and then stabbed into a medium¹ designed for maintenance of *Enterobacteriaceae* and related organisms. The cultures are stored at room temperature in a dark, dry place. Cultures may be also preserved by freeze-drying. Freeze-drying is the best procedure for preservation of pigmented strains.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Procedures for carbon source utilization (using Biotype-100 strips), glucose oxidation, gluconate- and 2-ketogluconate dehydrogenase, Voges-Proskauer (Richard's modification), and β -xylosidase tests are given in the chapter on *Serratia* in this volume.

DIFFERENTIATION OF THE GENUS *ENTEROBACTER* FROM OTHER GENERA

Because the genus is not monophyletic and may be split further, differentiation of the genus should be at the species level.

TAXONOMIC COMMENTS

The history of some species of the genus *Enterobacter* can be confusedly traced to the end of the nineteenth century. "*Bacillus lactis aerogenes*" was isolated by Escherich (1885) from milk and renamed "*Bacillus aerogenes*" by Kruse (1896) and "*Aerobacter aerogenes*" by Beijerinck (1900b). Differentiation of this organism from Friedländer's bacillus (now *Klebsiella pneumoniae*) was not clear before 1955, and most authors considered "*B. lactis aerogenes*" or "*Aerobacter aerogenes*" as nonmotile, or as containing motile and nonmotile strains (Grimbert and Legros, 1900; Edwards and Fife, 1955). This led Edwards and Fife (1955) to state that "*A. aerogenes*" strains were in fact *Klebsiella* strains.

"*Bacterium cloacae*" was described by Jordan (1890) and transferred to a new genus "*Cloaca*" as "*Cloaca cloacae*" by Castellani and Chalmers (1920). *Bergey's Manual* (Bergey et al., 1923) transferred this species to the genus "*Aerobacter*" as "*A. cloacae*". Because "*Aerobacter aerogenes*" was indistinguishable (at that time) from *Klebsiella pneumoniae*, it was proposed that the species "*A. aerogenes*" disappear (Edwards and Fife, 1955) although disappearance of the type species ("*A. aerogenes*") implied disappearance of the genus ("*Aerobacter*").

A significant step forward occurred when Møller (1955) devised simple methods for testing amino-acid decarboxylases. The "*Cloaca*" group, being arginine-positive, could now easily be distinguished from the *Klebsiella* group (arginine-negative) by biochemical tests. This led to the finding of motile strains of the "*Cloaca*" group which were arginine-negative and produced gas from inositol and glycerol (Hormaeche and Munilla, 1957). These were called "*Cloaca B*" (arginine-positive strains forming the "*Cloaca A* group"). After reexamination of many cultures with decarboxylase tests, Hormaeche and Edwards (1958) redefined the genus "*Aerobacter*" to include two species, "*A. aerogenes*" ("*Cloaca B*") and "*A. cloacae*" ("*Cloaca A*"). The type species was reaffirmed to be "*A. aerogenes*".

In an attempt to avoid confusion resulting from the reclassification in the genus *Klebsiella* of many nonmotile strains previously labeled "*A. aerogenes*", Hormaeche and Edwards (1960a, b) proposed a new genus, *Enterobacter*, as a substitute for "*Aerobacter*". This genus was then composed of *E. cloacae* (type species) and *E. aerogenes*. The Judicial Commission of the International Committee on Nomenclature of Bacteria placed the name *Enterobacter* on the list of conserved names (Judicial Commission, 1963).

The genus *Erwinia* has long been a depository for plant-associated members of the *Enterobacteriaceae*. Several species of this genus were found phenotypically similar to *Enterobacter* species (*Erwinia herbicola*, *Erwinia ananas**, *Erwinia uredovora*, *Erwinia millettiae*, *Erwinia dissolvens*, *Erwinia nimipressuralis*) (Lelliott, 1974; Lelliott and Dickey, 1984) or highly related to *Enterobacter* species by DNA hybridization (*Erwinia dissolvens*, *Erwinia nimipressuralis*) (Steigerwalt et al., 1976).

The other *Enterobacter* species have been described after delineation by DNA-DNA hybridization. DNA-DNA hybridization has allowed bacteriologists to modify the circumscription of some species and to find synonyms.

Enterobacter taylorae was described based on clinical strains closely related to *E. cloacae* (Farmer et al., 1985b). The species was later found synonymous to *Enterobacter cancerogenus* by both DNA relatedness and phenotypic characters (Grimont and Ageron, 1989).

Early DNA relatedness studies on *E. cloacae* (Steigerwalt et al., 1976) showed the genomic heterogeneity of this nomen-species. In work done in our laboratory (Grimont and Grimont, 1992; and unpublished) on 49 strains previously identified as *E. cloacae*, 45 strains fell into five DNA relatedness groups (1 to 5). Group 1 contained only three strains, including the type strains of *E. cloacae* and *E. dissolvens*. DNA from these type strains showed some divergence (ΔT_m values of 4.5°C). In spite of a search for similar strains in our large laboratory collection, no other strain of group 1 could be found. Group 2 contained seven strains, including

1. Maintenance medium (g/liter): Bacto-peptone (Difco), 10.0; NaCl, 5.0; Bacto-agar (Difco), 10.0; pH 7.4. The medium should be dispensed into small (9.5–10 × 90 mm) screw-capped tubes.

*Editorial Note: Mergaert et al. (1993) transferred *Erwinia ananas* to the genus *Pantoea* as *P. ananas*; the name was corrected to *Pantoea ananatis* by Trüper and De' Clari (1997).

reference strain CDC 1347-71 (Steigerwalt et al., 1976). Group 3 contained 15 strains, including the type strain of *Enterobacter hormaechei*. This group was slightly heterogeneous with ΔT_m values ranging from 0.0–4.0°C. Group 4 contained 17 strains and could be split into three subgroups based on ΔT_m values (0.0–2.5°C within subgroup and 4.0–6.5°C between subgroups). Subgroup 4a contained the type strain of *Enterobacter asburiae*. Group 5 (earlier referred to as group 6 (Bouvet et al., 1989)) contained three nitrogen-fixing strains.

Three problems are raised by this study. The first problem involves *E. hormaechei*. A group of similar strains (Enteric group 75), which differ phenotypically from *E. cloacae*, was genomically distinct from the type strain of *E. cloacae* and thus described as a new species, *E. hormaechei* (O'Hara et al., 1989). In our group 3, we found only one strain phenotypically similar to *E. hormaechei*. All other strains of group 3 fit the classical definition of *E. cloacae* (Richard, 1984). Group 3 strains (including *E. hormaechei*) produce an active glucose dehydrogenase, and this property differentiates group 3 from the other groups in the *E. cloacae* complex (Bouvet et al., 1989; P.A.D. Grimont, unpublished observations). Group 3 should be named *E. hormaechei*, and the species definition should be emended.

The second problem involves *E. asburiae*. Enteric group 17 differed from *E. cloacae* by some phenotypic features and DNA hybridization showed Enteric Group 17 (*E. asburiae*) to be distinct from the type strain of *E. cloacae*. Our subgroup 4a is genomically and phenotypically identical with *E. asburiae*. However, subgroups 4b and 4c are less differentiable from *E. cloacae* by phenotypic characteristics. Group 4 (including subgroups 4a, 4b, and 4c) should be called *E. asburiae*, and the species definition should be amended.

The third problem is in fact the cause of the other two. The choice of the neotype strain for *E. cloacae* was unfortunate. This strain was isolated from cerebrospinal fluid, which is not the usual habitat of what is currently known as *E. cloacae*. There are two possible solutions: either the nomenclatural rules are followed, in which case *E. cloacae* (our group 1) will disappear from routine clinical laboratory work (because most strains presently labeled *E. cloacae* are in fact *E. hormaechei* or *E. asburiae*), or the type strain can be changed and a new one designated from our group 2 (which can be encountered in clinical microbiology, although less frequently than *E. hormaechei* and *E. asburiae* genomic species).

Some species that were included in the genus *Enterobacter* are now excluded; these are discussed below.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *ENTEROBACTER*

Tables BXII.γ.220 and BXII.γ.221 give the characteristics differentiating the species and unnamed genomic groups in the genus *Enterobacter*.

List of species of the genus *Enterobacter*

1. ***Enterobacter cloacae*** (Jordan 1890) Hormaeche and Edwards 1960b, 72^{AL}. Nom. Cons. Opin. 28, Jud. Comm. 1963, 38 (*Bacillus cloacae* Jordan 1890, 836.)
clo.a' cae. L. n. *cloaca* a sewer; L. gen. n. *cloacae* of sewer.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Tables BXII.γ.220, BXII.γ.221 and BXII.γ.222. Several DNA hybridization groups compose this species. Some of these DNA groups can be identified to named species (*E. dissolvens*, *E. asburiae*, or *E. hormaechei*) with dif-

The group of strongly proteolytic strains named "*Aerobacter liquefaciens*" by Grimes and Hennerty (1931) was included in the genus *Enterobacter* by Edwards and Ewing (1972). The species was transferred to the genus *Serratia* as *S. liquefaciens* by Bascomb et al. (1971).

A group of strains named *Hafnia* by Møller (1954) was included in the genus *Enterobacter* as *E. hafniae* (Ewing and Fife, 1968). This species appeared in the 8th edition of the *Bergey's Manual* (Buchanan and Gibbons, 1974) as *Hafnia alvei*. Steigerwalt et al. (1976) favored the separation of *Hafnia alvei* from the genus *Enterobacter* based on DNA relatedness studies.

E. aerogenes is closer to *Klebsiella pneumoniae* (about 55% DNA relatedness) than to *E. cloacae* (about 45% relatedness) (Brenner et al., 1972c; Steigerwalt et al., 1976). In a numerical taxonomy study (Bascomb et al., 1971), *E. aerogenes* was so similar to the genus *Klebsiella* that transfer of this species to the genus *Klebsiella* was proposed. Since the name "*K. aerogenes*" had been used for bacteria that are indistinguishable from *K. pneumoniae* by DNA relatedness, the name *K. mobilis* was proposed for *E. aerogenes* (Bascomb et al., 1971). The species is treated in the chapter on the genus *Klebsiella* as *K. mobilis*.

E. agglomerans is a very complex group of environmental bacteria that may cause opportunistic infections. The name covers many (20–40) genomic groups (Brenner et al., 1988b) or phenons (Gavini et al., 1983b; Verdonck et al., 1987). In addition to this diversity, strains of the *E. agglomerans* complex are not closely related to *E. cloacae* (the type species of the genus *Enterobacter*) by DNA relatedness. Some groups in this complex have been placed in new genera (*Rahnella aquatilis*, *Ewingella americana*, *Lecclercia adecarboxylata*). A new genus, *Pantoea*, has been proposed for several groups of the *E. agglomerans* complex. These are treated in the chapter on the genus *Pantoea*.

Enterobacter intermedius (originally named *E. intermedium*; corrected by von Graevenitz (1990)) was a Voges-Proskauer positive species found in water and unpolluted soil (Izard et al., 1980). However, it has an *rrs* sequence identical to that of *Khuyvera cochleae*. Furthermore, DNA–DNA hybridization has shown the type strains of *E. intermedius* and *K. cochleae* to be 99% related with a thermal instability of hybridized molecules of 1.5°C (D.J. Brenner, personal communication).

FURTHER READING

Janda, J.M. and S.L. Abbott. 1998. The Enterobacteria, Lippincott-Raven, Philadelphia, pp. 387.

ferent circumscriptions than originally published. Occurs in water, sewage, soil, meat, and hospital environments and on the skin and in the intestinal tracts of man and animals as a commensal. May cause nosocomial infections

The mol% G + C of the DNA is: 52–54 (T_m).

Type strain: ATCC 13047, CIP 60.85, DSM 30054, JCM 1232, LMG 2783, NCTC 10005.

GenBank accession number (16S rRNA): AJ417484.

This strain is very close by DNA–DNA hybridization to the type strain of *E. dissolvens* and may not properly represent the strains routinely identified as *E. cloacae*.

TABLE BXII.γ.220. Characteristics of the species of the genus *Enterobacter*^a

Characteristic	<i>E. cloacae</i> complex ^b	<i>E.</i> <i>amnigenus</i>	<i>E.</i> <i>cancerogenus</i>	<i>E.</i> <i>cowanii</i>	<i>E.</i> <i>gergoviae</i>	<i>E.</i> <i>kobei</i>	<i>E.</i> <i>nimipressuralis</i>	<i>E.</i> <i>pyrinus</i>	<i>E.</i> <i>sakazakii</i>
Motility (36°C)	d	+	+	+	+	+	+	+	+
Yellow pigment	—	—	—	d	—	—	—	—	+
Urea hydrolyzed	—	—	—	—	+	d	—	+	—
Indole production	—	—	—	—	—	—	—	—	d
β-Xylosidase test	+	+	—	+	—	+	+	ND	+
Methyl red	d	—	—	+	—	ND	+	ND	—
Voges-Prokauer	d	+	+	+	+	—	+	+	+
Growth in KCN	+	+	+	+	—	+	+	+	+
Gelatin hydrolysis at 22°C	(d)	—	—	—	—	—	—	—	—
Deoxyribonuclease (25°)	—	—	—	—	—	—	—	—	(+)
Lysine decarboxylase	—	—	—	—	+	—	—	d	—
Arginine dihydrolase	+	+	+	—	—	+	+	—	+
Ornithine decarboxylase	+	+	+	—	+	d	+	+	+
Phenylalanine deaminase	—	—	—	—	—	—	—	—	d
Glucose dehydrogenase	D	—	—	+	+	ND	—	ND	+
Gluconate dehydrogenase	—	—	—	—	+	ND	—	+	—
Growth at 41°C	+	—	ND	+	+	+	—	ND	+
Esculin hydrolysis	d	+	—	+	+	d	+	+	+
Acetate	d	—	—	+	+	d	ND	ND	+
<i>Acid from:</i>									
Adonitol	d	—	—	—	—	—	—	—	—
L-Arabinose	+	+	+	+	+	+	+	+	+
D-Arabitol	d	—	—	—	+	—	ND	ND	—
Cellobiose	+	+	+	+	+	+	+	+	+
Dulcitol	D	—	—	+	—	d	—	—	—
meso-Erythritol	—	—	—	—	—	ND	—	ND	—
Glycerol	d	—	d	+	+	d	ND	+	—
myo-Inositol	d	—	—	—	—	d	d	+	(+)
Lactose	d	d	—	+	d	+	+	—	+
Maltose	+	+	+	+	+	+	+	+	+
D-Mannitol	+	+	+	+	+	+	+	+	+
Melibiose	d	+	—	+	+	+	+	+	+
α-Methylglucoside	+	d	d	—	—	+	+	—	+
Mucate	d	+	d	d	—	d	+	ND	—
Raffinose	+	+	—	+	+	+	d	—	+
L-Rhamnose	d	+	+	+	+	+	+	+	+
Salicin	D	+	+	+	+	+	+	+	+
D-Sorbitol	+	—	—	+	—	+	+	—	—
Sucrose	+	+	—	+	+	+	—	+	+
Trehalose	+	+	+	+	+	+	+	+	+
D-Xylose	+	+	+	+	+	+	+	+	+
<i>Utilization of:</i>									
cis-Aconitate	d	—	+	+	d	—	—	(d)	+
trans-Aconitate	d	—	+	+	—	—	—	(d)	d
Adonitol	D	—	—	—	—	—	—	—	—
4-Aminobutyrate	—	—	—	—	+	—	—	—	+
5-Aminovalerate	—	—	—	—	+	—	—	—	—
D-Arabitol	D	—	—	—	+	—	—	+	—
Benzoate	—	—	—	—	d	—	—	—	—
Citrate	+	+	+	+	+	+	+	—	+
m-Coumarate	—	—	—	—	+	—	—	—	—
Dulcitol	D	—	—	+	—	+	—	—	d
L-Fucose	D	—	+	—	d	—	—	—	—
Gentisate	—	—	—	(+)	d	—	—	—	—
Histamine	—	—	—	—	+	—	—	—	—
3-Hydroxybenzoate	—	—	—	—	d	—	—	—	—
4-Hydroxybenzoate	—	—	—	—	+	—	—	—	—
3-Hydroxybutyrate	d	—	—	—	d	—	—	—	—
myo-Inositol	D	—	—	—	d	—	—	—	d
5-Ketogluconate	—	—	—	—	+	—	—	+	—
2-Ketoglutarate	d	—	d	—	d	—	—	d	—
Lactose	(+)	+	(d)	(+)	(+)	+	+	—	+
Lactulose	d	d	—	(+)	d	+	—	—	+
D-Lyxose	D	+	—	+	—	—	+	—	—
D-Malate	d	—	d	(+)	d	+	d	—	(d)
Malonate	d	—	(d)	—	d	d	—	—	(d)
Maltitol	+	+	—	—	—	+	+	—	+
D-Melibiose	D	+	—	+	+	+	+	—	+
1-O-Methyl-α-galactoside	D	+	—	+	+	+	+	—	+
3-O-Methyl-D-glucose	D	—	+	—	—	v	—	—	—
1-O-Methyl-α-D-glucoside	+	d	—	—	—	+	+	—	+

(continued)

TABLE BXII.γ.220. (cont.)

Characteristic	<i>E. cloacae</i> complex ^b	<i>E.</i> <i>amnigenus</i>	<i>E.</i> <i>cancerogenus</i>	<i>E.</i> <i>cowanii</i>	<i>E.</i> <i>gergoviae</i>	<i>E.</i> <i>kobei</i>	<i>E.</i> <i>nimipressuralis</i>	<i>E.</i> <i>pyrinus</i>	<i>E.</i> <i>sakazakii</i>
Mucate	d	+	+	+	—	+	+	—	—
Palatinose	+	+	—	—	—	+	+	—	+
Phenylacetate	D	—	+	—	+	—	+	—	—
L-Proline	+	d	+	+	+	+	+	—	+
Protocatechuate	—	—	—	—	+	—	—	+	—
Putrescine	D	—	+	—	d	—	—	+	+
Quinate	—	—	—	—	+	—	—	—	—
D-Raffinose	d	+	—	+	+	+	d	—	+
L-Rhamnose	D	+	+	+	+	+	+	+	+
D-Saccharate	+	+	+	+	—	+	+	+	—
D-Sorbitol	+	—	—	+	—	+	+	—	—
Sucrose	+	+	—	+	+	+	—	+	+
D-Tagatose	d	—	—	d	—	—	d	+	—
meso-Tartrate	—	—	—	—	d	—	—	—	—
Tricarballoylate	—	—	—	—	d	—	—	—	—
Tryptamine	—	—	—	—	—	—	—	—	—
D-Turanose	d	(d)	—	—	—	+	d	—	d
L-Tyrosine	d	—	—	—	d	—	—	—	—

^a +, 90–100% strains positive in 1–2 days; (+), 90–100% strains positive in 1–4 days; —, 90–100% strains negative in 4 days; d, positive or negative in 1–4 days; (d), positive or negative in 3–4 days; D, test used to differentiate species within a complex; ND, no data.

^bThe *E. cloacae* complex includes *E. cloacae*, *E. dissolvens*, *E. hormaechei*, and *E. asburiae*.

TABLE BXII.γ.221. Phenotypic properties of genomic groups within the *Enterobacter cloacae* complex^a

Characteristic	Genomic group or subgroup ^b						
	1	2	3	4a	4b	4c	5
Glucose dehydrogenase	—	—	+	—	—	—	—
Motility	+	+	+	—	d	+	+
Malonate test	+	+	+	—	—	d	+
Esculin hydrolyzed	d	(d)	(d)	+	+	+	+
Utilization of:							
Adonitol	—	—	d	—	—	—	—
D-Arabitol	—	—	d	—	—	—	—
Dulcitol	—	d	d	—	—	d	+
Fucose	—	—	d	—	—	—	—
D-Galacturonate	+	d	+	+	+	+	+
myo-Inositol	+	+	d	+	+	+	+
Lyxose	d	—	+	+	+	d	+
D-Melibiose	+	+	d	—	+	+	+
3-Methylglucose	—	—	d	—	—	—	—
Phenylacetate	d	+	+	—	+	+	+
Putrescine	d	+	—	+	—	+	—
D-Raffinose	+	+	d	d	d	+	+
L-Rhamnose	+	+	+	—	d	—	+
D-Sorbitol	+	+	d	+	+	+	+
Xylitol	—	—	(d)	—	—	—	—

^a +, 90–100% strains positive in 1–2 days (utilization tests) or in 1 day (other tests); (+), 90–100% strains positive in 1–4 days; —, 90–100% strains negative in 4 days; d, positive or negative in 1–4 days; (d), positive or negative in 3–4 days.

^bThe type strain of *E. dissolvens* and the present type strain of *E. cloacae* are in genomic group 1; the type strain of *E. hormaechei* is in genomic group 3, and the type strain of *E. asburiae* is in genomic group 4, subgroup 4a.

2. *Enterobacter amnigenus* Izard, Gavini, Trinel and Leclerc 1981b, 37^{VP}

am.ni'ge.nus. L. adj. *amnigenus* coming from water.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Table BXII.γ.220. Isolated from water and rarely from clinical samples.

The mol% G + C of the DNA is: 60 (Bd).

Type strain: ATCC 33072, CCUG 14182, CIP 103169, DSM 4486, JCM 1237, LMG 2784, NCTC 12124..

GenBank accession number (16S rRNA): AB004749.

TABLE BXII.γ.222. Utilization of substrates by biogroups of genomic group 3 of the *E. cloacae* complex^a

Utilization of:	Biogroup ^b						
	3a	3b	3c	3d	3e	3f	3g
Adonitol	—	—	—	—	—	+	+
D-Arabitol	—	—	—	—	—	+	+
Fucose	d	+	+	—	+	+	+
Methyl-α-D-galactoside	—	+	+	+	—	+	+
3-Methylglucose	+	—	+	—	—	+	—
D-Melibiose	— ^b	+	+	+	—	+	+
D-Raffinose	—	+	+	+	+	+	+
D-Sorbitol	—	+	+	+	+	+	+

^a +, all strains positive in 1–2 days; —, strains negative in 4 days; d, positive or negative in 1–4 days.

^bFive strains representing *E. hormaechei* (including the type strain) and received from the Centers for Disease Control corresponded to biogroup 3a.

3. *Enterobacter asburiae* Brenner, McWhorter, Kai, Steigerwalt and Farmer III 1988b, 220^{VP} (Effective publication: Brenner, McWhorter, Kai, Steigerwalt and Farmer III 1986, 1117.)

as.bur'i.ae. N.L. gen. n. *asburiae* named in honor of Marie Alyce Fife-Asbury, an American bacteriologist.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are included in Tables BXII.γ.220 and BXII.γ.221.

The type strain belongs in DNA group 4a in the *E. cloacae* complex.

The mol% G + C of the DNA is: 55 (*T_m*).

Type strain: ATCC 35953, CIP 103358, JCM 6051, NCTC 12123.

GenBank accession number (16S rRNA): AB004744.

4. *Enterobacter cancerogenus* (Urosevic 1966) Dickey and Zumoff 1988, 373^{VP} (*Erwinia cancerogena* Urosevic 1966, 500.)

can.cer.o'ge.nus. L. *cancer* crab, the disease cancer; L. v. *gigno* to produce; L. masc. adj. *cancerogenus* cancer-inducing.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Table BXII.γ.220.

Isolated from cankers in poplars (*Populus* species) in

Czechoslovakia and in clinical specimens. *Enterobacter taylorae* (Farmer et al. 1985b; type strain ATCC 35317 = CDC 2126-81) is a junior synonym of *E. cancerogenus* (Grimont and Ageron, 1989).

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 33241, CCUG 25231, CFBP 4167, CIP 103787, ICMP 5706, LMG 2693, NCBBP 2176.

GenBank accession number (16S rRNA): Z96078.

5. **Enterobacter cowanii** Inoue, Sugiyama, Kosako, Sakazaki and Yamai 2001, 1619^{VP} (Effective publication: Inoue, Sugiyama, Kosako, Sakazaki and Yamai 2000, 419.) *co.wa'ni.i.* M.L. gen. n. *cowanii* of Cowan, named after S.T. Cowan, British bacteriologist.

The cell morphology is as given for the genus. Colonies on nutrient agar are often yellow pigmented. Physiological and nutritional characteristics are presented in Table BXII.γ.220. Isolated from diverse clinical specimens but clinical significance is unknown. May be found in foods.

The mol% G + C of the DNA is: 53 (HPLC).

Type strain: 88-76, CIP 107300, JCM 10956.

6. **Enterobacter dissolvens** (Rosen 1922) Brenner, McWhorter, Kai, Steigerwalt and Farmer III 1988b, 220^{VP} (Effective publication: Brenner, McWhorter, Kai, Steigerwalt and Farmer III 1986, 1119 (*Phytomonas dissolvens* Rosen 1922, 497; *Erwinia dissolvens* (Rosen 1922) Burkholder 1948a, 472.) *dis.sol'vens.* L. part. adj. *dissolvens* dissolving.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are included in Tables BXII.γ.220 and BXII.γ.221. Presently, it is not possible to phenotypically differentiate *E. dissolvens* from *E. cloacae*.

The type strain belongs in DNA group 1 in the *E. cloacae* complex. Originally isolated from diseased corn.

The mol% G + C of the DNA is: 54 (Bd).

Type strain: ATCC 23373, CIP 105586, ICMP 1570, JCM 6049, LMG 2683, NCBBP 1850.

GenBank accession number (16S rRNA): AJ417485, Z96079.

The type strain belongs in the *Enterobacter cloacae* complex.

7. **Enterobacter gergoviae** Brenner, Richard, Steigerwalt, Asbury and Mandel 1980b, 1^{VP} *ger.go'vi.ae.* M.L. gen. n. *gergoviae* of Gergovie Highland, France.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Table BXII.γ.220. Occurs in various environmental sources such as cosmetics or water. Has also been recovered from clinical specimens.

The mol% G + C of the DNA is: 60 (Bd).

Type strain: ATCC 33028, CIP 76.1, DSM 9245, JCM 1234, LMG 5739, NCTC 11434.

GenBank accession number (16S rRNA): AB004748.

8. **Enterobacter hormaechei** O'Hara, Steigerwalt, Hill, Farmer, Fanning and Brenner 1990, 105^{VP} (Effective publication: O'Hara, Steigerwalt, Hill, Farmer, Fanning and Brenner 1989, 2048.) *hor.ma.e'che.i.* M.L. gen. n. *hormaechei* of Hormaeche, named after Estenio Hormaeche, Uruguayan microbiologist.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are pre-

sented in Tables BXII.γ.220 and BXII.γ.221. Oxidation of glucose into gluconate in the absence of PQQ is a characteristic of the *E. hormaechei* genomic group. Isolated from clinical specimens.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 49162, CCUG 27126, CIP 103441.

GenBank accession number (16S rRNA): AJ417450.

9. **Enterobacter kobei** Kosako, Tamura, Sakazaki and Miki 1997, 915^{VP} (Effective publication: Kosako, Tamura, Sakazaki and Miki 1996, 264.) *kobe.i.* M.L. gen. n. *kobei* pertaining to the city of Kobe, Japan.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Table BXII.γ.220. Isolated from diverse clinical specimens but clinical significance is unknown. May be found in foods.

The mol% G + C of the DNA is: 53 (*T_m*).

Type strain: ATCC BAA-260, CIP 105566, DSM 13645, JCM 8580.

10. **Enterobacter nimipressuralis** (Carter 1945) Brenner, McWhorter, Kai, Steigerwalt and Farmer 1988b, 220^{VP} (Effective publication: Brenner, McWhorter, Kai, Steigerwalt and Farmer III 1986, 1119 (*Erwinia nimipressuralis*) Carter 1945, 423; Dye 1969a, 83.)

ni.mi.pres.su.ra'lis. L. adv. *nimis* overmuch; L. n. *pressura* pressure; M.L. adj. *nimipressuralis* with excessive pressure.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Table BXII.γ.220. Reported as the causative agent of "wetwood" disease in elm trees (Carter, 1945).

The mol% G + C of the DNA is: 55 (Bd).

Type strain: ATCC 9912, CIP 104980, ICMP 1577, JCM 6050, NCPPB 2045.

GenBank accession number (16S rRNA): Z96077.

11. **Enterobacter pyrinus** Chung, Brenner, Steigerwalt, Kim, Kim and Cho 1993b, 161^{VP} *pyr'i.nus.* L. n. *pyrus* pear; L. suff. *-inus* belonging to; L. adj. *pyrinus* from pears.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Table BXII.γ.220. Isolated from brown leaf spot lesions on pear trees in Korea.

The mol% G + C of the DNA is: 57–61 (*T_m*).

Type strain: ATCC 49851, CFBP 4168, CIP 104019, DSM 12410, ICMP 12530.

12. **Enterobacter sakazakii** Farmer, Asbury, Hickman and Brenner 1980a, 575^{VP} *sa.ka.za'ki.i.* M.L. gen. n. *sakazakii* of Sakazaki; named after the Japanese bacteriologist, Riichi Sakazaki.

The cell and colonial morphology are as given for the genus. Physiological and nutritional characteristics are presented in Table BXII.γ.220. Occurs in the environment and in foods. May contaminate milk powder and subsequent nutritional preparations and then cause generalized infection (septicemia, meningitis) in newborns.

The mol% G + C of the DNA is: 57 (*T_m*).

Type strain: ATCC 29544, CCUG 14558, CIP 103183, DSM 4485, JCM 1233, LMG 5740, NCTC 11467.

GenBank accession number (16S rRNA): AB004746.

Genus XIII. Erwinia Winslow, Broadhurst, Buchanan, Krumwiede, Rogers and Smith 1920, 209^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1

LYSIANE HAUBEN AND JEAN SWINGS

Er. wī'ni.a. M.L. fem. n. *Erwinia* named after Erwin F. Smith.

Straight rods, 0.5–1.0 × 1.0–3.0 µm; occur singly or in pairs. Gram negative. Motile by peritrichous flagella. Facultatively anaerobic, but anaerobic growth by some species is weak. Optimum temperature, 27–30°C; maximum temperature for growth is 40°C. Oxidase negative. Catalase positive. **Pectinase negative.** Acid is produced from fructose, D-galactose, D-glucose, and sucrose but not from adonitol, arabitol, dextrin, dulcitol, inulin, maltose, starch, and tagatose. Utilize fumarate, D-galactose, gluconate, D-glucose, glycerol, β-methylglucoside, malate, and succinate, but not L-arabitol, benzoate, butanol, methanol, oxalate, propionate, or sorbose as carbon- and energy-yielding sources. Utilize L-alanine, L-glutaminic acid, glycylglycine, and L-serine, but not kynureninic acid and trigonelline, as nitrogen sources. Sensitive to chloramphenicol, furazolidone, nalidixic acid, oxytetracycline, and tetracycline. Do not possess arginine dihydrolase, caseinase, pectinase, phenylalanine deaminase, or urease. Associated with plants as pathogens, saprophytes, or constituents of the epiphytic flora.

The species of the genus *Erwinia* comprise a distinct phylogenetic group, as determined by 16S rRNA gene sequence comparisons, and are characterized by 14 generic signature nucleotides (Table BXII.γ.223).

Erwinia species cause plant diseases that include mainly blights and wilts. The pathogen usually starts to cause damage in the vascular tissue and then spreads throughout the plant. Ingress by the pathogen generally occurs through natural openings and wounds.

The mol% G + C of the DNA is: 49.8–54.1.

Type species: Erwinia amylovora (Burrill 1882) Winslow, Broadhurst, Buchanan, Krumwiede, Rogers and Smith 1920, 209 emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Micrococcus amylovorus* Burrill 1882, 134.)

FURTHER DESCRIPTIVE INFORMATION

Gas production is comparatively weak or absent. Decarboxylases for arginine, lysine, or ornithine cannot be detected by Møller's method (Møller, 1955). Formation of putrescine occurs when the amino acids are decarboxylated under aerobic conditions (Zherebilo and Gvozdyak, 1976). Glutamic acid is not decarboxylated. Lipases are rarely produced. Additional characteristics of the species of the genus are given in Tables BXII.γ.223, BXII.γ.224, and BXII.γ.225.

Fermentation end products from glucose are CO₂ and different combinations of succinate, lactate, formate, and acetate; some strains form 2,3-butanediol and some ethanol (White and Starr, 1971). Starch is not hydrolyzed beyond dextrins.

Sequence analyses on 16S rDNA of the species of the genus *Erwinia* by Kwon et al. (1997) and Hauben et al. (1998a) are in good agreement and place the genus *Erwinia* within the *Enterobacteriaceae*, closely related to the genera *Pantoea*, *Pectobacterium*, *Brenneria*, and *Enterobacter* (Fig. BXII.γ.201; Table BXII.γ.223).

Burrill (1882) was the first to attribute the cause of a plant disease to a bacterium, which he named "*Micrococcus amylovorus*". In successive taxonomic studies, the bacterium was renamed "*Ba-*

cillus amylovorus" (Trevisan, 1889a), "*Bacterium amylovorus*" (Chester, 1897), "*Bacterium amylovorum*" (Serbinov, 1915), and finally *Erwinia amylovora* (Winslow et al., 1920). *Erwinia* species cause plant diseases that include blights, die back, leaf spots, wilts, and crown rot. *Erwinia tracheiphila* overwinters in the bodies of cucumber beetles (*Diabrotica vittata* Fabr. and *D. duodecimpunctata* Oliv.) (Leach, 1964).

Phytopathogenicity of *Erwinia rhapontici* and *Erwinia persicina** may be due to the release of proferrosamines (Feistner et al., 1997).

ENRICHMENT AND ISOLATION PROCEDURES

In general, *Erwinia* can be easily isolated. The affected plant material is washed in tap water, followed by sterile water, and dried by paper toweling. Surface sterilization (3 min in 1:10 dilution of 5.25% active sodium hypochlorite) is sometimes detrimental for isolation. Affected tissue is removed from a young lesion or the edge of older necrotic areas by a sterile scalpel; the tissue is crushed in sterile water, saline, or buffer solution and is streaked onto a solid medium, such as nutrient agar or yeast dextrose chalk agar (YDC) (0.5% yeast extract, 1% dextrose, 3% calcium carbonate) (Dye, 1968). The isolation of *E. tracheiphila* is more easily accomplished by aseptically cutting the affected stem, placing the two cut stem surfaces together, and then gently pulling apart, removing a portion of the threads of bacteria and placing them in nutrient broth or onto a solid medium (Burkholder, 1960b). The delicate growth of *E. tracheiphila* will appear in 3–4 d; frequent transfer is necessary, but virulence may be reduced or lost after repeated transfers.

The isolation of some *Erwinia* species can be facilitated by use of selective-differential media, but such media are usually not necessary. *E. amylovora* will grow on MS medium (Miller and Schroth, 1972) and produce characteristic colonies. Later, Schroth and Hildebrand (1980) substituted D-mannitol by D-sorbitol in the MS medium for the isolation of *E. amylovora*. The medium of Crosse and Goodman (1973)¹ or CCT medium² can also be used for *E. amylovora*. A soluble pink pigment is produced by *E. persicina* and *E. rhapontici* (Hao et al., 1990) on medium containing 1% casitone, 0.3% yeast extract, 0.5% NaCl, and 0.7% agar.

MAINTENANCE PROCEDURES

Stock cultures of *Erwinia* species should be grown on standard media of choice at 25–30°C until good growth occurs. The cul-

*Editorial Note: *Erwinia persicina* was originally known as *Erwinia persicinus*; the name was corrected by Euzéby (1998).

1. Crosse and Goodman's medium consists of sucrose, 160 g; nutrient agar, 12 g; 0.1% crystal violet in absolute ethanol, 0.8 ml; 0.1% cycloheximide, 20 ml; 380 ml distilled water.

2. CCT medium consists of sucrose, 100 g; D-sorbitol, 10 g; 1% tergitol anionic 7, 30 ml; 0.1% crystal violet in absolute ethanol, 2 ml; nutrient agar (Difco), 23 g; 970 ml distilled water; after autoclaving add 2 ml 1% (w/v) thallium nitrate and 50 mg cycloheximide (Ishimaru and Klos, 1984).

TABLE BXII.γ.223. Diagnostic characteristics of the genera *Erwinia*, *Brenneria*, *Enterobacter*, *Pantoea*, and *Pectobacterium*^a

Characteristic	<i>Erwinia</i>	<i>Brenneria</i>	<i>Enterobacter</i>	<i>Pantoea</i>	<i>Pectobacterium</i>
<i>Biochemical characteristics:</i>					
Citrate (Simmons)	nd	nd	+	nd	D
Production of acetoin	d	d	D	+	+
Arginine dihydrolase	—	—	nd	nd	D
Arginine decarboxylase	nd	—	nd	nd	—
Lysine decarboxylase	nd	—	D	nd	—
Ornithine decarboxylase	nd	—	+	nd	—
Esculine hydrolase	D	D	nd	nd	+
Caseinase	—	—	nd	nd	D
Pectinase	—	D	nd	nd	D
Phenylalanine deaminase	—	—	—	nd	D
Urease	—	D	nd	nd	—
mol% G + C of DNA	49–54	50–56	52–60	55–57	50–56
<i>Acid production from:</i>					
Adonitol	—	—	D	nd	—
N-acetylglucosamine	D	d	nd	nd	+
Inulin	—	nd	nd	nd	D
D-Lyxose	D	nd	nd	nd	—
Maltose	—	D	nd	+	D
D-Mannitol	D	+	nd	+	—
D-Mannose	D	+	nd	+	+
L-Rhamnose	D	+	nd	nd	+
D-Ribose	D	+	nd	+	+
Salicin	D	+	nd	nd	+
Trehalose	D	D	nd	+	D
D-Xylose	D	D	nd	+	—
<i>Utilization of carbon sources:</i>					
Acetate	D	nd	nd	nd	+
4-Aminobutyrate	nd	nd	nd	+	—
Arbutin	D	+	nd	nd	+
Citrate	+	D	+	nd	+
Maltose	nd	D	nd	nd	D
Mannose	D	+	nd	nd	+
L-Proline	D	nd	nd	nd	nd
Sucrose	nd	d	nd	nd	+
Trehalose	nd	D	nd	nd	d
<i>16S rDNA signature nucleotides:</i>					
<i>E. coli</i> numbering position:					
379	C	G	C	C	C
384	G	C	S	G	G
408	A	R	A	A	G
434	Y	Y	T	T	C
593	Y	T	C	T	Y
598	C	Y	C	C	T
599	R	R	G	A	G
638	Y	Y	C	T	C
639	G	R	G	G	A
646	G	R	G	G	A
839	C	G	C	C	C
847	G	C	G	G	G
848	H	Y	M	A	C
987	G	A	R	A	R
988	G	C	S	C	C
989	C	T	T	T	T
1216	G	A	A	A	A
1217	C	R	S	G	G
1218	C	T	Y	T	T
1308	C	T	T	C	T
1329	G	A	A	G	A

^aFor symbols see standard definitions; nd, not determined.

tures can be maintained for a maximum of 3 weeks in a refrigerator (4–5°C).

For long-term preservation, erwiniae can be successfully stored as lyophilized cultures usually suspended in a filter sterilized mixture of 200 ml horse serum (Oxoid SR035C) to which

1675 mg nutrient broth (Oxoid) and 20 g glucose in 67 ml distilled water is added. Strains have also been stored in liquid nitrogen and in glycerol at –80°C (broth + 15% glycerol). Regular viability controls are recommended.

TABLE BXII.γ.224. Diagnostic characteristics of *Erwinia* species^a

Characteristic	1. <i>E. amylovora</i>	2. <i>E. billingiae</i>	3. <i>E. mallotivora</i>	4. <i>E. persicina</i>	5. <i>E. psidii</i>	6. <i>E. rhapontici</i>	7. <i>E. tracheiphila</i>
Pink diffusible pigment	—	nd	—	+	—	+	—
Mucoid growth	+	nd	+	nd	nd	+	—
Growth in KCN broth	—	nd	—	nd	nd	+	—
Growth in 5% NaCl	nd	nd	—	nd	nd	+	—
Growth factors required	+	nd	+	nd	nd	—	+
NO ₃ [−] → NO ₂ [−]	—	+	—	+	—	+	—
Levan	+	nd	+	nd	+	nd	+
H ₂ S from cysteine	—	nd	v	nd	+	+	+
Production of acetoin	+	+	+	+	—	v	d
Esculin hydrolase	+	nd	—	nd	nd	nd	nd
Gelatinase	+	—	v	nd	v	—	—
<i>Acid production from:</i>							
N-Acetylglucosamine	nd	+	nd	—	nd	nd	+
Amygdalin	nd	—	nd	—	nd	nd	+
L-Arabinose	d	+	—	—	+	+	v
D-Cellobiose	—	—	—	+	—	+	—
Dulcitol	—	nd	v	—	+	d	—
Esculin	—	nd	—	nd	nd	+	—
Gentiobiose	+	+	—	—	nd	nd	—
D-Gluconate	nd	+	nd	—	nd	nd	nd
α-Methylglucoside	—	—	—	—	—	d	v
β-Methylglucoside	+	nd	+	+	+	+	—
Glycerol	—	—	—	—	—	+	v
Myo-inositol	—	nd	v	—	+	+	—
Inulin	—	nd	—	—	nd	+	—
Lactose	—	—	—	+	—	+	—
Lyxose	nd	—	nd	—	nd	+	nd
Maltose	—	nd	—	nd	nd	+	—
D-Mannose	—	+	+	nd	+	+	v
Melezitose	—	—	—	—	nd	+	—
Melibiose	—	—	—	+	nd	+	—
Raffinose	—	—	v	+	—	+	—
L-Rhamnose	—	+	—	+	+	+	v
D-Ribose	+	nd	+	+	—	+	—
Salicin	—	nd	v	nd	+	+	—
Starch	—	nd	—	—	nd	+	—
Trehalose	+	+	v	+	—	+	—
Xylitol	nd	—	nd	—	nd	+	nd
D-Xylose	—	+	v	—	—	d	—
<i>Utilization of carbon sources:</i>							
Acetate	+	—	+	+	—	nd	+
L-Arabinose	—	nd	—	+	+	nd	v
D-Arabitol	—	+	—	—	—	—	—
Arbutin	nd	+	—	+	nd	nd	+
D-Cellobiose	nd	nd	—	+	nd	nd	—
Citrate	+	—	+	+	nd	+	+
Formate	+	nd	—	nd	nd	+	d
Galacturonic acid	—	nd	—	nd	nd	d	—
L-Glutamate	—	nd	—	+	+	nd	—
Glycerol	nd	nd	nd	+	nd	nd	—
L-Histidine	—	nd	—	+	—	nd	—
DL-Lactate	+	nd	—	+	nd	+	—
Lactose	nd	nd	nd	+	nd	nd	—
Malonate	—	—	—	nd	—	+	—
D-Mannose	—	nd	+	+	+	nd	v
Melibiose	+	nd	—	+	—	nd	—
Proline	—	nd	—	+	—	nd	—
Raffinose	—	nd	—	+	—	nd	—
Tartrate	—	nd	—	nd	—	d	v
D-Xylose	—	nd	+	nd	—	nd	—
<i>Utilization of nitrogen sources:</i>							
L-Isoleucine	+	nd	—	nd	nd	nd	nd
Threonine	+	nd	—	nd	nd	nd	nd
Tryptamine	nd	nd	—	nd	nd	+	—
Xanthin	nd	nd	+	nd	nd	nd	—
<i>Sensitivity toward:</i>							
Furazolidone	+	nd	—	nd	nd	nd	nd

^aFor symbols see standard definitions; nd, not determined.

TABLE BXII.γ.225. Additional characteristics of *Erwinia* species^a

Characteristic	1. <i>E. amylovora</i>	2. <i>E. billingiae</i>	3. <i>E. mallotivora</i>	4. <i>E. persicina</i>	5. <i>E. psidii</i>	6. <i>E. rhapontici</i>	7. <i>E. tracheiphila</i>
Levan	+	nd	+	nd	+	nd	+
Production of indole	—	—	—	nd	—	—	—
Oxidase	—	—	—	—	—	—	—
Catalase	+	+	+	+	+	+	+
Alkaline phosphatase	nd	nd	nd	+	nd	nd	nd
Arginine dihydrolase	—	—	—	—	—	—	—
Caseinase	—	nd	—	—	—	—	—
Arylsulfatase	nd	nd	nd	—	nd	nd	nd
Esterase	nd	nd	nd	—	nd	nd	nd
β-Galactosidase	nd	+	nd	nd	nd	nd	+
β-Glucosidase	nd	nd	nd	+	nd	nd	nd
β-Glucuronidase	nd	nd	nd	—	nd	nd	nd
Lysine decarboxylase	nd	—	nd	nd	nd	nd	—
Nucleoside phosphotransferase	nd	nd	nd	—	nd	nd	nd
Pectinase	—	—	—	—	—	—	—
Phenylalanine deaminase	—	nd	—	—	—	—	—
Starch hydrolase	—	nd	—	nd	—	nd	—
Tetrathionate reductase	nd	—	nd	—	nd	nd	nd
Tween 80 hydrolase	nd	—	nd	—	nd	nd	nd
Urease	—	—	—	—	—	—	—
<i>Acid production from:</i>							
Adonitol	—	—	—	—	—	—	—
Dextrin	—	nd	—	—	—	—	—
Dulcitol	—	—	—	—	—	—	—
Erythritol	nd	—	nd	nd	nd	nd	—
Fructose	+	+	+	+	+	+	+
Fucose	nd	—	nd	—	nd	nd	—
D-Galactose	+	+	+	+	+	+	+
D-Glucose	+	+	+	+	+	+	+
Inulin	—	—	—	—	—	—	—
2-Ketogluconate	nd	nd	nd	—	nd	nd	nd
5-Ketogluconate	nd	nd	nd	—	nd	nd	nd
Maltose	—	nd	—	—	—	—	—
Mannitol	v	+	+	nd	+	+	v
α-Methylmannoside	nd	—	nd	—	nd	nd	—
β-Methylxyloside	nd	—	nd	nd	nd	nd	—
Sorbitol	d	nd	v	+	+	+	v
Sorbose	nd	—	nd	—	nd	nd	—
Starch	—	—	—	—	—	—	—
Sucrose	+	nd	+	+	+	+	+
Tagatose	—	—	—	—	—	—	—
D-Turanose	nd	—	nd	—	nd	nd	nd
<i>Utilization of carbon sources:</i>							
N-Acetylglucosamine	nd	nd	nd	+	nd	nd	nd
Aconitate	nd	nd	nd	+	nd	nd	nd
Adipate	nd	nd	nd	—	nd	nd	—
Adipinic acid	nd	nd	nd	nd	nd	nd	—
Adonitol	nd	nd	nd	nd	nd	nd	—
DL-α-Alanine	nd	nd	nd	+	nd	nd	nd
Amygdalin	nd	nd	nd	+	nd	nd	nd
D-Arabinose	nd	nd	nd	—	nd	nd	nd
L-Arabitol	—	nd	—	—	—	—	—
L-Aspartate	nd	+	nd	+	nd	nd	nd
Benzoate	—	nd	—	—	—	—	—
Betaine	nd	nd	nd	nd	nd	nd	—
Butanol	—	nd	—	—	—	—	—
Butyrate	nd	nd	nd	—	nd	nd	nd
Caprylate	nd	nd	nd	—	nd	nd	nd
Cysteine	nd	nd	nd	—	nd	nd	nd
Dextrin	nd	nd	nd	nd	nd	nd	—
Dulcitol	nd	nd	nd	—	nd	nd	—
Erythritol	nd	nd	nd	+	nd	nd	nd
Esculin	nd	nd	nd	+	nd	nd	nd
Ethanol	nd	nd	nd	nd	nd	nd	—
Ethylene glycol	nd	nd	nd	nd	nd	nd	—
Fructose	nd	nd	nd	+	nd	nd	nd
Fucose	nd	nd	nd	—	nd	nd	nd
Fumarate	+	nd	+	+	+	+	+
Galactose	+	nd	+	+	+	+	+
Gluconate	+	nd	+	+	+	+	+
Glucose	+	nd	+	+	+	+	+
DL-Glycerate	nd	nd	nd	+	nd	nd	nd

(continued)

TABLE BXII.γ.225. (cont.)

Characteristic	1. <i>E. amylovora</i>	2. <i>E. billingiae</i>	3. <i>E. mallotivora</i>	4. <i>E. persicina</i>	5. <i>E. psidii</i>	6. <i>E. rhapontici</i>	7. <i>E. tracheiphila</i>
Glycerol	+	nd	+	+	+	+	+
L-Glycine	nd	nd	nd	—	nd	nd	nd
L-Histamine	nd	nd	nd	—	nd	nd	nd
myo-Inositol	nd	nd	nd	+	nd	nd	nd
Inulin	nd	nd	nd	—	nd	nd	nd
2-Ketogluconate	nd	nd	nd	+	nd	nd	nd
5-Ketogluconate	nd	nd	nd	+	nd	nd	nd
Lactulose	nd	nd	—	nd	nd	nd	—
L-Leucine	nd	nd	nd	—	nd	nd	nd
L-Lysine	nd	nd	nd	—	nd	nd	nd
Lyxose	nd	nd	nd	—	nd	nd	—
Malate	+	+	+	+	+	+	+
Malonic acid	nd	nd	nd	nd	nd	nd	—
Maltose	nd	nd	+	+	nd	nd	+
D-Mannitol	nd	nd	nd	+	nd	nd	nd
Melezitose	nd	nd	nd	—	nd	nd	nd
Methanol	—	nd	—	—	—	—	—
β-Methylglucoside	+	nd	+	+	+	+	+
Naphthalene	nd	nd	nd	nd	nd	nd	—
L-Ornithine	nd	nd	nd	—	nd	nd	nd
Oxalate	—	nd	—	—	—	—	—
Pectinic acid	nd	nd	nd	nd	nd	nd	—
Propanol	nd	nd	nd	nd	nd	nd	—
Propionate	—	nd	—	—	—	—	—
Propionic acid	nd	nd	nd	nd	nd	nd	—
L-Rhamnose	nd	nd	nd	+	nd	nd	nd
Ribose	nd	nd	nd	+	nd	nd	+
Salicin	nd	nd	nd	+	nd	nd	+
L-Serine	nd	nd	nd	+	nd	nd	nd
Sorbinic acid	nd	nd	nd	nd	nd	nd	—
Sorbitol	+	nd	nd	+	nd	nd	+
Sorbose	—	nd	—	—	—	—	—
Spermine	nd	nd	nd	—	nd	nd	nd
Starch	nd	nd	nd	—	nd	nd	—
Succinate	+	nd	+	+	+	+	+
Sucrose	nd	nd	nd	+	nd	nd	+
D-Tagatose	nd	nd	nd	—	nd	nd	nd
L-Threonine	nd	nd	nd	—	nd	nd	nd
Trehalose	nd	nd	nd	+	nd	nd	nd
Tryptamine	nd	nd	nd	—	nd	nd	nd
D-Turanose	nd	nd	nd	—	nd	nd	nd
Urea	nd	nd	nd	—	nd	nd	nd
L-Valine	nd	nd	nd	—	nd	nd	nd
Xylitol	nd	nd	nd	nd	nd	nd	—
<i>Utilization of nitrogen sources:</i>							
L-Alanine	+	nd	+	+	+	+	+
Allantoin	nd	nd	nd	nd	nd	nd	+
Ammonium chloride	nd	nd	nd	nd	nd	nd	+
Anthranilic acid	nd	nd	—	nd	nd	nd	nd
Arginine	nd	nd	nd	nd	nd	nd	+
Betaine	nd	nd	nd	nd	nd	nd	—
Choline	nd	nd	nd	nd	nd	nd	—
Citrulline	nd	nd	nd	nd	nd	nd	+
Cysteamine	nd	nd	nd	nd	nd	nd	—
Glucosamine	nd	nd	nd	nd	nd	nd	+
L-Glutaminic acid	+	nd	+	+	+	+	+
Glutathion	nd	nd	nd	nd	nd	nd	+
Glycylglycine	+	nd	+	+	+	+	+
Hydroxyproline	nd	nd	nd	nd	nd	nd	—
Kynureninic acid	—	nd	—	—	—	—	—
Leucine	nd	nd	nd	nd	nd	nd	+
L-Methionine	+	nd	nd	nd	nd	nd	nd
Phenylalanine	nd	nd	nd	nd	nd	nd	+
Quinolinic acid	nd	nd	nd	nd	nd	nd	—
Sarcosine	nd	nd	nd	nd	nd	nd	—
L-Serine	+	nd	+	+	+	+	+
Spermidine	nd	nd	nd	nd	nd	nd	—
Spermine	nd	nd	nd	nd	nd	nd	—
Thymine	nd	nd	—	nd	nd	nd	—
Trigonelline	—	nd	—	—	—	—	—
L-Tryptophan	nd	nd	+	nd	nd	nd	nd
Tyramine	nd	nd	nd	nd	nd	+	nd

(continued)

TABLE BXII.γ.225. (cont.)

Characteristic	1. <i>E. amylovora</i>	2. <i>E. billingiae</i>	3. <i>E. mallotivora</i>	4. <i>E. persicina</i>	5. <i>E. psidii</i>	6. <i>E. rhapontici</i>	7. <i>E. tracheiphila</i>
Tyrosine	nd	nd	nd	nd	nd	nd	+
Valine	nd	nd	—	nd	nd	nd	nd
<i>Sensitivity to antibiotics:</i>							
Amikacin	nd	nd	nd	+	nd	nd	nd
Amoxycillin	nd	nd	nd	nd	nd	+	+
Ampicillin	nd	nd	+	nd	nd	+	+
Carbenicillin	nd	nd	+	nd	nd	nd	+
Cephalexin	nd	nd	+	nd	nd	nd	+
Cephaloridine	nd	nd	+	nd	nd	nd	+
Cephalothine	nd	nd	+	nd	nd	nd	+
Chloramphenicol	+	nd	+	+	+	+	+
Clindamycin	nd	nd	nd	+	nd	nd	nd
Erythromycin	nd	nd	nd	+	nd	+	nd
Framycetine	nd	nd	nd	nd	nd	nd	+
Furazolidone	+	nd	+	+	+	+	+
Fusidinic acid	nd	nd	—	nd	nd	nd	—
Gentamicin	nd	nd	nd	+	nd	nd	nd
Kanamycin	nd	nd	+	nd	nd	nd	+
Lincomycin	nd	nd	nd	+	nd	nd	nd
Methicillin	nd	nd	nd	nd	nd	nd	—
Minocyclin	nd	nd	nd	+	nd	nd	nd
Nalidixinic acid	+	nd	+	+	+	+	+
Nitrofurantoin	nd	nd	nd	+	nd	nd	nd
Novobiocin	nd	nd	—	nd	nd	nd	nd
Oxytetracycline	+	nd	+	+	+	+	+
Penicillin G	nd	nd	nd	—	nd	nd	nd
Polymyxin B	nd	nd	nd	+	nd	nd	nd
Spectinomycin	nd	nd	+	nd	nd	nd	nd
Streptomycin	v	nd	+	+	nd	nd	nd
Sulfafurazol	nd	nd	—	nd	nd	nd	—
Tetracycline	+	nd	+	+	+	+	+

^aFor symbols see standard definitions; nd, not determined.

DIFFERENTIATION OF THE GENUS *ERWINIA* FROM OTHER GENERA

Characteristics that differentiate *Erwinia* from the genera *Pectobacterium*, *Pantoea*, and *Brenneria* are given in Table BXII.γ.223. As it is very difficult to differentiate them phenotypically; genomic methods are recommended for differentiation.

TAXONOMIC COMMENTS

The genus *Erwinia*, named after the phytobacteriologist Erwin F. Smith, was created in 1920 to unite all Gram-negative, fermentative, nonsporulating, peritrichously flagellated plant-pathogenic bacteria (Winslow et al., 1920). The taxonomy of the genus *Erwinia* and designation of species in the genus has been complicated by the heterogeneity of the strains included in the taxon. It has been suggested in the past that members of the genus be placed into new groupings with other members of the *Enterobacteriaceae* (Starr and Mandel, 1969; White and Starr, 1971). This concept also was supported by studies of DNA–DNA relatedness (Gardner and Kado, 1972), DNA relatedness (Brenner et al., 1974a), and DNA–DNA segmental relatedness (Murata and Starr, 1974).

Waldee (1945) suggested that *Erwinia* should be limited to pathogens (*E. amylovora*, *Erwinia salicis*, and *E. tracheiphila*) that cause necrotic or wilt diseases, utilize a restricted range of carbon compounds, and usually require organic nitrogen compounds for growth, and that the biochemically more active soft rotting pathogens (*Erwinia carotovora* and *Erwinia chrysanthemi*) should be placed in a separate genus *Pectobacterium*. Although some work-

ers have supported this suggestion (Brenner et al., 1973b, 1974a), it was not generally accepted because some species are taxonomically intermediate between these two groups, i.e., they resemble *E. carotovora* in most of their characteristics but do not cause rots.

Taxonomic changes since the previous edition of *Bergey's Manual of Systematic Bacteriology* The species *Erwinia herbicola*, *Erwinia milletiae* (Gavini et al., 1989b), *Erwinia ananas*, *Erwinia ure-dovora*, and *Erwinia stewartii* (Mergaert et al., 1993) have been transferred to the genus *Pantoea*. The species *incertae sedis* *Erwinia dissolvens*, *Erwinia nimipressuralis* (Brenner et al., 1986), and *E. cancerogena* (Dickey and Zumoff, 1988) have been transferred to the genus *Enterobacter*. Five new *Erwinia* species and two new subspecies were created: *Erwinia alni* (Surico et al., 1996), *Erwinia billingiae* (Mergaert et al., 1999), *Erwinia cacticida* (Alcorn et al., 1991), *Erwinia persicina* (Hao et al., 1990), *Erwinia psidii* (Rodrigues-Neto et al., 1987), *E. carotovora* subsp. *odorifera* (Gallois et al., 1992), and *E. carotovora* subsp. *wasabiae* (Goto and Matsumoto, 1987).

Confusion exists about the organisms named *Erwinia carnegieana*. None of the original isolates are available and no strains have been isolated corresponding to the original description (Alcorn and Orum, 1988). The type strain of *E. carnegieana* NCPPB 439 is nonpectinolytic and non-plant pathogenic and was identified as a *Klebsiella pneumoniae* (Graham, 1964; Edwards and Ewing, 1972; Alcorn and Orum, 1988).

Recently obtained sequence data allowed estimations of phylogenetic relationships, shed a new light on these earlier viewpoints, and raised new perspectives for the taxonomy of the genus

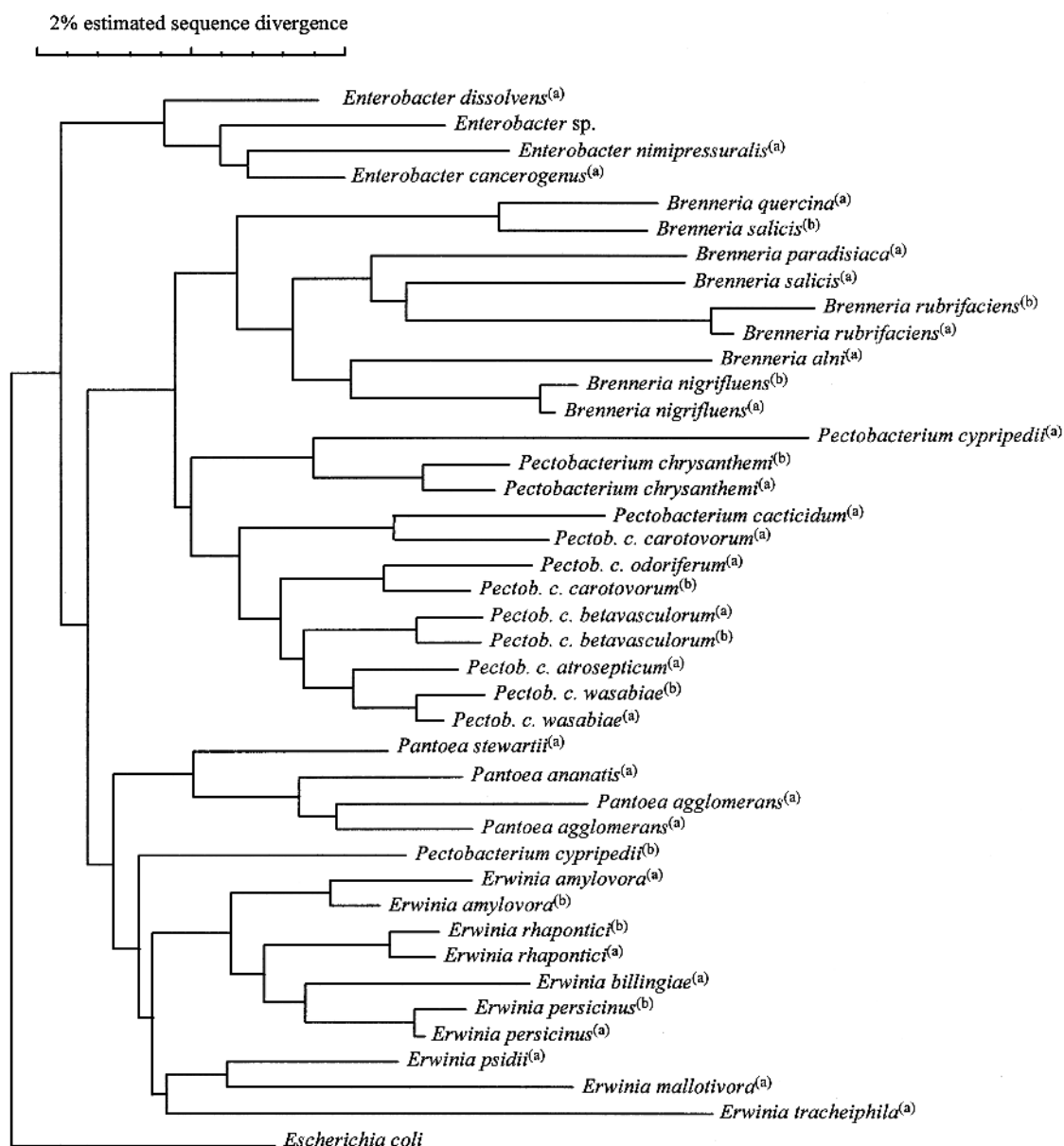


FIGURE BXII.γ.201. Neighbor-joining dendrogram depicting the estimated phylogenetic relationships among the genera *Erwinia*, *Pectobacterium*, *Brenneria*, *Pantoea*, and *Enterobacter*, based on pairwise comparisons of nearly complete 16S rDNA sequences calculated with the software package GeneCompar (Applied Maths, Kortrijk, Belgium), after removal of all unknown bases and gaps. The distance between two species is obtained by summing the lengths of the connecting horizontal branches using the scale on the top. *Pectob. c.* = *Pectobacterium carotovorum* subspecies; (a): 16S rDNA sequence obtained from Hauben et al. (1998a); (b): 16S rDNA sequence obtained from Kwon et al. (1997).

Erwinia. After comparison of the 16S rDNA sequences of representatives of the genus *Erwinia* with other members of the *Enterobacteriaceae*, the phylogenetic positions of the *Erwinia* spe-

cies were determined, and led to the emended description of the genera *Erwinia* and *Pectobacterium* and to the description of the new genus *Brenneria* (Hauben et al., 1998a).

DIFFERENTIATION OF THE SPECIES OF THE GENUS *ERWINIA*

The characteristics of the species of *Erwinia* are given in Tables BXII.γ.223, BXII.γ.224, and BXII.γ.225. Only four strains of *E.*

tracheiphila were studied, and data for this species should be treated with caution.

List of species of the genus *Erwinia*

1. ***Erwinia amylovora*** (Burrill 1882) Winslow, Broadhurst, Buchanan, Krumwiede, Rogers and Smith 1920, 209^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Micrococcus amylovorus* Burrill 1882, 134.)

a.my.lo'vo.ra. Gr. n. *amylum* starch; L. v. *voro* to devour; M.L. fem. adj. *amylovora* starch-destroying.

The characteristics are as given for the genus and as listed in Tables BXII.γ.224 and BXII.γ.225.

The virulence of *E. amylovora* is often correlated with its capacity to produce extracellular polysaccharides (or EPS, e.g., amylovoran), lipopolysaccharide, a high-molecular-mass levan, a low-molecular-mass glucan (Roberts and Coleman, 1991), and levansucrase (Gross et al., 1992). Levansucrase-negative mutants show retarded development of necrotic symptoms on inoculated pear seedlings (Geier and Geider, 1993). RcsA and RcsB are two regulator proteins that interact with the promoter of the operon for amylovoran biosynthesis (Bereswill and Geider, 1997; Kelm et al., 1997). The genes *amsA* to *amsI* are required for exopolysaccharide synthesis (Bugert and Geider, 1995, 1997).

After 2–3 days at 27°C, colonies on 5% sucrose nutrient agar are typically white, domed, shining, mucoid (levan type) with radial striations, and a dense flocculent center or central ring. Nonlevan forms are isolated rarely.

Several regions of the bacterial DNA of *E. amylovora* are species specific and can be used for identification by (1) a PCR assay with primers AMSbL (5'-GCTACCAGCAGGGT-GAG-3') and AMSbR (5'-TCATCACGATGGTGTAG-3') derived from the *ams* region, with an annealing temperature of 49°C (Bereswill et al., 1995); (2) a typical banding pattern derived from arbitrarily primed PCR with oligonucleotide APT1 (5'-CAGGACGCTACTTGTGT-3'), annealing temperature 32°C, from the sequence of transposon Tn5 (Bereswill et al., 1995); (3) a PCR assay with primers, primer A (5'-CGGTTTTTAACGCTGGG-3') and primer B (5'-GGGCAAATACTCGGATT-3'), annealing at 52°C, derived from a 29-kb plasmid (pEA29) that modulates the development of fire blight (Falkenstein et al., 1989; Bereswill et al., 1992); (4) nested PCR with primers A (5'-CGGTTT-TTAACGCTGGG-3'), B (5'-GGGCAAATACTCGGATT-3'), AJ75 (5'-CGTATTCACGGCTTCGCAGAT-3'), and AJ76 (5'-ACCCGCCAGGATAGTCGCATA-3') with an annealing temperature of 52°C, and PCR-dot-blot hybridizations derived from pEA29 (McManus and Jones, 1995); and (5) a PCR assay with primers EaF (5'-GCGCAGTAAAGGGTGACAGCCCCGTACACAAAAAGGCACT-3') and EaR (5'-CCCTAGCCGAAACAGTGTCTCTACCCCCGG-3') derived from the 23S rDNA sequence, with an annealing temperature of 72°C (Maes et al., 1996).

Repetitive element PCR (rep-PCR) (McManus and Jones, 1994) and RAPD fingerprinting (Momol et al., 1997) can distinguish *E. amylovora* strains isolated from Pomaceae from strains isolated from *Rubus* (subfamily Rosaceae). Pulsed field gel electrophoresis (PFGE) can distinguish *E. amylovora* strains of different geographical origin (Zhang and Geider, 1997).

E. amylovora produces harpins, bacterial protein elicitors that induce hypersensitive response-like necrosis when infiltrated into nonhost plants such as tobacco (Wei et al., 1992). Harpin activates a myelin basic protein kinase in

tobacco leaves (Popham et al., 1995; Adam et al., 1997) and elicits active oxygen production in suspension cells (Baker et al., 1993). The secretion of harpin occurs via a type III pathway, similar to the export system of pathogenic proteins of *Yersinia* spp. (Bogdanove et al., 1996). These proteins are encoded by the *hrp* gene cluster, consisting of seven transcriptional units (Wei and Beer, 1995; Kim et al., 1997b). The *dsp* genes also affect pathogenicity but not the eliciting of a hypersensitive response (Roberts and Coleman, 1991). The expression of the *hrp* and *dsp* genes is temperature-sensitive.

Plasmid pULB113 was found in *E. amylovora*, mediating chromosomal mobilization and R-prime formation (Chatterjee et al., 1985). The self-transmissible 56-kb plasmid Ea322 is not involved in pathogenicity (Steinberger et al., 1990). pEa8.7, an 8.7-kb plasmid was found in streptomycin- and sulfonamide-resistant strains (Palmer et al., 1997). The genes *strA* and *strB* were found in streptomycin-resistant strains (Chiou and Jones, 1991, 1995; McManus and Jones, 1994).

Cells infected with bacteriophage ERA103 produce an enzyme that degrades the extracellular polysaccharide of noninfected cells (Vandenburgh et al., 1985). *E. amylovora* cells are sensitive to bacteriophage Mu, which can be used as a genetic tool (Vanneste et al., 1990).

Several monoclonal antibodies have been produced against *E. amylovora* strains (Schaad, 1979) that can be used for the detection of the bacterium in suspensions as well as in contaminated twigs, plant fluids, and fruit tissues. The following methods can be applied: (1) Slide agglutination tests: agglutination with *E. amylovora* antiserum is a common and accurate method of determination (Lelliott, 1968; Zielke et al., 1993; Arsenijevic et al., 1994); the species is serologically homogeneous and has few agglutinogens in common with related species or with the saprophytes found in diseased material. (2) Immunofluorescence techniques (IFT) (Lin et al., 1987; Zielke et al., 1993). (3) Enzyme-linked immunosorbent assays (double-antibody sandwich [DAS]-ELISA) (Laroche and Verhoyen, 1984; Lin et al., 1987; Zielke et al., 1993). The sensitivity of the three methods is approximately the same, i.e., 10⁵ cells/ml.

Pantoea agglomerans and *Pseudomonas fluorescens* are antagonistic toward *E. amylovora* (Wilson et al., 1992; Wilson and Lindow, 1993; Wodzinski et al., 1994; Kearns and Hale, 1995, 1996; Mercier and Lindow, 1996).

E. amylovora causes a necrotic disease (fire blight) of most species of the Pomaceae and of some species in other subfamilies of the Rosaceae. Early in the season blossom and tips of twigs die back. Leaves shrivel and blacken and remain attached to the blackening stem that frequently crooks characteristically at the tip. The disease may progress into larger branches or even the trunk and form a canker. A *forma specialis* has been described from raspberry (*Rubus idaeus*) by Starr and Folsom (1951).

The mol % G + C of the DNA is: 53.6–54.1 (Bd).

Type strain: ATCC 15580, ICMP 1540, LMG 2024, NCPPB 683.

GenBank accession number (16S rRNA): Z96088, U80195.

2. ***Erwinia aphidicola*** Harada, Oyaizu, Kosako and Ishikara 1998, 1083^{VP} Harada, Oyaizu, Kosako and Ishikara 1997, 354

a.phi' di.co.la. L. *aphid* the aphid; L. suff. *-cola* dweller; M.L. n. *aphidicola* aphid dweller.

Cells are Gram-negative, oxidase-negative, catalase-positive, fermentative rods ($0.5\text{--}0.6 \times 1.6\text{--}2.0 \mu\text{m}$), and motile by means of peritrichous flagella. All strains grow well on nutrient agar and common laboratory media. Colonies on peptone-yeast extract agar are circular and smooth. These organisms grow well at 25, 30, and 35°C, but not at 42°C. Cells are motile. Voges–Proskauer test is positive, and cells can reduce nitrate. Cells can utilize citrate and malonate. Pigment is not produced. Indole is not produced. No gelatin liquefaction or tryptophan deaminase activity. H_2S production from sodium thiosulfate. No arginine hydrolase activity, no lysine decarboxylase activity, no ornithine decarboxylase activity, no deoxyribonuclease activity, and no urease activity. Acid is produced from the following carbohydrates: glycerol, erythritol, L-arabinose, D-ribose, D-xylose, adonitol, D-galactose, D-glucose, D-fructose, D-mannose, L-rhamnose, inositol, D-mannitol, N-acetyl-D-glucosamine, amygdalin, arbutin, esculin, salicin, cellobiose, maltose, melibiose, sucrose, trehalose, raffinose, D-xylitol, β -gentiobiose, D-lyxose, D-fucose, D-gluconate, 2-keto-D-gluconate, and 5-keto-D-gluconate.

The mol% G + C of the DNA is: unknown.

Type strain: IAM 14479.

3. ***Erwinia billingiae*** Mergaert, Hauben, Cnockaert and Swings 1999, 382^{VP}

bil.ling' i.ae. L. gen. n. *billingiae* of Billing, named after Eve Billing, who first isolated these organisms.

The characteristics are as given for the genus and are further listed in Tables BXII.γ.224 and BXII.γ.225.

Antisera prepared against live or heat-killed cells, non-purified or purified immunogens have been used for the differentiation or identification of *E. tracheiphila* (Elrod, 1946).

Strains were isolated from stem cankers, diseased blossoms, and immature fruits mainly of rosaceous trees, often in association with plant pathogens, and are considered as secondary invaders rather than primary pathogens.

The mol% G + C of the DNA is: 54.1–55.1 (T_m , Bd)

Type strain: LMG 2613, NCPPB 661.

GenBank accession number (16S rRNA): Y13249.

4. ***Erwinia mallotivora*** Goto 1976, 472^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1.

mal.lo.ti' vo.ra. M.L. n. *Mallotus* a genus of trees; L. v. *voro* to devour; M.L. adj. *mallotivora* *Mallotus*-destroying.

The characteristics are as described for the genus and are further listed in Tables BXII.γ.224 and BXII.γ.225.

Colonies on nutrient agar without sucrose are 1.5 mm in diameter, white, raised, transparent, and circular with smooth surfaces and entire margins after 2 d. Colonies on nutrient agar with 5% sucrose are flat, white, circular with entire margins and smooth surfaces, butyrous, and transparent after 1 d; after 4 d colonies are domed, circular, white, mucoid, and translucent, and sometimes possess radial striations.

Causes a leaf spot of Akamegashiwa, Mongolian oak (*Mallotus japonicus*). Initially, dark green spots appear on

the leaves next to the veins. As the disease evolves, these spots grow and turn dark brown with a yellow border. Eventually, the leaf dies and twigs wilt.

The mol% G + C of the DNA is: 49.8–51.0 (Bd).

Type strain: ATCC 29573, ICMP 5705, LMG 2708, NCPPB 2851.

GenBank accession number (16S rRNA): Z96084.

5. ***Erwinia persicina*** Hao, Brenner, Steigerwalt, Kosaka and Komagata 1990, 382^{VP} (“*Erwinia nulandii*”)

per.si' cin.a. M.L. adj. *persicina* peach color, because of the pinkish or peach-colored pigment of this organism.

The characteristics are as described for the genus and are further listed in Tables BXII.γ.224 and BXII.γ.225.

Strains vary in their ability to utilize adonitol, L-arginine, β -gentiobiose, and glucosamine. Resistance to ampicillin, carbenicillin, cefazolin, cephalothin, colistin, kanamycin, mydecamycin, nalidixic acid, sulfisoxazole, and tobramycin varies.

Strains were isolated from, but are not pathogenic for, tomato, banana, and cucumber. Strains are pathogenic to bean pods and seeds. One *E. persicina* strain was recently isolated from a human (O'Hara et al., 1998).

The mol% G + C of the DNA is: 52–54 (T_m).

Type strain: ATCC 35998, CDC 9108-82, LMG 11254.

GenBank accession number (16S rRNA): Z96086, U80205.

Additional Remarks: The type strain was isolated from a tomato by Komagata and Okada in 1962.

6. ***Erwinia psidii*** Rodrigues-Neto, Robbs and Yamashiro 1988, 328^{VP} (Effective publication: Rodrigues-Neto, Robbs and Yamashiro 1987, 348.)

psi' di.i. L. n. *Psidium* generic name.

The characteristics are as described for the genus and are further listed in Tables BXII.γ.224 and BXII.γ.225.

Pathogenic to guava (*Psidium guajava* L.) causing a die-back of twigs and branches, characterized by a collapse of the vascular tissue. In artificial inoculations, the bacterium was also pathogenic to strawberry guava (*Psidium cattleianum* Lam.), jambolana (*Eugenia jambolana* Lam.), and “mela-leuca” (*Melaleuca viridiflora* Brogn. and Gris.), and to all members of the Myrtaceae family.

The mol% G + C of the DNA is: 51.7–52.2 (T_m).

Type strain: ATCC 49406, ICMP 8426.

7. ***Erwinia pyrifoliae*** Kim, Gardan, Rhim and Geider 1999, 905^{VP}

py.ri' fo' liae. L. gen. fem. n. *pyrifoliae* of *pyrifolia*, the species name of the host plant, the Nashi pear, *Pyrus pyrifolia*.

Cells are Gram-negative, nonsporeforming, peritrichous, straight rods. The strains are facultatively anaerobic; oxidase is not produced. Nitrates are not reduced. This species conforms to the definition of the family *Enterobacteriaceae*. Strains grow on YPDA medium (1% yeast extract, 2% peptone, 2% glucose, and 20 $\mu\text{g}/\text{ml}$ adenine hemisulfate), producing colonies that are 2 mm after 48 h at 28°C. Colonies are circular, white, well-domed, and opaque. Glucose (dextrose) is fermented without gas production. Voges–Proskauer test is (weakly) positive. Polypectate gel is not acidified or liquefied. Arginine dihydrolase, lysine decarboxylase and ornithine decarboxylase are not present. Indole is not produced from tryptophan. Reducing compounds are produced from sucrose. Acid is not produced

from inulin, lactose, methyl- α -glucoside, melibiose, D-arabitol, D-arabinose, or raffinose. Acid is produced from mannitol, sorbitol, and saccharose. There is no alkalization of malonate, citrate, or D-tartrate. β -Galactosidase is not produced. The following substrates are utilized as sole sources of carbon and energy: D-fructose, D-galactose, D-trehalose, saccharose, methyl-D-glucopyranoside, D-ribose, L-arabinose, glycerol, *myo*-inositol, D-mannitol, D-sorbitol, L-malate, N-acetyl-D-glucosamine, D-gluconate, succinate, fumarate, L-glutamate, and L-proline. The following substances are not utilized: D-mannose, D-sorbose, α -D-melibiose, D-raffinose, maltotriose, maltose, α -lactose, lactulose, methyl- β -galactopyranoside, D-cellobiose, β -gentiobiose, esculin, D-xylose, palatinose, α -L-rhamnose, α -L-fructose, D-arabitol, L-arabitol, xylitol, dulcitol, D-tagatose, maltitol, D-turanose, adonitol, hydroxyquinoline β -glucoside, 3-methyl-D-glucopyranoside, D-saccharate, mucate, L-tartrate, D-tartrate, *meso*-tartrate, D-malate, *trans*-aconitate, tricarballoylate, D-glucuronate, D-galacturonate, keto-D-gluconate, L-tryptophan, phenylacetate, protocatechuate, *p*-hydroxybenzoate, quinate, *m*-hydroxybenzoate, benzoate, phenylpropionate, *m*-coumarate, trigonelline, betaine, putrescine, aminobutyrate, histamine, DL-lactate, caprate, caprylate, L-histidine, glutarate, DL-glycerate, aminovalerate, ethanolamine, tryptamine, itaconate, hydroxybutyrate, D-alanine, malonate, propionate, L-tyrosine, and ketoglutarate.

The pathogen was isolated from necrotic symptoms of leaves and branches of *Pyrus pyrifolia* cv. "Shingo". It is closely related to the fire blight pathogen *Erwinia amylovora* by the DNA sequence of the 16S rDNA, but distantly by its intergenic transcribed spacer region (ITS). It does not contain plasmid pEA29 (Rhim et al., 1999).

The mol% G + C of the DNA is: 52 (T_m).

Type strain: Ep16/96, CFBP 4172, CIP 106111, DSM 12163.

GenBank accession number (16S rRNA): AJ009930.

8. *Erwinia rhapontici* (Millard 1924) Burkholder 1948a, 475^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Phytomonas rhapontica* (sic) Millard 1924, 11; *Pectobacterium rhapontici* (Millard 1924) Patel and Kulkarni 1951b, 80; *Erwinia carotovora* biovar *rhapontici* (Millard 1924) Dye 1969a, 93.)

rha.pon'ti.ci. M.L. n. *rhaponticum* specific epithet of *Rheum rhaponticum*, rhubarb; M.L. gen. n. *rhapontici* of rhubarb.

The characteristics are as described for the genus and are further listed in Tables BXII. γ .224 and BXII. γ .225. Produces proferrosamine A.

Causes a crown rot of rhubarb (*Rheum rhaponticum*) that extends into the center of the root, pink grain of wheat that will show a cavity below the hilum (Roberts, 1974) and internal browning of hyacinth bulbs, and occurs epiphytically and saprophytically in lesions caused by other bacteria (Sellwood and Lelliott, 1978; Kokoskova, 1992). Rots potato, onion, and cucumber slices slowly, weakly, and erratically (Sellwood and Lelliott, 1978).

The mol% G + C of the DNA is: 51.0–53.1 (Bd).

Type strain: ATCC 29283, ICMP 1582, LMG 2688, NCPPB 1578.

GenBank accession number (16S rRNA): Z96087, U80206.

9. *Erwinia tracheiphila* (Smith 1895) Bergey, Harrison, Breed, Hammer and Huntoon 1923, 173^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Bacillus tracheiphilus* Smith 1895, 364.) *tra.che.i'phi.la.* L. n. *trachia* the windpipe; Gr. adj. *philus* loving; M.L. adj. *tracheiphila* trachea-loving, i.e., growing in the tracheiphila of the vascular bundles.

The characteristics are as given for the genus and are further listed in Tables BXII. γ .224 and BXII. γ .225.

Antisera prepared against live or heat-killed cells, non-purified or purified immunogens have been used for the differentiation or identification of *E. tracheiphila* (Elrod, 1946).

Grows very poorly on nutrient agar but moderately well on yeast extract glucose chalk agar (YCC) (10 g glucose, 5 g yeast extract, 30 g CaCO₃, 1.5% agar, 1 l distilled water) or glucose nutrient agar.

Causes a vascular wilt of *Cucurbita* species, *Cirullus lanatus*, and *Cucumis melo*. At first dull wilted areas appear on leaves. Later whole leaves and stems wilt, shrivel, and die. Infection is systemic and bacterial ooze is usually obvious at the cut ends of vascular tissues.

The mol% G + C of the DNA is: 50–52 (T_m , Bd).

Type strain: ATCC 33245, LMG 2906, NCPPB 2452.

GenBank accession number (16S rRNA): Y13250.

Genus XIV. *Ewingella* Grimont, Farmer, Grimont, Asbury, Brenner and Deval 1984a, 91^{VP}
(Effective publication: Grimont, Farmer, Grimont, Asbury, Brenner and Deval 1983a, 41)

CAROLINE M. O'HARA AND J.J. FARMER III

Ewing.el'la. M.L. dim. ending *-ella*; M.L. fem. n. *Ewingella* named to honor William H. Ewing an American bacteriologist who made many contributions to the nomenclature and classification of the families *Enterobacteriaceae* and *Vibrionaceae*.

Small rod-shaped cells, 0.6–0.7 $\times$ 1–1.8 μ m, conforming to the general definition of the family *Enterobacteriaceae*. Contains the enterobacterial common antigen. Motile by 3–10 peritrichous flagella. Facultatively anaerobic. Gram negative, oxidase negative, catalase positive. **Some strains grow faster and are more active biochemically at 25°C than 36°C. Positive reactions for methyl red, Voges-Proskauer, citrate utilization (Simmons), ONPG test, nitrate reduction to nitrite, and acid production from D-glucose, D-mannitol, salicin, trehalose, D-arabitol, D-mannose, and D-ga-**

lactose. Utilize 35 of 156 carbon sources. Negative reactions for indole production, H₂S production (TSI), urea hydrolysis, phenylalanine deaminase, lysine decarboxylase, arginine dihydrolase, ornithine decarboxylase, gelatin hydrolysis (22°C), growth in KCN, malonate utilization, acetate utilization, lipase (corn oil), DNase production, pigment production, and fermentation of sucrose, dulcitol, adonitol, *myo*-inositol, D-sorbitol, L-arabinose, raffinose, maltose, D-xylose, cellobiose, α -methyl-D-glucoside, erythritol, melibiose, and mucate. No visible gas is produced during

fermentation. Susceptible to colistin, nalidixic acid, sulfadiazine (most strains), gentamicin, streptomycin, kanamycin, tetracycline, chloramphenicol, and carbenicillin (disk diffusion method on Mueller-Hinton agar); **resistant to penicillin and cephalothin**; variable susceptibility to ampicillin.

Isolated from human clinical specimens that are normally sterile such as blood and urine. Also isolated from other clinical specimens such as throat, sputum, and wounds. **Isolated from outbreaks of bacteremia, and from outbreaks of pseudobacteremia traced to contaminated collection tubes.** Probably an **opportunistic pathogen** with a low capacity to cause extraintestinal infections in humans. Occurs in water, food, and mollusks. Implicated as the cause of internal stipe necrosis of cultivated mushrooms. **A rarely isolated genus of Enterobacteriaceae.**

The mol% G + C of the DNA is: 53.6–55.2.

Type species: **Ewingella americana** Grimont, Farmer, Grimont, Asbury, Brenner and Deval 1984a, 91 (Effective publication: Grimont, Farmer, Grimont, Asbury, Brenner and Deval 1983a, 41.)

FURTHER DESCRIPTIVE INFORMATION

Ewingella has been isolated from a variety of clinical specimens. In CDC's culture collection, blood and wound isolates account for 22 of 60 isolates. The blood isolates were usually from pneumonia or surgery patients. Five were from an outbreak of bacteremia in the intensive care unit of a community hospital (Pien and Bruce, 1986). All patients had undergone either cardiovascular or peripheral vascular surgery prior to infection. The probable source was a contaminated ice bath used to cool syringes for cardiac output determinations. *E. americana* was cultured from the bath. Twenty "blood isolates" and 15 environmental isolates were related to an outbreak of pseudobacteremia in a pediatric hospital that was traced to cross contamination from nonsterile citrated blood collection tubes (McNeil et al., 1987a). Additional case reports of bacteremia have been described by Pien et al. (1983) and DeVreese et al. (1992). In these reports, blood isolates were from immunocompetent patients who had undergone coronary bypass surgery and cholecystectomy, respectively.

Wounds of thumb, toe, hand, finger, and leg accounted for nine of 60 isolates. One isolate was thought to be a colonizer (Bear et al., 1986) of a leg wound, but there was insufficient information to evaluate the clinical significance of the other strains. Heizmann and Michel (1991) described a patient with conjunctivitis, and *E. americana* was isolated from both eyes. Strains of *E. americana* have also been isolated from respiratory specimens, including sputum, throat, and tracheal aspirate, and from urine and ear.

Clinical significance The clinical significance of *E. americana* and its ability to actually cause infections is still being evaluated. Systematic study and more good case reports are needed. The blood cultures and case reports suggest that *Ewingella* is an opportunistic pathogen. Only one isolate has come from feces, and there is no evidence that *E. americana* can actually cause diarrhea or intestinal infections.

Other isolates Environmental isolates have been from aqueous solutions in the hospital environment, water, mollusks (Müller et al., 1995a), and mushrooms (Inglis and Peberdy, 1996; Inglis et al., 1996).

Internal stipe necrosis of mushrooms In 1996 Inglis and Peberdy found that *E. americana* was often present on the com-

mercially cultivated mushroom *Agaricus bisporus*. In a second paper Inglis et al. (1996) presented evidence that *E. americana* caused this browning reaction known as internal stipe necrosis. This is a potentially serious disease to the mushroom industry in the United Kingdom, which has sales worth over 400 million dollars.

Isolation, identification, culture preservation, phenotypic characterization Strains of *E. americana* are not difficult to grow, and are typical *Enterobacteriaceae* in most respects. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, computer programs, and antibiotic susceptibility. Some strains of *E. americana* grow better, and are more active biochemically at 25°C than at 35°C.

Biochemical reactions and differentiation from other Enterobacteriaceae Table BXII.γ.193 in the introductory chapter on *Enterobacteriaceae* gives the results for *E. americana* in 47 biochemical tests normally used for identification (Farmer et al., 1985a; Farmer, 1999). There are no genus- or species-specific tests or sequences for identification. The best approach is to do a complete set of phenotypic tests (growth on plating media, biochemical tests, antibiogram, etc.) and compare the results with all the other named organisms in the family *Enterobacteriaceae*. This approach is described in more detail in the introductory chapter on *Enterobacteriaceae*. The biochemical results of a test strain can be compared to all the organisms listed in Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*. Several computer programs greatly facilitate analyzing the results. Table BXII.γ.226 may be helpful in differentiation.

TAXONOMIC COMMENTS

Discovery and DNA–DNA hybridization After the work had been completed on the characterization and description of the genus *Cedecea* (Grimont et al., 1981a), there remained a group of 10 strains that were phenotypically similar to, but distinct from, *Cedecea*. The group was named "Enteric Group 40" (Farmer, unpublished data; mentioned in Grimont et al., 1983a), and it was differentiated from *Cedecea* strains by a negative lipase (corn oil) reaction and several other tests. DNA–DNA hybridization indicated that Enteric Group 40 was less than 21% related to the other *Enterobacteriaceae*. Based on these data, Grimont et al. (1983a) proposed that Enteric Group 40 be reclassified as a new genus *Ewingella*, with a single species *Ewingella americana*.

Another genus named Ewingella In literature searches a source of possible confusion is the genus of radiolarians also named *Ewingella* (Pessagno, 1969).

Phylogeny based on 16S rDNA sequencing An unpublished sequence of 1398 bases has been deposited by P.W. Inglis for *E. americana* strain NCPPB 3905 (PI 98) described by Inglis and Peberdy (1996) and Inglis et al. (1996), which is a mushroom strain, not the type strain. A 16S rRNA tree that includes *E. americana* can be found in the section on *Budvicia* in this *Manual*. These 16S rDNA sequencing data agree with data from DNA–DNA hybridization, and both lead to the conclusion that *Ewingella* is distinct from other genera of *Enterobacteriaceae*. In the tree, *Rahnella* is the closest relative, in agreement with the tree published by Spröer et al. (1999), which shows these two genera on a distinct branch. However, the two trees differ in the placement of other organisms.

TABLE BXII.γ.226. Differentiation of *Ewingella americana* from several other *Enterobacteriaceae*^a

Test	<i>Ewingella americana</i>	<i>Cedecea davisae</i>	" <i>Enterobacter agglomerans</i> complex" ^b	<i>Rahnella aquatilis</i>	<i>Serratia marcescens</i>
Lipase (corn oil)	0 ^b	91	0	0	98
DNase at 25°C	0	0	0	0	98
Malonate utilization	0	91	65	73	3
Gas from D-glucose	0	70	20	98	55
Gelatin liquefaction at 22°C	0	0	0	0	90
<i>Fermentation of:</i>					
L-Arabinose	0	0	95	100	0
D-Arabitol	87	100	50	6	0
D-Sorbitol	0	0	30	96	99
Yellow pigment production at 25°C	0	0	75	0	0

^aEach number gives the percentage positive after 2 d incubation at 36°C (unless a different temperature is indicated) and is based on the data summarized by Farmer (2003). Most positive reactions occur within 24 h.

^bSee Farmer (2003) for a more detailed description of this "vernacular name". It is a term defined for practical identification in the clinical microbiology laboratory. It includes over a dozen DNA-DNA hybridization groups that were originally included in the species *Enterobacter agglomerans*, which is now known to be a heterogeneous species.

FURTHER READING

Grimont, P.A., J.J. Farmer III, F. Grimont, M.A. Asbury, D.J. Brenner and C. Deval. 1983. *Ewingella americana* gen. nov., sp. nov., a new *Enterobacteriaceae* isolated from clinical specimens. Ann. Microbiol. Inst. Pasteur (Paris) 134A: 39–52.

McNeil, M.M., B.J. Davis, S.L. Solomon, R.L. Anderson, S.T. Shulman, S. Gardner, K. Kabat and W.J. Martone. 1987. *Ewingella americana*: recurrent pseudobacteremia from a persistent environmental reservoir. J. Clin. Microbiol. 25: 498–500.

Pien, F.D. and A.E. Bruce. 1986. Nosocomial *Ewingella americana* bacteremia in an intensive care unit. Arch. Intern. Med. 146: 111–112.

List of species of the genus *Ewingella*

1. ***Ewingella americana*** Grimont, Farmer, Grimont, Asbury, Brenner and Deval 1984a, 91^{VP} (Effective publication: Grimont, Farmer, Grimont, Asbury, Brenner and Deval 1983a, 41.)
a.mer.i.can' a. M.L. adj. *americana* to denote that the original 10 strains were isolated in the United States of America.

The characteristics are as described for the genus and as listed in Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*. Occurs in human clinical specimens and the environment. May cause internal stipe necrosis of the mushroom species *Agaricus bisporus*, which is a potentially serious disease to the mushroom industry in the United Kingdom.

In the original study, Grimont et al. (1983a) noted that all 10 strains of *E. americana* were highly related by DNA-DNA hybridization. The relative binding percentages at both 60°C and 75°C were greater than 73%. For this reason a single species was defined and named. However, strain 0679-79 had a ΔT_m value of 5.5 at 60°C, which was much higher than all the other strains, which had values of 1.2–2.5.

Grimont et al. (1983a) defined two biogroups of *E. amer-*

icana based on three phenotypic differences. Biogroup 1 was defined to include nine strains, and does not ferment L-rhamnose and D-xylose, and is β -xylosidase negative. Biogroup 2 included strain 0679-79 described above, and was positive for these three tests. Phenotypically, this strain differed in other ways (Table II of Grimont et al., 1983a).

The original 10 strains (Grimont et al., 1983a) were very active biochemically at 36°C. However, other strains studied since 1983 have been biochemically more active at 25°C than at 36°C. These strains should be studied at both temperatures.

Additional studies are needed to determine if the unusual strains described above and the newly described strains from different habitats are genetically different from the type strain of *E. americana*.

The mol % G + C of the DNA is: 53.6–55.2 (T_m).

Type strain: ATCC 33852, CCUG 14506, CDC 1468-78, CIP 8194, DSM 4580, NCTC 12157.

Additional Remarks: The American Type Culture Collection includes four other strains from human clinical specimens: ATCC 33850, blood; ATCC 338514, sputum; ATCC 33853, gallbladder; ATCC 33854, thumb wound.

Genus XV. *Hafnia* Møller 1954, 272^{AL}

RIICHI SAKAZAKI

Haf ni.a. L. fem. n. *Hafnia* the old name for Copenhagen.

Straight rods, ~1.0 × 2.0–5.0 µm, conforming to the general definition of the family *Enterobacteriaceae*. Not encapsulated. Gram negative. **Motile by peritrichous flagella at 30°C**, but nonmotile strains may occur. Facultatively anaerobic. Oxidase negative. Catalase positive. **The majority of strains utilize citrate as a sole carbon source at 30°C** after 3–4 d of incubation. Nitrate is reduced to nitrite. H₂S is not produced in the butt of Kligler iron

agar. Gelatinase, lipase, and deoxyribonuclease are not produced. Alginate is not utilized. Pectate is not decomposed. Phenylalanine deaminase is not produced. **Lysine and ornithine decarboxylase tests are positive**, but the arginine dihydrolase test is negative. D-Glucose is fermented with the production of acid and gas. **Acid is not produced from D-sorbitol, raffinose, melibiose, D-adonitol, D-arabitol, and myo-inositol.** The methyl red

test is usually positive at 35°C and negative at 22°C. Acetyl-methylcarbinol is usually produced from D-glucose at 22–28°C but may not be produced at 35°C. Occur in the feces of humans and a wide range of animals including birds; also occur in sewage, soil, water, and dairy products. A member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 48–49.

Type species: *Hafnia alvei* Møller 1954, 272.

FURTHER DESCRIPTIVE INFORMATION

Members of *Hafnia* are able to grow at 35°C, but many of their physiological and biochemical activities at this temperature are irregular. The majority of strains are motile at 25–30°C, but may be nonmotile at 35°C. Although most strains fail to produce acetyl-methylcarbinol from D-glucose at 35°C, they give a positive Voges–Proskauer reaction when incubated at 22–28°C. They produce gas from D-glucose and about 60% of them grow on Simmons citrate agar after incubation for 3–4 d at 25°C, but in many strains these reactions may be negative at 35°C. Lactose is not fermented, but plasmid-mediated lactose-positive strains may occur (Le Minor and Coynault, 1976).

Members of *Hafnia* are defined as H₂S-negative organisms. Møller (1954) and Kauffmann (1954) defined the *Hafnia* group of *Bacteria* as H₂S-producing organisms, since the majority of strains of *Hafnia alvei* slightly darken ferric chloride-gelatin medium (Kauffmann, 1954) and SIM medium (Difco). However, they fail to blacken the butt of Kligler iron agar and of triple sugar iron agar. Ewing (1960) suggested, for the H₂S test of the family *Enterobacteriaceae*, that sensitivity of the test should be placed at a certain level, and that either Kligler iron agar or triple sugar iron agar serves this purpose because each easily permits group differentiation within the family.

The serology of *Hafnia* was first studied by Stuart and Rustigian (1943), who divided their cultures of biotype 32011, the majority of which are now classified in *Hafnia* (Sakazaki and Namioka, 1957), into eight serovars. Eveland and Faber (1953) studied serologically 58 strains of biotype 32011 and reported 21 somatic and 22 flagellar antigens. Deacon (1952) also carried out a serological study on 17 cultures of “*Aerobacter cloacae*” including biotype 32011 and recognized 12 somatic and 6 flagellar antigens among the cultures. Serology of cultures that were biochemically defined as members of the genus *Hafnia* was studied by Sakazaki (1961) and Matsumoto (1963, 1964). They established an antigenic scheme of the *Hafnia* group consisting of 68 O groups and 34 H antigens. However, the scheme is no longer available, because many test strains for O and H antigens have been lost. Independent of the scheme of previous workers, Baturo and Raginskaya (1978) have published another antigenic scheme including 39 O and 35 H antigens of *H. alvei*. Intergeneric relationships of O antigens were recognized between *Hafnia* cultures and other members of the family *Enterobacteriaceae* (Stamp and Stone, 1944; Eveland and Faber, 1953; Harada et al., 1957; Sakazaki, 1961; Matsumoto, 1963, 1964; Sedláč and Slajsova, 1966). Deacon (1952) reported the diphasic variation in the H antigens of the *Hafnia* strains he studied, but Sakazaki (1961) and Matsumoto (1963) failed to confirm such variation. Some *Hafnia* strains may have K antigen and alpha antigen of Stamp and Stone (1944) (Deacon, 1952; Emslie-Smith, 1961). Sakazaki (1961) suggested that the K antigen that inhibits the O agglutination was a slime antigen.

The majority of strains of *H. alvei* are susceptible to carbenicillin, streptomycin, gentamicin, kanamycin, tetracycline, polymyxin B, and nalidixic acid, but resistant to cephalosporins and

ampicillin. Washington et al. (1971) noted a definite difference between *H. alvei* and *Serratia liquefaciens* with respect to susceptibility testing of ampicillin and polymyxin B. They reported that most of the isolates of *Serratia liquefaciens* were susceptible to ampicillin at 20 µg/ml, while none of the *Hafnia* strains were. On the other hand, all of the *Hafnia* strains tested were susceptible to polymyxin B at 10 µg/ml, whereas only 6% strains of *Serratia liquefaciens* were.

A *Hafnia*-specific bacteriophage 1672 that provides a reliable tool for the identification of *Hafnia* strains was described by Guinée and Valkenburg (1968). They reported that the phage 1672 lysed all 100 strains of *H. alvei* tested, whereas it did not lyse strains of *Enterobacter*, *Klebsiella*, *Citrobacter*, *Serratia*, and *Salmonella*.

H. alvei occurs in humans and animals, including birds, and in natural environments such as soil, sewage, and water. McClure et al. (1957) described occurrence of the organisms in wild and caged birds, none of which appeared sick. Occasionally, the organisms cause illness. Kume (1962) described a case of equine abortion in which *H. alvei* was isolated from a fetus and lochia in pure culture. Gelev et al. (1990) reported epizootic hemorrhagic septicemia in rainbow trout associated with *H. alvei*.

Although it is difficult to assign a clear-cut clinical significance, *H. alvei* has been reported to cause septicemia (Englund, 1969; Mobley, 1971), respiratory tract infections (Washington et al., 1971; Klapholz et al., 1994; Fazal et al., 1997), meningitis (Mojtabae and Siadati, 1978), abscesses (Washington et al., 1971; Agustin and Cunha, 1995), urinary tract infections (Whitby and Muir, 1961), and wound infections (Washington et al., 1971; Berger et al., 1977). In most cases, however, it has been found in mixed culture and seems to be an opportunistic pathogen that produces infections in patients with some underlying illness or predisposing factors such as diabetes, chronic renal failure, chronic obstructive pulmonary disease, malignancy, and HIV infection. Washington et al. (1971), who reviewed the epidemiology of *H. alvei* isolated in their laboratory, concluded that the majority of the isolates originated in the respiratory tract and were considered to be commensals, while a few were secondary invaders. They also reported that previous administration of ampicillin or cephalosporins was a common feature of patients who acquired the organisms nosocomially.

There has been a controversy on whether *H. alvei* is enteropathogenic. Kauffmann (1954) suggested that the organisms of the *Hafnia* group are probably nonpathogenic for humans. Matsumoto (1963) reported the isolation of this organism from 13% of stool specimens from apparently healthy individuals. Ewing (1986a) reported that “members of this species are not known to be incitants of gastroenteritis.” On the other hand, earlier investigators (Stuart and Rustigian, 1943; Stuart et al., 1943; Deacon, 1952; Eveland and Faber, 1953) incriminated biotype 32011 of paracolon bacteria or non-lactose-fermenting “*Aerobacter cloacae*”, which are now classified in *Hafnia alvei*, as a causative agent of intestinal disorder. Harada et al. (1957) reported the isolation of *H. alvei* from sporadic cases of diarrheal diseases. Emslie-Smith (1961) suggested a possible role for *H. alvei* in gastroenteritis. In addition, Ratnam et al. (1979) reported the incrimination of *H. alvei* in a nosocomial outbreak of gastroenteritis. However, no conclusive evidence on enteropathogenicity of the *Hafnia* has been obtained. Albert et al. (1991, 1992) reported that strains of *H. alvei* isolated from diarrheal stools produced the attaching-effacing (AE) lesion in rabbits and HEp-2 cells resembling enteropathogenic *Escherichia coli* (EPEC), and that the virulence-associated gene *eaeA* was shared between EPEC and *H. alvei*. In addition, Ridell et al. (1995) found that *eaeA*-positive strains of

H. alvei have some characteristic properties in which they give negative reactions in the 2-ketogluconate and histidine assimilation tests and a positive reaction in the 3-hydroxybenzoate assimilation test in contrast with *eaeA*-negative strains. On the other hand, Ismaili et al. (1996) examined 11 strains of *H. alvei* isolated from children with diarrhea in Canada, and a Bangladesh *H. alvei* strain for AE lesion formation, presence of *eaeA* gene, profile of outer membrane protein extracts, chromosomal macrorestriction fragments, and plasmids. They found that, in contrast with the Bangladesh strain, which possessed the *eaeA* gene and forms the AE lesion, none of the Canadian strains had the *eaeA* gene, nor did they form the AE lesion. The outer membrane protein profile of all of the Canadian strains were identical to each other but differed from that of the Bangladesh strain. In addition, pulsed-field gel electrophoresis and plasmid profile analyses of the Canadian strains differed substantially from those of the Bangladesh strain. Their results indicate that there is heterogeneity among *H. alvei* strain factors associated with enteropathogenicity. Ridell et al. (1994) found an epidemiological association of *H. alvei* with diarrhea. *H. alvei* was isolated from 16% of adult Finnish tourists who visited Morocco and contracted diarrhea, but from 0% of tourists without diarrhea. *H. alvei* was isolated only from 2% of another group of adult Finnish patients with diarrhea. However, the authors found that those Finnish strains from diarrheal patients were negative for AE lesion formation and did not have the *eaeA* gene of EPEC. These recent papers suggest that some strains of *H. alvei* have the potential to cause diarrhea and that, although AE lesion formation is a virulence factor, mechanisms other than AE lesion formation may also be involved in the association of *H. alvei* with diarrhea.

ENRICHMENT AND ISOLATION PROCEDURES

H. alvei can grow on less selective isolation media for enterobacteria such as eosin-methylene blue, MacConkey, xylose-lysine-deoxycholate, and Hektoen enteric agars. On highly selective isolation media, such as Salmonella-Shigella and deoxycholate-citrate agars, many of the *Hafnia* cultures would be inhibited. The *Hafnia* organisms may not grow on bismuth sulfite agar. Colonies of *H. alvei* on less inhibitory plating agar media are relatively large, translucent, circular, low convex, and colorless, with a smooth surface and entire edge and resemble those of *Salmonella* (*Hafnia* strains are sometimes misidentified as *H₂S*-negative *Salmonella*). Some strains may produce pink colonies on media containing sucrose. Rarely does a strain produce mucoid colonies. For the differential isolation of *H. alvei*, MacConkey agar containing 1% sorbitol may be useful, on which colorless colonies of *H. alvei* can be distinguished from other intestinal inhabitants, because the former is lactose- and sorbitol-negative but the latter are usually positive for these sugars.

There are no selective enrichment broth media for the isolation of *H. alvei*. Many strains of *H. alvei* may grow in selenite and tetrathionate broth media, but some strains fail to grow in those broths. Ability of *H. alvei* to grow at lower temperature may be applied for cold enrichment, which is a useful procedure for the isolation of *Yersinia* strains.

MAINTENANCE PROCEDURES

Stock cultures may be maintained at room temperature in a semisolid medium consisting of 1.0% Bacto-casitone (Difco), 0.3% yeast extract, 0.5% NaCl, and 0.3% agar (pH 7.0). *Hafnia* strains remain viable for up to several years without subculture if the culture is sealed with a rubber stopper or a cork soaked

in hot paraffin wax. Strains may also be preserved by lyophilization.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

The specific bacteriophage 1672 was isolated from surface water with *H. alvei* 1672 as the propagating strain using the following method (Guinée and Valkenburg, 1968). A well-dried nutrient agar plate is surface-inoculated with a fresh broth culture of the strain to be tested. After decantation of the plate at room temperature for 15 min, a drop of the undiluted phage 1672 is spotted on the plate with a Pasteur pipette and the plate is again allowed to dry. Readings are made after 16–20 h of incubation at 35°C. Clear plaques with a diameter of 1–5 mm may be produced. The phage preparation is obtained after the usual purification and will contain around 10⁹ plaque-forming units/ml. The phage is not inactivated by heating at 60°C for 30 min.

The L-pyrrolidonyl aminopeptidase test is performed by dissolving 10 mg of L-pyrrolidonyl-β-naphthylamide (Sigma) in 10 ml of 95% ethanol. Dip sterile cotton swabs into the solution and rotate each swab several times with firm pressure on the inside wall of the tube to remove excess fluid. Dry the swabs overnight at 37°C and store at –20°C. Make a heavy suspension (MacFarland no. 1) of the test culture into 0.2–0.3 ml of 0.1 M phosphate buffer (pH 7.4), then dip the reagent swab into the suspension and incubate at 37°C for 10 min. Finally, add a drop of cinnamaldehyde solution.¹ A positive result is shown by a pink or red color.

The β-glucuronidase test is performed by dissolving 20 mg of 4-methylumbelliferyl-β-D-glucuronide (Sigma) in a small volume of dimethyl sulfoxide and bringing the volume to 10 ml with 0.1 M phosphate buffer (pH 7.4). Dip sterile cotton swabs into the solution and rotate each swab several times with firm pressure on the inside wall of the tube to remove excess fluid. Dry the swabs overnight at 37°C and store at –20°C. For testing, make a heavy suspension (MacFarland no. 1) of the test culture in 0.2–0.3 ml of 0.1 M phosphate buffer (pH 7.4). Dip the reagent cotton swab in the suspension and incubate at 37°C for 30 min. A positive reaction is indicated by development of blue fluorescence under long wave (360 μm) ultraviolet lamp.

DIFFERENTIATION OF THE GENUS *HAFNIA* FROM OTHER GENERA

Hafnia are often misidentified as members of *Enterobacter* or *Serratia*. Members of the genus *Hafnia* give a positive reaction in lysine and ornithine decarboxylase tests but are negative in arginine dihydrolase test. In differentiating *Hafnia* from *Enterobacter* and *Serratia*, in which some species show the same reactions in those tests, the failure of *Hafnia* to ferment raffinose, D-sorbitol, adonitol, and *D*-inositol may be valuable. Moreover, members of *Hafnia* have no activity of pyrrolidonyl aminopeptidase, although *Enterobacter* and *Serratia* possess this enzyme. Stuart et al. (1943) and Eveland and Faber (1953) described a close similarity between paracolon biotype 32011, which is now classified in the *Hafnia*, and *Salmonella* in biochemical reactions. Like *Salmonella*, *Hafnia* cultures are positive in tests of lysine and ornithine decarboxylases and negative in tests of indole, urease, and fermentation of lactose and sucrose, as well as in the Voges-Proskauer

1. Cinnamaldehyde solution contains *p*-dimethylaminocinnamaldehyde (Sigma), 0.5 g; glacial acetic acid, 3.5 ml; ethylene glycol-1-methylester, 6.0 ml; sodium lauryl sulfate, 3.5 g; and distilled water, 90 ml. Store in a screw-capped brown container in a refrigerator.

reaction at 35°C, although *Hafnia* cultures fail to produce H₂S. In addition, Eveland and Faber (1953) and Harada et al. (1957) reported that the *Hafnia* cultures are often agglutinated by *Salmonella* O antisera. *Hafnia* cultures may also be misidentified with some non-lactose-fermenting *Escherichia coli* strains, because the former gives a negative reaction in tests of Voges–Proskauer and citrate utilization at 35°C. However, *E. coli* produces indole and β-glucuronidase, while *Hafnia* is negative in these two tests. On the other hand, *Hafnia* is occasionally misidentified with Shiga toxin-producing *E. coli* O157:H7, because both organisms fail to ferment sorbitol and to produce β-glucuronidase. In addition, some strains of *H. alvei* are agglutinated by *E. coli* O157 antiserum (Aleksić et al., 1992). *Yokenella regensburgei*, which was first considered to be *Hafnia*, is another organism to be differentiated from the *Hafnia*. *Y. regensburgei* is different from *Hafnia* in its ability to utilize citrate, and to produce acid from cellobiose, melibiose, and D-arabitol. *Y. regensburgei* cannot be lysed by the *Hafnia* specific bacteriophage. Table BXII.γ.227 indicates the differential characteristics between *Hafnia* and biochemically similar taxa.

TAXONOMIC COMMENTS

The bacteria of the genus *Hafnia* have been described under several names: “*Bacillus asiaticus*” and “*Bacillus asiaticus*” biovar “mobilis” (Castellani, 1912, cited by Ewing and Fife, 1968), “*Bacterium cadaveria*” (Gale and Epps, 1943), biotype 32011 of the genus “*Aerobacter*” (Stuart et al., 1943), “*Enterobacter alvei*” (Sakazaki, 1961), *Enterobacter aerogenes* subsp. “*hafniae*” (Ewing, 1963), and “*Enterobacter hafniae*” (Ewing and Fife, 1968). However, *Hafnia alvei* (Møller, 1954) is the only correct name for these organisms. Møller (1954) found a new group of organisms, in which a supposedly authentic strain of “*Bacillus paratyphi-alvei*” of Bahr (1919) was included. He proposed the name *Hafnia alvei* for this group of bacteria because he considered that Bahr’s strain ought to be regarded as the type strain of this group. Sakazaki (1961) suggested the new combination “*Enterobacter alvei*” for *H. alvei*, because of its biochemical similarity to *Enterobacter*. Ewing and Fife (1968) pointed out that Bahr’s strain, which had been designated as the type strain of *H. alvei* by Møller (1954), was not an authentic strain of this species, since biochemical reactions of the strain were not the same as those described by Bahr (1919). They considered therefore that the specific epithet “*alvei*” was illegitimate, and proposed the name “*Enterobacter hafniae*” for *H. alvei*. However, there is no doubt that Bahr’s strain studied by Møller (1954) was a new bacterium at that time. In addition, numerical taxonomy studies by Johnson

et al. (1975) and Gavini et al. (1976a) indicated that *Hafnia* strains retained a position separate from *Enterobacter*. In DNA–DNA hybridization studies, Steigerwalt et al. (1976) reported 11–26% relatedness of *H. alvei* with *Enterobacter* and *Klebsiella*. These data appear to justify the status of the *Hafnia* as a separate genus rather than including it in the genus *Enterobacter*.

Although only a single species, *Hafnia alvei*, has been designated, Steigerwalt et al. (1976) indicated in DNA reassociation studies that there were two genomospecies that were about 50% interrelated within the genus *Hafnia*. Overall phenotypic profiles may serve to separate the second subspecies from the named species *H. alvei*; however, it has not been named. Farmer et al. (1985a) referred to this second genomospecies as *Hafnia alvei* biogroup 1.

Priest et al. (1973) proposed that *Obesumbacterium proteus* Shimwell 1964, a common brewery contaminant, should be placed in the genus *Hafnia* as “*H. protea*”. They described two groups in this species by numerical analysis of phenotypic characteristics. Brenner (1981) determined DNA relatedness in both groups and found that one group appears to be a biovar of *H. alvei*, whereas the other is a new species that does not belong to the genus *Hafnia*.

Another group resembling *H. alvei*, *Yokenella regensburgei*, was first considered as a species of the genus *Hafnia*. Kosako et al. (1984), who studied the phenotypic and genotypic characteristics of both groups, confirmed that the former was distinct from members of the *Hafnia* and suggested the name *Yokenella regensburgei* for this group.

ACKNOWLEDGMENTS

Riichi Sakazaki, who died in 2002, spent more than forty years at the Nippon Institute of Biological Sciences. His illustrious career focused on the classification and epidemiology of human and fish pathogens in Japan, particularly pathogenic bacteria in the families *Vibrionaceae* and *Enterobacteriaceae*. His contributions to Japanese bacteriology paralleled those of Drs. Edwards and Ewing in the U.S. He discovered and described many species in the genera *Enterobacter*, *Edwardsiella*, and *Vibrio*, and he developed serotyping schemes for *V. cholerae* and *V. parahaemolyticus*, as well as for several important species of *Enterobacteriaceae*. His research was always careful and comprehensive. He was one of the first Japanese scientists to publish in English language journals, thereby making his impressive accomplishments available to the world. He was recognized as one of the world’s foremost experts on the genus *Vibrio* and on many genera in *Enterobacteriaceae*. He was a member of the WHO Subcommittees on the Taxonomy of *Vibrionaceae* and on the Taxonomy of *Enterobacteriaceae* for more than thirty years. He authored chapters on these organisms in three editions of *Bergey’s Manual*. He was a good friend to the Bergey’s Trust, as well as to all who knew him.

TABLE BXII.γ.227. Differential characteristics of the genus *Hafnia* and biochemically related genera^a

Characteristic	<i>Hafnia</i>	<i>Enterobacter</i>	<i>Escherichia coli</i>	<i>Escherichia</i> strain STEC 0157 ^b	<i>Salmonella</i>	<i>Serratia</i>	<i>Yokenella</i>
H ₂ S production	—	—	—	—	+	—	—
Indole production	—	—	+	+	—	D	—
β-Galactosidase	D	+	+	+	D	+	+
β-Glucuronidase	—	—	+	—	D	—	—
L-Pyrrolidonyl aminopeptidase	—	+	—	—	—	+	+
Sorbitol fermentation	—	D	+	—	+	D	—
<i>Hafnia</i> specific bacteriophage lysis ^c	+	—	—	—	—	—	—
Mol% G + C of DNA	48–49	52–60	48–52		50–53	52–60	58–59

^aSymbols: +, 90–100% of strains are positive; —, 90–100% of strains are negative; D, different reactions given by different species or biogroups.

^bShiga toxin-producing *Escherichia coli* O157:H7.

^cData from Guinée and Valkenburg (1968).

DIFFERENTIATION OF THE SPECIES OF THE GENUS *HAFNIA*

Table BXII.γ.228 presents the biochemical characteristics of *Hafnia* species, which are also useful for differentiation between *H. alvei* and unnamed genospecies (genomospecies 2).

List of species of the genus *Hafnia*1. *Hafnia alvei* Møller 1954, 272^{AL}

al' ve.i. L. n. *alveus* a beehive; L. gen. n. *alvei* of a beehive.

The morphology is as given for the genus. Motility is most pronounced at 30°C. Nonmotile strains may be encountered occasionally.

Grows readily on ordinary media. Colonies on nutrient agar are generally 2–4 mm in diameter, smooth, moist, translucent, and gray, with a shiny surface and entire edge. Rare strains may produce mucoid colonies. Physiological and biochemical characteristics are presented in Table BXII.γ.228.

Found in the feces of humans and other animals, including birds. Also found in sewage, soil, water, and dairy products. It may be isolated in association with some pathological processes in patients with underlying illness. Some strains may cause diarrheal diseases.

The mol% G + C of the DNA is: 48.0–48.7 (*T_m*).

Type strain: ATCC 13337, DSM 30163, NCTC 8106.

GenBank accession number (16S rRNA): M59155.

TABLE BXII.γ.228. Characteristics of *Hafnia alvei* and genospecies 2^a

Characteristic	<i>H. alvei</i>	Genomospecies 2
Indole production	–	–
Voges–Proskauer test (25°C)	+	+
Voges–Proskauer test (35°C)	d	d

(continued)

TABLE BXII.γ.228. (cont.)

Characteristic	<i>H. alvei</i>	Genomospecies 2
Citrate (Simmons) (25°C)	d	–
Citrate (Simmons) (35°C)	–	–
Lysine decarboxylase	+	+
Arginine dihydrolase	–	–
Ornithine decarboxylase	+	d
H ₂ S production (Kligler)	–	–
Phenylalanine deaminase	–	–
Urease (Christensen)	–	–
β-Galactosidase	+	d
β-Glucuronidase	–	–
Malonate utilization	d	d
L-Pyrrolidonyl aminopeptidase	–	–
Gelatinase	–	–
Lipase (Tween 80)	–	–
Deoxyribonuclease	–	–
Growth in KCN medium	+	–
Esculin hydrolysis	–	–
Gas from D-glucose	+	–
Acid from carbohydrates:		
L-Arabinose, maltose, L-rhamnose, D-xylose	+	–
Lactose, melibiose, raffinose, sucrose ^b , adonitol, D-arabitol, dulcitol, myo-inositol, D-sorbitol, α-methyl-D-glucoside	–	–
Cellobiose	d	–
D-Mannitol	+	d
Trehalose	+	d
Salicin	d	d

^aFor symbols see standard definitions.

^bLate positive reactions are given by approximately 50% of the strains of *H. alvei*.

Genus XVI. *Klebsiella* Trevisan 1885, 105^{AL} emend. Drancourt, Bollet, Carta and Rousselier 2001, 930

PATRICK A.D. GRIMONT AND FRANCINE GRIMONT

Kleb.si.el' la. M.L. dim. ending -ella; M.L. fem. n. *Klebsiella* named after Edwin Klebs (1834–1913), a German bacteriologist.

Straight rods, 0.3–1.0 × 0.6–6.0 μm, arranged singly, in pairs or short chains; often surrounded by a **capsule**. Conform to the general definition of the family *Enterobacteriaceae*. Gram negative. **Nonmotile** (except *K. mobilis*). Facultatively anaerobic, having both a respiratory and a fermentative type of metabolism. Grow on meat extract media (except *K. granulomatis*, which has not been cultured), producing more or less dome-shaped, glistening colonies of varying degrees of stickiness depending on the strain and the composition of the medium. **Oxidase negative**. Glucose is fermented with the production of acid and gas (more CO₂ is produced than H₂), but anaerogenic strains occur. Most strains produce 2,3-butanediol as a major end product of glucose fermentation. The **Voges–Proskauer test is usually positive**. Lactic, acetic, and formic acids are formed in smaller amounts and ethanol in larger amounts than in a mixed acid fermentation. All

strains utilize L-arabinose, D-arabitol, D-cellobiose, citrate, D-fructose, D-galactose, D-glucose, 2-ketogluconate, maltose, D-mannitol, D-melibiose, D-raffinose, D-trehalose, and D-xylose as sole carbon sources. With the exception of some *K. pneumoniae* subsp. *ozaenae* strains, all *Klebsiella* strains **utilize myo-inositol, L-rhamnose, and sucrose as sole carbon sources**. With the exception of some *K. pneumoniae* subsp. *ozaenae* and *K. pneumoniae* subsp. *rhinoscleromatis* strains, all strains utilize lactose and D-sorbitol as sole carbon sources. No strain utilizes betaine, caprate, caprylate, glutarate, itaconate, 3-phenylpropionate, and propionate. H₂S is not produced, β-glucuronides are not hydrolyzed, and L-tryptophan and L-histidine are not deaminated. Ornithine is not decarboxylated by klebsiellae strains except *K. mobilis*, *K. ornithinolytica*, and rare strains of *K. pneumoniae*. Most strains hydrolyze urea and β-galactosides. Some strains fix nitrogen. Occur in intestinal

contents, clinical specimens from humans and animals (e.g., horses, swine, monkeys), soil, water, or on plants.

The mol% G + C of the DNA is: 53–58.

Type species: *Klebsiella pneumoniae* (Schroeter 1886) Trevisan 1887, 94 (*Hyalococcus pneumoniae* Schroeter 1886, 1952.)

FURTHER DESCRIPTIVE INFORMATION

Phylogenetic treatment The agent of donovanosis, *Calymmatobacterium granulomatis*, was found to be so closely related to *Klebsiella pneumoniae* by *rrs* (16S rRNA gene) sequence comparison that the species was transferred to the genus *Klebsiella* as *Klebsiella granulomatis* (Carter et al., 1999; Kharsany et al., 1999). However, it is not known whether *K. granulomatis* is a species distinct from *K. pneumoniae*.

The phylogenetic position of klebsiellae has been addressed by *rrs*, *rpoB* (Drancourt et al., 2001), *gyrA*, and *parC* (Brisse and Verhoef, 2001) sequence comparison.

The genus *Klebsiella* was found to be polyphyletic by Drancourt et al. (2001). Cluster I contained *Klebsiella pneumoniae* subsp. *pneumoniae*, *K. pneumoniae* subsp. *ozaenae*, *K. pneumoniae* subsp. *rhinoscleromatis*, and *K. granulomatis*. Cluster II contained *K. planticola*, *K. ornithinolytica*, and *K. terrigena*. Cluster III contained *K. oxytoca*. As a consequence, cluster II was proposed to constitute a new genus, *Raoultella* (Drancourt et al., 2001). The position of *Klebsiella mobilis* (*Enterobacter aerogenes*) was very close to cluster II, although the proposal by Drancourt et al. (2001) did not include that species.

A somewhat different structure was found by Brisse and Verhoef (2001), who uncovered two groups. The first group contained *K. pneumoniae* with its three subspecies, and the second group contained *K. oxytoca*, *K. planticola*, *K. ornithinolytica*, *K. terrigena*, and *K. mobilis*. Furthermore, three clusters (KpI to KpIII) were evidenced in *K. pneumoniae*, which did not correlate with the named subspecies. These clusters may have different habitats and different physiological properties (D-adonitol fermentation, for example). *K. oxytoca* was composed of two clusters, the significance of which is unknown.

Cell morphology and cell wall composition The outermost layer of *Klebsiella* bacteria consists of a large polysaccharide capsule, a characteristic that distinguishes members of this genus from most other bacteria in the family (*Escherichia coli* strains with a heat-stable K antigen may form similar capsules). The cell wall is structured similarly to other *Enterobacteriaceae*. Above the cytoplasmic membrane is the peptidoglycan layer and the outer membrane containing lipopolysaccharide (LPS). In addition, *Klebsiella* strains may possess fimbriae (pili) (Duguid, 1959).

The production of the large capsules gives rise to large mucoid colonies of a viscid consistency. The capsular material also diffuses freely into the surrounding liquid medium as extracellular capsular material.

Nutrition and growth conditions Best results with biochemical tests are obtained when *Klebsiella* cultures are incubated at 30–35°C. *Klebsiella* strains (except *K. granulomatis*) grow readily on all kinds of media commonly used to isolate *Enterobacteriaceae*, such as nutrient agar, tryptic casein soy agar, bromocresol purple lactose agar, Drigalski agar, MacConkey agar, eosin-methylene blue (EBM) agar, and bromothymol blue (BTB) agar. *K. pneumoniae* and *K. oxytoca* colonies are lactose positive, more or less dome-shaped, 3–4 mm in diameter after overnight incubation at 37°C or 30°C, with a mucoid aspect and sometimes stickiness depending on the strain and the composition of the medium.

K. planticola, *K. terrigena*, *K. ornithinolytica*, and *K. mobilis* (*Enterobacter aerogenes*) colonies are also lactose positive, 1.5–2.5 mm in diameter, dome-shaped, with a weakly mucoid aspect. *K. pneumoniae* subsp. *ozaenae*, *K. pneumoniae* subsp. *rhinoscleromatis*, and occasionally *K. pneumoniae* K1 strains grow more slowly on the same media, yielding voluminous, rounded, very mucoid, translucent, and confluent colonies in 48 h at 30°C or 37°C (Ørskov, 1981; Richard, 1982). Similar colonies, indistinguishable from those of *Klebsiella*, may be formed by other genera of the *Enterobacteriaceae*, particularly *E. coli* mucoid varieties with capsular K antigens.

The use of a carbohydrate-rich medium, such as bromothymol blue lactose agar or Hajna or Worfel-Ferguson medium, produces a better formed capsule than a carbohydrate-poor medium.

Almost all *Klebsiella* strains grow in minimal medium with ammonium ions or nitrate as sole nitrogen source and a carbon source, without growth factor requirement. Some *K. pneumoniae* K1 isolates require arginine or adenine or both as growth factors. *K. pneumoniae* subsp. *rhinoscleromatis* requires arginine and uracil but growth factor requirements of *K. pneumoniae* subsp. *ozaenae* are not fully determined (leucine, cysteine, and methionine are stimulatory). Ornithine can replace arginine for these requirements (Grimont et al., 1991).

K. granulomatis has not been grown axenically in artificial media. Cultures have been achieved *in vivo* in the yolk sac of developing chick embryo and in the developing chick embryo brain. Growth can be obtained in cell cultures, fresh mononuclear cells (Kharsany et al., 1997) or in Hep-2 human epithelial cell line (Carter et al., 1997).

Metabolism The characteristics useful for the identification of *Klebsiella* species are given in Table BXII.γ.229.

The Voges-Proskauer (VP) test is usually positive in *Klebsiella*, meaning that acetoin and 2,3-butanediol are produced during glucose fermentation. Some strains of *K. pneumoniae* subsp. *rhinoscleromatis*, do not form acetoin and 2,3-butanediol. In some strains, 2,3-butanediol is utilized and acetoin will disappear before the VP reaction is tested. Richard's modification of the VP test (Le Minor and Richard, 1993) most often gives positive results with *Klebsiella* strains (except *K. pneumoniae* subsp. *ozaenae* and subsp. *rhinoscleromatis*).

Carbon source utilization tests are conveniently obtained with Biotype-100 strips (BioMérieux, La Balme-les-Grottes, France) used with a minimal medium (Biotype Medium 1) containing 16 growth factors. *K. pneumoniae* subsp. *rhinoscleromatis* and some strains of *K. pneumoniae* subsp. *ozaenae* may need Biotype medium 2, which contains all known growth factors for enteric bacteria. The strips are examined for visual growth after 2 and 4 d (Grimont et al., 1991). Almost all *Klebsiella* strains utilize the following compounds as sole carbon and energy sources (provided growth factor requirements of some strains are met): *N*-acetyl-D-glucosamine, L-alanine, L-arabinose, D-arabitol, L-aspartate, D-cellobiose, citrate, D-fructose, L-fucose, fumarate, D-galactose, gentiobiose, D-gluconate, D-glucosamine, D-glucose, DL-glycerate, glycerol, *myo*-inositol, 2-ketogluconate, DL-lactate, lactose, D-malate, L-malate, maltose, maltotriose, D-mannitol, D-mannose, D-melibiose, 1-*O*-methyl-β-glucoside, L-proline, D-raffinose, D-ribose, L-serine, D-trehalose, and D-xylose. Esculin is hydrolyzed. H₂S, β-glucuronidase, phenylalanine deaminase, and tryptophan deaminase are not produced; DNA and tributyrin are not hydrolyzed. Almost no strain of *Klebsiella* can utilize the following com-

TABLE BXII.γ.229. Differentiation characteristics of the species and subspecies of the genus *Klebsiella*^a

Characteristics	1a. <i>K. pneumoniae</i> subsp. <i>pneumoniae</i>	1b. <i>K. pneumoniae</i> subsp. <i>ozaenae</i>	1c. <i>K. pneumoniae</i> subsp. <i>rhinoscleromatis</i>	3. <i>K. mobilis</i>	4. <i>K. oxytoca</i>	5. <i>K. planticola</i>	6. <i>K. terrigena</i>	Species <i>incertae sedis</i> <i>K. ornithinolytica</i>
Motility	—	—	—	+	—	—	—	—
Growth at:								
5°C	—	—	—	+	—	+	+	+
41°C	+	+	+	+	+	d	—	+
44.5°C	+	nd	nd	nd	d	—	—	nd
Urea hydrolyzed	+	d	—	—	+	+	+	+
Pectate hydrolyzed	—	—	—	—	+	—	—	—
ONPG test	+	+	—	+	+	+	+	+
Indol produced	—	—	—	—	+	d	—	+
Voges-Proskauer test	+	—	—	+	+	+	+	+
Malonate test	+	—	+	+	+	+	+	+
Lysine decarboxylated	+	d	—	+	+	+	+	+
Ornithine decarboxylated	—	—	—	+	—	—	—	+
Glucose dehydrogenase:								
Without added	+	—	—	+	—	—	—	nd
pyrroloquinoline quinone								
With added	+	—	—	+	+	+	+	nd
pyrroloquinoline quinone								
Gluconate dehydrogenase	+	—	—	+	—	+	+	nd
Utilization of:								
Adonitol	d	+	(+)	+	+	+	+	+
D-Alanine	+	+	—	+	+	+	+	+
L-Arabinol	—	—	—	—	d	—	—	—
Benzoate	d	—	—	+	d	+	d	+
m-Coumarate	d	d	—	—	+	+	—	+
Dulcitol	d	—	—	d	d	d	—	d
±Erythritol	—	—	—	—	—	(d)	—	d
D-Galacturonate	+	+	—	+	+	+	+	+
Gentisate	—	—	—	+	+	—	d	—
D-Glucuronate	+	+	—	+	+	+	+	+
Histamin	—	—	—	(+)	—	+	d	d
3-Hydroxybenzoate	—	—	—	+	+	—	(+)	—
4-Hydroxybenzoate	+	—	—	d	+	+	+	+
5-Ketogluconate	d	—	—	d	+	+	+	+
Lactulose	+	d	—	(+)	+	+	(+)	+
Malonate	d	—	d	d	(d)	(d)	—	d
Maltitol	+	+	—	+	+	+	+	+
D-Melezitose	—	—	—	—	d	—	+	—
1-O-Methyl-β-galactoside	+	+	—	+	+	+	+	+
3-O-Methyl-D-glucose	—	—	—	—	—	+	+	+
1-O-Methyl-α-D-glucoside	d	+	—	+	+	+	+	+
Mucate	+	d	—	+	+	+	+	(+)
Palatinose	+	+	—	+	+	+	+	+
Phenylacetate	d	d	—	+	+	+	+	+
Protocatechuate	+	d	—	+	+	+	+	+
Putrescine	d	d	—	d	(+)	+	(+)	+
Quinate	+	d	—	+	+	+	+	+
L-Rhamnose	+	d	+	+	+	+	+	+
D-Saccharate	+	d	d	+	+	+	+	(+)
D-Sorbitol	+	d	d	+	+	+	+	+
L-Sorbose	d	d	—	—	+	+	+	+
Sucrose	+	d	+	+	+	+	+	+
D-Tagatose	d	—	—	d	d	d	—	d
D-Tartrate	d	—	—	—	—	(d)	—	d
Tricarballoylate	d	—	—	+	+	—	+	—
Trigonelline	—	—	—	d	—	—	—	—
Tryptamine	—	—	—	d	—	—	—	—
D-Turanose	d	d	—	d	d	(d)	—	d
L-Tyrosine	—	—	—	d	—	—	—	—
D-Xylitol	d	—	—	d	(d)	d	—	d

^aSymbols: +, 95–100% strains positive in 1–2 d (utilization tests) or in 1 d (other tests); (+), 95–100% strains positive in 1–4 d; —, 95–100% strains negative in 4 d; d, different reactions; nd, not determined.

pounds as sole carbon and energy sources after 4-d incubation: betaine, caprate, caprylate, glutarate, itaconate, 3-phenylpropionate, and propionate. It is obvious from Table BXII.γ.229 that *K. pneumoniae* subsp. *ozaenae* and *K. pneumoniae* subsp. *rhinoscle-*

romatis are biotypes of *K. pneumoniae* with less nutritional versatility. There is no substrate utilized by subsp. *ozaenae* and subsp. *rhinoscleromatis* that is not utilized by subspecies subsp. *pneumoniae*. For this reason, it is necessary to check the identification

of these less active subspecies by capsular typing. Carbon source utilization tests are essential for the precise identification of *Klebsiella* species (Grimont et al., 1991). In Biotype-100 strips, *K. oxytoca* often produces a soluble yellow compound in all cupules showing some growth, and this may be analogous to the pigment produced by this species on ferric gluconate (Ørskov, 1984b). Growth on gentisate and 3-hydroxybenzoate never occurs with *K. pneumoniae* and *K. planticola*. Growth on histamine never occurs with *K. pneumoniae* and *K. oxytoca*. Growth on D-sorbose is always negative with *K. mobilis* (*E. aerogenes*) and growth on tri-carballylate is always negative with *K. planticola*. Only *K. terrigena* and some strains of *K. oxytoca* can grow on D-melezitose, and only *K. planticola* and *K. terrigena* can grow on 3-O-methyl-D-glucose. These carbon sources are often sufficient for the identification of all *Klebsiella* species (Grimont et al., 1991).

Oxidation of glucose to gluconate (mediated by glucose dehydrogenase) in the absence of added pyrroloquinoline quinone is a distinctive property of *K. pneumoniae* subsp. *pneumoniae* and *K. mobilis* (Bouvet et al., 1989). The other species require the addition of cofactor pyrroloquinoline quinone to express glucose dehydrogenase activity. Gluconate is oxidized into 2-ketogluconate (by a gluconate dehydrogenase) by *K. pneumoniae* subsp. *pneumoniae*, *K. planticola*, *K. terrigena*, and *K. mobilis*, but not by *K. oxytoca*, *K. pneumoniae* subsp. *ozaenae*, and *K. pneumoniae* subsp. *rhinoscleromatis*. *Klebsiella* species do not oxidize 2-ketogluconate to 2,5-ketogluconate (Bouvet et al., 1989).

K. pneumoniae subsp. *pneumoniae* can grow fermentatively on glycerol thanks to a glycerol dehydrogenase type I (induced by glycerol and dihydroxyacetone) and 1,3-propanediol dehydrogenase, which are typical enzymes of the anaerobic glycerol dissimilation pathway (Bouvet et al., 1995a). Other species of *Klebsiella* cannot grow fermentatively on glycerol and possess a glycerol dehydrogenase, but no 1,3-propanediol dehydrogenase (Bouvet et al., 1995a).

Some *K. oxytoca* and *K. planticola* strains are capable of fixing nitrogen and are classified as associative nitrogen fixers (Ladha et al., 1983). Nitrogen fixation is catalyzed by the enzyme nitrogenase in the absence of combined nitrogen, and under anaerobic conditions. Nitrogen fixation in the facultative anaerobe *K. pneumoniae* occurs only under anaerobic or microaerobic conditions. Low O₂ concentrations enhance nitrogenase synthesis; higher O₂ concentrations inhibit both synthesis and activity (Hill et al., 1984).

Genetic recombination has been reported in *Klebsiella* (Matsumoto and Tazaki, 1970), and *K. pneumoniae* has been used by several workers for detailed genetic analysis of the genes involved in N₂ fixation (*nif* genes). These genes are clustered near the *his* region on the chromosome but can be mobilized and transferred to other organisms. The genetics and regulation of nitrogen fixation in *Klebsiella* have been reviewed by Gussin et al. (1986).

Klebsiella strains may harbor a *lac* plasmid giving a stronger lactose positive phenotype (Reeve and Braithwaite, 1973).

Klebsiella strains may be lysogenic, but phages used by some workers for phage typing have been isolated from stools or sewage (Slopek et al., 1967; Slopek, 1978). Many *Klebsiella* strains produce bacteriocin (klebecin) and typing sets of such producers can be selected (Slopek and Maresz-Babczyzyn, 1967; Edmondson and Cooke, 1979).

Antigenic structure The outer membrane contains the lipopolysaccharide (LPS) that forms the O-antigen. The oligosac-

charide repeating units of the O-antigen consists of only a few carbohydrate components. Some of the O-antigens are homopolysaccharides such as galactans or mannans. There are only 12 different O-antigens. O-antigen determination is difficult because it is hampered by the stable K-antigens (Ørskov and Ørskov, 1984b).

Klebsiellae are enveloped by a polysaccharide capsule of considerable thickness. This constitutes the capsular or K-antigen. The capsular polysaccharide also diffuses freely into the liquid medium as extracellular capsular material. The *Klebsiella* capsular polysaccharides are acidic and composed of repeating units. Only a few carbohydrate constituents are found. Most K-antigens contain charged monosaccharide constituents such as glucuronic and hexoses, occasionally 6-deoxyhexoses. Noncarbohydrate constituents such as formyl, acetyl, or pyruvate groups are found. Following the pioneer work of Julianelle (1926), who determined the first three capsular types, a total of 82 K-antigens (K:1 to K:82) were described, although K:73, K:75, K:76, and K:78 were invalidated. There are thus only 78 available types. Cross-reactions among established K-antigens are numerous. K-antigen determination is traditionally by the capsular swelling reaction. When bacteria and the corresponding serum are mixed on a glass slide and observed under the microscope, the capsule becomes highly refractile and easily visible. Other methods include slide or tube agglutination, indirect immunofluorescence, and countercurrent immunoelectrophoresis (Ørskov and Ørskov, 1984b).

Some K-antigens are correlated with pathogenicity. *K. pneumoniae* subsp. *rhinoscleromatis* strains have capsular antigen K:3, whereas *K. pneumoniae* subsp. *ozaenae* strains are mostly K:4 and occasionally K:3 or K:5. *K. pneumoniae* subsp. *pneumoniae* strains recovered from respiratory tract infections have generally antigens K:1 to K:6, whereas strains from other nosocomial infections (e.g., urinary tract) have often antigens K:2, K:21, K:55, K:10, or K:24. Other K-types (17, 25, 22, 43, 1, 3, 33) can also be encountered (Le Minor and Richard, 1993). Strains found in the sperm of stallions and that cause metritis and abortion in mares contaminated with such sperm are often K:1, K:2, K:7, or K:30. Nosocomial *K. oxytoca* strains have been found with K:26, K:68, K:43, and less frequently K:26, K:21, K:18, or K:47. *K. mobilis* (*E. aerogenes*) strains most of the time can show capsular swelling with *Klebsiella* antisera. K-types associated with this species are often K:68 or K:26 and occasionally K:42, K:59, K:11, or K:4 (Le Minor and Richard, 1993).

Pathogenicity The genus *Klebsiella* is ubiquitous. Strains of this genus are found in surface water, sewage, soil, and on plants. They also colonize the mucosal surfaces of humans, horses, and swine (Podschun and Ullmann, 1998). In humans, *K. pneumoniae* is present as a commensal in the nasopharynx (detection rate, 1–6%) and in the intestinal tract (detection rate, 5–38%). In hospitals, colonization rates increase with the length of stay (detection rates, 77% in the stool, 19% in the pharynx, and 42% on the hands of patients) and hospital personnel can carry *K. pneumoniae* (Podschun and Ullmann, 1998). The high rate of colonization in patients is associated with the use of antibiotics.

The genus *Klebsiella* can be associated with different sorts of infections. *Klebsiella pneumoniae* is a cause of community-acquired bacterial pneumonia (Friedländer's pneumonia), occurring particularly in chronic alcoholics, and showing characteristic radio-

graphic abnormalities. The fatality rate is high if untreated (Podschun and Ullmann, 1998).

K. pneumoniae subsp. *rhinoscleromatis* is the causative agent of rhinoscleroma, a chronic infection that can involve the nasal cavity (most often) or the upper airways (pharynx, larynx, trachea). Typically, a granulomatous destructive and disfiguring process results in airway obstruction. Ultimately, extensive fibrosis and scarring occur. Histological features include mixed diffuse infiltrate of the upper and lower dermis with numerous plasmocytes, lymphocytes, neutrophils, and characteristic large macrophages with foamy cytoplasm (Mikulicz cells) and that contain *Klebsiella*.

K. pneumoniae subsp. *ozaenae* has been implicated in ozena, a chronic atrophic rhinitis giving off a very bad smell. However, the pathogenic role of the bacterium in this syndrome is less clear.

Klebsiella granulomatis (formerly *Calymmatobacterium granulomatis*) is the presumed causative agent of granuloma inguinale (donovanosis), a genital ulceration that is sexually transmissible. Histological features include dense dermal infiltrate of plasmocytes, neutrophils, and large macrophages with vacuolated cytoplasm that contain intracellular bacilli (Donovan bodies). The similarity of donovanosis and rhinoscleroma histological lesions has been pointed out (Carter et al., 1999).

Klebsiella spp. (with the exception of *K. pneumoniae* subsp. *rhinoscleromatis*, *K. pneumoniae* subsp. *ozaenae*, and *K. granulomatis*) are mostly considered nosocomial pathogens. *Klebsiella pneumoniae* and *K. mobilis* (*Enterobacter aerogenes*) are most frequently involved, although *K. oxytoca* and *K. planticola*, and rarely *K. terrigena*, can be found.

The urinary tract is the most common site of infection, and *K. pneumoniae* accounts for 6–17% of all nosocomial urinary tract infections (Podschun and Ullmann, 1998). *K. pneumoniae* is also a frequent cause of bacteremia. In premature infants, *K. pneumoniae* is often involved in neonatal sepsis.

The principal reservoir of *K. pneumoniae* in the hospital is the gastrointestinal tract of patients. The principal vectors are the hands of personnel. Occasional sources are contaminated medical equipment and blood products (Podschun and Ullmann, 1998).

Renewed interest in *Klebsiella* infections stemmed from the spread of multiresistant strains. *K. pneumoniae* is naturally susceptible to aminoglycosides. However, plasmids mediating aminoside-modifying enzymes have spread in the 1970s. In a French survey, 78% of *Klebsiella* isolates were susceptible to gentamicin, tobramycin, netilmycin, and amikacin. Different phenotypes of aminoside resistance were observed, involving resistance to gentamicin, tobramycin, netilmycin, and/or amikacin (Lemozy et al., 1985). The most frequent aminoside inactivating enzymes found produced by *Klebsiella* isolates were aminoglycoside-3'-N-acetyltransferases and aminoglycoside -2''-O-nucleotidyltransferase (Lemozy et al., 1985).

Strains of *K. pneumoniae* and *K. oxytoca* that have not acquired any resistance determinant are naturally resistant to aminopenicillins (ampicillin) and carboxypenicillins (carbenicillin) and susceptible to other β -lactam antibiotics. This is due to the production of a chromosomal penicillinase, which is inhibited by clavulanic acid. A small zone of inhibition around 100-mg carbenicillin disks is typical of this phenotype (Jarlier, 1985). Acquired resistance arose from the production of a plasmid-determined penicillinase. The strains showed a higher resistance to carbenicillin (no zone around carbenicillin disks), and resistance

to ureidopenicillins, cefalothin, cefamandole, and cefuroxime. Inactivation of amoxycillin was not blocked by clavulanic acid. However, these strains remained susceptible to third-generation cephalosporins (e.g., cefotaxime) (Jarlier, 1985). Since 1982, strains producing extended spectrum β -lactamases (ESBL) encoded by a plasmid have spread. *K. pneumoniae* strains resistant to ceftazidime were commonly producing a SHV-5 type β -lactamase in Europe and a TEM-10 or TEM-12 type in the United States. These strains account for 5% (USA) to 16% (France) of the *K. pneumoniae* tested (Podschun and Ullmann, 1998). These ESBL-producing strains have been susceptible to carbapenems (imipenem, meropenem). However, ESBL-producing *K. pneumoniae* resistant to imipenem have been isolated. They produce a β -lactamase of the AmpC type (Podschun and Ullmann, 1998). *Klebsiella mobilis* (*Enterobacter aerogenes*) was rarely encountered as a nosocomial pathogen until strains carried a plasmid encoding an ESBL. In France and Belgium, β -lactamases of type SHV-4, TEM-24, and TEM-3 were characterized in *K. mobilis*. It was shown that TEM-24 producing isolates found in several epidemics in different countries constituted mostly a single clone (Galdbart et al., 2000; De Gheldre et al., 2001; Mammeri et al., 2001).

Pathogenesis *Klebsiella* species are surrounded by a hydrophilic polysaccharide capsule, which is the first virulence factor described in klebsiellae (Toennissen, 1914; Cryz et al., 1984). Simoons-Smit et al. (1984) showed that strains with K1, K2, K4, and K5 capsular antigens were more virulent for mice (skin model) than strains with K6 and K above 6. The loss of K antigen by K⁺ strains resulted in the reduced virulence of their K⁻ variants. This reduced virulence may be explained by a higher degree of phagocytosis as measured by chemiluminescence response of human polymorphonuclear leukocytes (PMNLs) and by enhanced killing by either human PMNLs or human serum or both (Simoons-Smit et al., 1986). Other studies (mouse peritonitis model) found strains with antigens K1 and K2 more virulent than strains with other K antigens (Podschun and Ullmann, 1998). The K antigen plays a crucial role in protecting the bacterium from opsonophagocytosis in the absence of specific antibodies (Williams et al., 1983, 1986b; Simoons-Smit et al., 1986). The antiphagocytic function consists of inhibiting the activation or uptake of complement components, especially C3b. In addition, the K antigen inhibits the differentiation and functional capacity of macrophages *in vitro*. The degree of virulence conferred by a particular K antigen might be connected to the mannose content of the capsular polysaccharide (Podschun and Ullmann, 1998). K2 serotype is among the most common K-type isolated from patients with urinary tract infection, pneumonia, or bacteremia. K2 is the predominant serotype of human isolates worldwide, whereas K2 strains are very rarely encountered in the environment (Podschun and Ullmann, 1998).

Strains of *K. pneumoniae*, *K. oxytoca*, *K. planticola*, and *K. terrigena* may produce thick, channeled (type-1) fimbriae associated with mannose-sensitive hemagglutination (MS-HA). *K. mobilis* (*E. aerogenes*) produces antigenically similar fimbriae. Type 1 fimbriae mediate the attachment of *K. pneumoniae* to uroepithelial cells (Williams and Tomas, 1990) and to ciliated tracheal cells *in vitro* (Fader et al., 1988).

Strains of *K. pneumoniae*, *K. oxytoca*, *K. planticola*, and *K. terrigena* may also produce thin, nonchanneled (type-3) fimbriae associated with mannose-resistant hemagglutinins of the *Klebsiella* type (MR/K-HA). These fimbriae are efficient in promoting adherence to the roots of various grasses and cereals (Haahtela and

Korhonen, 1985). They are capable of binding to endothelial cells, respiratory tract epithelial cells, and uroepithelial cells. In the kidney, they mediate bacterial adhesion to tubular basement membranes, Bowman's capsules, and renal vessels (Podschun and Ullmann, 1998). Three new putative colonization factors have also been described (Podschun and Ullmann, 1998).

The most important role of the O antigen is to protect *K. pneumoniae* from complement-mediated killing (Williams and Tomas, 1990). For this protection, O antigen chain length seems to be important (McCallum et al., 1989).

Iron is essential for bacterial growth. Brewer et al. (1982) observed that virulence was enhanced by hyperferremi. Since in the human body iron is complexed to carrier molecules such as transferrin (in the serum) or lactoferrin (in milk and other secretions), or sequestered within cells (in heme proteins), potentially pathogenic *Enterobacteriaceae* produce high-affinity systems (siderophores) to solubilize and import the required iron. The iron-chelating compounds produced are mostly of two sorts, phenolates (e.g., enterochelin) and hydroxamates (aerobactin) (Payne, 1988). Almost all strains of *Klebsiella* produce enterochelin whereas only a few produce aerobactin (Williams et al., 1987). Nassif and Sansonetti (1986) correlated the virulence of *K. pneumoniae* serotypes K1 or K2 with the presence of a 180-kb plasmid encoding the hydroxamate siderophore aerobactin. Aerobactin is an essential factor of pathogenicity and the 180-kb plasmid carries additional genes encoding other virulence factors (Nassif and Sansonetti, 1986). Some strains of *K. pneumoniae* express a ferric aerobactin uptake system without making the chelator itself. This may confer a selective advantage in mixed infections in competition with other aerobactin-producing bacteria (Williams and Tomas, 1990).

The production of cytotoxins, enterotoxins, and hemolysins have been sporadically described (Podschun and Ullmann, 1998).

Ecology *Klebsiellae* have been recovered from aquatic environments receiving industrial wastewaters, plant products, fresh vegetables, food with a high content of sugars and acids, frozen orange juice concentrate, sugar cane wastes, living trees, plants, and plant by-products. They are commonly associated with wood, saw dust, and waters receiving industrial effluents from pulp and paper mills and textile finishing plants. Isolates have been described in forest environments, degrading kraft-lignin, tannic acid, pine bark, and condensed tannin, or from living or decaying wood and bark or composted wood.

Klebsiella can frequently be isolated from the root surfaces of various plants. *K. pneumoniae*, *K. oxytoca*, or *K. planticola* are capable of fixing nitrogen and are classified as associative nitrogen fixers.

Strains from plants certainly need to be reidentified in the light of present taxonomic schemes. Strains of *K. pneumoniae sensu stricto* that are associated with plants differ from those associated with serious human infections. These environmental *K. pneumoniae* strains are most often able to utilize 5-ketogluconate as sole carbon source and never have capsular types K1 to K6. Strains involved in serious infection do not utilize 5-ketogluconate and may have capsular types K1 to K6 as well as other capsular types (Grimont et al., 1991).

Capsular types K1, K2, and K5 were major causes of epidemic metritis in mares in England, whereas type K7 was associated with sporadic, opportunistic genital infection. Outbreaks of metritis in mare were due to type K2 in the United States and in France.

The stallion plays an important role in the transmission of *K. pneumoniae*. Type K7 was found on the preputial skin of stallions and may be part of the normal bacterial flora in this location. Thus it is important to type *Klebsiella* isolated in the genital tract of horses to detect a stallion carrying an epidemic strain among other stallions carrying less pathogenic *K. pneumoniae* (Grimont et al., 1991).

Klebsiella have been frequently associated with bovine mastitis, and causes serious infections in other animals including rhesus monkeys, guinea pigs, or muskrats. Epidemics of fatal generalized infections among captive squirrel monkeys (*Saimiri sciureus*) in French Guyana and lemurs in a French zoo were due to *K. pneumoniae* K5 and K2, respectively. Immunization of the monkeys with the corresponding capsular polysaccharide is efficient in stopping the epidemic (Grimont et al., 1991).

ENRICHMENT AND ISOLATION PROCEDURES

The detection, isolation, and enumeration in clinical, industrial, and natural environments can be facilitated by using a selective medium.

On agar plates, although colonies develop overnight, the characteristic elevated and mucoid appearance is observed after incubation for 48 h.

The ability to utilize citrate (Cooke et al., 1979) or *myo*-inositol (Legakis et al., 1976) has been applied to the formulation of selective media. Resistance of *Klebsiella* spp. to methyl violet (Campbell and Roth, 1975), double violet (Campbell et al., 1976), potassium tellurite (Tomas et al., 1986), and carbenicillin (Thom, 1970) has been used in selective media.

Thom (1970) developed a medium based on the MacConkey agar in which lactose is replaced by inositol (1% w/v), with the addition of 100 µg of carbenicillin per ml. Bagley and Seidler (1978) devised a similar medium with only 50 µg/ml carbenicillin. On this medium, about 95% of pink-to-red colonies were verified to be *Klebsiella* spp., whereas only 1% of yellow background colonies were *Klebsiella*.

Since about 10% of *Klebsiella* strains are susceptible to 50 µg/ml of carbenicillin, the antibiotic was replaced in the above formula by tellurite (K_2TeO_3 , 3 µg/ml) (Tomas et al., 1986), which is a strong inhibitor of phosphate transport in *E. coli*. Minimal inhibitory concentrations of K_2TeO_3 were 100 or 200 µg/ml for *K. pneumoniae* subsp. *pneumoniae*, *K. oxytoca*, *K. planticola*, and *K. terrigena*, 10 µg/ml for *K. pneumoniae* subsp. *ozaenae*, and 1–3 µg/ml for other *Enterobacteriaceae* (Tomas et al., 1986). In a field test, 77% of pink-to-red colonies on MacConkey–inositol–potassium tellurite agar were confirmed as *Klebsiella* spp.; however, the efficiency of plating was about 1% (Dutka et al., 1987).

Bruce et al. (1981) devised an agar medium combining Koser citrate and raffinose (carbon sources) and ornithine and low pH (for ornithine decarboxylase). On acidic Koser citrate agar with ornithine and raffinose, *Klebsiella* strains grow as yellow mucoid colonies. Other *Enterobacteriaceae* either do not grow, or produce small colorless, pink, red, or orange colonies.

For the isolation of *K. pneumoniae* and *K. oxytoca* from human feces, Van Kregten et al. (1984) have developed a medium based on the presence of two carbon sources, citrate and inositol, without inhibitor. The medium consists of Simmons citrate agar with 1% inositol. *Klebsiella* spp. appear as yellow, dome-shaped, often mucoid colonies, whereas *E. coli* appears as tiny, watery colonies. Apart from some *Enterobacter* strains, no other of bacteria grow on the medium.

Wong et al. (1985) devised a minimal medium in which the carbon source is lactose and the nitrogen source is potassium nitrate. Inhibitory compounds were deoxycholate, neutral red, and crystal violet. On this medium, *K. pneumoniae* and *K. oxytoca* grow as convex, and 1–2 mm in diameter, rather mucoid pink-to-red colonies, or larger, more watery pale red colonies with a dark red center. Non-klebsiellae either fail to grow or form colorless colonies.

MAINTENANCE PROCEDURES

Klebsiella strains can be easily maintained at room temperature in meat extract semisolid agar, at -80°C in a broth medium with 10–50% (v/v) glycerol, or freeze-dried. *K. granulomatis* will not remain viable when stored at 5° or 37°C . At 25°C , viability of egg yolk cultures is maintained for 8–10 d.

DIFFERENTIATION OF THE GENUS *KLEBSIELLA* FROM OTHER GENERA

See Table BXII.γ.193 of the family *Enterobacteriaceae* for characteristics that can be used to distinguish *Klebsiella* from other genera of the family. No strain of *Klebsiella* other than *K. mobilis* is motile.

Classically, a Gram-negative capsulated isolate that is non-motile, aerogenic, Voges–Proskauer positive, lysine decarboxylase positive, ornithine decarboxylase negative, and arginine dihydrolase negative belongs in the genus *Klebsiella*. However, inclusion of *K. mobilis* and *K. ornithinolytica* makes this classical identification more difficult at the genus level. In the following discussion, the genus *Enterobacter* will be considered without *E. aerogenes*.

Carbon source utilization tests are essential. All *Klebsiella* strains (with the exception of *K. pneumoniae* subsp. *ozaenae* and subsp. *rhinoscleromatis*) utilize the following six substrates: D-arabitol, myo-inositol, palatinose, quinate, D-sorbitol, and sucrose. No *Enterobacter*, *Pantoea*, or *Erwinia* species utilizes all six of these substrates. In the genus *Serratia*, only *S. ficaria* would utilize all six substrates and also L-arabitol and D-erythritol, which no *Klebsiella* has the simultaneous ability to utilize. Furthermore, D-adonitol is utilized by all *Klebsiella* strains except environmental strains of *K. pneumoniae*. In the genus *Enterobacter*, D-adonitol is utilized only by some strains of *Enterobacter cloacae*. Some strains of *Pantoea* also utilize D-adonitol.

TAXONOMIC COMMENTS

The first organism of the genus *Klebsiella* described was a capsulated bacillus isolated from patients with rhinoscleroma (von Frisch, 1882). Then Friedländer (1882) described a bacterium from lungs of a patient who had died of pneumoniae. The organism was subsequently named "*Hyalococcus pneumoniae*" (Schroeter, 1889). The genus *Klebsiella* was coined by Trevisan (1885) to honor the German microbiologist Edwin Klebs (1834–1913). In 1887 (Trevisan, 1887), the genus contained *Klebsiella pneumoniae* (Friedländer bacillus) and *K. rhinoscleromatis* (von Frisch's bacillus). "*Bacillus mucosus ozaenae*", which Abel (1893) observed in nasal secretion of patients with ozena, was later included in the genus as *K. ozaenae* (Bergey et al., 1925).

The genus *Klebsiella* with three species lasted up to the eighth edition of the *Bergey's Manual of Determinative Bacteriology*.

Flügge (1886) described "*Bacillus oxytocus pernicius*" from old milk. This organism was named "*Aerobacter oxytocus*" (Bergey et al., 1923) and *Klebsiella oxytoca* (Lautrop, 1956). For many years, the existence of this indole-positive species was questioned and

it was often considered a biogroup of *K. pneumoniae* (Edwards and Ewing, 1972; Ørskov, 1974). DNA relatedness studies showed *K. oxytoca* to be clearly distinct from *K. pneumoniae* (Jain et al., 1974; Brenner et al., 1977).

K. planticola (Bagley et al., 1981) was proposed for strains isolated primarily from botanical and soil environments (Bagley et al., 1981). *K. planticola* is distinct from other *Klebsiella* species based on DNA relatedness (Woodward et al., 1979; Izard et al., 1981a).

K. terrigena (Izard et al., 1981a), was proposed for strains isolated mainly from aquatic and soil environments (Izard et al., 1981a). It is a distinct species according to DNA relatedness studies (Woodward et al., 1979; Izard et al., 1981a).

The species described as *K. trevisanii* (Ferragut et al., 1983) was shown to be synonymous with *K. planticola* by DNA–DNA hybridization (Gavini et al., 1986).

In the first edition of the *Bergey's Manual of Systematic Bacteriology*, four species were recognized: *K. pneumoniae*, *K. oxytoca*, *K. planticola*, and *K. terrigena*.

K. ozaenae and *K. rhinoscleromatis* cannot be separated from *K. pneumoniae* by DNA relatedness (Brenner et al., 1972c). For this reason, *K. ozaenae* and *K. rhinoscleromatis* were treated as subspecies of *K. pneumoniae* in the first edition of the *Bergey's Manual of Systematic Bacteriology* (Ørskov, 1974).

For many years, *K. pneumoniae* could not be objectively separated from an organism observed by Escherich (1885), named "*Bacterium lactis aerogenes*" (Kruse, 1896), transferred to genus "*Aerobacter*" as "*A. aerogenes*", and finally as *Enterobacter aerogenes* (Hormaeche and Edwards, 1960b). Confusion occurred since both *K. pneumoniae* and "*Aerobacter aerogenes*" fermented many carbohydrates (often with gas production), gave a positive Voges–Proskauer reaction, and reacted with *Klebsiella* capsular antisera. When Møller (1955) introduced the decarboxylase tests, *K. pneumoniae* was defined as nonmotile and ornithine decarboxylase negative, whereas "*A. aerogenes*" was defined as motile or nonmotile and ornithine decarboxylase positive (Hormaeche and Edwards, 1958). Later, a new genus *Enterobacter* was formed (Hormaeche and Edwards, 1960b) to which "*A. aerogenes*" was transferred as *E. aerogenes*.

Nomenclatural confusion occurred when a biogroup of *K. pneumoniae* was named "*Klebsiella aerogenes*" (Taylor et al., 1956). Cowan et al. (1960) subdivided *K. pneumoniae sensu lato* into *K. pneumoniae sensu stricto*, "*K. aerogenes*", "*K. edwardsii* subsp. *edwardsii*", and "*K. edwardsii* subsp. *atlantae*". Authentic or typical strains of *K. pneumoniae*, *K. edwardsii*, and "*K. aerogenes*" cannot be differentiated by DNA relatedness or protein electrophoresis, and thus belong to a single species, *K. pneumoniae* (Brenner et al., 1972c; Jain et al., 1974; Ferragut et al., 1989).

These problems around the epithet *aerogenes* were due to a lack of authentic cultures. Neotypes have since been designated when no original type strain was available.

Enterobacter aerogenes is much closer to *Klebsiella* species than to *Enterobacter cloacae* (the type species of that genus) based on phenotypic traits and DNA relatedness (Bascomb et al., 1971; Brenner et al., 1972c; Steigerwalt et al., 1976; Izard et al., 1980). The transfer of *E. aerogenes* to the genus *Klebsiella* has been proposed (Bascomb et al., 1971). However, since the name "*Klebsiella aerogenes*" had been used for another organism, a new name, *Klebsiella mobilis*, was coined (Bascomb et al., 1971).

Nonmotile, encapsulated, ornithine decarboxylase positive strains were described under the name *Klebsiella ornithinolytica*

(Sakazaki et al., 1989a). Apart from ornithine decarboxylase test, these strains are biochemically undistinguishable from *K. planticola*, and are identical to *K. planticola* by DNA hybridization (Farmer et al., 1985a; D.J. Brenner, personal communication). Farmer et al. (1985a) questioned the validity of *K. ornithinolytica* as a species distinct from *K. planticola*. Some ornithine decarboxylase positive strains have been shown to belong in *K. pneumoniae* by DNA relatedness criteria (Lindh and Frederiksen, 1990).

Finally, *Calymmatobacterium granulomatis*, which has not been cultured on bacteriological media, has been transferred to the genus *Klebsiella* as *Klebsiella granulomatis*, mostly based on *rrs* sequences (Carter et al., 1999). Partial *rpoB* sequencing suggested that *K. granulomatis* is very close to *K. pneumoniae* (Drancourt et al., 2001) and *phoE* sequencing showed *K. granulomatis* close to *K. pneumoniae* subsp. *rhinoscleromatis* (Carter et al., 1999). Thus, there is presently no proof that *K. granulomatis* is a species distinct

from *K. pneumoniae*, and instead there are arguments to make it part of *K. pneumoniae*.

Sequence data are increasingly used to suggest or support nomenclatural changes and some of these changes can be anticipated. The taxonomic significance of the three sequence clusters observed in *K. pneumoniae* or the two sequence clusters in *K. oxytoca* (Brise and Verhoef, 2001) is still unknown, and DNA-DNA hybridization data are needed before a proposal can be made to design new species or subspecies. Since the phylogenetic tree obtained by *rrs* gene (encoding 16S rRNA) sequence comparison showed the genus *Klebsiella* to be split in at least two groups, a new genus, *Raoultella*, was proposed to contain *R. planticola* (type species), *R. ornithinolytica*, and *R. terrigena* (Drancourt et al., 2001). More sequencing work needs to be done to evaluate this proposal since partial *rpoB* sequence failed to group *Raoultella* species in a single cluster and these sequences were only 512 nucleotides long.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *KLEBSIELLA*

Table BXII.γ.229 presents the characteristics differentiating the species and subspecies of *Klebsiella*.

List of species of the genus *Klebsiella*

1. ***Klebsiella pneumoniae*** (Schroeter 1886) Trevisan 1887, 94^{AL} (*Hyalococcus pneumoniae* Schroeter 1886, 1952.) *pneu.mō'ni.æ*. Gr. n. *pneumonia* pneumonia, inflammation of the lungs; M.L. gen. n. *pneumoniae* of pneumonia.

The characteristics are as described for the genus and as listed in Table BXII.γ.229. *K. pneumoniae* can be divided into three sequence clusters (Brise and Verhoef, 2001) unrelated to the present subdivision in subspecies. *K. pneumoniae* is normally found in the intestinal tract of humans and animals. It may be isolated in association with several pathological processes in humans, e.g., community-acquired pneumonia or nosocomial urinary tract infection. Serotype K2 is the most common K-type isolated from patients with urinary tract infection, pneumonia, or bacteremia. In animals, *K. pneumoniae* may be isolated from metritis in mares, bovine mastitis, or generalized infections in captive monkeys. Environmental strains generally utilize more carbon sources than clinical strains, but often fail to utilize D-adonitol. Intraspecies DNA relative reassociation values among strains is ~80–90% (Brenner et al., 1972c) or 73–100% (Woodward et al., 1979).

The mol% G + C of the DNA is: 56–58 (T_m) (Seidler et al., 1975).

Type strain: ATCC 13883, CIP 82.9, DSM 30104, JCM 1662.

GenBank accession number (16S rRNA): X87276, Y17656, AB004753, AF130981.

- a. ***Klebsiella pneumoniae* subsp. *pneumoniae*** (Schroeter 1886) Trevisan 1887, 94^{AL} (*Hyalococcus pneumoniae* Schroeter 1886, 1952.)

Distinguished from the subspecies subsp. *ozaenae* and

subsp. *rhinoscleromatis* by the characteristics listed in Table BXII.γ.229. Clinical strains may produce an extended-spectrum β-lactamase.

The mol% G + C of the DNA is: 56–58 (T_m) (Seidler et al., 1975).

Type strain: ATCC 13883, CIP 82.9, DSM 30104, JCM 1662.

GenBank accession number (16S rRNA): X87276, Y17656, AB004753, AF130981.

- b. ***Klebsiella pneumoniae* subsp. *ozaenae*** (Abel 1893) Ørskov 1984c, 355^{VP} (Effective publication: Ørskov 1984b, 463) (*Klebsiella ozaenae* (Abel 1893) Bergey, Harrison, Breed, Hammer and Huntoon 1925, 266; *Bacillus mucosus ozaenae* Abel 1893, 167; *Bacillus ozaenae* (Abel 1893) Lehmann and Neumann 1896, 204.) *o.za'e'nae*. L. fem. n. *ozaena* ozena; L. gen. n. *ozaenae* of ozena.

Distinguished from the subspecies subsp. *pneumoniae* and subsp. *ozaenae* by the characteristics listed in Table BXII.γ.229. Occurs in ozena and other chronic diseases of the respiratory tract.

The mol% G + C of the DNA is: not available.

Type strain: ATCC 11296, CIP52.211, JCM 1663, LMG 3113.

GenBank accession number (16S rRNA): Y17654, AF130982.

- c. ***Klebsiella pneumoniae* subsp. *rhinoscleromatis*** (Trevisan 1887) Ørskov 1984c, 355^{VP} (Effective publication: Ørskov 1984b, 464) (*Klebsiella rhinoscleromatis* Trevisan 1887, 95; *Bacterium rhinoscleromatis* (Trevisan 1887) Migula 1900, 352.)

rhi.no.scle.ro' ma.tis. M.L. adj. *rhinoscleromatis* pertaining to rhinoscleroma.

Distinguished from the subspecies subsp. *pneumoniae* and subsp. *ozaenae* by the characteristics listed in Table BXII.γ.229. Found in patients with rhinoscleroma.

The mol% G + C of the DNA is: not available.

Type strain: ATCC 13884, CIP 52.210, JCM 1664, LMG 3184.

GenBank accession number (16S rRNA): Y17657, AF130983.

2. **Klebsiella granulomatis** (Aragão and Vianna 1913) Carter, Bowden, Bastian, Myers, Sriprakash and Kemp 1999, 1698^{VP} (*Calymmatobacterium granulomatis* Aragão and Vianna 1913, 221.)

gran.u.lo' ma.tis. L. dim. n. *granulum* a small grain; Gr. suff. *-oma* a swelling or tumor; M.L. n. *granuloma* a granuloma; M.L. gen. n. *granulomatis* of a granuloma.

Occurs in granuloma inguinale (donovanosis) lesions. The bacilli (Donovan bodies) are seen in large vacuolated macrophages found in dermal infiltrate of genital ulcerations. Grow in the yolk sac of developing embryo or in cell cultures (fresh mononuclear cells or Hep-2 cell line). No growth in bacteriological media. *Type strain*: No type culture is currently available.

The mol% G + C of the DNA is: not available.

GenBank accession number (16S rRNA): AF009171, AF010251, AF010252, AF010253.

3. **Klebsiella mobilis** Bascomb, Lapage, Willcox and Curtis 1971, 279^{AL*}

mo' bi.lis. L. adj. *mobilis* movable, motile.

The characteristics are described in Table BXII.γ.229. This is the only motile species in the genus. Strains producing extended-spectrum β-lactamase occur as nosocomial pathogens. Occur in water, sewage, soil, dairy products, and the feces of humans and animals. Also an opportunistic pathogen.

The mol% G + C of the DNA is: 53–54 (Bd).

Type strain: ATCC 13048, CIP 60.86, DSM 30053, JCM 1235, LMG 2094.

GenBank accession number (16S rRNA): AB004748.

4. **Klebsiella oxytoca** (Flügge 1886) Lautrop 1956, 375^{AL} (*Bacillus oxytocus perniciosus* Flügge 1886, 268.)

oxy.to' ca. Gr. *oxys* sour, acid; Gr. suff. *-tokos* bearer, producer; M.L. n. *oxytocus* acid-producer, spurious; M.L. adj. *oxytoca* (sic) acid-producing.

The characteristics are as described for the genus and as listed in Table BXII.γ.229. Present in the intestinal tract of humans and animals. Can be isolated from various pathological processes and from botanical and aquatic environments. *K. oxytoca* can be divided into two sequence clusters (Brisse and Verhoef, 2001). Encapsulated; typable with *Klebsiella* K antisera. The intraspecies DNA relative reassociation values were 75% in the study by Brenner et al. (1972c) and 95% (average value) in the study by Woodward et al. (1979).

The mol% G + C of the DNA is: 55–58 (*T_m*).

Type strain: ATCC 13182, CIP 103434, JCM 1665, LMG 3055.

GenBank accession number (16S rRNA): Y17655, AB004754, AF129440.

5. **Klebsiella planticola** Bagley, Seidler and Brenner 1982, 266^{VP} (Effective publication: Bagley, Seidler and Brenner 1981, 109.)*

plan.ti' co.la. L. fem. n. *planta* a plant; L. suff. *-cola* dweller; M.L. fem. n. *planticola* plant-dweller.

The characteristics are as described for the genus and as listed in Table BXII.γ.229. Isolated mainly from botanical, aquatic, and soil environments. Three biovars have been described (Naemura et al., 1979). Encapsulated; typable with *Klebsiella* K antisera. The average intraspecies DNA relative reassociation value is above 75% (Woodward et al., 1979).

The mol% G + C of the DNA is: 53.9–55.4 (*T_m*) (Seidler et al., 1975).

Type strain: ATCC 33531, ATCC 33558, CIP 100751, DSM 3069, IFO 14939.

GenBank accession number (16S rRNA): X93215, Y17659, AB004755, AF129443.

6. **Klebsiella terrigena** Izard, Ferragut, Gavini, Kersters, De Ley and Leclerc 1981a, 125^{VP}

ter.ri.ge' na. L. n. *terra* soil; L. suff. *gena* origin; M.L. n. *terrigena* from soil.

The characteristics are as described for the genus and as listed in Table BXII.γ.229. Isolated mainly from aquatic and soil environments. The average intraspecies DNA relative reassociation value is above 86% (Izard et al., 1981a).

The mol% G + C of the DNA is: 56.7 (*T_m*) (Izard et al., 1981a).

Type strain: ATCC 33257, CIP 80-07, DSM 2687, JCM 1687, LMG 3222.

GenBank accession number (16S rRNA): Y17658, AF129442.

Species Incertae Sedis

1. **Klebsiella ornithinolytica** Sakazaki, Tamura, Kosako and Yoshizaki 1989b, 495^{VP} (Effective publication: Sakazaki, Tamura, Kosako and Yoshizaki 1989a, 205.)

or.ni.thi.no.ly' ti.ca. M.L. *ornithinum* ornithine, an amino acid; Gr. adj. *lyticus* dissolving; M.L. fem. adj. *ornithinolytica* ornithine dissolving.

The CDC group called *Klebsiella* Group 47 corresponds to this species (Sakazaki et al., 1989a). Physiological and nutritional characteristics are presented in Table BXII.γ.229. Isolated from clinical materials, including urine, sputa, stool, and pus. Can be found in food. Clinical significance unknown.

The mol% G + C of the DNA is: 57–58 (*T_m*) (Sakazaki et al., 1989a).

Type strain: ATCC 31898, CIP 103576, DSM 7464, JCM 6096.

GenBank accession number (16S rRNA): AF129441, AJ251467.

*Editorial Note: *Klebsiella mobilis* Bascomb et al. 1971 and *Enterobacter aerogenes* Hormaeche and Edwards 1960b have the same type strain and therefore are homotypic synonyms.

*Editorial Note: *Klebsiella trevisanii* Ferragut et al. 1983 (phenon K of Gavini et al., 1977) is a junior subjective synonym of *K. planticola* (Gavini et al., 1986).

Genus XVII. *Kluyvera* Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981b, 382^{VP} (Effective publication: Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981a, 927)

J.J. FARMER III

Kluy' ve.ra. M.L. fem. n. *Kluyvera* named by Asai et al. (1956) to honor the Dutch microbiologist A.J. Kluyver, who made many contributions to microbial physiology and taxonomy.

Small rod-shaped cells, $0.5\text{--}0.7 \times 2\text{--}3 \mu\text{m}$, conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. Motile, with peritrichous flagella. Contains the enterobacterial common antigen. Facultatively anaerobic. Catalase positive (weak). Oxidase negative. Nonpigmented. Ferment, rather than oxidize, D-glucose and other carbohydrates. Reduce nitrate to nitrite. **Positive for indole production, methyl red, citrate utilization (Simmons), ornithine decarboxylase, motility at 36°C, growth in the presence of cyanide (KCN test), malonate utilization, and esculin hydrolysis. Ferment D-glucose, lactose, sucrose, D-mannitol, salicin, L-arabinose, raffinose, L-rhamnose, maltose, D-xylose, trehalose, cellobiose, α -methyl-D-glucoside, melibiose, mucate, D-mannose, and D-galactose. Produce visible gas during fermentation. Produce large amounts of α -ketoglutaric acid during the fermentation of D-glucose.** Negative for Voges-Proskauer, H₂S production (TSI), urea hydrolysis, phenylalanine deaminase, arginine dihydrolase, gelatin hydrolysis (22°C), lipase (corn oil), DNase, and the fermentation of dulcitol, adonitol, *myo*-inositol, erythritol, and D-arabitol. Susceptible to colistin, sulfadiazine, gentamicin, kanamycin, tetracycline, and chloramphenicol (disk diffusion method on Mueller-Hinton agar); **resistant to carbenicillin; penicillin, ampicillin, and cephalothin**; variable susceptibility to nalidixic acid and streptomycin.

Some strains produce kluyveramycin, unusual reddish blue crystals of unknown composition. **Occur in human clinical specimens, food, soil and sewage. Probably an occasional opportunistic pathogen and a cause of extraintestinal infections of humans.**

The genus *Kluyvera* includes *Kluyvera ascorbata*, *K. cryocrescens*, and *K. georgiana*. There is also a fourth species with standing in nomenclature, *Kluyvera cochleae*, but it appears to be a synonym of *Enterobacter intermedius* (originally named *E. intermedius*; corrected by von Graevenitz, 1990).

The mol% G + C of the DNA is: 55.1–56.6.

Type species: ***Kluyvera ascorbata*** Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981b, 382 (Effective publication: Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981a, 927.)

FURTHER DESCRIPTIVE INFORMATION

Literature Since *Kluyvera* was described in 1956, there have been many reports in the literature; over 40 have been cataloged in MEDLINE and many others in BIOSIS. In addition to the genus and species names, "Enteric Group 8", "*K. citrophila*", and "*K. noncitrophila*" should also be included as a search terms, although they are rarely used today.

The National Library of Medicine's Internet site Entrez (<http://www.ncbi.nlm.nih.gov>), which covers most of the sequence databases, includes full sequences of the 16S rRNA gene for *K. ascorbata*, *K. georgiana*, and *K. cochleae* (*Enterobacter intermedius*), and a partial sequences for *K. cryocrescens*. A search for the term *Kluyvera* yielded 29 sequences.

Problems in evaluating the literature Computer literature searches often yield articles such as "First case of bacteremia due

to xx" or "Wound infections due to yy"; where "xx" and "yy" are newly described bacterial species. The reader should evaluate these reports very critically, with particular attention to the way that cultures were identified. Over half the cultures of new or unusual *Enterobacteriaceae* in these reports have been misidentified (Farmer unpublished). *Kluyvera* strains from two published studies were subsequently identified as *Serratia fonticola* (Enteric Reference Laboratory at CDC, unpublished).

Sources, strains, collections The original collection described by Farmer et al. (1981a) included isolates from body sites that are normally sterile such as blood (2 isolates) and urine (13). Strains were also from nonsterile body sites: intestine or feces (8), throat or sputum (40), and other or unspecified (10). Other isolates were from nonhuman sources or culture collections: cow (1), water (1), sewage (4), food (1), milk (1), soil (1), hospital sink (1), and unspecified or unknown (18). These other sources suggest ways that humans and animals might come in contact with the organism. The description of the genus was based on an unweighted tabulation of 193 *Kluyvera* strains from the Enteric Reference Laboratory's collection; 149 of *K. ascorbata*, 38 of *K. cryocrescens*, and six of *Kluyvera* species 3.

Occurrence in human feces The original report of Farmer et al. (1981a) listed eight isolates from human feces. All these presumably came from cases of diarrhea or intestinal infections, and were identified as *Kluyvera ascorbata*. Although clinical information that accompanied the cultures was limited, there was no suggestion of an etiological role for *Kluyvera*. Fainstein et al. (1982a) isolated strains of *Kluyvera* from patients with and without diarrhea, and in the title of their paper suggested that *Kluyvera* strains might have had a role in some of the diarrhea cases. Kay et al. (1990) isolated 28 *Kluyvera* strains during a 2-yr prospective case-control study of acute gastroenteritis in children younger than 2 years old, and concluded that *Kluyvera* did not appear to be a diarrheal pathogen. In summary, there is no strong evidence that *Kluyvera* strains can actually cause diarrhea or intestinal infections, but this issue warrants further study. The presence of *Kluyvera* in food and water is a possible source of these intestinal isolates.

Occurrence in extraintestinal human specimens: clinical significance Most of the *Kluyvera* strains have been from the respiratory tract. Forty strains from the original collection (Farmer et al., 1981a) were from throat or sputum. Clinical significance and the ability to actually cause infections are still being evaluated. Systematic study and additional case reports are needed. Schwach (1979) reported three isolates of Enteric Group 8 from upper respiratory tract specimens. The strains were in mixed culture and not detected in subsequent specimens; thus they were of doubtful clinical significance. Braunstein et al. (1980) reported two cases that yielded cultures identified as Enteric Group 8 (subsequently, both were identified as *K. ascorbata*). One of these was from the sputum of a 6-year-old boy with pulmonary tuberculosis, and was not considered clinically significant. A second isolate was from gallbladder drainage fluid of a 63-year-old

woman with acute pancreatitis. Based on chart review, this isolate was considered clinically significant. Of CDC's current collection of 144 *Kluyvera* strains, none has been isolated from spinal fluid, but five strains have been isolated from blood: three strains (two from France) of *K. ascorbata*, one strain of *K. cryocrescens* (a 3-month-old, at autopsy), and one of *Kluyvera* species group 3. Most of these were submitted with little or no patient information, so it was impossible to evaluate clinical significance. The five blood isolates and the growing number of literature reports (Aldová et al., 1985b; Wong, 1987; Luttrell et al., 1988; Thaller et al., 1988; Tristram and Forbes, 1988; Dollberg et al., 1990; Sierra-Madero et al., 1990; Yogev and Koszowski, 1990; Sezer et al., 1996; Padilla et al., 1997) suggest that *Kluyvera* is more than a benign saprophyte. Most new species of *Enterobacteriaceae* have at least attained the status of "infrequent opportunistic pathogen". Based on present knowledge, this status also seems appropriate for *Kluyvera*. The respiratory tract has been the most common source for *Kluyvera*, but there is no strong evidence that it is clinically significant at this site (however, one isolate of *K. ascorbata* was from a lung at autopsy). The respiratory tract (particularly sputum) is notoriously difficult to evaluate for clinical significance. The urinary tract has been the next most common source, but it has also been difficult to document clinical significance (Tristram and Forbes, 1988).

Occurrence in other animals Bangert et al. (1988) identified strains as *Kluyvera* species along with other *Enterobacteriaceae* in a fecal culture survey of 47 captive raptors (the bird orders *Falconiformes* and *Strigiformes*).

Reporting of *Kluyvera* species Originally, Farmer et al. (1981a) were pessimistic that most clinical and public health laboratories could identify *Kluyvera* strains to species. They suggested that reference laboratories could attempt complete identification, but that clinical laboratories should not attempt the additional phenotypic testing with the addition of special tests. Normal identification methods should allow an identification to the genus level, with a report of "*Kluyvera* species". The addition of one new *Kluyvera* species, and seven new *Buttiauxella* species has made it even more difficult to identify cultures of *Kluyvera*–*Buttiauxella* to species. Further work is needed to find better tests for differentiation. In the meantime, a conservative approach would be to use terms such as "*Kluyvera* species", "*Buttiauxella* species", and "*Kluyvera*–*Buttiauxella* group" for cultures that cannot be definitively assigned a species name.

Accumulation of α -ketoglutaric acid Asai et al. (1956) assigned considerable importance to the fact that *Kluyvera* strains accumulated extremely large amounts of α -ketoglutaric acid during D-glucose fermentation. However, few other *Enterobacteriaceae* have been investigated for this property.

ENRICHMENT AND ISOLATION PROCEDURES

Strain of *Kluyvera* are not difficult to grow and are typical *Enterobacteriaceae* in most respects. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, computer programs, and antibiotic susceptibility.

DIFFERENTIATION OF THE GENUS *KLUYVERA* FROM OTHER GENERA

Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* gives the results for *K. ascorbata*, *K. cryocrescens*, and *K. georgiana*

in 47 biochemical tests normally used for identification (Farmer, 1999). There are no genus- or species-specific tests or sequences for the identification of *Kluyvera*. The best approach is to do a complete set of phenotypic tests (growth on plating media, biochemical tests, antibiogram, etc.) and compare the results with all the other named organisms in the family *Enterobacteriaceae*. This approach is described in more detail in the section on the family *Enterobacteriaceae*. The biochemical results of a test strain can be compared to all the organisms listed in Table BXII.γ.193 of that section. Several computer programs greatly facilitate analyzing the results.

TAXONOMIC COMMENTS

Kluyvera is a genus with a turbulent history. Although the name was used in the literature from 1956 to 1980, the genus name, and the two original species names ("*K. citrophila*" and "*K. noncitrophila*") did not appear on the 1980 Approved Lists of Bacterial Names. The reasons for these omissions were never stated. Thus, *Kluyvera*, "*K. citrophila*", and "*K. noncitrophila*" lost standing in nomenclature. To resolve this nomenclatural problem, Farmer et al. (1981a) proposed a redefined genus *Kluyvera*.

The history of *Kluyvera* Asai et al., 1956 In 1956 and 1957, Asai and co-workers in Japan proposed the genus name *Kluyvera* for a group of "polarly flagellated" bacteria that produced large amounts of α -ketoglutaric acid during the fermentation of D-glucose. Five strains were originally studied, one from soil and four from sewage (Table BXII.γ.230). Two species names were proposed, "*K. citrophila*" and "*K. noncitrophila*", which were based on the difference among the strains in utilizing citrate as the sole source of carbon and energy (Asai et al., 1956). Asai and co-workers state (based on their interpretation of Kluyver and van Niel, 1936) that the genus was named to honor Professor A.J. Kluyver, who, with C.B. van Niel, in 1936 postulated that there may be a group of polarly flagellated organisms in the tribe *Pseudomonadeae* that have a mixed acid type of fermentation similar to *Escherichia* (called *Bacterium* in the paper). If such a group were to be discovered, it could be a separate genus, which would differentiate it from the genus *Aeromonas*, which has a butylene glycol fermentative pathway rather than a mixed acid pathway. Asai and co-workers thought they had discovered this postulated group of polarly flagellated organisms and named the group *Kluyvera* in honor of A.J. Kluyver for his many contributions to microbial metabolism and physiology. *Kluyvera* was classified in the tribe *Pseudomonadeae*, which at that time included nonfermentative genera, but also included the fermenters of the genus *Aeromonas*. Today, the family *Pseudomonadaceae* is restricted to bacteria that do not ferment glucose.

***Kluyvera* Asai et al., 1956 was "abolished" in 1962** In 1962, Asai and co-workers confirmed the observations of J.M. Shewan and Rudolph Hugh that all five of their *Kluyvera* strains actually had peritrichous rather than polar flagella (Asai et al., 1962). Thus, they proposed an alternative classification; the two species "*K. citrophila*" and "*K. noncitrophila*" were "transferred" to the genus *Escherichia* in the family *Enterobacteriaceae*. Thus, this alternative classification would result in the creation of two "new combinations"—"*Escherichia citrophila*" and "*Escherichia noncitrophila*"; however, these names were rarely used and did not appear on the 1980 Approved Lists of Bacterial Names.

History of the redefined genus *Kluyvera*: *Kluyvera* Farmer et al., 1981a The Enteric Reference Laboratory's interest in this group of organisms began in 1977 when the laboratory's

TABLE BXII.γ.230. Current nomenclature and classification (Farmer et al., 1981a) for strains originally proposed as "*Kluyvera citrophila*" and "*K. noncitrophila*"

Original name given by Asai et al. (1962)	Current (proposed) classification	ATCC number	Original strain designation	Source
" <i>K. citrophila</i> "	<i>K. ascorbata</i>	14236	6,β	Sewage, Japan
" <i>K. citrophila</i> "	<i>K. cryocrescens</i>	14237	84C,α	Soil, Tokyo, Japan
" <i>K. citrophila</i> "	<i>K. cryocrescens</i>	14238	11,γ	Sewage, Keihin District, Japan
" <i>K. noncitrophila</i> "	<i>K. cryocrescens</i>	14239	4	Sewage, Keihin District, Japan
" <i>K. noncitrophila</i> "	<i>K. cryocrescens</i>	14240	10	Sewage, Keihin District, Japan

STRAIN MATCHER computer programs listed over a dozen strains with almost identical biochemical reactions that had been reported as "unidentified". During this period, groups of unidentified *Enterobacteriaceae* were being defined, and this was the eighth group. Thus, the vernacular name "Enteric Group 8" was given to the group, and strains were then reported with this designation and included a comment that additional strains would be welcomed so the group could be further studied. Soon after Enteric Group 8 was defined, Holmes of the Computer Identification Laboratory, National Collection of Type Cultures (NCTC), England, reported that the NCTC collection had strains that were almost identical to our Enteric Group 8, and they had been cataloged under the name *Kluyvera* (unpublished). The NCTC stains of *Kluyvera* were then studied at CDC and the conclusions of Holmes were confirmed. Since then, Enteric Group 8 was thought of as a synonym of *Kluyvera*. The DNA-DNA hybridization studies of Fanning et al. (1979) defined two large species groups in *Kluyvera* and defined a third group of strains that became "*Kluyvera* species group 3", and was thought to represent one or more additional *Kluyvera* species. Although *Kluyvera* was "abolished" in 1962 (Asai et al., 1962), reports in the literature continued to use the name *Kluyvera*, probably because strains had been deposited in the American Type Culture Collection, NCTC, and other culture collections. The name *Kluyvera* was proposed as a redefined genus by Farmer et al. (1981a) because this was considered a better alternative than proposing a new genus. The main consideration was that the name *Kluyvera* was still being used in the literature, and that the genus *Kluyvera* was well represented in culture collections by established strains of long standing.

Phylogenetic position Based on both phenotype and DNA-DNA hybridization, Farmer et al. (1981a) thought that *Kluyvera* was a tight, well-defined genus of *Enterobacteriaceae*. Two 16S rDNA trees (Figure BXII.γ.198 in the section on *Budvicia* and Figure BXII.γ.189 in the chapter on the family *Enterobacteriaceae*) include *Kluyvera*. Unfortunately, the species of *Kluyvera* and *Buttiauxella* were not included in the 16S rDNA tree published by Spröer et al. (1999).

Additional species of *Kluyvera* There are probably several additional *Kluyvera* species. The Enteric Reference Laboratory's collection has 21 strains that were reported as "*Kluyvera* species", three as "possible-probable" *Kluyvera ascorbata*, three as *Kluyvera cryocrescens*, and two as "*Kluyvera-Buttiauxella* Group" (Enteric Reference Laboratory, 2000, unpublished). These 29 and the following strains need to be studied by techniques that measure evolutionary relatedness: the four additional strains included in *Kluyvera georgiana* by Müller et al. (1996); the three strains of *K. ascorbata* and five strains of *K. cryocrescens* that had divergence values in DNA-DNA hybridization experiments in the range 5–9 (see Table 4 of Farmer et al., 1981a); and all other "atypical strains" of *Kluyvera* and *Buttiauxella*. Detailed study of this collection would probably result in the definition of many new species.

Further studies needed New phenotypic and molecular tests are needed for the routine identification and differentiation of *Kluyvera*, *Buttiauxella*, and related organisms. The 16S rDNA sequences of all *Kluyvera* and *Buttiauxella* species need to be completed or repeated and new trees drawn. This could provide a useful way to identify strains and clarify relationships of the genera and species.

FURTHER READING

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- Luttrell, R.E., G.A. Rannick, J.L. Soto-Hernandez and A. Verghese. 1988. *Kluyvera* species soft tissue infection: case report and review. *J. Clin. Microbiol.* 26: 2650–2651.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *KLUYVERA*

The original two named species of *Kluyvera*, *K. ascorbata* and *K. cryocrescens*, had almost identical results in the 47 tests commonly used to identify cultures of *Enterobacteriaceae* (Farmer et al., 1981a; see Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*). However, these species can be differentiated by several

simple phenotypic tests (Table BXII.γ.231). *K. georgiana* is also almost identical to *K. ascorbata* and *K. cryocrescens* in the 47 tests commonly used to identify cultures of *Enterobacteriaceae* (Farmer et al., 1981a), which is one of the reasons it was not originally named. At present there are no simple ways to differentiate it.

TABLE BXII.γ.231. Tests useful for differentiating the three *Kluyvera* species; and between *Kluyvera* and *Buttiauxella*^a

Test or property	<i>K. ascorbata</i>	<i>K. cryocrescens</i>	<i>K. georgiana</i>	<i>Buttiauxella</i>
Ascorbate test	97	0	nd	nd
Growth and D-glucose fermentation at 5°C within 21 d	3	100	nd	nd
Zone of inhibition around carbenicillin and cephalothin (see Figure 4 of Farmer et al., 1981a)	Small	Large	nd	nd
Growth on cefsulodin-irgasan-novobiocin agar ^b	100	0	nd	nd
Minimum inhibitory concentration (MIC) for irgasan (μg/ml) ^b	>128	≤0.25	nd	nd
Lysine decarboxylase (Møller)	97	23	100	—
Gas production during the fermentation of D-glucose	93	95	17	+
Dulcitol fermentation	25	0	33	0
Indole production	92	90	100	—
Sucrose fermentation	98	81	100	0
Citrate utilization	96	80	100	d
Raffinose fermentation	98	100	100	d
α-Methyl-D-glucoside fermentation	98	95	100	—
Melibiose fermentation	99	100	100	d
Acetate utilization	50	86	83	0

^aSymbols: +, most strains are positive; —, most strains are negative; d, species to species variation, but many strains are negative. Each number is the percentage positive after 2 d incubation at 36°C.

^bThe data for growth on CIN agar and the irgasan MIC are from Altwegg et al. (1986); the other data are from Farmer et al. (1981a) and Farmer (1999).

List of species of the genus *Kluyvera*

1. ***Kluyvera ascorbata*** Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981b, 382^{VP} (Effective publication: Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981a, 927.)

a.scor.ba'ta. ascorbate, a modern chemical term, a salt of ascorbic acid; M.L. fem. adj. *ascorbata* pertaining to ascorbate.

The characteristics are as described for the genus. See Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* for the results of 47 biochemical tests normally used for identification (Farmer, 1999). Occurs in human clinical specimens, water, sewage, and food. Probably an opportunistic pathogen of humans that is occasionally encountered in clinical microbiology laboratories. The type strain was isolated from human sputum.

The mol% G + C of the DNA is: 56.1–56.6 (Bd).

Type strain: ATCC 33433, CDC 0648-74. (holotype).

GenBank accession number (16S rRNA): AF176560.

Additional Remarks: The ATCC includes two other strains; ATCC 33434 and ATCC 14236 (Table BXII.γ.230).

2. ***Kluyvera cryocrescens*** Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981b, 382^{VP} (Effective publication: Farmer, Fanning, Huntley-Carter, Holmes, Hickman, Richard and Brenner 1981a, 927.)

cry.o.cres'cens. Gr. n. *kryos* cold; L. fem. pres. part. *crescens* growing; M.L. fem. adj. *cryocrescens* growing in the cold; referring to the fact the cultures grow at 4–5°C.

The characteristics are as described for the genus. See Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* for the results of 47 biochemical tests normally used

for identification (Farmer, 1999). Occurs in human clinical specimens, soil, water, sewage, and the hospital environment. Probably an opportunistic pathogen of humans that is rarely encountered in clinical microbiology laboratories.

The mol% G + C of the DNA is: 55.1 (Bd).

Type strain: ATCC 33435, CDC 2065-78 (holotype).

Additional Remarks: The ATCC includes six other strains, four of which are described in Table BXII.γ.230.

3. ***Kluyvera georgiana*** Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 63^{VP}

geor.gi.d'na. M.L. fem. adj. *georgiana* pertaining to Georgia, U.S.A., where the characterization and redefinition of *Kluyvera* was done.

Synonym: *K. georgiana* and *Kluyvera* species group 3 of Farmer et al. (1981a) are similar in several ways, but are different in circumscription as described below.

The characteristics are as described for the genus. See Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* for the results of 47 biochemical tests normally used for identification (Farmer, 1999). Currently, it is not possible to completely differentiate *K. georgiana* from *K. ascorbata* and *K. cryocrescens* with simple tests (Table BXII.γ.231). The type strain was isolated from human sputum.

Phenotypic properties as originally given by Müller et al. (1996). The species includes the type strain but no other strains. Other strains named *K. georgiana* include CDC 2774-70 (ATCC 51703), CDC 2065-76 (ATCC 51702), CDC 4246-74, and CDC 3108-76 (Müller et al., 1996). However, no data are presented to show that these four strains are highly related to the type strain by either DNA–DNA hybridization

or phenotype (see the discussion below of *Kluyvera* species 3). Thus, there is a change in circumscription between the two species. In the future, additional strains that are highly related to the type strain should be included.

The mol% G + C of the DNA is: not determined.

Type strain: 189, ATCC 51603, CDC 2891-76, DSM 9409 (holotype).

Other Organisms

1. *Kluyvera cochleae* Müller, Brenner, Fanning, Grimont and Kämpfer 1996, 63^{VP}

coch'le. ae. L. fem. gen. n. *cochleae* of a snail.

Müller et al. (1996) defined a fourth *Kluyvera* species, *Kluyvera cochleae*, which now appears to be a junior subjective synonym of *Enterobacter intermedius*. *K. cochleae* included two strains isolated from snails in Germany and one strain from a slug in Great Britain. In DNA–DNA hybridization experiments (Müller et al., 1996), the type strain of *K. cochleae* S3/1-49^T was labeled and was highly related (86–92% at 60°C with divergence values of 0.5–1.0) to the two other strains. It was 19–24% related to strains of *Buttiauxella*, but no relatedness values were given for its relationship to the type strains of the three *Kluyvera* species or to other *Enterobacteriaceae*.

Farmer (unpublished, see Farmer, 2003) studied the three strains of *K. cochleae* and found that they apparently metabolize D-glucose via the butanediol pathway (i.e., they are Voges–Proskauer positive). This is a very unlikely pathway to occur in *Kluyvera*, which was defined to be methyl red positive (well documented to accumulate large amounts of α-ketoglutaric acid) and Voges–Proskauer negative. Computer analysis of the phenotypic data indicated that the strains of *K. cochleae* were essentially identical to 10 reference strains of *Enterobacter intermedius* (Farmer, unpublished). *Enterobacter* is a genus that is well known to have the butanediol pathway of D-glucose metabolism (Voges–Proskauer positive). Later it was shown that the type strains of these two species have almost identical published 16S rRNA sequences. These observations led to further laboratory studies (Brenner and Steigerwalt, unpublished; Farmer, unpublished). By DNA–DNA hybridization, the type strains of the two species are very highly related (D.J. Brenner, personal communication).

To summarize, all current data indicate that *Kluyvera cochleae* and *Enterobacter intermedius* are different names for the same organism. The two species have different type

strains, so they should be considered as subjective synonyms, with *Enterobacter intermedius* being the senior subjective synonym, since it was validly published first.

2. “*Kluyvera fluvialis*”

In an abstract, Gadaleta et al. (1996) from Buenos Aires, Argentina, described their work with “strain 21g” isolated from a polluted river. They determined its 16S rDNA sequence, compared it to the sequences of *K. ascorbata* and *K. cryocrescens*, and concluded “. . . the 21g strain is a member of an until now undescribed species of *Kluyvera* that we propose to be named *K. fluvialis*”. Since the name “*Kluyvera fluvialis*” has not been effectively published or validly published, it lacks standing in nomenclature.

The vernacular name *Kluyvera* species group 3 was originally given by Farmer et al. (1981a) to five strains (2774-70, 2065-76, 2891-76, 4246-74, and 3108-76) that were 60–68% related (with divergence values of 9–12%) by DNA–DNA hybridization to strains of *K. ascorbata* and *K. cryocrescens* (see Table 4 in Farmer et al., 1981a). None of these strains were labeled and tested against the others to determine if they all belonged to the same species. However, the authors concluded that the five strains represented one or more additional species of *Kluyvera*. Because of its uncertain circumscription and the lack of simple tests to differentiate it from the two named *Kluyvera* species, they did not give it a scientific name, designate a type strain, or deposit strains in the American Type Culture Collection.

One of the strains (2891-76) was later designated as the type strain of *Kluyvera georgiana* by Müller et al. (1996) as described above. *K. georgiana* of Müller et al. (1996) also include the other four strains that Farmer et al. (1981a) included in *Kluyvera* species 3. However, no information is included in the paper of Müller et al. (1996) to show that these latter four strains are highly related to the type strain of *K. georgiana* by DNA–DNA hybridization or phenotype. The assignment of these four strains to *K. georgiana* or a new *Kluyvera* species awaits further study.

Genus XVIII. *Leclercia* Tamura, Sakazaki, Kosako, and Yoshizaki 1987, 179^{VP} (Effective publication: Tamura, Sakazaki, Kosako, and Yoshizaki 1986, 183)

CAROLINE M. O'HARA AND J.J. FARMER III

Le. clerc' i. a. M.L. fem. n. *Leclercia* named to honor H. Leclerc, a French bacteriologist, who first described and named this organism *Escherichia adecarboxylata* in 1962, and who made many other contributions to enteric bacteriology.

Small rod-shaped cells, conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. Motile at 36°C and 25°C with peritrichous flagella. Facultatively anaerobic. Catalase positive. Oxidase negative. **Many strains produce a nondiffusible yellow pigment that may be weak and may be lost on storage and subculture.** Ferment, rather than oxidize, D-glucose and other carbohydrates. Reduce nitrate to nitrite. **Positive for indole production, methyl red, growth in the presence of cyanide, malonate utilization, esculin hydrolysis, ONPG, and the fermentation of**

lactose, D-mannitol, dulcitol, salicin, adonitol, L-arabinose, L-rhamnose, maltose, D-xylose, trehalose, cellobiose, erythritol, esculin, melibiose, D-arabitol, mucate, D-mannose, and D-galactose. Produce visible gas during fermentation. Negative for Voges–Proskauer, citrate utilization (Simmons), H₂S production (TSI), phenylalanine deaminase, **lysine decarboxylase, arginine dihydrolase, ornithine decarboxylase**, gelatin hydrolysis (22°C), lipase (corn oil) production, DNase production, and the fermentation of *myo*-inositol, D-sorbitol, α-methyl-D-glucoside, and erythritol.

Most strains are susceptible to colistin, nalidixic acid, sulfadiazine, gentamicin, streptomycin, kanamycin, tetracycline, chloramphenicol, carbenicillin, cephaloridine, and ampicillin (disk diffusion method on Mueller–Hinton agar); **resistant to penicillin**.

Isolated from human clinical specimens that are normally sterile such as blood and urine. Also isolated from other clinical specimens such as sputum and wounds, usually in mixed culture. Probably an **opportunistic pathogen that occasionally causes extraintestinal infections** in humans. Occasionally occurs in human feces, but there is **no evidence that it causes diarrhea or intestinal infections**. Natural reservoirs and ecological niches are not completely defined, but has been isolated from water, food, milk, and other environmental samples. Appears to be important in the processing of “dry-cured” hams. A rarely isolated genus of *Enterobacteriaceae*.

The mol% G + C of the DNA is: 52–55.

Type species: *Leclercia adecarboxylata* (Leclerc 1962) Tamura, Sakazaki, Kosako, and Yoshizaki 1987, 179 (Effective publication: Tamura, Sakazaki, Kosako, and Yoshizaki 1986, 183) (*Escherichia adecarboxylata* Leclerc 1962, 737.)

FURTHER DESCRIPTIVE INFORMATION

Literature Since the genus was described in 1986, there have been a few reports in the literature; seven cataloged in MEDLINE and 18 cataloged in BIOSIS. These reports include reviews of new *Enterobacteriaceae*; human clinical specimens and infections; occurrence in animals, plants, food, water, and the environment; taxonomic studies; and studies on basic biology such as physiology-metabolism, biochemistry, and genetics.

Leclercia strains in reference laboratory collections The original report by Tamura et al. (1986) described 58 isolates from human clinical specimens including blood, urine, sputum, and feces, and 27 isolates from nonhuman sources, including food, water, and the environment. However, they did not specify the number of isolates from each of these sources. Farmer et al. (1985a) described their studies with the type strain (isolated from water), and concluded that their collection also contained 6–10 isolates that were very similar. The CDC collection now includes 18 human isolates: blood (5), urine (1), wound (1), sputum (1), hand wound (1), leg wound (2), finger (1), ear (1), feces (3), and unspecified (2). It also includes 11 isolates from nonhuman sources or culture collections: intravenous fluid bottle (1), milk (1), environmental (1), and unspecified or unknown (8).

Occurrence in human clinical specimens Strains of *L. adecarboxylata* have been from a wide variety of human clinical specimens, but there have been few systematic clinical studies or case reports. Temesgen et al. (1997) found only five strains during 1984–1995 at the Mayo Clinic in Rochester, Minnesota, USA. In three cases it was isolated from wounds of the lower extremities, all in mixed culture. Another case was a pneumonia patient whose sputum yielded *L. adecarboxylata* along with other organisms. One patient had a culture of *Leclercia adecarboxylata* that was clinically significant. A 35-year-old woman with acute nonlymphocytic leukemia had undergone bone marrow transplantation. She developed bacteremia and fever while she was neutropenic, and *L. adecarboxylata* was isolated in pure culture from one of two sets of blood cultures.

Other cases of bacteremia have also been described. Daza et al. (1993) reported the case of a 45-year-old alcoholic who had severe abdominal pain, hypotension, tachycardia, and diapho-

resis 6 hours after undergoing paracentesis for the removal of ascitic fluid. Four days after undergoing a laparotomy, he developed fever and had three blood cultures positive for *L. adecarboxylata*. Otani and Bruckner (1991) reported a case in an 8½-month-old with a history of congenital gastroschisis and intestinal atresia. He came to the emergency department with shaking chills and fever about 1 hour after total parenteral nutrition through a central line catheter. The blood culture from the central line was positive for *L. adecarboxylata*. Dudkiewicz and Szezyk (1993) described one isolate of *L. adecarboxylata* in their series of 72 cases of bacterial endocarditis.

When *L. adecarboxylata* is isolated in mixed culture from specimens such as sputum and wounds, it is difficult to determine clinical significance. Martinez et al. (1998) reported its isolation from an ulcer exudate. However, the isolation of this organism from blood cultures suggests that it is at least an opportunistic pathogen. Additional studies and case reports are needed to better define its role as a human pathogen.

Occurrence in human feces Several collections have included isolates from feces (Tamura et al., 1986), which usually are from patients with diarrhea. Cai et al. (1992) described three patients in China with diarrhea whose fecal cultures were positive for *L. adecarboxylata*, but the authors could not demonstrate an etiological role. Although it has occasionally been isolated from feces, there is no evidence that *L. adecarboxylata* can actually cause diarrhea or infections of the intestinal tract.

Occurrence in animals, plants, food, water, and the environment In addition to sources previously mentioned, *L. adecarboxylata* (referred to as Enteric Group 41 by Bangert et al., 1988) also has been isolated from captive raptors (the bird order *Falconiformes*; the falcons) (Bangert et al., 1988). *L. adecarboxylata* appears to be important in the processing of “dry-cured” hams. It was the only species present at the end of the fast-curing process, and predominated along with some other *Enterobacteriaceae* at the end of the slow-curing process (Marin et al., 1996). These environmental reservoirs suggest additional ways that hospital and other patients can come in contact with *L. adecarboxylata*. If a blood culture from a patient on total parenteral nutrition is positive for *L. adecarboxylata*, all the liquids should be cultured as possible reservoirs.

Isolation, identification, culture preservation, phenotypic characterization Strains of *Leclercia adecarboxylata* grow well on media normally used in enteric bacteriology, and are typical *Enterobacteriaceae* in most respects. The typical yellow pigment is probably present in most cultures on initial isolation, but is easily lost. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, and antibiotic susceptibility. Table BXII.γ.232 below summarizes the phenotypic characteristics of *L. adecarboxylata*. Also see Table BXII.γ.193 of the chapter on the family *Enterobacteriaceae* that has the tabulated biochemical reactions of *L. adecarboxylata* and other *Enterobacteriaceae*.

TAXONOMIC COMMENTS

History Leclerc (1962) recognized a group that he referred to as “yellow-pigmented coliform bacteria” that was similar to *E. coli* in its IMViC reactions, but differed by producing a yellow pigment and fermenting many sugars. He proposed that it be recognized as a separate species in the genus *Escherichia*, *Escher-*

TABLE BXII.γ.232. Phenotypic characteristics of *Leclercia adecarboxylata* based on data from three different laboratories

Test or property	Study 1 ^a	Study 2	Study 3
Number of strains studied	20	85	33
Incubation temperature ^b	30°C	35°C	36°C
Final reading for the test ^b	2 d	2 d	2 d
Yellow pigment production, 2 d	60 ^c (30°C)	nd	37 ^c (25°C)
Yellow pigment production, 7 d	nd	14 (25°C)	63 (25°C)
Indole production	90	100	100
Methyl red	nd	100	100
Voges-Proskauer	0	0	0
Citrate utilization (Simmons)	0	0	0
H ₂ S production	0	0 (KIA)	0 (TSI)
Urea hydrolysis	70	10	48
Phenylalanine deaminase	0	0	0
Lysine decarboxylase	0	0	0
Arginine dihydrolase	0	0	0
Ornithine decarboxylase	0	0	0
Motility, 35–37°C	nd	99	79
Motility, 22–25°C	nd	nd	60
Motility, 30°C	100	nd	nd
Gelatin hydrolysis	0	0	0 (22°C)
KCN test (% resistant)	100	100	97
Malonate utilization	100	100	93
D-Glucose, acid production	100	100	100
D-Glucose, gas production	100	99	97
Lactose fermentation	85	100	93
Sucrose fermentation	20	52	66
D-Mannitol fermentation	100	100	100
Dulcitol fermentation	70	87	86
Salicin fermentation	100	100	100
Adonitol fermentation	55	98	93
myo-Inositol fermentation	nd	0	0
D-Sorbitol fermentation	8	9	0
D-Arabinose fermentation	100	100	100
Raffinose fermentation	30	59	66
L-Rhamnose fermentation	100	100	100
Maltose fermentation	100	100	100
D-Xylose fermentation	100	100	100
Trehalose fermentation	100	100	100
Cellobiose fermentation	100	100	100
α-Methyl-D-glucoside fermentation	0	0	0
Erythritol fermentation	0	0	0
Esculin hydrolysis	100	100	100
Melibiose fermentation	100	100	100
D-Arabitol fermentation	nd	96	96
Glycerol fermentation	100 (30 d)	3	3
Mucate fermentation	90	65	93
Tartrate fermentation (Jordan)	70	nd	83
Acetate utilization	nd	0	28
Lipase (corn oil)	nd	0	0
DNase production	0	0 (25°C)	0 (25°C)
Nitrate reduction to nitrite	100	99	100
Oxidase	nd	0	0
ONPG test	100	100	100
D-Mannose fermentation	100	100	100
Tyrosine hydrolysis	nd	nd	0
D-Galactose fermentation	100	100	100
MacConkey agar, growth	nd	nd	100
Catalase	nd	nd	+
Citrate, Christensen	nd	0	0
H ₂ S on peptone iron agar	nd	nd	0
<i>Kauffmann-Peterson tests:</i>			
Citrate	nd	0	nd
D-Tartrate	0	0	nd
L-Tartrate	30	nd	nd
meso-Tartrate	0	nd	nd
Growth at 4–5°C	15 (30 d)	49 (14 d)	nd
Growth at 35–37°C	nd	nd	+

(continued)

TABLE BXII.γ.232. (cont.)

Test or property	Study 1 ^a	Study 2	Study 3
Growth at 41°C	85	nd	nd
Growth at 42°C	nd	0	nd
Growth at 50°C	0	nd	nd
β-Glucuronidase	0	0	nd
Pectinase	nd	0	nd
Tetrathionate reductase	25	54	nd
β-Xylosidase	95	100	nd
<i>Carbon source utilization:</i>			
Number of compounds tested	123	122	nd
Number (%) utilized	26 (21.1)	31 (25.4)	nd
Amygdalin fermentation	nd	0	nd
D-Arabinose fermentation	nd	0	nd
L-Arabitol fermentation	nd	0	nd
Arbutin fermentation	nd	100	nd
D-Fructose fermentation	100	nd	nd
D-Fucose fermentation	nd	0	nd
L-Fucose fermentation	nd	0	nd
Gentiobiose fermentation	nd	100	nd
D-Gluconate fermentation	nd	100	nd
2-Ketogluconate fermentation	nd	100	nd
5-Ketogluconate fermentation	nd	0	nd
Glycogen fermentation	5	nd	nd
Inulin fermentation	nd	0	nd
D-Levulose fermentation	nd	100	nd
D-Lyxose fermentation	nd	0	nd
L-Lyxose fermentation	nd	0	nd
Melizitose fermentation	0	0	nd
Methylmannoside fermentation	nd	0	nd
D-Ribose fermentation	100	100	nd
L-Sorbose fermentation	0	0	nd
D-Tagatose fermentation	nd	0	nd
D-Turanose fermentation	nd	0	nd
Xylitol fermentation	nd	0	nd
Tween 80 hydrolysis	nd	0	nd

^aStudy 1, Izard et al. (1985); study 2, Tamura et al. (1986); study 3, Farmer (2003).^bThe standard conditions used in each study; a different temperature or time for the final reading is indicated in parentheses. Abbreviations: TSI, triple sugar iron agar; KIA, Kligler iron agar; ONPG, o-nitrophenyl-β-D-galactopyranoside.^cEach number is the percentage of strains that are positive. Most positive reactions occur during the first 24 h. For a few tests in study there were not enough quantitative data, so these results are represented by symbols with the usual meaning stated in this *Manual*; nd, not determined.

ichia adecarboxylata. Ewing and Fife (1972) studied the type strain of *Escherichia adecarboxylata* and proposed a different classification. They classified it in their redefined species (which is now known to be heterogeneous) *Enterobacter agglomerans* as “a typical strain of biogroup G3”.

Escherichia adecarboxylata* as a distinct group of *Enterobacteriaceae*; proposal of *Leclercia Several other investigators considered the taxonomic position of *E. adecarboxylata* and concluded that it is different from *Escherichia*, the *Enterobacter agglomerans*–*Erwinia* complex, and from other *Enterobacteriaceae* (see Table BXII.γ.233) (Gavini et al., 1983b; Sakazaki et al., 1983; Izard et al., 1985; Verdonck et al., 1987). The situation was clarified when Tamura et al. (1986) used DNA–DNA hybridization to study their collection of strains. They showed that *E. adecarboxylata* was different from other taxa of *Enterobacteriaceae*. The type strain of *E. adecarboxylata* was more than 70% related to eight other strains and was 64–66% related to two additional strains. It was only 32% related to the type strain of *Enterobacter agglomerans* and only 26% related to the type strain of *Escherichia coli*. Because it was distinct by both DNA–DNA hybridization and phenotype, they proposed a new genus *Leclercia* with one species *Leclercia adecarboxylata* to

include strains formerly classified as *Escherichia adecarboxylata*. Their proposal is an alternative classification that has been well accepted by the scientific community.

Phylogeny based on 16S rDNA sequencing Recently a complete 16S rDNA sequence of "strain LBV 449" (which is not the type strain) was deposited (GenBank accession number AJ276393) (De Baere et al., 2001). A 16S rRNA tree that includes *L. adecarboxylata* can be found in the chapter on the family *Enterobacteriaceae*. The 16S rDNA sequencing data agree with data from DNA–DNA hybridization experiments that *Leclercia* is distinct from other genera of *Enterobacteriaceae* (Tang et al., 1998). Unfortunately, *L. adecarboxylata* was not included in the tree published by Spröer et al. (1999).

Several sequences related to mercury resistance have also been deposited.

FURTHER READING

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- Izard, D., J. Mergaert, F. Gavini, A. Beji, K. Kersters, J. De Ley and H. Leclerc. 1985. Separation of *Escherichia adecarboxylata* from the *Erwinia herbicola*-*Enterobacter agglomerans* complex and from the other enterobacteriaceae by nucleic acid and protein electrophoretic techniques. Ann. Inst. Pasteur Microbiol. 136B: 151–168.
- Tamura, K., R. Sakazaki, Y. Kosako and E. Yoshizaki. 1986. *Leclercia ade-*

TABLE BXII.γ.233. Differentiation of *Leclercia adecarboxylata* from its closest phenotypic relatives in *Enterobacteriaceae*

Test	<i>Leclercia adecarboxylata</i>	<i>Escherichia coli</i>	" <i>Enterobacter agglomerans</i> complex" ^a
Indole production	+	+	d
Methyl red	+	+	d
Lysine decarboxylase	–	+	–
Arginine dihydrolase	–	d	–
Ornithine decarboxylase	–	d	–
Yellow pigment	d	–	d
KCN	+	–	d
D-Sorbitol fermentation	–	+	d
Cellobiose fermentation	+	–	d
D-Glucose, gas	+	+	d
Dulcitol fermentation	+	d	d
D-Arabitol fermentation	+	–	–

^aSee Farmer (2003) for a more detailed description of this "vernacular name". It is a term defined for practical identification in the clinical microbiology laboratory. It includes over a dozen DNA–DNA hybridization groups that were originally included in the species *Enterobacter agglomerans*, which is now known to be a heterogeneous species.

carboxylata, gen. nov., comb. nov., formerly known as *Escherichia adecarboxylata*. Curr. Microbiol. 13: 179–184.

Temesgen, Z., D.R. Toal and F.R. Cockerill, III. 1997. *Leclercia adecarboxylata* infections: case report and review. Clin. Infect. Dis. 25: 79–81.

List of species of the genus *Leclercia*

- 1. *Leclercia adecarboxylata*** (Leclerc 1962) Tamura, Sakazaki, Kosako, and Yoshizaki 1987, 179^{VP} (Effective publication: Tamura, Sakazaki, Kosako, and Yoshizaki 1986, 183) (*Escherichia adecarboxylata* Leclerc 1962, 737; *Enterobacter agglomerans* biogroup G3 Ewing and Fife 1972, 10.) *a.de.car.box.y.la'ta*. Gr. adj. *a* without; M. Fr. n. *decarboxyl* removal of a molecule of carbon dioxide from an organic compound; *adecarboxylata* without decarboxylase activity; because it has negative reactions in lysine decarboxylase, ornithine decarboxylase and arginine dihydrolase; i.e., "triple decarboxylase negative".

The characteristics are as described for the genus; a more complete description is given in Table BXII.γ.232. Isolated from human clinical specimens, environmental samples, food, and water. Its clinical significance is not fully documented but its potential role as a pathogen is suggested

by isolates from blood and similar specimens that are normally sterile. However, it may be colonizing rather than infecting nonsterile body sites. The isolates from food, drinking water, feces, and an intravenous fluid bottle suggest ways that humans come in contact with it. There is no evidence that it can cause diarrhea or intestinal infections. It should be considered a rarely isolated species of *Enterobacteriaceae*, and a possible opportunistic pathogen (extraintestinal infections only) for humans. The type strain was isolated from drinking water by Leclerc, 1962.

The mol% G + C of the DNA is: 52–55 (*T_m*).

Type strain: 1783, ATCC 23216, CIP 82.92, DSM 5077, HAMBI 1696, JCM 1667, LMG 2803.

Additional Remarks: The American Type Culture Collection includes 7 other strains of *L. adecarboxylata*, including four strains from "apple fruit".

Other Organisms

Currently *Leclercia* has only one species, but it will be interesting to see if future studies will change or clarify the current concept or circumscription of the genus. Tamura et al. (1986) included only one species in *Leclercia*; however, Table 2 of their paper includes strains 1523 and 363, which were only 66.2% and 64.2% related to the type strain by DNA–DNA hybridization; 70% is the usual cutoff point for the level of species. They did not include divergence values (ΔT_m), so it is more difficult to deter-

mine if these two strains are as related to the type strain as the other eight strains. The CDC collection contains five strains that were identified as "possible-probable" *Leclercia adecarboxylata* because their phenotypic characteristics were similar to, but different from, strains now included in the species. These five strains, the two of Tamura et al. (1986), and similar strains should be investigated as possible additional species of *Leclercia*.

Genus XIX. *Leminorella* Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985c, 375^{VP} (Effective publication: Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985b, 235)

J.J. FARMER III AND FRANCES W. BRENNER

Le.mi.no.rel'la. M.L. dim. -ella ending; M.L. fem. n. *Leminorella* named to honor Leon Le Minor, a French microbiologist, for his many contributions to enteric bacteriology including the nomenclature, classification, and serotyping of *Salmonella*; lysogeny; metabolic plasmids; and new and rapid biochemical tests. The name also honors Simone Le Minor, who also made many contributions to enteric bacteriology as head of the National *Salmonella* Centre of France and for her research on *Serratia* serotyping.

Small rod-shaped cells, conforming to the general definition of the family *Enterobacteriaceae*. Gram-negative. **Nonmotile at 36°C and 25°C.** Contain the enterobacterial common antigen. Facultatively anaerobic. Catalase positive (strong and rapid). Oxidase negative. Nonpigmented. Ferment, rather than oxidize, D-glucose. Reduce nitrate to nitrite. **Inactive biochemically. Positive for H₂S production (TSI and PIA), tyrosine hydrolysis, and fermentation of L-arabinose, D-xylose, and L-tartrate.** Negative for indole production, Voges-Proskauer, urea hydrolysis, phenylalanine deaminase, lysine decarboxylase, arginine dihydrolase, ornithine decarboxylase, growth in the presence of cyanide (KCN test), malonate utilization, esculin hydrolysis, gelatin hydrolysis (22°C), lipase (corn oil), DNase, and the fermentation of lactose, sucrose, D-mannitol, salicin, adonitol, myo-inositol, D-sorbitol, raffinose, L-rhamnose, maltose, trehalose, cellobiose, α-methyl-D-glucoside, erythritol, melibiose, D-arabitol, glycerol, D-mannose, and D-galactose. Susceptible to colistin, nalidixic acid, sulfadiazine, gentamicin, kanamycin, tetracycline, chloramphenicol; **resistant to streptomycin, penicillin, ampicillin, carbenicillin, and cephalothin** (disk diffusion method on Mueller-Hinton agar).

Usually isolated from feces, but there is no evidence that strains cause diarrhea or intestinal infections. Some strains agglutinate in *Salmonella* diagnostic antisera, and **can be misidentified as being *Salmonella*.** **Rarely isolated from other clinical specimens.** Ecological niche may be the human intestinal tract since there are only a few isolates from other clinical specimens, and no reported isolates from animals, food, water or the environment. **A very rarely isolated genus of *Enterobacteriaceae*.**

The mol% G + C of the DNA is: not reported.

Type species: *Leminorella grimontii* Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985c, 375 (Effective publication: Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985b, 235.)

FURTHER DESCRIPTIVE INFORMATION

Literature Since the genus was described in 1985, there have been only a handful of reports in the literature, six cataloged in MEDLINE and 11 cataloged in BIOSIS. Except for the original report (Hickman-Brenner et al., 1985b), none of these has dealt specifically with *Leminorella*. The literature reports that have mentioned *Leminorella* include reviews and taxonomic studies of new *Enterobacteriaceae* (Schindler, et al., 1992; Gilchrist, 1995; Aleksic and Bockemühl, 1999); evaluation of commercial identification products or "kits" for *Enterobacteriaceae* (Kitch et al., 1994); and studies on physiology, metabolism, or biochemistry that included *Leminorella* strains (Bouvet and Grimont, 1987a; Satta et al., 1988; Bouvet et al., 1989; Grimont and Bouvet, 1989; Hodinka et al., 1991; Thaller et al., 1995; Hamana, 1996). In addition to the genus and species names, "Enteric Group 57" should be included as a search term in computerized literature searches, although it is rarely used today.

Sources and clinical significance Nine of the original *Leminorella* isolates were from feces, and two were from urine. Since the original paper in 1985, the CDC Enteric Reference Laboratory has received and identified nine additional human isolates, seven from feces and one each from urine and a decubitus wound. The clinical significance of *Leminorella* strains in these extraintestinal specimens is unknown, and further study and case reports are needed. Although strains of *Leminorella* have occasionally been isolated from feces of people with diarrhea, there is no evidence that it can actually cause diarrhea or intestinal infections. Its interest for clinical microbiology and public health appears to be more as a nuisance because *Leminorella* strains might be misidentified as *Salmonella* unless more complete biochemical and serological testing is done.

Original misidentification as *Salmonella* Several of the isolates from feces had been referred to the CDC's *Salmonella* reference laboratory because the strains agglutinated one or more *Salmonella* antisera. Presumably these tests were done with commercial antisera and with "live", rather than alcohol-treated, cultures, and the referring laboratory could not completely serotype the strain or confirm an identification as *Salmonella*. Four *L. grimontii* strains and one *L. richardii* strain have been reported to cross-react when tested at CDC with alcohol-treated antigens and CDC reference antisera. One of the strains agglutinated very weakly in O group B serum and in single factor O27 serum. The other strains did not agglutinate.

Isolation, identification, culture preservation, phenotypic characterization Strains of *Leminorella* grow on media normally used in enteric bacteriology, but they grow more slowly and are slower or more inactive biochemically than typical *Enterobacteriaceae*. One strain of *L. grimontii* and all four strains of *L. richardii* were negative for D-glucose fermentation after incubation for 24 h, but all became positive at 48 h (Hickman-Brenner et al., 1985b). On MacConkey agar at 24 h, strains appear as small colonies that are colorless (lactose negative). See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, computer programs, and antibiotic susceptibility.

Biochemical reactions Table BXII.γ.234 summarizes the biochemical reactions of the two *Leminorella* species. The tests useful for differentiation of *L. grimontii* and *L. richardii* are given in Table BXII.γ.235. Compared to other species of *Enterobacteriaceae*, strains of *Leminorella* are weak and inactive biochemically. One important characteristic is hydrogen sulfide production. Within 48 h on triple sugar iron agar, the species give an alkaline slant and a weak acid reaction in the butt with H₂S production.

Antibiotic susceptibility In their original description, Hickman-Brenner et al. (1985b) gave the zone sizes for each of the

TABLE BXII.γ.234. Biochemical reactions of *Leminorella grimontii* and *L. richardii* based on strains studied by DNA–DNA hybridization

Characteristic	<i>Leminorella grimontii</i>	<i>Leminorella richardii</i>
Number of strains	6 ^a	4 ^a
Indole production	0 ^b	0
Methyl red	100	0
Voges–Proskauer	0	0
Citrate utilization (Simmons)	100	0
H ₂ S production (TSI ^c)	100	100
H ₂ S production (PIA)	100	100
Urea hydrolysis	0	0
Phenylalanine deaminase	0	0
Lysine decarboxylase	0	0
Arginine dihydrolase	0	0
Ornithine decarboxylase	0	0
Motility at 36°C	0	0
Gelatin hydrolysis (22°C)	0	0
Growth in KCN	0	0
Malonate utilization	0	0
D-Glucose, acid production at 24 h	83	0
D-Glucose, acid production at 48 h	100	100
D-Glucose, gas production	33 (100)	0
Lactose fermentation	0	0
Sucrose fermentation	0	0
D-Mannitol fermentation	0	0
Dulcitol fermentation	83	0
Salicin fermentation	0	0
Adonitol fermentation	0	0
myo-Inositol fermentation	0	0
D-Sorbitol fermentation	0	0
L-Arabinose fermentation	100	100
Raffinose fermentation	0	0
L-Rhamnose fermentation	0	0
Maltose fermentation	0	0
D-Xylose fermentation	83	100
Trehalose fermentation	0	0
Cellobiose fermentation	0	0
α-Methyl-D-glucoside fermentation	0	0
Erythritol fermentation	0	0
Esculin hydrolysis	0	0
Melibiose fermentation	0	0
D-Arabitol fermentation	0	0
Glycerol fermentation	17 (33)	0
Mucate fermentation	100	50 (75)
Tartrate fermentation (Jordan's)	100	100
Acetate utilization	0	0
Lipase (corn oil)	0	0
DNase production (25°C)	0	0
Nitrate reduction to nitrite	100	100
Oxidase	0	0
ONPG test	0	0
Yellow pigment (25°C)	0	0
D-Mannose fermentation	0	0
Tyrosine hydrolysis	83 (100)	75 (100)
D-Galactose fermentation	0	0

^aThese are the original strains that were studied by DNA–DNA hybridization and used by Hickman-Brenner et al. (1985b) to define the species.

^bEach number gives the percentage positive after 2 d incubation at 36°C (unless a different time or temperature is indicated; phenylalanine, oxidase, and nitrate reduction are done only at 1 d). The cumulative % positive between 3 and 7 d is given for some tests in parentheses. Some of the percentages are slightly different from those given in the original descriptions (Hickman-Brenner et al., 1985b) and are based on tests results that were repeated after the original publication.

^cTSI, triple sugar iron agar; PIA, peptone iron agar; ONPG, o-nitrophenyl-β-D-galactopyranoside.

11 strains with the 12 different antibiotics that have been used for many years in their reference laboratory as a aid in identification. There was little variation among the three *Leminorella* species, and all 11 strains were resistant to streptomycin, penicillin, ampicillin, carbenicillin, and cephalothin. Although this

TABLE BXII.γ.235. Differentiation of *Leminorella grimontii* and *L. richardii*^a

Test	Incubation time, days	<i>L. grimontii</i>	<i>L. richardii</i>
Methyl red	2	100	0
Citrate utilization (Simmons)	2	100	0
D-Glucose, acid production	1	83	0
D-Glucose, gas production	7	100	0
Dulcitol fermentation	2	83	0
Glucosaminidase activity ^b		100	0

^aThese data are based on results in Table BXII.γ.234 which included only strains that were documented to belong to these two species by DNA–DNA hybridization (Hickman-Brenner et al., 1985b). Each number gives the percentage positive at the specified indicated period; all the biochemical tests were done at 36°C.

^bData from Hodinka et al. (1991).

is not a unique pattern, the antibiogram can be a useful additional way of differentiating *Leminorella* from other genera of *Enterobacteriaceae*.

DNA, RNA, and protein sequences There is a full sequence (bases 1 to 1482) of the 16S rRNA gene of *L. grimontii* listed in GenBank (Spröer et al., 1999).

A 16S rDNA tree that includes *Leminorella grimontii* can be found in the chapter on the family *Enterobacteriaceae* (Fig. BXII.γ.189). The 16S rDNA sequencing data agree with data from DNA–DNA hybridization that *Leminorella* is distinct from other genera of *Enterobacteriaceae*. In the tree, *Pragia fontium* is the closest relative, in agreement with the tree published by Spröer et al. (1999) with respect to these two genera being on a distinct branch. However, the two trees differ in the placement of other organisms.

DIFFERENTIATION OF THE GENUS *LEMINORELLA* FROM OTHER GENERA

Strains of *Leminorella* can be distinguished because they do not ferment D-mannose, but hydrolyze tyrosine, much like strains of *Proteus*; however, they are negative for urea hydrolysis and phenylalanine deaminase. They differ from *Salmonella* because they are lysine and ornithine decarboxylase negative and do not ferment D-sorbitol. Also see Table BXII.γ.193 of the chapter on the family *Enterobacteriaceae*, which gives the percentage positive for 47 biochemical tests done at the Enteric Reference Laboratories at CDC with standard media and methods. These tests have proved useful for identification and for differentiating *Leminorella* from other species in the family.

TAXONOMIC COMMENTS

History and discovery In 1980 one of us (JJF, unpublished) noticed two H₂S⁺ strains that seemed to be distinct from other *Enterobacteriaceae*, and coded these as a “possible new group of *Enterobacteriaceae*” in the laboratory’s master computer database. In 1981 diagnostic culture 1944-81 was received and studied in more detail (Hickman-Brenner et al., 1985b). It was from a stool culture of a 7-month-old patient in Hawaii with gastroenteritis. The culture had a low identification score for other *Enterobacteriaceae* in the computer program GEORGE (Farmer et al., 1985a). However, the computer program STRAIN MATCHER listed nine additional strains that had been reported as “unidentified” that were very similar biochemically. Several of these strains had been sent to CDC because they reacted weakly in *Salmonella* antisera. Five strains had been sent as *Salmonella*, or with a diagnosis of suspected salmonellosis or diarrhea; two were sent as “possible

Citrobacter"; and one was a "suspect H₂S⁺ *Shigella*" (Hickman-Brenner et al. 1985b). The vernacular name Enteric Group 57 was coined in 1981 (Hickman-Brenner and Farmer, unpublished; see Farmer et al., 1985a) for the original group of 10 strains, and the taxonomic position of the group was unknown. Enteric Group 57 was simply thought of as being a unique group of H₂S-producing strains that was being confused with *Salmonella* in some reference laboratories. An 11th strain was received in 1982.

Proposal of the genus *Leminorella* Hickman-Brenner et al. (1985b) used DNA–DNA hybridization (hydroxyapatite, ³²PO₄) to further characterize Enteric Group 58. Strain 1944-81^T was labeled and was highly related (77–97% at 60°C with divergence values of 0–0.5%) to five other Enteric Group 58 strains. Strain 1944-81 was only 3–16% related to other species of *Enterobacteriaceae*. Because it was distinct from all *Enterobacteriaceae* by both DNA–DNA hybridization and phenotype, it was given a scientific name. The group of six strains was named *Leminorella grimontii*, which was designated as the type species for the genus *Leminorella*. Five other strains were 32–60% related to 1944-81, so strain 0978-82 was chosen from this group and tested by DNA hybridization against all 11 strains. Three strains were 93–94% related to strain 0978-82 with divergence values of 0.5–1.0. This group of four strains was named *Leminorella richardii*. The remaining strain 3346-72 was 60% and 40% related to strains 1944-81 and 0978-82, respectively. It was considered to be a third *Leminorella* species, that was closer to *L. grimontii*, and was given the vernacular name "*Leminorella* species 3". Since there was only one strain, it was not given a scientific name. Although *Leminorella* species 3 was distinct by DNA hybridization, it could not be distinguished from *L. grimontii* by simple tests (Hickman-Brenner et al., 1985b). Since the original report in 1985, additional strains of *Leminorella* have been studied biochemically, but not by DNA–DNA hybridization. *Leminorella* and its two named species gained standing in no-

menclature in July 1985, when the names appeared on Validation List 18 (Hickman-Brenner et al., 1985c).

Problems in routine identification In the original publication DNA hybridization was used to divide the strains into two named species, and a description was written based on this clear separation (Tables BXII.γ.234 and BXII.γ.235). Unfortunately, *Leminorella* species 3 cannot be distinguished from *L. grimontii* with simple tests (Hickman-Brenner et al., 1985b). This strain illustrates the difficulty in identifying *Leminorella* cultures to the species level without the benefit of DNA–DNA hybridization. Since the original report in 1985, the Enteric Reference Laboratory has received 10 additional strains of *Leminorella*. Based on the differential reactions listed in Table BXII.γ.235, five of these were identified as *L. grimontii*; one as a "possible-probable" *L. grimontii* and four as *L. richardii*. Five of the strains identified as *L. grimontii* had at least one test that was in disagreement with the composite results in Table BXII.γ.234. The strains identified as *L. richardii* were more typical, and agreed with the definition given in Table BXII.γ.235, except that one strain had one atypical reaction. Thus, the identification of new strains of *Leminorella* to the genus level is probably correct, but the identification to the species level is more tentative and needs confirmation by DNA–DNA hybridization or other methods. Enzyme profiles (Hodinka et al., 1991) and whole-cell protein patterns (Schindler et al., 1992) appear promising for differentiation. Further study and better and simpler methods are needed to assist routine identification.

FURTHER READING

Hickman-Brenner, F.W., M.P. Vohra, G.P. Huntley-Carter, G.R. Fanning, V.A.I. Lowery, D.J. Brenner and J.J. Farmer III. 1985. *Leminorella*, a new genus of *Enterobacteriaceae*: identification of *Leminorella grimontii* sp. nov. and *Leminorella richardii* sp. nov. found in clinical specimens. J. Clin. Microbiol. 21: 234–239.

List of species of the genus *Leminorella*

1. ***Leminorella grimontii*** Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985c, 375^{VP} (Effective publication: Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985b, 235) *gr̄i.mon' ti.i.* M.L. gen. n. *grimontii* named to honor Patrick Grimont and Francine Grimont, French microbiologists at the Pasteur Institute for their many contributions to enteric bacteriology.

The characteristics are as given for the genus. The biochemical reactions are given in more detail in Table BXII.γ.234, and tests for the differentiation of *L. grimontii* and *L. richardii* are given in Table BXII.γ.235. Isolated from human feces, but there is no evidence that it can cause diarrhea or intestinal infections. Rarely isolated from other clinical specimens. Its clinical significance in these extra-intestinal specimens needs further study. One isolate was from mouse feces. The type strain was isolated from the feces of a 7-month-old child with gastroenteritis.

The mol% G + C of the DNA is: not reported.

Type strain: ATCC 33999, CDC 1944-8, DSM 5078.

GenBank accession number (16S rRNA): AJ233421.

Additional Remarks: The American Type Culture Collection includes four other strains documented to be *L. gri-*

montii by DNA–DNA hybridization: ATCC 43006 (CDC 3257-77), from human feces, California; ATCC 43005 (CDC 3244-76), from human urine of an 80-year-old woman with a urinary tract infection, Hawaii; ATCC 43007 (CDC 3595-77), from urine of a 9-year-old girl, Pennsylvania; and ATCC 43008 (CDC 0301-79), from mouse feces, Maryland.

2. ***Leminorella richardii*** Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985c, 375^{VP} (Effective publication: Hickman-Brenner, Vohra, Huntley-Carter, Fanning, Lowery, Brenner and Farmer 1985b, 235.) *ri.char' di.i.* M.L. gen. n. *richardii* named to honor Claude Richard, a French microbiologist at the Pasteur Institute for his many contributions to enteric bacteriology.

The characteristics are as given for the genus. The biochemical reactions are given in more detail in Table BXII.γ.234, and the differentiation of *L. richardii* from *L. grimontii* is given in Table BXII.γ.235. Isolated from human feces, but there is no evidence that it can cause diarrhea or intestinal infections. Nonhuman strains have not been reported. The type strain was isolated from the feces of a patient in Texas who had diarrhea. She also had systemic lupus erythematosus, and was on corticosteroid therapy. No other enteric pathogens were present.

The mol% G + C of the DNA is: not reported.

Type strain: ATCC 33998, CDC 0978-82.

Additional Remarks: The American Type Culture Collection includes three other strains documented to be *L. ri-*

chardii by DNA–DNA hybridization; all were from human feces in the USA: ATCC 43009 (CDC 598-78), Indiana; ATCC 43010 (CDC 2209-80), Pennsylvania; and ATCC 43011 (CDC 2502-80), Texas.

Other Organisms

1. "Leminorella species 3"

In addition to the two named species of *Leminorella*, Hickman-Brenner et al. (1985b) also described a third species. Since there was only one strain, and because it could not be differentiated from *L. grimontii* by phenotypic tests, they gave it the vernacular name "*Leminorella* species 3".

The reference strain, ATCC 43012, was isolated from the human feces, Georgia, USA. This is the only strain of species 3, and we suggest that it be designated the type strain if species 3 is given a scientific name.

Deposited strain: ATCC 43012, CDC 3346-72.

Strains studied since 1985 have been reported without the benefit of DNA hybridization. One strain (1201-84) was reported as a "possible-probable" *Leminorella grimontii*. Five were reported as *L. grimontii* because they were more active biochemically. However, four of these had one or more atypical test results for this species. All new strains of *Leminorella* should be studied by DNA hybridization or other methods to determine their correct taxonomic position.

Genus XX. *Moellerella* Hickman-Brenner, Huntley-Carter, Saitoh, Steigerwalt, Farmer and Brenner 1984c, 355^{VP} (Effective publication: Hickman-Brenner, Huntley-Carter, Saitoh, Steigerwalt, Farmer and Brenner 1984a, 462)

J.J. FARMER III AND FRANCES W. BRENNER

Moellerella. M.L. dim. ending *-ella*; M.L. fem. n. *Moellerella* named to honor Vagn Møller for his contributions to enteric bacteriology, especially for Moeller media for the determination of lysine decarboxylase, ornithine decarboxylase, and arginine dihydrolase, that are widely used for the identification of *Enterobacteriaceae*.

Small rod-shaped cells, conforming to the general definition of the family *Enterobacteriaceae*. Contain the enterobacterial common antigen. Gram negative. Nonmotile. Facultatively anaerobic. Catalase positive. Oxidase negative. Nonpigmented. Ferment, rather than oxidize, D-glucose and other carbohydrates. Reduce nitrate to nitrite. **Positive for methyl red, citrate utilization (Simmons), ONPG, and the fermentation of lactose, sucrose, adonitol, raffinose, melibiose, D-arabitol, D-mannose, and D-galactose.** Negative for indole production, Voges–Proskauer, H₂S production (TSI), urea hydrolysis, phenylalanine deaminase, lysine decarboxylase, arginine dihydrolase, ornithine decarboxylase, motility at 36°C, malonate utilization, esculin hydrolysis, gelatin hydrolysis (22°C), lipase (corn oil), and DNase, and the fermentation of dulcitol, salicin, *myo*-inositol, D-sorbitol, L-arabinose, L-rhamnose, D-xylose, trehalose, cellobiose, α-methyl-D-glucoside, erythritol, glycerol, and mucate. No visible gas produced during fermentation.

Most strains are susceptible to nalidixic acid, gentamicin, streptomycin, kanamycin, chloramphenicol, and cephalothin (disk diffusion method on Mueller-Hinton agar); **resistant to colistin, penicillin, ampicillin, and carbenicillin.** Variable susceptibility to sulfadiazine and tetracycline. Rarely isolated from human clinical specimens that are normally sterile. Most isolates have been from human feces, but there is **no evidence that it actually causes diarrhea or intestinal infections.** Natural reservoirs and

ecological niches are not known. Can occur in water and food. **A rarely isolated genus of *Enterobacteriaceae* that is probably an opportunistic pathogen.**

The mol% G + C of the DNA is: not determined.

Type species: *Moellerella wisconsensis* Hickman-Brenner, Huntley-Carter, Saitoh, Steigerwalt, Farmer and Brenner 1984c, 355 (Effective publication: Hickman-Brenner, Huntley-Carter, Saitoh, Steigerwalt, Farmer and Brenner 1984a, 462.)

FURTHER DESCRIPTIVE INFORMATION

Literature Since *Moellerella* was described in 1984, there have been only a few reports in the literature, 11 cataloged in MEDLINE and 12 cataloged in BIOSIS. In addition to the genus and species names, "Enteric Group 46" should also be included as a search term in computerized literature searches, although it is rarely used today. The literature reports with *M. wisconsensis* include reviews and taxonomic studies of new *Enterobacteriaceae* (Farmer et al., 1985a; Richard, 1989; Gilchrist, 1995; Pokhil, 1996; Aleksic and Bockemühl, 1999); evaluation of commercial identification products or "kits" for *Enterobacteriaceae* (Gonzalez et al., 1986; Kitch et al., 1994); and surveys or comparisons of the family *Enterobacteriaceae* for metabolic pathways (Bouvet et al., 1989), acid phosphatases (Thaller et al., 1995), siderophores (Rabsch and Winkelmann, 1991), and polyamines (Hamana, 1996).

Occurrence in human feces Strains of *M. wisconsensis* were first recognized because of six strains submitted to CDC from Wisconsin; all were from human feces. Eight of the nine original cultures described by Hickman-Brenner et al. (1984b) were from feces and the other one was from water. Most isolates of *M. wisconsensis* have been from people with diarrhea, but there is no evidence that the strains are actually causing diarrhea or intestinal infections (Hickman-Brenner et al., 1984a). Marshall et al. (1986) reported the isolation of three strains of *M. wisconsensis* from 400 stool specimens screened for this organism with the aid of a new selective medium, developed in their laboratory. There was one report of *M. wisconsensis* isolation in the U.K., but its etiological role was not established in any of the diarrhea cases.

The interest of *M. wisconsensis* to clinical microbiology and public health laboratories may be more as a possible nuisance because strains have been picked as "suspect *Salmonella*" or "suspect *Yersinia*". More complete biochemical and serological testing should easily rule out these incorrect identifications.

Occurrence in extraintestinal human specimens Wallet et al. (1994) reported the isolation of *M. wisconsensis* from a bronchial aspirate of a patient with inhalation pneumonia. It was from an autopsy culture of a patient admitted in a deep coma following cardiac arrest. They concluded that the origin of the organism may have been from digestive secretions that inoculated the lower respiratory tract. This was the first isolate of *M. wisconsensis* from a human source other than stool or gallbladder. Wittke et al. (1985) reported the isolation of *M. wisconsensis* from the infected gallbladder of a 71-year-old man with typical signs of acute cholecystitis. Equal numbers of enterococci were also isolated. Serum from the patient taken on the 24th day after cholecystectomy did not contain agglutinating antibodies to heated or unheated suspensions of *M. wisconsensis*, providing negative evidence for its pathogenic role. Ohanessian et al. (1987) reported the isolation of *M. wisconsensis* along with *Hafnia alvei* from the infected gallbladder of a 77-year-old woman with coronary cardiac failure and gallstones. In 1985 the CDC's Enteric Reference Laboratory received the first isolate from human blood.

Clinical significance of human isolates The clinical significance and the ability of *M. wisconsensis* to actually cause infections should be carefully evaluated in cases that yield this new organism. Systematic study and good case reports are needed. The isolates described above suggest that it is probably an opportunistic pathogen with only a slight capacity to cause extraintestinal infections in humans.

Occurrence in other animals Bangert et al. (1988) isolated *M. wisconsensis* and other *Enterobacteriaceae* in a fecal culture survey of 47 captive raptors (the bird orders *Falconiformes* and *Strigiformes*). Giordano-Dias et al. (1997) reported isolates from the sandfly *Lutzomyia longipalpis* (*Diptera: Psychodidae*) maintained in the laboratory.

Occurrence in food and water One of the original nine strains reported by Hickman-Brenner et al. (1984a) was from a routine drinking water sample from a South Dakota (USA) town that did not chlorinate its water. Cabadajova and Kudrna (1988) of the Czech Republic isolated five strains of *M. wisconsensis* from food, but could not show an etiological role in relation to the corresponding human diarrhea cases.

Isolation, identification, culture preservation, phenotypic characterization Strains of *Moellerella* are not difficult to grow,

and are typical *Enterobacteriaceae* in most respects. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, computer programs, and antibiotic susceptibility. Strains grow on media normally used in enteric bacteriology. On MacConkey agar, colonies of *M. wisconsensis* are bright red with precipitated bile around them, and thus are indistinguishable from *Escherichia coli* colonies (Hickman-Brenner et al., 1984a). Marshall et al. (1986) reported the enhanced isolation of *M. wisconsensis* from stool specimens with a selective medium developed in their laboratory that contained bacitracin and polymyxin.

Biochemical reactions and differentiation from other *Enterobacteriaceae* Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* gives the results for *Moellerella* in 47 biochemical tests normally used for identification (Farmer, 1999). There are no genus- or species-specific tests or sequences for the identification of *Moellerella*. The best approach is to do a complete set of phenotypic tests (growth on plating media, biochemical tests, antibiogram, etc.) and compare the results with all the other named organisms in the family *Enterobacteriaceae*. This approach is described in more detail in the section on the family *Enterobacteriaceae*. The biochemical results of a test strain can be compared to all the organisms listed in Table BXII.γ.193 in that chapter. Several computer programs greatly facilitate analyzing the results.

Antibiotic susceptibility In their original description, Hickman-Brenner et al. (1984a) gave the zone sizes for each of the nine strains around 12 different antibiotics (the standard "antibiogram" that has been used for many years in their reference laboratory as an aid in identification). There was some variation among the strains, but all were resistant to colistin and penicillin. Although this is not a unique pattern, the antibiogram can be a useful additional way of differentiating *M. wisconsensis* from other species of *Enterobacteriaceae*.

TAXONOMIC COMMENTS

Discovery and DNA-DNA hybridization In 1980 the vernacular name Enteric Group 46 was applied (Hickman-Brenner et al., 1984b) to a group of strains that had been studied at the Centers for Disease Control, and most had originally been sent from the Wisconsin State Laboratory of Hygiene. The strains were characterized biochemically and were phenotypically distinct from all of the described organisms in the family *Enterobacteriaceae*. By DNA-DNA hybridization the strains were 78–97% related to the type strain, with divergence values (ΔT_m) of 0–1.5. Other *Enterobacteriaceae* were only 2–32% related. Because Enteric Group 46 was distinct by both DNA-DNA hybridization and phenotype, a new genus *Moellerella* with a single species, *Moellerella wisconsensis*, was proposed (Hickman-Brenner et al., 1984a).

FURTHER READING

- Hickman-Brenner, F.W., G.P. Huntley-Carter, Y. Saitoh, A.G. Steigerwalt, J.J. Farmer, III and D.J. Brenner. 1984. *Moellerella wisconsensis* a new genus and species of *Enterobacteriaceae* found in human stool specimens. *J. Clin. Microbiol.* 19: 460–463.
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- Wittke, J.W., S. Aleksic and H.H. Wuthe. 1985. Isolation of *Moellerella wisconsensis* from an infected human gallbladder. *Eur. J. Clin. Microbiol.* 4: 351–352.

List of species of the genus Moellerella

1. **Moellerella wisconsensis** Hickman-Brenner, Huntley-Carter, Saitoh, Steigerwalt, Farmer and Brenner 1984c, 355^{VP} (Effective publication: Hickman-Brenner, Huntley-Carter, Saitoh, Steigerwalt, Farmer and Brenner 1984a, 462.)

wis.con.sen'sis. M.L. fem. adj. *wisconsensis* pertaining to the state of Wisconsin, U.S.A., where most of the original strains were isolated.

The characteristics are as given for the genus and are

summarized in Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*. Isolated from human feces but rarely from extraintestinal specimens, also isolated from animals, water, and food. The type strain was isolated from the stool culture of a 16-year-old girl in Wisconsin.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 35017, CDC 2896-78, DSM 5076.

Additional Remarks: The American Type Culture Collection includes four other strains, all from human feces.

Genus XXI. Morganella Fulton 1943, 81^{AL}

J. MICHAEL JANDA AND SHARON L. ABBOTT

Mor.ga.nel'la. M.L. dim. ending *-ella*; M.L. fem. n. *Morganella* named after H. de R. Morgan, who first studied the organism.

Straight rods, $0.6\text{--}0.7 \times 1.0\text{--}1.7 \mu\text{m}$. Gram negative. Motile by means of peritrichous flagella. Facultatively anaerobic. **Oxidative deamination of various amino acids including L-phenylalanine and L-tryptophan** (Singer and Volcani, 1955). **Urease positive**. **Indole positive** (Penner, 1984). Relatively few carbohydrates fermented. **Ornithine decarboxylase positive**. **Acid produced from D-mannose but not D-xylose**. **Gelatin not degraded**. The genus *Morganella* presently resides in the class *Gammaproteobacteria* in the family *Enterobacteriaceae*, based upon DNA hybridization data and not rDNA sequencing (Stackebrandt et al., 1988). 16S rDNA sequence data are presently unavailable.

The mol% G + C of the DNA is: 50 (Falkow et al., 1962).

Type species: **Morganella morganii** (Winslow, Kliger and Rothberg 1919) Fulton 1943, 81 (*Bacillus morgani* (sic) Winslow, Kliger and Rothberg 1919, 481; *Proteus morganii* (Winslow, Kliger and Rothberg 1919) Yale 1939b, 435.)

FURTHER DESCRIPTIVE INFORMATION

M. morganii displays morphologic characteristics typical of members of the *Enterobacteriaceae* appearing as short, straight rods. Capsules not produced (McKell and Jones, 1976). Peritrichous flagella with normal and curly curvature have been described (Leifson et al., 1955). Normal flagella have a wavelength of $2.09\text{--}2.36 \mu\text{m}$ with an amplitude of $0.51 \mu\text{m}$. Curly flagella possess a shorter periodicity ($1.14\text{--}1.18 \mu\text{m}$) and amplitude ($0.37\text{--}0.40 \mu\text{m}$). Strains often express multiple fimbriae consisting of morphologically thin (outside diameter [o.d.] $4\text{--}5 \text{ nm}$) and thick forms (o.d. $7\text{--}8 \text{ nm}$) on the same cell (Old and Adegbola, 1982). The ultrastructure of *M. morganii* has not been investigated to any extent.

The peptidoglycan of *M. morganii* is O-acetylated at the C-6 hydroxyl group of N-acetylmuramyl residues (Clarke, 1993). The degree of O-acetylation ranges from 43.0–49.6% and appears responsible for resistance to muramidase (lysozyme) activity. The major fatty acids present in *M. morganii* lipopolysaccharide are 3-hydroxytetradecanoic ($\text{C}_{14:0} 3\text{OH}$), tetradecanoic ($\text{C}_{14:0}$), hexa-

decanoic ($\text{C}_{16:0}$), and dodecanoic ($\text{C}_{12:0}$) acids (Vasyurenko and Chernyavskaya, 1990). Cellular fatty acid analysis of morganellae reveals hexadecanoic acid as the dominant peak with octadecanoic acid ($\text{C}_{18:1}$) and methylenehexadecanoic acid ($\text{C}_{16:0 \text{ ante}}$) as additional major peaks (Vasyurenko and Chernyavskaya, 1990). The presence of detectable amounts of dodecanoic acid separates morganellae from *Proteus* and *Providencia*. Histamine, cadaverine, and diaminopropane are the predominant polyamines synthesized under defined conditions (Hamana, 1996).

Half of all morganellae grown in broth culture show uniform turbidity with ring or pellicle formation (McKell and Jones, 1976). Growth occurs between 4 and 45°C . On nutrient agar, *M. morganii* produces smooth transparent colonies with an entire edge. Pigmentation not observed (McKell and Jones, 1976). On nonselective media, most morganellae appear as nonhemolytic, buff-colored, convex colonies, 2–3 mm in diameter after overnight incubation at $35\text{--}37^\circ\text{C}$ (Janda and Abbott, 1998a). Hemolysis on blood agar may be detected by prolonged incubation (48–72 h). On media containing an aromatic amino acid such as phenylalanine agar an almond-like odor may be emitted (Müller, 1986a). Because most strains are sucrose- and lactose-negative they appear as colorless colonies on selective media such as MacConkey's, xylose-lysine-desoxycholate, eosin-methylene blue, and Salmonella-Shigella agar. Some strains are highly pleomorphic producing multiple morphovars on media such as Salmonella-Shigella agar (Janda and Abbott, 1998a).

Morganellae possess a number of metabolic features almost exclusively associated with members of the tribe *Proteeae* (*Morganella*, *Proteus*, and *Providencia*). Oxidative deamination of certain amino acids (e.g., phenylalanine deaminase) to form keto acids that then react with ferric compounds yielding chromogenic products is a unique characteristic of this tribe (Singer and Volcani, 1955; Ewing, 1986a). On DL-tryptophan agar, *Morganella* produces a reddish-brown melanin-like compound with a molecular weight $\leq 12,000$ (Polster and Svobodová, 1964; Müller, 1986a). *M. morganii* also degrades L-tyrosine crystals incorporated

into solid media within 24 h, presumably mediated via a tyrosine phenol-lyase (Sheth and Kurup, 1975). All proteae, including *M. morganii*, produce extracellular bacteriolytic enzymes capable of degrading cell wall components of *Escherichia coli* and *Pseudomonas aeruginosa*. This reaction presumably occurs through extracellular secretion of a peptidoglycan-hydrolase (Branca et al., 1996). Like *Proteus*, morganellae elaborate a type II glycerol dehydrogenase that distinguishes these taxa from *Providencia* (type III); 1,3-propanediol dehydrogenase is not produced (Bouvet et al., 1995a).

The genus is biochemically homogeneous. Indole formed. Nitrates reduced. The methyl red test is positive. The Voges-Proskauer test is negative. Lysine decarboxylase activity is variable. Growth in KCN broth. Citrate, acetate, malonate, and mucate not utilized. Hydrolysis of urea. No H₂S produced on triple sugar iron (TSI) agar slants, although upon prolonged incubation (>24 h) a slight blackening at the junction may occur due to small amounts of a reddish-brown pigment being produced (Janda and Abbott, 1998a).

Acid with gas from D-glucose fermentation. Anaerobic fermentation of D-mannose. Fermentation of glycerol and trehalose is variable (Siboni, 1976). No fermentation of adonitol, L-arabinose, cellobiose, dulcitol, α -methyl-D-glucoside, *m*-inositol, lactose, maltose, D-mannitol, melibiose, raffinose, L-rhamnose, salicin, sucrose, and D-xylose.

Deoxyribonuclease, ribonuclease, alkylsulfatase, arylsulfatase, lipase, lecithinase, hyaluronidase, and protease activities not produced. Chitin, elastin, pectin, mucin, and fibrin are not degraded (Janda et al., 1996b). Alkaline and acid phosphatase, leucine arylamidase, and naphthol-AS-BI-phosphohydrolase positive.

Morganella exhibits 20% relatedness to core members of the *Enterobacteriaceae* in DNA hybridization studies (Brenner et al., 1978). All *M. morganii* studied constituted a single DNA relatedness group. The average relatedness of 18 strains was 90% in 60°C reactions and 85% in 75°C reactions with <1% divergence (Brenner et al., 1978). Biochemically aberrant morganellae, exhibiting either the lysine decarboxylase-positive or ornithine decarboxylase-negative phenotype, were subsequently found to be highly related to classic *M. morganii* using the lysine decarboxylase-positive strain, CDC 1274-75, as the source of labeled DNA (Hickman et al., 1980). Hybridization studies indicated 73–92% relatedness in 60°C reactions and 79–96% relatedness in 75°C reactions.

Morganella strains harbor between one and four extrachromosomal elements ranging from 35–60 MDa (Cornelis et al., 1981; Janda et al., 1996b). Plasmids in *Morganella* that confer antibiotic resistance belong to compatibility groups N, FI, and FII (Hedges et al., 1973). Additionally, lactose fermentation and lysine decarboxylation are attributable to plasmid carriage (Le Minor and Coynault, 1976; Cornelis et al., 1981). A bacteriophage typing scheme using seven phages defined 14 lytic patterns, but plaques produced are poorly developed (Schmidt and Jeffries, 1974). Phages are lytic only for *Morganella* and have not been found to attack *Proteus* or *Providencia* (Coetzee, 1963). A bacteriocin (morganocin) system detecting both morganocin production and sensitivity revealed 33 types in 45 serologically distinct strains of *Morganella* (Senior and Vörös, 1989).

The antigenic schema of *Morganella* consists of 77 serotypes with 44 somatic (O), 4 capsular (K), and 38 flagellar (H) antigens (Vörös and Senior, 1990). In 1990, 11 additional O types were added, extending the number of somatic antigens to 55 (Vörös

and Senior, 1990). Passive hemagglutination can be used to determine the O antigen (Penner and Hennessy, 1979b).

Morganella is generally resistant to ampicillin, extended-spectrum penicillins, first-generation cephalosporins, and cefoxitin. Antimicrobial agents to which strains are susceptible include third-generation cephalosporins, aztreonam, quinolones, tobramycin, and chloramphenicol. Variable resistance has been demonstrated for gentamicin, amikacin, imipenem, and tetracycline. Disk diffusion testing may be unreliable for some cepheids (Biedenbach et al., 1993).

Although isolated more often from urine than other sources, *Morganella* is an uncommon cause of urinary tract infection (UTI) (Janda and Abbott, 1998a). It is often isolated as a colonizer in cases of bacteriuria in patients undergoing long-term catheterization (Mobley and Warren, 1987). Bacteremia is rare, usually occurring in immunocompromised patients; 70% of cases are acquired nosocomially (Janda and Abbott, 1998a). Surgical patients are affected most often, with the focus of infection being wounds. Urinary tract infections are not a common source for bacteremias (McDermott and Mylotte, 1984).

M. morganii produce mannose-sensitive hemagglutinins (MSHA) (Coetzee et al., 1962); electron micrographs of MSHA strains show peritrichia but with denser polar aggregations. Catecholate- or hydroxamate-type siderophores used for iron acquisition are not produced (Drechsel et al., 1993). However, α -keto acids generated by L-amino acid deaminases produced by *M. morganii* can form ferric complexes that are sufficiently stable to transport iron (Drechsel et al., 1993). *M. morganii* can also use exogenous siderophores (ferrichromes, rhizoferrin, and citrate) for iron acquisition. *M. morganii* strains producing heat labile enterotoxin have been reported in a group of Swedish travelers with diarrhea (Jertborn and Svennerholm, 1991). Approximately 50% of morganellae produce a cell-free hemolysin related to the α -hemolysin of *E. coli* (Koronakis et al., 1987). The cytoplasmic location of urease and its low activity optimum of pH 5.5 in *M. morganii* allows the organism to survive under acidic conditions when other urease-positive, Gram-negative rods die (Young et al., 1996a). However, *M. morganii* grows more slowly and is less efficient in producing alkaline conditions in the urine than *P. mirabilis*, which is a more frequent cause of UTIs (Senior, 1983).

ENRICHMENT AND ISOLATION PROCEDURES

Morganellae may be recovered from specimens from sterile body sites using routine enteric isolation media. Both smooth and rough morphovars may be present. For fecal specimens, MacConkey agar with methyl blue and phenolphthalein diphosphate may be helpful (Janda and Abbott, 1998a). When testing for human intestinal carriage, Rustigian and Stuart (1945) reported increases in *Morganella* recovery of 1.8–10% using tetrathionate or selenite enrichment broth prior to plating on media.

MAINTENANCE PROCEDURES

Morganella strains may be maintained at room temperature in agar deeps, especially motility agar, for several months. For long-term storage cultures should be frozen and maintained at –70°C.

DIFFERENTIATION OF THE GENUS *MORGANELLA* FROM OTHER GENERA

See the genus *Proteus*, Table BXII.γ.254, for characteristics that can be used to differentiate *Morganella* from other related genera of *Enterobacteriaceae*.

List of species of the genus *Morganella*

1. ***Morganella morganii*** (Winslow, Kliger and Rothberg 1919) Fulton 1943, 81^{AL} (*Bacillus morganii* (sic) Winslow, Kliger and Rothberg 1919, 481; *Proteus morganii* (Winslow, Kliger and Rothberg 1919) Yale 1939b, 435.)

morga'ni.i. M.L. gen. n. *morganii* of Morgan; named after H. de R. Morgan, a British bacteriologist who first studied the organism.

The description is the same as that for the genus. Trehalose negative, susceptible to tetracycline See Tables BXII.γ.236 and BXII.γ.237 for other characteristics.

Occurs in the feces of humans, dogs, other mammals, and reptiles. Opportunistic human pathogens.

The mol% G + C of the DNA is: 50 (T_m).

Type strain: ATCC 25830, DSM 30164, IFO 3848, NCIB 235.

- a. ***Morganella morganii* subsp. *morganii* subsp. nov.** (Winslow, Kliger and Rothberg 1919) Fulton 1943, 81^{AL} (*Bacillus morganii* [sic] Winslow, Kliger and Rothberg 1919, 481; *Proteus morganii* (Winslow, Kliger and Rothberg 1919) Yale 1939b, 435.)

Distinguished from subsp. *sibonii* by positive trehalose reaction and resistance to tetracycline. See Table BXII.γ.236 for other characteristics.

The mol% G + C of the DNA is: 50 (T_m).

Type strain: ATCC 25830, DSM 30164, IFO 3848, NCIB 235.

- b. ***Morganella morganii* subsp. *sibonii* subsp. nov.** Jensen, Frederiksen, Hickman-Brenner, Steigerwalt, Riddle and Brenner 1992, 619^{VP} *si.bo'ni.i.* L. gen. n. *sibonii* of Siboni, named after Knud Siboni, a Danish microbiologist who first recognized trehalose-fermenting *Morganella morganii*.

Distinguished from subsp. *morganii* by characteristics in Table BXII.γ.236. The above subspecies designations were created in keeping with the recommendation that "genetically close organisms that diverge in phenotype" be given subspecies status; DNA relatedness data for these strains indicated the creation of subspecies as opposed to species.

The mol% G + C of the DNA is: 50.

Type strain: 8103-85, ATCC 49948.

TABLE BXII.γ.236. Characteristics of *Morganella morganii*^{a,b}

Characteristic	<i>Morganella morganii</i> subsp. <i>morganii</i>	<i>Morganella morganii</i> subsp. <i>sibonii</i>
Indole	+	d
Voges-Proskauer	—	—
Utilization of Simmons citrate, malonate, mucate, acetate	—	—
Urease	+	+
H ₂ S production (triple sugar iron agar)	d	—
Phenylalanine deaminase	+	+
o-Nitrophenyl-β- galactopyranoside (ONPG) hydrolysis	D	—
<i>Møller amino acid decarboxylases:</i>		
Lysine decarboxylase	d	d
Arginine dihydrolase	—	—
Ornithine decarboxylase	d	d
Motility	+	d
Gelatin liquefaction 22°C	—	—
Growth in potassium cyanide (KCN)	+	d
NO ₃ [−] reduction to NO ₂ [−]	+	+
Corn oil lipase	—	—
Deoxyribonuclease	—	—
<i>Acid production from:</i>		
Glucose, D-mannose	+	+
Trehalose, adonitol, D-arabitol	—	+
L-Arabinose, cellobiose, dulcitol, erythritol, <i>myo</i> - inositol, lactose, maltose, mannitol, α-methylglucoside, melibiose, raffinose, rhamnose, salicin, D-sorbitol, sucrose, D-xylose	—	—
Tetracycline susceptibility	+	—

^aFor symbols see standard definitions; temperature of reactions, 36 ± 1°C. All reactions are for 48 h.

^bAdapted from Farmer (1995) and Jensen et al. (1992).

TABLE BXII.γ.237. Identification of *Morganella morganii* biogroups^{a,b}

Characteristic	Biogroup						
	A	B	C	D	E	F	G
<i>Acid production from:</i>							
Trehalose	—	—	—	—	+	+	+
Glycerol	+ ^c	d	+ ^c	—	—	d ^c	d ^c
<i>Møller amino acid decarboxylase:</i>							
Lysine decarboxylase	—	+	—	+	+	d	—
Ornithine decarboxylase	+	+	—	—	+	—	+
Motility	+ ^c	d	d	—	+	d	+
Tetracycline susceptibility	+	+	d	+	—	—	d

^aFor symbols see standard definitions. Temperature of reactions 36 ± 1°C. All reactions are for 48 h.

^bAdapted from Jensen et al., 1992.

^cPositive at 3–7 d.

Genus XXII. *Obesumbacterium* Shimwell 1963, 759^{AL}

J.J. FARMER III AND DON J. BRENNER

O.be'sum.bac.te'ri.um. L. neut. adj. *obesum* fat; L. neut. n. *bacterium* rod; M.L. neut. n. *Obesumbacterium* a fat, rod-shaped bacterium.

Pleomorphic rods 0.8–2.0 × 1.5–100 µm (short, “fat” rods predominate when grown in beer wort with live yeasts; long pleomorphic rods usually predominate when grown in most bacteriological media; some strains have been reported to display a branching cell morphology), conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. **Nonmotile**. Facultatively anaerobic. Slow growing, forming colonies <0.5 mm in diameter on ordinary plating media at 24 h. Optimal growth temperature is 25–32°C; **growth at 37°C is comparatively poor**. **Acid formed from D-glucose and D-mannose; very few other carbohydrates are fermented**. Gas formation during fermentation appears to be variable (original description says gas is produced, but none of the strains studied produced gas). **Lysine decarboxylase is positive**. Nitrate is reduced to nitrite. **Many biochemical tests normally used for differentiation of *Enterobacteriaceae* are negative or delayed positive (3–7 d at 36°C)**. **One of only three genera** (along with *Hafnia* and *Pragia*) **in *Enterobacteriaceae* to contain only gluconate dehydrogenase** in its D-glucose oxidation pathway (Bouvet et al., 1989). **Occurs as a brewery contaminant** that can survive and grow in the presence of live yeasts during beer production. Not isolated from human clinical specimens; no evidence of pathogenicity for humans or animals. The genus has a single species, *O. proteus*, with two defined biogroups (1 and 2). The description of the genus above is a “composite description” based on data for both biogroups. However, the two biogroups are actually distinct species that are phenotypically different and only distantly related by DNA–DNA hybridization (Brenner, 1981). *O. proteus* biogroup 1 is actually a biogroup of *Hafnia alvei* (Brenner, 1981), and will be referred to as *H. alvei* biogroup 1 (Farmer et al., 1985a) in this chapter.

The mol% G + C of the DNA is: 48–49.

Type species: *Obesumbacterium proteus* (Shimwell and Grimes 1936) Shimwell 1963, 759 (*Flavobacterium proteum* (sic) Shimwell and Grimes 1936, 348.)

FURTHER DESCRIPTIVE INFORMATION

Obesumbacterium, as a member of *Enterobacteriaceae*, belongs to the *Gammaproteobacteria*. Shimwell's original description of an organism he called “*Flavobacterium proteum*” centered on its cellular morphology (Shimwell, 1936; Shimwell and Grimes, 1936). He noted that it appeared as plump rods 0.8–1.2 × 1.5–4 µm when grown in wort media or when taken directly from breweries during fermentation. This morphology had no doubt led to the term “short fat rod of pitching yeasts”, which had been used in breweries for many years. Shimwell (1936) also noted much pleomorphism when the organism was grown in laboratory media that were alkaline or neutral. Chains of up to 100 µm were observed under these conditions.

SDS-PAGE protein fingerprints differentiate *H. alvei* biogroup 1 (*O. proteus* biogroup 1) from *O. proteus* biogroup 2 (Fernandez et al., 1993), as do plasmid profiles, ribotyping, enteric repetitive intergenic consensus (ERIC)-PCR (Prest et al., 1994), and random amplified polymorphic DNA (RAPD) profiles (Savard et al., 1994). Plasmid profiling, ERIC-PCR, and RAPD profiles successfully differentiated between isolates of *O. proteus* biogroup 2 (Prest et al., 1994; Savard et al., 1994).

A number of reports discuss the ecology of *Obesumbacterium* in breweries (Shimwell, 1936, 1948, 1963, 1964; Shimwell and Grimes, 1936; Strandskov et al., 1953; Case, 1965). Unfortunately, no distinction has been made between biogroups 1 and 2, so it is usually impossible to determine whether these two biogroups (which are really distinct species, one of which does not belong *Obesumbacterium*; see Taxonomic Comments, below), are different in their ecology, distribution, and other factors. Future studies in breweries should resolve this problem, but only if the two distinct biogroups are differentiated. This approach was used by van Vuuren (1978) in South African breweries.

Changes in brewery practices have eliminated many bacterial contaminants that were once carried along with pitching yeasts; thus *Obesumbacterium* is isolated much less frequently than in the past (F.G. Priest, personal communication). This fact may hamper future studies on the organism.

ENRICHMENT AND ISOLATION PROCEDURES

Quantitative recovery of *O. proteus* from samples of ale yeast was best after 3 d incubation at 25°C on universal beer agar and Wallerstein Laboratories' differential medium, containing cycloheximide to inhibit yeast growth, intermediate on wort agar and yeast mannitol (YM) (Difco) agar, and unsatisfactory on MacConkey agar and membrane lauryl sulfate agar (Fernandez et al., 1993).

MAINTENANCE PROCEDURES

Long-term storage can be accomplished by quick freezing at –80°C in yeast extract peptone dextrose broth supplemented with 15% glycerol (Fernandez et al., 1993), or in 10% skim milk.

TAXONOMIC COMMENTS

Obesumbacterium was first proposed as a genus by Shimwell in 1963 (Shimwell, 1963, 1964) to accommodate the organism known as “*Flavobacterium proteus*” (Shimwell and Grimes, 1936), which was called “*Flavobacterium proteum*” in the original proposal (Shimwell and Grimes, 1936). The specific epithet “*proteus*” was chosen because the organism has a very pleomorphic cell morphology depending upon the particular growth conditions (Shimwell, 1936, 1948, 1964). The organism was first named in 1936 when Shimwell was doing studies on the “short fat rods of pitching yeasts”. He gave an adequate description (based on the available techniques) of the organism, but much of his description is not helpful in identifying it. Shimwell deposited a pure culture of “*F. proteus*” (isolated from yeast of Beamish and Crawford's Brewery, Cork, Ireland) in the National Collection of Type Cultures (NCTC), England. However, when Shimwell's own culture was “lost”, he wrote the NCTC and found they had also “lost” this culture (unpublished letter of 27 February 1964 from J.L. Shimwell to E.F. Lessel of the ATCC). Thus it appears that no culture has survived from those originally studied by Shimwell in writing his description of “*F. proteus*”. The strain most studied is apparently ATCC 12841 (NCIB 8771; strain 42 of Strandskov and Bockelmann), which, until 1980 was only a reference strain (*Bergey's Manual of Determinative Bacteriology*, 8th ed.). This strain was isolated from lager and ale yeasts and deposited by the Shaefer

Brewing Co. (Strandskov and Bockelmann, 1955). This strain was given status, without comment, as the type strain (neotype) of *O. proteus* when the Approved Lists of Bacterial Names were issued in 1980.

There is confusion whether the current type strain of *O. proteus* is the same organism that Shimwell studied and named "*F. proteus*". In 1956, the NCTC sent three cultures (numbers 42, 2, and 41), isolated and described by Strandskov and Bockelmann (1955), to Shimwell to examine and determine whether he thought they were "*F. proteus*". For no. 42 (the current type strain of *O. proteus*) he concluded: "This is almost certainly an authentic strain. Its morphology is almost exactly that of my original isolation, namely thick (up to 2 or more μm), long (up to 100 μm or more) filaments etc. together with the usual short fat rods in fair numbers". The properties of these strains include indole, acetylmethylcarbinol, H_2S and starch, all negative; nitrite from nitrate, positive. However, probably hundreds of Gram-negative bacteria would answer to this description. Indeed, the species is very poorly characterized biochemically, its main characteristic being its extremely large cell size, and its almost incredible pleomorphism, by means of which (taken in conjunction with its presence in a brewery fermentation) it can readily be identified. In cataloguing any of the strains I suggest that no. 42 could be safely named *F. proteus*; no. 2 a little doubtfully, and no. 41 very doubtful indeed" (unpublished letter of 7 June 1956 from J.H. Shimwell to W.S. Greaves of the NCTC). Thus the current concept of *O. proteus* based on its type strain ATCC 12841 (no. 42) is not incompatible with "*F. proteus*" as defined by Shimwell based on strains that no longer exist. Unfortunately, Shimwell's original description of "*F. proteum*" fits both *H. alvei* biogroup 1 (*O. proteus* biogroup 1) and *O. proteus* biogroup 2, so it is uncertain which of these latter two organisms (or perhaps both) was originally studied by Shimwell (1936). It is even possible that it was neither since Shimwell described "*F. proteum*" as a producer of gas during fermentation of carbohydrates, but neither *H. alvei* biogroup 1 nor *O. proteus* biogroup 2 produce gas (Priest et al., 1973; see also Table BXII. γ .193 in the chapter on the family *Enterobacteriaceae*).

"*F. proteus*" had been known for many years as a brewery contaminant that can survive and grow in the presence of live yeasts during beer production. Because it fermented D-glucose and other carbohydrates, it was incompatible with a redefined genus *Flavobacterium* that was limited to oxidative rather than fermentative bacteria. Its removal from *Flavobacterium* was subsequently confirmed by Bauwens and De Ley (1981), who used DNA-rRNA hybridization to show that "*F. proteus*" was not closely related to other *Flavobacterium* species. Shimwell (1963, 1964) formed the new genus *Obesumbacterium* for "*F. proteus*" because its phenotypic properties and ecological niche differed from those of other genera. He did not assign *Obesumbacterium* to a family, but its properties (see also Table BXII. γ .193 in the chapter on the family *Enterobacteriaceae*) are compatible with those of the family *Enterobacteriaceae*.

Priest et al. (1973) determined the phenotypic properties and did DNA-DNA relatedness studies on 19 strains of *O. proteus*, including 16 brewery isolates. They defined two biogroups, which had the same mol% G + C content of DNA, 48.0–48.5. They also proposed that *O. proteus* be reclassified in the genus *Hafnia* where its citation would be "*H. protea*" (Shimwell and Grimes, 1936) Priest, Somerville, Cole and Hough 1973. The new combination could have been proposed as "*Hafnia proteus*", since

"*proteus*" is a substantive that need not agree in gender with its genus. This change in classification was accepted to some extent, but in reality both *O. proteus* and "*H. protea*" have, until recently, been rarely used in the literature. Most of the existing citations have been in journals related to brewing. The name "*Hafnia protea*" lost standing in nomenclature on 1 January 1980, because it did not appear on the Approved Lists of Bacterial Names; however, *Obesumbacterium* and *O. proteus*, with its type strain ATCC 12841, have standing in nomenclature since they did appear on the lists.

The classification of *Obesumbacterium* was clarified by Brenner and co-workers (1981), who used DNA-DNA hybridization to determine the relatedness of *O. proteus* biogroups 1 and 2 to each other and to other *Enterobacteriaceae*. *O. proteus* biogroup 1 was very highly related to *Hafnia alvei*, and it was concluded that this biogroup is a synonym of *H. alvei*. It can best be thought of as the pleomorphic, KCN-negative, nonmotile, non-gas-producing, salicin-positive, L-arabinose-negative, L-rhamnose-negative, maltose-negative, D-xylose-negative, β -galactosidase-negative biogroup of *H. alvei* that has adapted to the brewery environment. This adaptation to the brewery environment was noted by Shimwell and Grimes (1936) in the original description of "*F. proteus*": "The organism sometimes failed to grow in dilute media, probably owing to its having become accustomed to the more concentrated nature of beer-wort in the brewery." This adaptation has presumably made the organism very "sluggish" in its metabolic activities, as is reflected in its slow growth rate and diminished activity in the tests normally done for identification of *Enterobacteriaceae*. The classification of *O. proteus* biogroup 1 as *H. alvei* is further strengthened by the fact that strains of *O. proteus* biogroup 1 (but not biogroup 2) are lysed (Farmer, 1984a; Table BXII. γ .238) by the *Hafnia*-specific bacteriophage 1672 described by Guinée and Valkenburg (1968).

Van Vuuren et al. (1981) studied 10 cultures of "*Hafnia alvei* *Obesumbacterium proteus*" isolated from South African lager-beer breweries. One of their five distinct biogroups (based on API 20E biochemical profiles) was very active biochemically and more like typical cultures of *H. alvei*. The other four biogroups were progressively less biochemically active with the least active group more like *O. proteus*. Based on phenotypic and DNA relatedness data (Brenner, 1981), Farmer et al. (1985a) proposed that *O. proteus* biogroup 1 and biochemically inactive strains of *H. alvei* isolated from breweries be classified in a single taxon. They proposed the names "*Hafnia alvei* biogroup 1" and "*Hafnia alvei* brewery biogroups" to aid in the recognition and identification of these strains adapted to the brewery environment. *H. alvei*, *H. alvei* biogroup 1, and *O. proteus* biogroup 2 have been listed as separate taxa in the CDC's master biochemical chart of *Enterobacteriaceae* since 1985. Based on these data, it appears that biogroups intermediate between typical *H. alvei* and *O. proteus* biogroup 1 may occur in breweries. A complete set of biochemical tests (Table BXII. γ .238, also see Table BXII. γ .193 in the chapter on the family *Enterobacteriaceae*) and lysis by the *Hafnia*-specific bacteriophage are needed to characterize isolates from breweries that resemble *Hafnia/Obesumbacterium*.

O. proteus biogroup 2 is different biochemically from *H. alvei* biogroup 1 and from *H. alvei* (Table BXII. γ .238). In DNA-DNA relatedness studies *O. proteus* biogroup 2 was only 25–30% related to *H. alvei* biogroup 1 (Brenner, 1981). In fact, its closest relative is *Escherichia blattae* to which it was 60–65% related. Thus *O. proteus*

TABLE BXII.γ.238. Differentiation of *Obesumbacterium proteus* biogroup (BG) 2, *Hafnia alvei*, and *Hafnia alvei* biogroup 1 (*Obesumbacterium proteus* biogroup 1)^a

Characteristic	Incubation time (d)	<i>O. proteus</i> BG 2	<i>Hafnia alvei</i> wildtype	<i>Hafnia alvei</i> BG 1 ^b
Lysis by the <i>Hafnia</i> -specific bacteriophage of Guinée and Valkenburg (1968)	1	—	+	+
Strong rapid catalase ^c		—	+	+
Voges-Proskauer (22°C)	4	—	+	+
Acid production from:				
D-Mannitol	10	—	+	+
Salicin	7	—	[—]	+
D-Xylose	7	+	+	—
Esculin hydrolysis	7	—	[—]	+

^aSymbols: +, positive for 90–100% of strains; [—], positive for 11–25% of strains; —, positive for 0–10% of strains.
^bFormerly known as *O. proteus* biogroup 1.
^cStrains of *O. proteus* biogroup 2 are catalase positive but the reaction is weak and takes 10–30 seconds. Strains of wild-type *H. alvei* and of *H. alvei* biogroup 1 give a very strong and rapid (1 second) catalase reaction. This is a very simple way to differentiate between strains of *H. alvei* and *O. proteus* biogroup 2.

biogroup 2 should not be included in the same species or in the same genus as *H. alvei* biogroup 1.

The type strain of *O. proteus* is ATCC 12841 (NCIB 8771). It was listed as “*H. protea*” in the 14th edition of the ATCC catalog, and has been listed as *O. proteus* in the 15th to 19th editions. This culture is lysed by the *Hafnia*-specific bacteriophage 1672, is methyl red negative, D-mannitol negative, salicin positive (6 d), maltose negative, D-xylose negative, esculin positive (7 d), and ONPG negative. Thus from Table BXII.γ.238 it clearly belongs to *H. alvei* biogroup 1, although it is somewhat more inactive or slower in its biochemical reactions than other strains. *H. alvei* biogroup 1 can be logically considered as a series of inactive biogroups of *H. alvei*. However, if it were classified as a named subspecies it could cause serious nomenclatural problems in *Hafnia* because it has priority over *H. alvei*. Since the name *Obesumbacterium* has traditionally been well known in the brewing industry, which seems to be the ecological niche for this organism, the most practical solution at present is to continue use of the names *H. alvei* biogroup 1 and *O. proteus* biogroup 2 until all the taxonomic problems are resolved and a “final” nomenclature and classification can be proposed after careful analysis.

Since *O. proteus* is now known to be a heterogeneous species, it is essential to append “biogroup 2” to the name that will correspond to the second distinct species shown by DNA–DNA hybridization and biochemical tests. Unless this is done, the intended meaning of *O. proteus* is unclear. We recommend that the term *O. proteus* biogroup 1 be discontinued, and replaced by *H. alvei* biogroup 1.

O. proteus biogroup 2 is a unique genomospecies (DNA hybridization group); however, since its type strain is in a different hybridization group than that of *H. alvei*, the species and the genus (since it is the type species of the genus) would be invalid because it is a junior synonym of *H. alvei*. To rectify this, a new type strain would have to be designated. A Request for an Opinion to do this would have to be made to the Judicial Commission. If the request were granted, the genus *Obesumbacterium* and the species *O. proteus* would again be valid under the rules of nomenclature of the Bacteriologic Code. Other alternatives remain since, based on its DNA–DNA relatedness to *Escherichia blattae*, *O. proteus* could be classified in the genus *Escherichia*. Placing it in this genus would be a simple matter—designating a new type strain and describing a new combination: “*Escherichia proteus*”. An alternative would be to classify *E. blattae* and *O. proteus* biogroup 2 together, either in a redefined genus *Obesumbacterium* or as a new genus. The results from 16S rDNA sequencing (not currently available), should be helpful in deciding on the most logical classification.

FURTHER READING

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Priest, F.G., H.J. Somerville, J.A. Cole and J.S. Hough. 1973. The taxonomic position of *Obesumbacterium proteus*, a common brewery contaminant. *J. Gen. Microbiol.* 75: 295–307.

Shimwell, J.L. 1936. A study of the common rod bacteria of brewers' yeast. *J. Inst. Brew.* 42: 119–127.

List of species of the genus Obesumbacterium

- 1. **Obesumbacterium proteus** (Shimwell and Grimes 1936) Shimwell 1963, 759^{AL} (*Flavobacterium proteum* (sic) Shimwell and Grimes 1936, 348.)
pro'te.us. Gr. masc. n. *proteus* the ancient Greek sea-god noted for being able to change his form at will; Gr. masc. n. *proteus* pleomorphic.
Obesumbacterium proteus is really two different species, which can be differentiated by phenotypic tests (Table

BXII.γ.238) and DNA–DNA relatedness tests. The type strain has the properties of *H. alvei* (*O. proteus*) biogroup 1 (see Taxonomic Comments, above). The description of *O. proteus* biogroup 2 is as given for the genus and as listed in Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*. Occurs in breweries where it grows in beer-wort along with yeasts early in the fermentation. There is no evidence that it is pathogenic for humans or animals.

The mol% G + C of the DNA is: 48–49 (Bd).

Type strain: 42 of Strandkov and Bockelmann (1955), ATCC 12841, DSM 2777, NCIB 8771.

GenBank accession number (16S rRNA): AJ233422.

Additional Remarks: Strain No. 42 belongs to *H. alvei* (*O. proteus*) biogroup 1. Therefore, at present there is no valid

type strain for *O. proteus* biogroup 2—the true representative of the species. Strains 520, 531, and 580 of Priest et al. (1973) belong to *O. proteus* biogroup 2 by DNA–DNA hybridization and are phenotypically typical (Table BXII.γ.238). Thus, they should be considered as the candidates for the proposed type strain.

Genus XXIII. *Pantoea* Gavini, Mergaert, Beji, Mielcarek, Izard, Kersters and De Ley 1989b, 343^{VP} emend. Mergaert, Verdonck and Kersters 1993, 171

PATRICK A.D. GRIMONT AND FRANCINE GRIMONT

*Pan.toe'*a. Gr. adj. *pantoios* of all sorts and sources; M.L. fem. n. *Pantoea* [bacteria] from diverse [geographical and ecological] sources.

Straight rods, 0.5–1.3 × 1.0–3.0 µm. Nonencapsulated. Non-sporeforming. Some strains form symplasmata. **Most strains are motile and are peritrichously flagellated.** Gram negative. Colonies on nutrient agar are smooth, translucent, and more or less convex with entire margins or heterogenous in consistency and adhering to the agar. **Colonies are yellow, pale beige to pale reddish yellow, or nonpigmented. Facultatively anaerobic. Oxidase negative. Glucose dehydrogenase and gluconate dehydrogenase are produced** and are active without an added cofactor. Acid is produced from the fermentation of D-fructose, D-galactose, trehalose, and D-ribose. Most strains are **Voges–Proskauer positive. Lysine and ornithine are not decarboxylated.** Urease negative. Pectate is not degraded. H₂S is not produced from thiosulfate. Optimum temperature 30°C. N-acetyl-D-glucosamine, L-aspartate, D-fructose, D-galactose, D-gluconate, D-glucosamine, D-glucose, L-glutamate, glycerol, D-mannose, D-ribose, and D-trehalose are utilized as sole sources of carbon and energy. 5-Aminovalerate, benzoate, caprate, caprylate, *m*-coumarate, ethanolamine, gentisate, glutarate, histamine, 3-hydroxybenzoate, 4-hydroxybenzoate, 3-hydroxybutyrate, itaconate, maltitol, D-melezitose, 1-O-methyl-α-D-glucoside, palatinose, 3-phenylpropionate, propionate, L-sorbose, tricarballoylate, tryptamine, D-turanose, and L-tyrosine are not utilized as sole sources of carbon and energy. *Pantoea* spp. are isolated from plants, seeds, fruits, soils, water, and from humans (urine, blood, wounds, internal organs), and animals. Some strains are (or have been thought to be) phytopathogenic.

The mol% G + C of the DNA is: 49.7–60.6.

Type species: ***Pantoea agglomerans*** (Ewing and Fife 1972) Gavini Mergaert, Beji, Mielcarek, Izard, Kersters and De Ley 1989b, 343 (*Enterobacter agglomerans* (Beijerinck 1888) Ewing and Fife 1972, 10.)

FURTHER DESCRIPTIVE INFORMATION

Much of our knowledge about the characteristics of the genus *Pantoea* was acquired when all *Pantoea*, *Leclercia* and related bacteria were called *Enterobacter agglomerans*. Therefore, in this chapter, we will refer to *Pantoea agglomerans* when this precise species is meant and to the *Enterobacter agglomerans* complex when there is insufficient taxonomic information. Older data refer to *Erwinia herbicola* or the “herbicola group”. Strains in the *Enterobacter agglomerans* complex (or *Erwinia herbicola* or the “herbicola group”) may or may not belong to the genus *Pantoea*.

Phylogenetic treatment Of the 124 strains of the *Enterobacter agglomerans* complex analyzed by DNA–DNA hybridization by

Brenner et al. (1984a), 90 formed 13 distinct DNA groups. The remaining 34 strains did not fit any of these groups. The synonymy between these groups and named species is given in Table BXII.γ.239.

Comparison of *rrs* gene (encoding 16S rRNA) sequences showed the genus *Pantoea* (represented by *Pantoea agglomerans*, *Pantoea ananatis*, and *Pantoea stewartii*) to constitute a monophyletic cluster distinct from clusters corresponding to the genera *Erwinia*, *Pectobacterium*, *Brenneria*, and other genera of the *Enterobacteriaceae* (Hauben et al., 1998a). When all DNA groups from Brenner et al. (1984a) and named *Pantoea* species are studied

TABLE BXII.γ.239. Correspondence between DNA relatedness groups, phenons, and nomenspecies in the *Enterobacter agglomerans* complex.

Gavini et al., 1983b	Brenner et al., 1984a	Verdonck et al., 1987	Nomenspecies (older synonyms)
phenon	DNA group	phenon	
—	I	—	<i>Pantoea</i> species
—	IV	—	<i>Pantoea</i> species
(B4) ^a	V	7B	<i>Pantoea</i> species
	XIII	8	<i>P. agglomerans</i> (<i>Enterobacter</i> <i>agglomerans</i> , <i>Erwinia</i> <i>herbicola</i> , <i>Erwinia</i> <i>milletiae</i>)
(B4)			
(B5)	II	9	<i>Pantoea</i> species
(B5)	III	10	<i>Pantoea dispersa</i>
(B9)	—	11	—
—	X	ungrouped	Close to the genus <i>Buttiauxella</i>
	VI	12	<i>Pantoea ananatis</i> (<i>Erwinia</i> <i>anas</i> , <i>Erwinia</i> <i>uredovora</i>)
B8			
	XII	16	Close to the <i>Enterobacter</i> <i>cloacae</i> complex
ungrouped			
C	—	(17)	<i>Rahnella aquatilis</i>
D1	—	(17)	Close to <i>Rahnella aquatilis</i>
ungrouped	IX	18	<i>Enterobacter cowanii</i>
	VIII	23	Close to <i>Enterobacter</i> <i>pescicaria</i> and <i>Erwinia</i> <i>rhapontici</i>
(B9)			
E2, E3, E5	XI	(26)	<i>Leclercia adacarboxylata</i>
	VII	(26)	Close to <i>Leclercia</i> <i>adacarboxylata</i> and <i>Enterobacter cloacae</i>
E4			

^a() : parentheses indicate partial correspondence.

by *rrs* and *rpoB* sequence comparisons, the following congruent clusterings are obtained:

- P. agglomerans* (DNA group XIII), *P. ananatis* (DNA group VI), *P. stewartii*, and DNA group V cluster together.
- DNA groups I and II and a cluster containing *Pantoea dispersa* (DNA group III) and DNA group IV branch with the above-mentioned cluster to constitute a monophyletic *Pantoea* cluster.
- Pantoea citrea*, *Pantoea terrea*, and *Pantoea punctata* constitute a discrete cluster that joins the *Pantoea* cluster at a lower level.
- E. agglomerans* DNA groups VII, VIII, IX, X, XI, and XII branch away from the *Pantoea* cluster (unpublished data).

More taxonomic work is needed to justify the assignment of *P. citrea*, *P. terrea*, and *P. punctata* to the genus *Pantoea*.

Cell morphology Symplasmata are sausage-shaped zoogloeal masses of bacteria observed by phase contrast microscopy in hanging drop preparations from the water of syneresis in glucose nutrient agar slopes after 24 h incubation (Graham and Hodgkiss, 1967). Symplasmata have been observed in bacteria now reclassified in the genus *Pantoea* ("*Bacterium herbicola*", *Erwinia lathyri*, *Erwinia uredovora*, and *Erwinia milletiae*) (Graham and Hodgkiss, 1967). The epithet "*agglomerans*" probably refers to symplasmata (Beijerinck, 1888).

Nutrition and growth conditions Best results are obtained when *Pantoea* cultures are incubated at 30°C. Most strains grow at 37°C on plating media, but some cultures fail to grow or only produce microcolonies. A few isolates can grow on brilliant green or bismuth sulfite agar media and can produce light growth of colonies that range from very small to 2.0 mm in diameter. Many strains produce mucoid colonies at 37°C, and most cultures do so when the media are incubated at room temperature, about 25°C.

On nutrient agar, anaerogenic strains of the *E. agglomerans* complex (*Pantoea* spp.) form mucoid colonies, smooth and irregularly round colonies, or rough and wrinkled colonies that are difficult to remove with a platinum wire. This latter aspect is what led to the description of "biconvex bodies", which can be observed with a low power stereoscopic binocular microscope in 2 or 3 day-old colonies. These bodies appear to be granular structures analogous to symplasmata or down-growths of the colonies into the medium (Graham and Hodgkiss, 1967).

P. agglomerans, *P. stewartii*, and *P. ananatis* colonies are usually yellow pigmented; *P. dispersa* is often pigmented; and *P. citrea*, *P. punctata*, and *P. terrea* are never pigmented. Yellow pigmentation of *P. agglomerans* may be weak at 37°C.

In Biotype-100 strips (BioMerieux, Craponne, France) or in a minimal medium containing ammonium sulfate as the nitrogen source, the following compounds serve as sole carbon sources for most strains: *N*-acetyl-D-glucosamine, L-aspartate, D-fructose, D-galactose, D-gluconate, D-glucosamine, D-glucose, L-glutamate, glycerol, D-mannose, D-ribose, and D-trehalose.

Most strains cannot utilize the following substrates as sole carbon and energy sources: 5-aminovalerate, benzoate, caprate, caprylate, *m*-coumarate, ethanolamine, gentisate, glutarate, histamine, 3-hydroxybenzoate, 4-hydroxybenzoate, 3-hydroxybutyrate, itaconate, maltitol, D-melezitose, 1-*O*-methyl- α -D-glucoside, palatinose, 3-phenylpropionate, propionate, L-sorbose, tricarballoylate, tryptamine, D-turanose, and L-tyrosine.

Metabolism Most cultures produce acid from sucrose and therefore yield acid throughout in tubes of triple sugar iron agar.

P. agglomerans gives negative reactions in all three decarboxylase tests.

Under aerobic conditions, all *Pantoea* species produce gluconate from D-glucose in the presence of iodoacetate due to a glucose dehydrogenase, without added pyrroloquinoline quinone (PQQ) (Bouvet et al., 1989). A reducing compound, 2-ketogluconate, is also produced from gluconate due to a gluconate dehydrogenase (Bouvet et al., 1989). 2,5-Diketogluconate is produced from 2-ketogluconate by *Pantoea citrea*, *P. punctata*, and *P. terrea* (Kageyama et al., 1992).

Pantoea species are negative for the following tests: H₂S production from thiosulfate, tetrathionate reduction, tributyrin and corn oil hydrolysis, and β -glucuronidase.

Mutants, plasmids, phages and phage typing, bacteriocins

Pantoea agglomerans strain Eh318 produces two antibiotics (pantocin A and B) that inhibit *Erwinia amylovora* *in vitro* (Wright et al., 2001). *Erwinia herbicola* produces two lipopeptide antibiotics, herbicolin A and B, which are active against sterol-containing fungi (Greiner and Winkelmann, 1991). Four antibiotics, agglomerins A, B, C, and D, were isolated from the culture broth of a strain identified as *Enterobacter agglomerans*; these antibiotics are active against a wide variety of anaerobic bacteria *in vitro* (Shoji et al., 1989).

The eight bacteriocin-producing *Enterobacter cloacae* strains used to type *E. cloacae* isolates by bacteriocin susceptibility are useful in typing isolates identified as *Enterobacter agglomerans* (Bauernfeind and Petermüller, 1984).

Antigenic structure The principal antigens in most strains studied under the name *E. agglomerans* (*Erwinia herbicola*) are uncharged capsular polysaccharides. These antigens are not removed from the cells by heating at 100°C for 30 min (Slade and Tiffin, 1984). Most members of the "herbicola group" are motile, but there is no report on the antigenic structure of the flagella (Slade and Tiffin, 1984). Most strains produce antigenically similar high molecular weight acidic polysaccharides.

Two schemes of serotyping have been proposed for *Erwinia herbicola*. Muraschi et al. (1965) used immunodiffusion and antigens extracted by aqueous ether and classified 55 isolates into 7 serotypes, although "some cultures were mixtures of more than one serotype". A second scheme was established by Slade (cited by Slade and Tiffin, 1984), using laboratory strains. This scheme has not been used to type new isolates.

Pathogenicity Strains of the *E. agglomerans* complex may occur in clinical samples (blood, wounds, sputum, urine), often with dubious clinical significance (von Graevenitz, 1970; Gilardi and Bottone, 1971; Pien et al., 1972). In some cases (Cooper-Smith and von Graevenitz, 1978; von Graevenitz and Palermo, 1980), clinical significance has been demonstrated. In an Ohio hospital, *E. agglomerans* accounted for four of 58 episodes of *Enterobacter* bacteremia and one of 42 cases of nosocomial bacteremia (Watanakunakorn and Weber, 1989). In 1970, *E. agglomerans* was implicated in a United States-wide and a Canadian outbreak of septicemia caused by contaminated closures on bottles of infusion fluids. Twenty-five hospitals were involved with 378 cases and 40 deaths (Maki et al., 1976).

Strains identified as *Pantoea agglomerans* have been isolated from joint fluid of patients with arthritis following injuries with plant thorns, wood splinters, or wooden splinters (Flatauer and Khan, 1978; Olenginski et al., 1991; de Champs et al., 2000).

A case of metastatic endophthalmitis caused by a strain identified as *Enterobacter agglomerans* was reported. One day following

internal hemorrhoidal ligation, the patient developed anterior uveitis, followed by panophthalmitis and loss of vision (Zeiter et al., 1989).

Enterobacter agglomerans was isolated from the blood of a patient with cotton fever, a benign febrile leukocytic syndrome seen in intravenous narcotic abusers. The strain was also isolated from cotton that the patient had used to filter heroin (Ferguson et al., 1993). *Enterobacter agglomerans* is known to heavily colonize cotton and cotton plants.

Antibiotic sensitivity Most *E. agglomerans* strains are naturally resistant to ampicillin and cephalothin and susceptible to many antibiotics including aminoglycosides, carbenicillin, cefamandole, cefuroxime, and cefoxitin. Resistance to carbenicillin may occur.

Pathogenesis Contaminated medicinal agents can be sources of outbreaks. Maki and Martin (1975) showed that strains identified as *E. agglomerans* (as well as strains of *E. cloacae* and *Serratia marcescens*) multiplied in 5% dextrose solution at 25°C better than did other members of the *Enterobacteriaceae*. A single tested strain of *E. agglomerans* was devoid of hemagglutinin (Adegbola and Old, 1983a). *E. agglomerans* strains produce aerobactin. Two of nine strains of *E. agglomerans* produced a hydroxamate compound other than aerobactin (Reissbrodt and Rabsch, 1988). In another study, strains identified as *Enterobacter agglomerans* produced hydroxamate siderophores identified as ferrioxamine E. The strains had also multiple siderophore receptors in the outer membrane (Berner et al., 1988).

The lipopolysaccharide from strains identified as *E. agglomerans* (commonly found in cotton dust) can bind to the pulmonary lipid-proteinaceous lining material (surfactant) and alter its surface tension properties (DeLucca et al., 1988). This binding in the lung may change the physiological properties of surfactant and be a possible mechanism for the pathogenesis of byssinosis, an occupational respiratory disorder caused by the inhalation of cotton dust (DeLucca et al., 1988).

Ecology The *Enterobacter agglomerans* complex is ubiquitous in the environment. This complex predominates on the leaf and bract of pre- and postsenescent cotton plants (DeLucca and Palmgren, 1986). The main species of Gram-negative bacteria in cotton dusts found in mills is *E. agglomerans* (Haglund et al., 1981). The following occurrences have been reviewed by Slade and Tiffin (1984). Strains of the *Enterobacter agglomerans* complex are found on the aerial surfaces of plants and within healthy plant tissues and seeds. Nitrogen fixing strains have been found in the rhizosphere of wheat and sorghum. In fact, these bacteria are typical of the innermost part of the rhizosphere of wheat (Kleeberger et al., 1983). Nitrogen-fixing strains identified as *Enterobacter agglomerans* have also been isolated from gut of the wood-eating termite *Coptotermes formosanus* (Potrikus and Breznak, 1977). Strains of this complex have been isolated from water, paper mill process water, soil and decaying wood. They are frequently isolated from damaged plant tissues and lesions, although they are rarely considered pathogenic. A pathogenic role has been shown in some instances. Strains named *Erwinia milletiae* produce β -indolyl acetic acid (a plant hormone) causing galls on some plants in Japan (reviewed by Slade and Tiffin, 1984). Some strains initiate freezing of buffer solutions (which would normally freeze at temperatures below -10°C) at about -4°C . This ice-nucleating property plays a critical role in causing frost damage on plants (Kozloff et al., 1983; Lindow et al., 1978).

Pink disease of pineapple, caused by *Pantoea citrea*, is characterized by a dark coloration on fruit slices after autoclaving (Cho et al., 1980). This coloration is initiated by the oxidation of glucose to gluconate, then the oxidation of gluconate to 2-ketogluconate, and finally the production of 2,5-diketogluconate. The latter appears to be responsible for the dark color characteristic of the pink disease of pineapple (Pujol and Kado, 2000).

Some strains of the *Enterobacter agglomerans* complex are associated with stalk and leaf necrosis of onion (Hattingh and Walters, 1981). Strains originally named *Erwinia uredovora* (now *Pantoea ananatis*) attack the uredia of rust (*Puccinia* spp.) on wheat, oats and rye (Pon et al., 1954; Dye, 1969b).

Stewart's bacterial wilt is a disease of corn (*Zea mays*) caused by *Pantoea stewartii* subsp. *stewartii*. On corn seedlings at the three- to five-leaf stage, the disease is characterized by water-soaked lesions on leaves, leading to stunted plants with severe yield reductions in susceptible corn hybrids (Dillard and Kline, 1989; Wilson et al., 1994). The corn flea beetle, *Chaetocnema pulicaria*, is the overwintering host and vector of *P. stewartii*, and the abundance of the primary inoculum is related to corn flea beetle populations. Many countries ban import of seed corn unless it is certified free of *P. stewartii*. A ligase chain reaction assay has been proposed for the detection of *P. stewartii* in infected plant and vector material (Wilson et al., 1994).

P. stewartii subsp. *stewartii* produces an extracellular heteropolysaccharide capsule, which plays several roles in the development of Stewart's wilt on sweet corn. The capsule provides a barrier that protects the bacterium against plant host defense factors (Braun, 1982; Beck von Bodman and Farrand, 1995). It also partially contributes to the induction of water-soaking symptoms early in the development of Stewart's wilt, and it obstructs the free flow of water in the host vascular system, causing necrosis and wilting during the systemic phase of the infection (Braun, 1982; Beck von Bodman and Farrand, 1995). *P. stewartii* subsp. *stewartii* synthesizes *N*-(3-oxohexanoyl)-L-homoserine lactone, which is an autoinducer for capsular polysaccharide biosynthesis and *P. stewartii* pathogenicity (Beck von Bodman and Farrand, 1995).

A strain identified as *Pantoea agglomerans* was isolated from necrotic spots in the leaves of a beach pea (*Lathyrus maritimus*) that grew on the shorelines of Newfoundland, Canada. The bacterium produced cellulase and amylase and was a plant wound parasite (Khetmalas et al., 1996).

Some strains of the *E. agglomerans* complex have been used for biological control of plant pathogens (such as *Erwinia amylovora* or *Xanthomonas oryzae*), either by competition for nutrients, acid production, bacteriocin production or phage production (reviewed by Slade and Tiffin, 1984). *Pantoea agglomerans* CFA-2 is effective for the biological control of postharvest pear diseases due to *Botrytis cinerea*, *Penicillium expansum*, and *Rhizopus stolonifer* (Nunes et al., 2001).

Pantoea agglomerans in fecal pellets of locusts produces large amounts of guaiacol and small amounts of phenol, both of which are components of the locust cohesion pheromone (Dillon et al., 2002).

ENRICHMENT AND ISOLATION PROCEDURES

All media designed for the isolation of *Enterobacteriaceae* can be used for the isolation of *Pantoea* species: MacConkey agar, Dri-galski lactose agar, Hektoen agar, deoxycholate lactose citrate agar, etc. *Pantoea* strains can also grow on media for general use, such as blood agar, nutrient agar, tryptic soy agar, bromocresol

purple lactose agar, etc. Media specifically selective for *Pantoea* strains are not available.

A differential medium, lysine-ornithine-mannitol agar containing vancomycin, was proposed for the isolation of *Pantoea agglomerans*. It yields colorless colonies from mannitol-negative strains, yellow colonies from mannitol-positive, ornithine- and lysine-decarboxylase negative strains, and greenish-blue colonies from mannitol-positive strains that produce one or both decarboxylases (Bucher and von Graevenitz, 1982).

MAINTENANCE PROCEDURES

Strains are initially grown on tryptic soy agar at their optimum temperature. They are then inoculated by stabbing a maintenance medium¹ designed for maintenance of *Enterobacteriaceae* and related organisms. The cultures are then stored at room temperature in a dark, dry place.

Cultures may be also preserved by freeze-drying. Freeze-drying is the best procedure for preservation of pigmented strains.

DIFFERENTIATION OF THE GENUS *PANTOEA* FROM OTHER GENERA

Enterobacter strains that are negative for lysine and ornithine decarboxylases and arginine dihydrolase and that may or may not produce a yellow pigment were formerly identified as *Enterobacter agglomerans*. The genus *Pantoea* is a subset of the *Enterobacter agglomerans* complex. Species of *Pantoea* are able to oxidize D-glucose to D-gluconate (glucose dehydrogenase activity) without added pyrroloquinoline quinone, and D-gluconate to 2-ketogluconate (gluconate dehydrogenase activity). In addition, they cannot utilize phenylacetate, and most strains can utilize myo-inositol.

TAXONOMIC COMMENTS

The genus *Erwinia* has long been a depository for plant-associated members of the family *Enterobacteriaceae*. Several species of this genus were found to be phenotypically similar to *Enterobacter* species (*Erwinia herbicola*, *Erwinia ananatis*, *Erwinia uredovora*, *Erwinia milletiae*, *Erwinia dissolvens*, and *Erwinia nimipressuralis*) (Lelliott, 1974; Lelliott and Dickey, 1984), or highly related to *Enterobacter* species by DNA hybridization (*Erwinia dissolvens* and *Erwinia nimipressuralis*) (Steigerwalt et al., 1976).

The (invalid) name "*Bacterium herbicola aureum*" was introduced by Duggeli (1904) and "*Bacterium herbicola*" is attributed to Löhnis (1911). The species was transferred to the genus *Erwinia* as *E. herbicola* by Dye (1964). It has always been considered a saprophyte associated with plants (epiphyte).

The earliest synonym of *Erwinia herbicola*, represented by an extant culture, is "*Pseudomonas trifolii*" (Hüss 1907) shown by Dye (1964) to be identical with *Erwinia herbicola*. Unfortunately, the epithet "*trifolii*" did not appear in the Approved Lists, although it had priority over the epithet *herbicola*.

Manns and Taubenhaus (1913) described "*Bacillus lathyri*" as the cause of streak disease of sweet peas (now attributed to a virus). The species was transferred to the genus *Erwinia* as *E. lathyri* by Holland (1920).

"*Bacillus milletiae*" was described by Kawakama and Yoshida (1920) as causing galls on millet. The species was transferred to the genus *Erwinia* as *E. milletiae* by Magrou (1937).

"*Bacillus ananas*" was isolated from bacterial fruitlet brown-rot of pineapple in the Philippines by Serrano (1928). It is given as *Erwinia ananas* Serrano 1928 on the Approved Lists of Bacterial Names (Skerman et al., 1989).

"*Xanthomonas uredovorus*" was isolated from uredia of cereal rusts (Pon et al., 1954) and reclassified in the genus *Erwinia* by Dye (1963c).

Graham (1958) reported that the pathogenic properties of *E. ananas* and *E. milletiae* are doubtful and that culturally and biochemically these species were indistinguishable from *E. lathyri*.

Dye (1969b) proposed a classification of the "herbicola group" containing *Erwinia herbicola* subsp. *herbicola* (18 species names were listed as synonyms including *E. herbicola*, *E. lathyri*, and *E. milletiae*), *Erwinia herbicola* subsp. *ananas*, *E. uredovora* and *E. stewartii*.

"*Pseudomonas stewartii*" was isolated by Smith (1898) from Stewart's sweet corn wilt and was transferred to *Erwinia* as *E. stewartii* by Dye (1963b).

After studying many cultures of the "herbicola-lathyri" group, including clinical isolates, Ewing and Fife (1972) exhumed an old work by Beijerinck (1888) describing "*Bacillus agglomerans*", a bacterium that formed what might have been symplasmata. Although no culture was available, a new combination, *Enterobacter agglomerans* (Beijerinck) Ewing and Fife 1972, was proposed for the "herbicola-lathyri" group on the argument of priority. However, the Approved Lists gave the name as *Enterobacter agglomerans* Ewing and Fife 1972, thus losing the connection with Beijerinck (1888). The other synonyms found in the Approved Lists are *Erwinia herbicola* (Löhnis 1911) Dye 1964 and *Erwinia milletiae* (Kawakama and Yoshida 1920) Magrou 1937.

E. agglomerans is a very complex group of bacteria, which may cause opportunistic infections. The name covers many (20 to 40) genomic groups (Brenner et al., 1984a) or phena (Gavini et al., 1983b; Verdonck et al., 1987). In addition to this diversity, strains of the *E. agglomerans* complex are not closely related to *E. cloacae* (the type species of the genus *Enterobacter*) by DNA relatedness. Some groups in this complex have been designated as new genera (*Rahnella aquatilis*, *Ewingella americana*, *Leclercia adacarboxylata*). There is an apparent confusion in the literature, and a close examination of published papers for the presence of commonly studied strains is needed to extract convergent pieces of information. A table relating phena and DNA groups has been created after such a literature analysis (Table BXII.γ.239).

The numerical study of Gavini et al. (1983b), based on 169 strains, yielded five phena (A to E). Phenon A corresponded to *Erwinia carotovora*. Phenon B, which included strains of the *Enterobacter agglomerans* complex, was split into nine smaller phena (B1 to B9). Phenon B4 contained the type strains of *Erwinia herbicola* and *Erwinia milletiae*. Phenon B8 contained the type strain of *Erwinia ananas* and a reference strain of *Erwinia uredovora*. Phenon C corresponded to *Rahnella aquatilis*. Phenon D was split into three smaller phena (D1 to D3), with phenon D2 corresponding to *Enterobacter sakazakii*. Phenon E was split into five smaller phena (E1 to E5), with phenon E5 containing strains previously identified as *Escherichia adacarboxylata* (now *Leclercia adacarboxylata*).

A larger numerical study (Verdonck et al., 1987), based on 529 strains and including many type and reference strains, distributed 66 strains of the *Enterobacter agglomerans* complex into 21 phena. The correspondence between phena from both studies

1. Maintenance medium (g/liter): Bacto-peptone (Difco), 10.0, NaCl, 5.0; Bacto-agar (Difco), 10.0; pH 7.4. The medium should be dispensed into small (9.5–10 × 90 mm) screw-capped tubes.

(Gavini et al., 1983b; Verdonck et al., 1987) is given in Table BXII.γ.239.

The DNA relatedness work of Brenner et al. (1984a) revealed the extreme genomic diversity of the *Enterobacter agglomerans* complex: of 124 strains studied, 90 fell into 13 DNA groups (I to XIII) and 34 strains did not fit into any group. Furthermore, four groups (V, XI, XII, and XIII) were heterogeneous with respect to $\Delta(T_m)$ values. An interesting finding was that aerogenic strains and anaerogenic strains were not found in the same hybridization group.

Lind and Ursing (1986) identified 52 of 86 clinical isolates with *Enterobacter agglomerans sensu strictu* by DNA hybridization. In the same study, they demonstrated the synonymy of *Enterobacter agglomerans*, *Erwinia herbicola*, and *Erwinia milletiae*. This synonymy was confirmed by Beji et al. (1988) who, in addition, identified DNA group XIII (Brenner et al., 1984a) with *Enterobacter agglomerans sensu strictu*.

To separate *Enterobacter agglomerans/Erwinia herbicola* from the genera *Enterobacter* and *Erwinia*, a new genus, *Pantoea*, was proposed with *P. agglomerans* as the type species (Gavini et al., 1989b). The epithet *agglomerans* was retained because it was believed to have priority over epithet *herbicola*. Strains of DNA group III (Brenner et al., 1984a) or phenon 8 (Verdonck et al., 1987) were proposed as a new species, *Pantoea dispersa* (Gavini et al., 1989b). The DNA groups closest to *Pantoea agglomerans* are DNA groups II, III, IV, V, and VI (Lind and Ursing, 1986). It is interesting that strains of group II to VI and XIII are characterized by the presence of a glucose oxidation pathway that produces 2-ketogluconate from glucose (Bouvet et al., 1989). This finding has been extended to DNA group I (P.A.D. Grimont, unpublished observations). Thus, the genus *Pantoea* can be envisioned to include DNA groups I, II, IV, V, and VI in addition to groups XIII (*P. agglomerans*) and III (*P. dispersa*).

Erwinia ananas (Serrano 1928) corresponds to group VI of Brenner et al. (1984a) and was transferred to the genus *Pantoea*

as *P. ananas* (Mergaert et al., 1993). The name was corrected to *P. ananatis* by Trüper and De' Clari, 1997). *Erwinia uredovora* belongs to that species.

Erwinia stewartii was included in the genus *Pantoea* as *P. stewartii* subsp. *stewartii* (Smith 1898) Mergaert et al. 1993. *P. stewartii* subsp. *indologenes* was a new subspecies composed of strains phenotypically different from *Erwinia stewartii* (and *P. stewartii* subsp. *stewartii*) and resembling *P. ananatis*. However, DNA hybridization and sequence studies indicate that both subspecies should belong to the same species.

The three species (*Pantoea punctata*, *P. citrea*, and *P. terrea*) described by Kageyama et al. (1992) have the ability to oxidize 2-ketogluconate to 2,5-diketogluconate. These species differ from the other *Pantoea* species in several biochemical or nutritional characteristics. Their phylogenetic position is as a branch on the border of the *Pantoea* cluster.

DNA group XI (Brenner et al., 1984a) was identified with *Escherichia adecarboxylata* by DNA relatedness (Izard et al., 1985). *Escherichia adecarboxylata* was transferred to a new genus *Leclercia*, as *L. adecarboxylata* (Tamura et al., 1986). DNA group VII (Brenner et al., 1984a) was close to (but distinct from) *Escherichia adecarboxylata* (*Leclercia adecarboxylata*) (Izard et al., 1985). In our laboratory, strains of DNA group VII were indistinguishable from strains of phenon E4 (Gavini et al., 1983b) by carbon source utilization tests (P.A.D. Grimont, unpublished observations). Thus, DNA group VII is a good candidate as a new species of *Leclercia*.

DNA group VIII (Brenner et al., 1984a) was 64% related to *Enterobacter persicina* (a close relative of *Erwinia rhapontici*).

DNA group IX (Brenner et al., 1984a) is identical with *Enterobacter cowanii* based on nutritional tests and *rpoB* sequence analysis (unpublished data).

More work is needed to characterize unclassified strains of the *E. agglomerans* complex.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *PANTOEAE*

The characteristics of named or unnamed species of *Pantoea* are given in Table BXII.γ.240. The results of carbon source utilization tests given in Table BXII.γ.240 are mostly unpublished observations (P.A.D. Grimont and E. Ageron).

The genus *Pantoea* can be divided into two groups of species: (1) the *Pantoea* core group with *P. agglomerans*, *P. dispersa*, *P. ananatis*, and *P. stewartii* and (2) the "Japanese" group with *P. citrea*, *P. punctata*, and *P. terrea*. Species of the core group utilize the following substrates as sole carbon sources which the "Japanese"

group cannot utilize: D-alanine, L-alanine, *myo*-inositol, DL-lactate, 1-*O*-methyl-β-D-glucoside, and L-serine. Furthermore, all species of the core group utilize L-arabinose and D-mannitol, whereas two species of the "Japanese" group fail to utilize these substrates. Species of the core group are negative for arginine dihydrolase and 2-ketogluconate dehydrogenase, whereas two species of the "Japanese" group are positive for arginine dihydrolase and all three species are positive for 2-ketogluconate dehydrogenase.

List of species of the genus *Pantoea*

1. ***Pantoea agglomerans*** (Ewing and Fife 1972) Gavini Mergaert, Beji, Mielcarek, Izard, Kersters and De Ley 1989b, 343^{VP} (*Enterobacter agglomerans* (Beijerinck 1888) Ewing and Fife 1972, 10.)
ag.glo' mer.ans. L. v. *agglomerare* to form into a ball; L. part. adj. *agglomerans* forming into a ball (referring to the occurrence of symplasmata bacteria in aggregates surrounded by a translucent sheath in anaerogenic strains).

The species has all the characteristics given for the genus. Cells may form symplasmata. Strains grow well on nutrient agar at 30°C but not at 44°C. Some strains grow slowly at 4 or 41°C. The biochemical and nutritional characteristics are shown in Table BXII.γ.240. Key characteristics are: alkaline reaction in malonate broth; utilization of D-glucuronate and D-tartrate as sole carbon sources; inability to utilize *meso*-erythritol, gentiobiose, 5-ketogluconate, D-me-

TABLE BXII.γ.240. Characteristics of the named and unnamed species of the genus *Pantoea*.^a

Characteristic	<i>Pantoea agglomerans</i>	<i>Pantoea ananatis</i>	<i>Pantoea citrea</i>	<i>Pantoea dispersa</i>	<i>Pantoea punctata</i>	<i>Pantoea stewartii</i> subsp. <i>stewartii</i>	<i>Pantoea stewartii</i> subsp. <i>indologenes</i>	<i>Pantoea terreia</i>	DNA group I	DNA group II	DNA group IV	DNA group V
Motility (36°C)	+	+	—	+	—	—	d	+	ND	d	d	d
Yellow pigment	d	+	—	d	—	+	+	—	ND	d	d	d
Indole production	—	+	—	—	—	—	+	—	—	—	—	—
Malonate (Leifson)	+	—	ND	—	ND	—	—	ND	—	d	—	d
β-Xylosidase test	d	—	ND	—	ND	ND	ND	ND	+	—	—	+
Voges-Prokauer	+	+	+	d	+	d	d	+	ND	d	d	d
Gelatin hydrolysis at 22°C	(+)	+	—	(d)	—	—	—	—	ND	d	d	+
Arginine dihydrolase	—	—	+	—	+	—	—	—	—	—	—	—
Phenylalanine deaminase	d	—	—	—	—	—	—	—	ND	+	d	d
Glucose dehydrogenase	+	+	+	+	+	ND	ND	+	+	+	+	+
Gluconate dehydrogenase	+	+	+	+	+	ND	ND	+	+	+	+	+
2-ketogluconate dehydrogenase	—	—	+	—	+	ND	ND	+	ND	—	—	—
Esculin hydrolysis	+	d	—	d	d	—	+	+	+	d	d	d
Nitrate reduced	+	d	+	d	+	—	—	+	ND	+	d	d
ONPG hydrolyzed	+	+	+	+	—	+	+	—	ND	ND	ND	ND
<i>Acid from:</i>												
L-Arabinose	+	+	ND	d	ND	+	+	ND	ND	+	+	+
D-Arabitol	+	+	d	+	—	—	+	—	ND	ND	ND	ND
Cellobiose	d	+	—	d	—	—	+	—	ND	d	d	d
Dulcitol	—	—	—	—	—	—	—	—	ND	—	—	d
meso-Erythritol	—	—	d	—	—	—	—	—	ND	ND	ND	ND
Glycerol	(d)	+	ND	(d)	ND	—	d	ND	ND	d	—	d
myo-Inositol	(d)	+	ND	d	ND	—	+	ND	ND	d	d	d
Lactose	d	+	+	d	—	—	+	—	+	d	d	d
Maltose	+	+	ND	+	ND	—	+	ND	ND	ND	ND	ND
D-Mannitol	+	+	+	+	—	+	+	—	ND	+	+	+
Melibiose	—	d	+	d	+	+	+	+	ND	ND	ND	ND
Raffinose	(d)	+	ND	—	ND	+	+	ND	ND	—	—	d
L-Rhamnose	+	d	—	+	—	—	d	—	ND	+	d	d
Salicin	+	+	d	(d)	—	—	+	+	ND	d	d	d
D-Sorbitol	—	+	d	—	—	—	—	d	ND	—	—	d
Sucrose	+	+	ND	+	ND	+	+	ND	ND	d	+	d
Trehalose	+	+	ND	+	ND	+	+	ND	ND	ND	ND	ND
D-Xylose	+	+	+	+	—	+	+	+	ND	+	d	+
<i>Utilization of:</i>												
trans-Aconitate	+	+	+	+	—	—	+	—	+	d	+	+
Adonitol	—	—	—	—	—	—	—	—	—	—	+	—
L-Arabinose	+	+	+	+	—	+	+	—	+	+	+	+
D-Arabitol	+	+	—	+	—	—	+	—	—	d	+	d
L-Arabitol	—	—	—	—	—	—	—	—	—	—	+	—
Betaine	—	—	—	—	—	—	+	—	—	—	—	—
Cellobiose	(d)	+	—	+	(+)	d	+	—	+	d	+	+
Citrate	d	+	+	+	(+)	—	+	+	+	d	+	+
Dulcitol	—	—	—	(d)	—	—	—	—	—	(d)	—	—
meso-Erythritol	—	—	+	+	—	—	—	—	—	—	d	—
L-Fucose	—	—	—	—	—	—	—	—	—	d	—	d
D-Galacturonate	(+)	+	—	+	—	d	+	—	+	+	d	+
Gentiobiose	—	+	—	+	+	—	d	—	+	d	d	—
D-Glucuronate	(+)	+	—	+	—	—	+	—	+	+	+	+
myo-Inositol	+	+	—	+	—	+	+	—	+	d	+	+
5-Ketogluconate	—	(+)	+	+	(+)	—	+	+	+	d	+	(d)
Lactose	—	+	(+)	—	—	d	+	—	+	d	—	(d)
Lactulose	—	+	+	—	—	d	+	—	+	d	—	d
D-Malate	(+)	d	+	(d)	—	+	+	—	(+)	(+)	—	(+)
Maltose	(+)	+	+	+	—	—	+	—	+	d	+	+
Maltotriose	(+)	+	+	+	—	—	+	—	+	d	+	+
D-Melibiose	—	+	—	—	—	+	+	—	—	d	—	—
1-O-Methyl-α-galactoside	—	(+)	—	—	—	+	+	—	—	d	—	—
3-O-Methyl-D-glucose	—	(d)	—	—	—	—	+	—	—	—	—	—

(continued)

TABLE BXII.γ.240. (cont.)

Characteristic	<i>Pantoeia agglomerans</i>	<i>Pantoeia ananatis</i>	<i>Pantoeia citrea</i>	<i>Pantoeia dispersa</i>	<i>Pantoeia punctata</i>	<i>Pantoeia stewartii</i> subsp. <i>stewartii</i>	<i>Pantoeia stewartii</i> subsp. <i>indologenes</i>	<i>Pantoeia terreia</i>	DNA group I	DNA group II	DNA group IV	DNA group V
1- <i>O</i> -Methyl-β-D-glucoside	(+)	+	—	+	—	+	+	—	+	+	+	+
Protocatechuate	—	+	—	—	—	—	+	—	—	—	—	—
Quinate	—	(+)	—	—	—	—	+	—	—	—	—	—
D-Raffinose	—	+	—	—	—	+	+	—	—	d	—	d
L-Rhamnose	+	d	—	+	—	d	d	—	+	+	d	+
D-Saccharate	(+)	(+)	—	+	—	—	+	—	+	+	+	+
D-Sorbitol	—	d	—	—	—	—	—	—	+	(d)	—	—
Sucrose	(+)	+	—	+	(+)	(+)	+	—	—	d	+	+
D-Tagatose	—	—	+	—	—	—	—	—	—	—	—	—
D-Tartrate	(+)	—	—	—	—	—	—	—	—	—	—	—
L-Tartrate	—	—	—	d	—	—	—	—	—	d	—	d
meso-Tartrate	(+)	d	—	d	—	+	+	—	—	d	—	(+)
Trigonelline	—	—	—	—	(d)	—	—	—	—	d	+	—
Xylitol	—	—	—	—	—	—	—	—	—	d	+	—
D-Xylose	+	+	—	+	—	+	+	—	+	+	—	+

^aSymbols: +, 90–100% of strains positive in 1–2 days; (+), 90–100% of strains positive in 1–4 days; —, 90–100% of strains negative in 4 days; d, positive in 1–4 days; (d), positive in 3–4 days; ND, no data.

libiose and D-raffinose as sole carbon sources; and no production of indole and 2-ketogluconate dehydrogenase.

Isolated from plants, flowers, seeds, vegetables, water, soil and foodstuffs. Some strains are of human (wounds, blood, urine, internal organs) and animal origin. Some strains (synonym, *Erwinia milletiae*) have been reported to cause galls on *Wisteria floribunda* and *Wisteria japonica*; some strains cause galls on *Gypsophyla paniculata*, and some strains cause stalk and leaf necrosis on onion plants.

The mol% G + C of the DNA is: 55.1–56.8 (T_m).

Type strain: ATCC 27155, CCUG 539, CFBP 3845, CIP 57.51, DSM 3493, ICMP 12534, JCM 1236, LMG 1286, NCTC 9381.

GenBank accession number (16S rRNA): AB004691, AJ233423.

- Pantoeia ananatis*** corrig. (Serrano 1928) Mergaert, Verdonck and Kersters 1993, 170^{VP} (*"Bacillus ananas"* Serrano 1928, 271; *Erwinia ananas* (sic) Serrano 1928; *Pantoeia ananas* (sic) Mergaert, Verdonck and Kersters 1993, 170.) *a' na. na. tis*. M.L. n. *ananis* generic name of the pineapple.

The species has all the characteristics given for the genus. The biochemical and nutritional characteristics are given in Table BXII.γ.240. Key characteristics are: indole production; no alkaline reaction in malonate broth; utilization of gentiobiose, D-glucuronate, 5-ketogluconate, lactose, lactulose, D-melibiose, 1-*O*-methyl-α-galactoside, protocatechuate, quinate and D-raffinose as sole carbon sources; utilization of D-sorbitol by most strains; and no utilization of meso-erythritol and D-tartrate.

Strains formerly classified as *Erwinia uredovora* show a stronger proteinase activity and produce a DNase.

The mol% G + C of the DNA is: 53.6–56.4 (T_m).

Type strain: ATCC 33244, CFBP 3612, CIP 105207, LMG 2665, NCPPB 1846.

GenBank accession number (16S rRNA): U80196.

- Pantoeia citrea*** Kageyama, Nakae, Yagi and Sonoyama 1992, 209^{VP}
ci' tre. a. M.L. adj. *citrea* of citrus.

The cell morphology and colonial morphology are as given for the genus. The cells are nonmotile. Either nicotinic acid or nicotinamide is required for growth. Good growth occurs at 20–34°C; no growth occurs at 41°C. Colonies grown on nutrient agar at 30°C for 2 d are pale beige to pale reddish yellow. Physiological and nutritional characteristics are presented in Table BXII.γ.240. Key characteristics are: production of 2-ketogluconate dehydrogenase; positive arginine dihydrolase reaction; hydrolysis of ONPG but not esculin; utilization of L-arabinose, meso-erythritol, maltose, and D-tagatose as sole carbon and energy sources; and no utilization of gentiobiose. Isolated from mandarin oranges.

The mol% G + C of the DNA is: 49.7 (HPLC).

Type strain: SHS 2003, ATCC 31623, CIP 105599, CCUG 30156, DSM 13699.

- Pantoeia dispersa*** Gavini, Mergaert, Beji, Mielcarek, Izard, Kersters and De Ley 1989b, 344^{VP}
dis. per' sa. L. v. *dispergere* to spread, to scatter; L. fem. part. adj. *dispersa* spread, scattered.

The species has all the characteristics of the genus. Strains grow well on nutrient agar at 30 and 41°C but not at 44°C or 4°C. Key characteristics are: no alkaline reaction in malonate broth; utilization of *meso*-erythritol, gentiobiose, D-glucuronate and 5-ketogluconate as sole carbon sources; no utilization of D-melibiose, D-raffinose, and D-tartrate as sole carbon sources; and no production of indole and 2-ketogluconate dehydrogenase. Isolated from plant surfaces, seeds, humans, and the environment.

The mol% G + C of the DNA is: 56.5–60.6 (T_m).

Type strain: ATCC 14589, CCUG 25232, CIP 103338, DSM 30073, LMG 2603.

5. ***Pantoea punctata*** Kageyama, Nakae, Yagi and Sonoyama 1992, 209^{VP}

punc.ta' ta. L. n. *punctum* a point; M.L. adj. *punctata* full of points.

The cell morphology and colonial morphology are as given for the genus. Cells are nonmotile. Either nicotinic acid or nicotinamide is required for growth. Good growth occurs at 20–34°C; no growth occurs at 41°C. Colonies grown on nutrient agar at 30°C for 2 d are pale beige to pale reddish yellow. Yellow pigment is not produced on nutrient agar. Physiological and nutritional characteristics are presented in Table BXII.γ.240. Key characteristics are: production of 2-ketogluconate dehydrogenase; positive arginine dihydrolase reaction; hydrolysis of esculin by some strains; no hydrolysis of ONPG; utilization of gentiobiose as a sole source of carbon and energy; and no utilization of L-arabinose, *meso*-erythritol, maltose, and D-tagatose. Isolated from mandarin oranges.

The mol% G + C of the DNA is: 50.0–50.3 (HPLC).

Type strain: SHS 2006, ATCC 31626, CIP 105598, DSM 13700.

6. ***Pantoea stewartii*** (Smith 1898) Mergaert, Verdonck and Kersters 1993, 170^{VP} (*"Pseudomonas stewartii"* Smith 1898, 422; *Erwinia stewartii* (Smith 1898) Dye 1963b, 504.) *stew.art' i.i.* M.L. en. n. *stewartii* of Stewart; named after F.C. Stewart.

The species has all the characteristics given for the genus, except that some strains produce a capsular polysaccharide. The species is divided into two subspecies that differ strongly in vigor. Previous information about the characteristics of *Erwinia stewartii* applies only to *P. stewartii* subsp. *stewartii*, not to *P. stewartii* subsp. *indologenes*, which can hardly be differentiated from *P. ananatis* by biochemical and nutritional properties. Common characteristics of the species are listed in Table BXII.γ.240. Key characteristics are: production of a yellow pigment; no alkaline reaction in malonate broth; no reduction of nitrate to nitrite; acid production from L-arabinose, sucrose, and D-mannitol but not from D-sorbitol; utilization of L-alanine, L-arabinose, *myo*-inositol, DL-lactate, D-mannitol, D-melibiose, 1-O-methyl-α-galactoside, 1-O-methyl-β-galactoside, 1-O-methyl-β-D-glucoside, D-raffinose, L-serine, sucrose, *meso*-tartrate, and D-xylose as sole sources of carbon and energy; and no utilization of *meso*-erythritol and D-sorbitol. Isolated from *Zea mays*, several other grasses, *Ananas comosus*, and beetles.

The mol% G + C of the DNA is: 53.6–56.4 (T_m).

Type strain: ATCC 8199, CIP 104005, DSM 30176, LMG 2715, NCPPB 2295.

a. ***Pantoea stewartii* subsp. *stewartii*** (Smith 1898) Mergaert, Verdonck and Kersters 1993, 170^{VP} (*"Pseudomonas stewartii"* Smith 1898, 422; *Erwinia stewartii* (Smith 1898) Dye 1963b, 504.)

The biochemical and nutritional characteristics are given in Table BXII.γ.240. A few strains grow at 4°C. Some strains grow at 37°C, but no strains grow at 41°C. Key characteristics are: lack of motility; no production of indole; no hydrolysis of esculin; no acid production from D-arabitol, cellobiose, *myo*-inositol, lactose, maltose, L-rhamnose and salicin; and no utilization of D-arabitol, betaine, citrate, D-glucuronate, 5-ketogluconate, maltose, 3-O-methyl-D-glucose, protocatechuate, quinate, and D-saccharate as sole carbon and energy sources.

Causative agent of Stewart's bacterial wilt of corn, a vascular disease of *Zea mays*. Also isolated from the insect vector, the corn flea beetle *Chaetocnema pulicaria*.

The mol% G + C of the DNA is: 54.6–55.1 (T_m).

Type strain: ATCC 8199, CIP 104005, DSM 30176, LMG 2715, NCPPB 2295.

GenBank accession number (16S rRNA): Z96080.

b. ***Pantoea stewartii* subsp. *indologenes*** Mergaert, Verdonck and Kersters 1993, 171^{VP}

in.do.lo' gen.es. M.L. n. *indolum* indole; Gr. n. *gennao* to produce; M.L. adj. *indologenes* indole producing.

The biochemical and nutritional characteristics are given in Table BXII.γ.240. Many strains grow at 4 and 37°C and some strains grow at 41°C, but few strains grow at 44°C. Key characteristics are: motility; production of indole; hydrolysis of esculin; acid production from D-arabitol, cellobiose, *myo*-inositol, lactose, maltose, D-rhamnose and salicin; and utilization of L-arabitol, betaine, citrate, D-glucuronate, 5-ketogluconate, maltose, 3-O-methyl-D-glucose, protocatechuate, quinate and D-saccharate as sole carbon and energy sources.

Thought to cause leaf spot on foxtail millet (*Setaria italica*) and pearl millet (*Pennisetum americanum*) and rot of *Ananas comosus*. Also isolated from cluster bean (*Cyamopsis tetragonolobus*).

The mol% G + C of the DNA is: 56.4 (T_m).

Type strain: ATCC 51785, CIP 104006, LMG 2632.

GenBank accession number (16S rRNA): Y13251.

7. ***Pantoea terreia*** Kageyama, Nakae, Yagi and Sonoyama 1992, 210^{VP}

ter' re.a. L. n. *terra* soil; L. adj. *terrea* of soil.

The cell morphology and colonial morphology are as given for the genus. The cells are motile by means of one or two lateral flagella. Colonies grown on nutrient agar at 30°C for 2 d are pale beige to pale reddish yellow. Yellow pigment is not produced on nutrient agar. Physiological and nutritional characteristics are presented in Table BXII.γ.240. Key characteristics are: production of 2-ketogluconate dehydrogenase; negative arginine dihydrolase reaction; hydrolysis of esculin but not ONPG; and no utilization of L-arabinose, *meso*-erythritol, gentiobiose, maltose, and D-tagatose as sole carbon and energy sources. Isolated from soil in Japan

The mol% G + C of the DNA is: 51.0–51.9 (HPLC).

Type strain: SHS 2008, ATCC 31628, CCUG 30161, CIP 105600, DSM 13701.

Genus XXIV. *Pectobacterium* Waldee 1945, 469^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1

LYSIANE HAUBEN, FREDERIQUE VAN GIJSEGEN AND JEAN SWINGS

Pec.to.bac.te'ri.um. Gr. dim. neut. n. *bakterion* a small rod; M.L. neut. n. *Pectobacterium* a pectolytic bacterium.

Straight rods, 0.5–1.0 × 1.0–3.0 µm, rounded ends; occur singly or in pairs. Gram-negative. Motile by peritrichous flagella. Fermentative. Facultatively anaerobic. Optimum growth temperature, 27–30°C; maximum temperature for growth is 40°C. Oxidase negative. Catalase positive. Acid is produced from **N-acetylglucosamine**, fructose, D-galactose, D-glucose, D-mannose, L-rhamnose, D-ribose, salicin, and sucrose but not from adonitol, L-arabitol, D-lyxose, α-methylmannoside, L-sorbose, starch, or D-tagatose. Utilize acetate, arbutin, fructose, fumarate, D-galactose, D-glucose, glycerol, malate, D-mannitol, mannose, β-methylglucoside, ribose, salicin, succinate, and sucrose but not adipate, benzoate, betaine, butanol, gallate, methanol, oxalate, propionate, or sorbose as carbon and energy-yielding sources. Utilize L-alanine, allantoin, γ-aminobutyric acid, ammonium chloride, arginine, asparagine, asparaginic acid, citrulline, glucosamine, glutamine, L-glutaminic acid, glutathione, glycine, glycylglycine, histidine, leucine, L-methionine, phenylalanine, L-serine, L-tryptophan, tyrosine, and urea, but not anthranilic acid, betaine, choline, cysteamine, hydroxyproline, kynureninic acid, quolinic acid, sarcosine, spermidine, spermine, trigonelline, or trimethylammonium as nitrogen sources. No decarboxylases are formed for arginine, lysine or ornithine. Do not possess tryptophan deaminase or urease. **Hydrolyze esculin but not starch.**

The species of the genus *Pectobacterium* comprise a distinct phylogenetic group, as determined by 16S rRNA gene sequence comparisons, and are characterized by 17 signature nucleotides (Table BXII.γ.223 in Genus *Erwinia*).

Pectobacterium species cause plant diseases that include blights, cankers, die back, leaf spots, wilts, discoloration of plant tissues, and especially soft rots variously described as stalk rot, crown rot, stem rot, or fruit collapse. Ingress by the pathogen generally occurs through natural openings and wounds. Soft rot is characterized by the breakdown of the plant cell wall, leading to maceration of the parenchyma, loss of electrolytes, and cell death. Pectinases and pectate lyases are mainly responsible for these symptoms.

The mol% G + C of the DNA is: 50.5–56.1.

Type species: ***Pectobacterium carotovorum*** (Jones 1901) Waldee 1945, 469 emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Erwinia carotovora* (Jones 1901) Bergey, Harrison, Breed, Hammer and Huntoon 1923, 171; *Bacillus carotovorus* Jones 1901, 12.)

FURTHER DESCRIPTIVE INFORMATION

Metabolic features are the same as for the genus *Erwinia*. Decarboxylases for arginine, lysine, or ornithine are not present except in a few (usually 5% or less) strains of *Pectobacterium carotovorum* and *Pectobacterium chrysanthemi*.

Pectate lyases are produced by all species except by *Pectobacterium cyprripedii*. Cellulases (Cx) are produced by strains of *P. carotovorum*, *P. carotovorum* subsp. *atrosepticum*, and *P. chrysanthemi* (El-Helaly et al., 1979). *P. carotovorum* and *P. chrysanthemi* produce pectinases, cellulases, hemicellulases, and proteases (Garibaldi and Bateman, 1973; Bertheau et al., 1984; Collmer and Keen, 1986; Ried and Collmer, 1986; Wandersman et al., 1986; Willis

et al., 1987). *P. carotovorum* produces an endopolygalacturonase (Lei et al., 1985; Willis et al., 1987) as well. The genes encoding these enzymes that have been mapped are located on the chromosome (Chatterjee et al., 1981; Hugouvieux-Cotte-Pattat et al., 1996). In most *Pectobacterium* species, production of the plant cell wall degrading enzyme pectin lyase (Pnl) is activated by DNA-damaging agents such as mitomycin C (MC), nalidixic acid, and UV light (Liu et al., 1994b).

Naturally occurring plasmids have been detected in strains of *P. carotovorum* and *P. chrysanthemi*, and plasmids from bacteria other than *Pectobacterium* have been introduced into strains of the foregoing *Pectobacterium* species (Lacy and Leary, 1979; Chatterjee and Starr, 1980). Plasmid-mediated transfer of chromosomal genes by conjugation also has been reported for strains of *P. carotovorum* and *P. chrysanthemi*.

Plasmid pULB113 (Van Gijsegem and Toussaint, 1982) mediates chromosomal mobilization and R-prime formation in *P. carotovorum* and *P. chrysanthemi* (Chatterjee et al., 1985).

Virulent or temperate phages have reported to be active against strains of *P. carotovorum* (Chapman et al., 1951; Faltus and Kishko, 1980; Pirhonen and Palva, 1988; Gross et al., 1991; Toth et al., 1993) and *P. chrysanthemi* (Paulin and Nassan, 1978). Bacteriocinogeny or production of bacteriocin-like substances has been noted for strains of *P. carotovorum* (Itoh et al., 1978), *P. chrysanthemi* (Echandi and Moyer, 1979), and *Pectobacterium* species from sugar beet (Stanghellini et al., 1977). Bacteriocin-resistant mutants of *P. chrysanthemi* have been isolated (Expert and Toussaint, 1985).

Lipopolysaccharide defective mutants of *P. carotovorum* (Pirhonen and Palva, 1988) and genetically engineered kanamycin resistant strains of *P. carotovorum* (Scanferlato et al., 1989; Orvos et al., 1990) have been isolated.

Antisera prepared against live or heat-killed cells, nonpurified or purified immunogens have been used for the differentiation or identification of all *Pectobacterium* species except *P. cyprripedii* (Schaad, 1979). Serogroups have been determined for *P. carotovorum* (De Boer et al., 1979) and *P. chrysanthemi* (Samson and Nassan-Agha, 1978; Yakus and Schaad, 1979).

More recently, a conductimetric assay was developed for automated detection of *P. carotovorum* subsp. *atrosepticum*, *P. carotovorum* subsp. *carotovorum*, and *P. chrysanthemi* (Fraaije et al., 1996a, 1997), and a luminescence-based assay is available for the detection of *P. carotovorum* (Grant et al., 1992; McLennan et al., 1992).

A strain of *P. carotovorum* subsp. *carotovorum* was found to produce the antibiotic 1-carbapen-2-em-3-carboxylic acid (Bainton et al., 1992b).

16S rDNA sequence analyses of the species of the genus *Pectobacterium* by Kwon et al. (1997) and Hauben et al. (1998a) agree very well, except for the type strain of *P. cyprripedii*, and situate the genus *Pectobacterium* within the *Enterobacteriaceae*, closely related to the genera *Pantoea*, *Erwinia*, *Brenneria*, and *Enterobacter* (Fig. BXII.γ.201 and Table BXII.γ.223 of the genus *Erwinia*). The sequence of the type strain of *P. cyprripedii*, sequenced by Kwon et al. (1997) (ATCC 29267) and Hauben et al. (1998a) (LMG 2657), differ in 53 nucleotides. The sequence reported by Hauben et al. (1998a) places *P. cyprripedii* in the *Pectobacterium* cluster.

ENRICHMENT AND ISOLATION PROCEDURES

The isolation procedure is the same as for the genus *Erwinia*.

The isolation of some *Pectobacterium* species can be facilitated by use of selective-differential media, but such media are usually not necessary. Selective media have been developed for the isolation of pectolytic bacteria (Kelman and Dickey, 1980). The crystal violet pectate (CVP)¹ medium is commonly used. Miller-Schroth (Miller and Schroth, 1972) medium modified by replacement of most of the agar by sodium polypectate selectively allows the growth of pectolytic species. NaOH (105%) and MOPS (3-(-N-morpholino)propanesulfonic acid; 0.4%) are added to raise the pH and buffer the medium (Pierce and McCain, 1992). *P. chrysanthemi* strains isolated from plants grow well on LB medium².

MAINTENANCE PROCEDURES

Stock cultures of *Pectobacterium* species should be grown on standard media of choice at 25–30°C. The cultures can be maintained for short-term storage in a refrigerator (4–5°C); some strains of *P. chrysanthemi* are nonviable after 3–4 weeks at 4°C, but remain viable for longer periods when stored at 12°C. Long-term preservation is the same as for the genus *Erwinia*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *PECTOBACTERIUM*

The differential characteristics of the species of *Pectobacterium* are given in Tables BXII.γ.241, BXII.γ.242, BXII.γ.243, and

DIFFERENTIATION OF THE GENUS *PECTOBACTERIUM* FROM OTHER GENERA

Characteristics that differentiate *Pectobacterium* from the genera *Erwinia*, *Pantoea*, and *Brenneria* are given in Table BXII.γ.223 of the genus *Erwinia*. Apart from their pectolytic nature, it is very difficult to differentiate them phenotypically; genomic methods are recommended for differentiation.

TAXONOMIC COMMENTS

The species of the genus *Pectobacterium* were formerly classified under *Erwinia*. We refer the reader to the taxonomic comments section of the genus *Erwinia* for a discussion on this issue.

Strains of *P. chrysanthemi* have been isolated from numerous plant species and cultivars (Dickey, 1981). Six pathovars (pathovar *chrysanthemi*, pathovar *dianthicola*, pathovar *dieffenbachiae*, pathovar *paradisiaca*, pathovar *parthenii*, and pathovar *zeae*) have been designated for *P. chrysanthemi* (Dye et al., 1980). The relationship between pathogenicity, phenotypic properties and serological reactions of strains of the pathovars is not clear (Samson and Nassan-Agha, 1978; Yakus and Schaad, 1979; Dickey, 1981; Thomson et al., 1981).

BXII.γ.244. Only a small number (eight) of strains of *P. cypripedii* have been studied.

List of species of the genus *Pectobacterium*

1. ***Pectobacterium carotovorum*** (Jones 1901) Waldee 1945, 469^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Erwinia carotovora* (Jones 1901) Bergey, Harrison, Breed, Hammer and Huntoon 1923, 171; *Bacillus carotovorus* Jones 1901, 12.) *ca.ro.to'vo.rum*. L. n. *carota* carrot; L. v. *voro* to devour; M.L. adj. *carotovorum* carrot-devouring.

The characteristics are as described for the genus and as listed in Tables BXII.γ.241, BXII.γ.242, BXII.γ.243, and BXII.γ.244. Causes rotting, particularly of storage tissues, of a wide variety of plants and causes a vascular and parenchymal disease (blackleg) of potato plants. The small diffusible autoinducer signal molecule *N*-(β-ketocaproyl) homoserine lactone (KHL) acts as a molecular control signal for the expression of genes controlling carbapenem antibiotic biosynthesis (Bainton et al., 1992a). The *pnl* gene, encoding pectin lyase (Ohnishi et al., 1991), and the *peh* gene, encoding polygalacturonase (Peh) (Lei et al., 1992) of *P. carotovorum*, have been located and characterized.

Subspecies can be differentiated by (1) PCR with primers Y1 (5'-TTACCGGACGCCGAGCTGTGGCGT-3') and Y2 (5'-CAGGAAGATGTCGTTATCGCGAGT-3'), annealing at 65°C, and (2) restriction fragment length polymorphism (RFLP) of a *pel* gene identifying *P. carotovorum* subsp. *atrosepticum* and *P. carotovorum* subsp. *wasabiae* strains (Darrasse et al., 1994b). The pectate lyase (*pel*) genes I and III were

characterized by Nikaido et al. (1985) and Yoshida et al. (1991), respectively. RAPD-PCR can differentiate *P. carotovorum* subsp. *carotovorum* and *P. carotovorum* subsp. *atrosepticum* (Maki-Valkama and Karjalainen, 1994; Parent et al., 1996) and a DNA probe, isolated from a genomic library, can differentiate the subspecies *P. carotovorum* subsp. *carotovorum*, *P. carotovorum* subsp. *atrosepticum*, and *P. carotovorum* subsp. *betavascuorum* (Ward and De Boer, 1990).

The bacteriocin carotovoricin was identified (Itoh et al., 1982) and the gene *recA*, which is required for its induction (Zink et al., 1985; Zhao and McEntee, 1990). Monoclonal antibodies were developed for detection of *P. carotovorum* subsp. *carotovorum*, *P. carotovorum* subsp. *atrosepticum*, and *P. carotovorum* subsp. *betavascuorum* (Ward and De Boer, 1989; Murray et al., 1990a; Vernon-Shirley and Burns, 1992). *Drosophila melanogaster* Meigen and *Drosophila rucksii* Coquillett can function as vectors for *P. carotovorum* subsp. *carotovorum* and *P. carotovorum* subsp. *atrosepticum* (Brewer et al., 1981).

The mol % G + C of the DNA is: 50.5–53.1 (*T_m*, Bd).

Type strain: ATCC 15713, NCPPB 312, ICMP 5702.

- a. ***Pectobacterium carotovorum* subsp. *carotovorum*** (Jones 1901) Waldee 1945, 469^{AL} (*Erwinia carotovora* subsp. *carotovora* Bergey, Harrison, Breed, Hammer and Huntoon 1923, 171.)

Characteristics distinguishing this subspecies from the other *Pectobacterium carotovorum* subspecies are indicated in Table BXII.γ.243.

The production of extracellular enzymes such as pectate lyase (Pel), polygalacturonase (Peh), cellulase (Cel), and protease (Prt) is activated by the cell density (quorum)-sensing signal, *N*-(3-oxohexanoyl)-L-homoserine lactone (HSL), and/or CarR (Barras et al., 1994). *RmsA*, a global repressor gene, controls the production of these

1. CVP medium consists of 1N NaOH, 4.5 ml; 10% CaCl₂·H₂O, 3 ml; Bacto-agar, 1.5 g; NaNO₃, 1 g; Bacto-yeast extract, 0.05 g; blended with 300 ml boiling distilled water for 15 s, to which 15 g sodium polypectate and 200 ml boiling distilled water is added and blended (Cuppels and Kelman, 1974; Woodward and Robinson, 1990).

2. LB medium consists of (per 1 distilled water) tryptone, 10 g; yeast extract, 5 g; NaCl, 10 g.

TABLE BXII.γ.241. Diagnostic characteristic of the *Pectobacterium* species^a

Characteristic	1. <i>P. carotovorum</i>	2. <i>P. cacticida</i>	3. <i>P. chrysanthemi</i>	4. <i>P. cypripedii</i>
Pectinase	+	+	+	—
Reducing sugars produced from sucrose	v	—	+	nd
Indole	—	—	+	—
Growth at 37°C	v	+	+	nd
Growth at 40°C	—	+	nd	nd
Production of acetoin	v	+	+	nd
Arginine dihydrolase	—	nd	d	nd
Caseinase	v	nd	+	—
Gelatinase	v	—	+	nd
Lecithinase	—	nd	+	—
Phenylalanine deaminase	—	nd	—	+
Phosphatase	—	d	+	nd
<i>Acid production from:</i>				
Lactose	+	nd	+	—
Malonate	—	nd	d	nd
Maltose	v	nd	—	+
Raffinose	v	nd	+	—
Trehalose	+	nd	—	+
<i>Utilization of carbon sources:</i>				
D-Arabinose	—	—	d	nd
L-Arabinose	+	—	+	nd
Cellobiose	+	—	+	nd
Gentiobiose	+	d	—	nd
α-Methylglucoside	d	nd	—	nd
myo-Inositol	+	—	nd	nd
Malonate	—	+	+	nd
Melibiose	v	—	+	nd
Raffinose	+	—	+	nd
Tartrate	—	—	—	+
<i>Sensitivity toward:</i>				
Erythromycin	—	—	+	+

^aFor symbols see standard definitions: nd, not determined.

enzymes and tissue macerating ability (Bainton et al., 1992b; Cui et al., 1995; McGowan et al., 1995; Mukherjee et al., 1996). The gene *aeoA* (activator of extracellular protein production) controls the production of pectolytic enzymes Pel, Peh, Cel, and Prt (Liu et al., 1993c; Murata et al., 1994), and the *out* gene cluster encoding the proteins of the type II or general secretory pathway (GSP) apparatus is required for secretion of pectinases and cellulases (Reeves et al., 1993; Thomas et al., 1997). The *pnlA* gene, encodes DNA damage-inducible pectin lyase (Chatterjee et al., 1991) and *rdgB* encodes a transcriptional factor that specifically interacts with the pectin lyase structural gene *pnlA* promoter/regulatory region (Liu et al., 1997e). The *pnlA* gene is activated by DNA-damaging agents like RecA, RdgA, and RdgB. RdgA and RdgB also control bacteriocin production, phage release, and cell lysis (Barras et al., 1994).

A pleiotrophic reduced virulence (Rvi-negative) mutant was found to be defective in flagella assembly proteins (Mulholland et al., 1993). Several monoclonal antibodies have been produced against *P. carotovorum* subsp. *carotovorum* strains that can be used for identification and detection by Ouchterlony double diffusion (ODD), indirect immunofluorescence (IIF), and enzyme-linked immunosorbent assay (ELISA) (Alarcon et al., 1995), and for detection by reverse passive hemagglutination (RPH) (Koehm and Eggers-Schumacher, 1995).

Pseudomonas fluorescens strains are antagonistic (biological control agent) to *P. carotovorum* subsp. *carotovorum* (El Hendawy et al., 1998). Causes rotting, particularly of

storage tissues, of a wide variety of plants, e.g., mushroom (*Agaricus* sp.), century plant (*Agave* sp.), mustard (*Brassica campestris*, *B. nigra*, and *B. juncea*), Chinese cabbage (*Brassica chinensis*), cauliflower (*Brassica oleracea* var. *botrytis*), cabbage (*Brassica oleracea* var. *capitata*), Brussels sprouts (*Brassica oleracea* var. *gemmifera*), broccoli (*Brassica oleracea* var. *italica*), turnip (*Brassica rapa* var. *rapa*), and ornamental plants. Larger fleshy organs are particularly susceptible and, once infected, they usually become softened to a pulp very quickly.

The mol% G + C of the DNA is: 50.5–53.1 (*T_m*, Bd).

Type strain: ATCC 15713, DSM 30168, LMG 2404.

GenBank accession number (16S rRNA): Z96089, U80197.

- b. ***Pectobacterium carotovorum* subsp. *atrosepticum*** (van Hall 1902) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 393) (*Erwinia carotovora* subsp. *atroseptica* (van Hall 1902) Dye 1969a, 81; *Bacillus atrosepticus* van Hall 1902, 134.)
at.ro.sep' ti.cum. L. adj. *ater* black; Gr. adj. *septicus* producing a putrefaction; M.L. adj. *atrosepticum* producing a black rot.

Characteristics distinguishing this subspecies from the other *P. carotovorum* subspecies are indicated in Table BXII.γ.243. Several regions of the bacterial DNA of *P. carotovorum* subsp. *atrosepticum* seem to be subspecies-specific and can be used for identification by (1) genomic subtraction of DNA probes (Darrasse et al., 1994a);

TABLE BXII.γ.242. Additional characteristics of the *Pectobacterium* species^a

Characteristic	1. <i>P. carotovorum</i>	2. <i>P. cacticida</i>	3. <i>P. chrysanthemi</i>	4. <i>P. cypripedii</i>
NO ₃ ⁻ → NO ₂ ⁻	+	+	+	nd
Production of aminopeptidase	nd	+	+	nd
Growth in 5%NaCl	v	nd	—	nd
Oxidase	—	—	—	—
Catalase	+	+	+	+
Citrate (Simmons)	v	d	+	nd
H ₂ S from cysteine	v	nd	v	—
β-Galactosidase	v	nd	+	nd
Arginine decarboxylase	—	—	—	—
Lysine decarboxylase	—	—	—	—
Ornithine decarboxylase	—	—	—	—
Esculin hydrolase	+	+	+	+
Starch hydrolase	—	—	—	—
Tryptophan deaminase	—	—	—	—
Urease	—	—	—	—
<i>Acid production from:</i>				
Adonitol	—	—	—	—
Amygdalin	nd	+	+	nd
D-Arabitol	v	nd	—	nd
L-Arabitol	—	—	—	—
L-Arabinose	nd	+	+	nd
Arbutin	nd	+	+	nd
Cellobiose	nd	+	+	nd
Citrate	v	nd	+	nd
Dulcitol	nd	—	—	—
Erythritol	nd	—	—	nd
Esculin	nd	+	+	nd
Fructose	+	+	+	+
DL-Fucose	nd	—	—	nd
Galactose	+	+	+	+
Gentiobiose	+	+	nd	nd
D-Glucose	+	+	+	+
N-acetylglucosamine	+	+	+	+
Glycerol	nd	+	+	+
Glycogen	nd	—	—	nd
5-Ketogluconate	—	—	nd	nd
Inulin	v	nd	d	—
D-Lyxose	—	—	—	—
D-Mannose	+	+	+	+
α-Methylmannoside	—	—	—	—
Melezitose	—	—	—	nd
Melibiose	nd	v	+	nd
α-Methylglucoside	nd	v	—	—
β-Methylxyloside	nd	—	—	nd
Palatinose	v	nd	—	nd
L-Rhamnose	+	+	+	+
D-Ribose	+	+	+	+
Salicin	+	+	+	+
Sorbitol	v	nd	—	nd
L-Sorbose	—	—	—	—
Starch	—	—	—	—
Sucrose	+	+	+	+
D-Tagatose	—	—	—	—
Xylitol	nd	—	—	nd
D-Xylose	nd	+	+	nd
L-Xylose	nd	—	—	nd
<i>Utilization of carbon sources:</i>				
Acetate	+	+	+	+
Adipate	—	—	—	—
Amygdalin	d	d	—	nd
L-Arabinose	nd	+	+	+
D-Arabitol	v	—	—	nd
Arbutin	+	+	+	+
Benzoate	—	—	—	—
Betaine	—	—	—	—
Butanol	—	—	—	—
Citrate	+	+	+	nd
Dulcitol	nd	—	—	—
Erythritol	nd	—	—	nd
Esculin	nd	+	+	nd
Fructose	+	+	+	+

(continued)

TABLE BXII.γ.242. (cont.)

Characteristic	1. <i>P. carotovorum</i>	2. <i>P. cacticida</i>	3. <i>P. chrysanthemi</i>	4. <i>P. cypripedii</i>
DL-Fucose	nd	—	—	nd
Fumarate	+	+	+	+
Galactose	+	+	+	+
Gallate	—	—	—	—
Gluconate	nd	+	+	+
Glucose	+	+	+	+
DL-Glycerate	nd	nd	nd	nd
Glycerol	d	+	+	+
Glycogen	—	nd	nd	nd
α-Ketoglutarate	nd	+	nd	nd
Inulin	v	—	d	nd
Lactose	d	+	d	nd
Lyxose	—	nd	nd	nd
Mallate	+	+	+	+
Maltose	d	d	—	nd
Mannitol	+	+	+	+
Mannose	+	+	+	+
Melezitose	—	nd	nd	nd
Methanol	—	—	—	—
β-Methylglucoside	+	+	+	+
Mucate	nd	nd	+	nd
Oxalate	—	—	—	—
Pectinic acid	+	+	+	nd
Propionate	—	—	—	—
L-Rhamnose	+	+	+	nd
Ribose	+	+	+	+
Saccharose	+	nd	nd	nd
Salicin	+	+	+	+
Sorbitol	v	—	—	nd
Sorbose	—	—	—	—
Starch	—	nd	nd	nd
Succinate	+	+	+	+
Sucrose	+	+	+	+
D-Tagatose	—	nd	nd	nd
meso-Tartrate	—	—	nd	nd
Trehalose	d	+	v	nd
Triacetine	ND	—	nd	nd
D-Turanose	v	—	—	nd
Xylitol	—	—	nd	nd
D-Xylose	d	ND	+	nd
<i>Utilization of nitrogen sources:</i>				
Adenine	nd	nd	+	nd
Alanine	+	+	+	+
Allantoin	+	+	+	+
Ammonium chloride	+	+	+	+
Anthranilic acid	—	—	—	—
Arginine	+	+	+	+
Asparagine	+	+	+	+
Asparginic acid	+	+	+	+
Betaine	—	—	—	—
γ-Aminobutanic acid	+	+	+	+
Carnosine	nd	nd	+	nd
Choline	—	—	—	—
Citrulline	+	+	+	+
Creatine	—	—	nd	nd
Cysteamine	—	—	—	—
Glucosamine	+	+	+	+
Glutamine	+	+	+	+
Glutaminic acid	+	+	+	+
Glutathion	+	+	+	+
Glycine	+	+	+	+
Glycylglycine	+	+	+	+
Guanine	—	—	nd	nd
Histidine	+	+	+	+
Hydroxyproline	—	—	—	—
Kynureninic acid	—	—	—	—
Leucine	+	+	+	+
Methionine	+	+	+	+
Nicotinic acid	—	—	nd	nd
Octopine	nd	nd	+	nd
Ornithine	nd	nd	+	nd

(continued)